Impacts of Mining Contamination in the Pilcomayo River Basin and Opportunities for Remediation

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Signature Page

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The Bren School of Environmental Science & Management produces professionals with unrivaled training in environmental science and management who will devote their unique skills to the diagnosis, assessment, mitigation, prevention, and remedy of the environmental problems of today and the future. A guiding principle of the Bren School is that the analysis of environmental problems requires quantitative training in more than one discipline and an awareness of the physical, biological, social, political, and economic consequences that arise from scientific or technological decisions.

The Group Project is required of all students in the Master of Environmental Science and Management (MESM) Program. The project is a year-long activity in which small groups of students conduct focused, interdisciplinary research on the scientific, management, and policy dimensions of a specific environmental issue. This Group Project Final Report is authored by MESM students and has been reviewed and approved by:

Dr. Arturo Keller	Date

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1. Introduction

1.1 Project Overview

The Pilcomayo River Basin covers an area of 272,000 km^2, connecting Bolivia, Argentina, and Paraguay and supporting an estimated 1.5 million people (Laboronti, 2011). The river serves as a critical source of agricultural water for the region, while indigenous communities depend on it more directly for fishing, subsistence agriculture, and in some cases drinking water (Stassen et al., 2012). For over 450 years, the basin has been heavily contaminated by mining waste originating from the Potosí region, including toxic heavy metals and acidic byproducts (Miller et al., 2004). High concentrations of arsenic, antimony, cadmium, lead, silver, mercury, thallium, and other hazardous substances have been recorded in upstream areas (Miller et al., 2007). Long-term exposure to these contaminants is associated with serious health outcomes including developmental delays in children, neurological disorders, and increased cancer risk (Stassen et al., 2012). Evidence also suggests that contaminants have leached into regional groundwater and aquifer systems, compounding risks to community health and ecological stability (Bundschuh et al., 2012).

Advanced Minerals Technology S. Korea, Inc. (AMTSK) is among the first organizations to attempt large-scale remediation of the Pilcomayo River. AMTSK plans to deploy its novel water remediation technology to treat the Pilcomayo Basin. For the deployment, AMTSK plans to install one water remediation system as a pilot, followed by 6-7 additional systems. Analysis was performed by the River Remedy team to support decision-making in siting AMTSK's treatment systems, with the aim of maximizing benefits to local communities. This report was produced in collaboration with the Bolivian government and Tomás Frías University to support AMTSK's remediation efforts, incorporating data analysis, GIS mapping, and scientific literature review to assess the environmental and human health impacts of historic and ongoing mining activity. A criteria-based hazard ranking methodology is developed to identify the most severely affected regions, the communities most at risk, and the relative hazard posed by specific pollutants. This report also presents targeted remediation strategies and recommendations for reducing future mining impacts across the basin.

1.2 Bolivia Site Visit

In early September 2025, the project team traveled to west-central Bolivia to visit the Pilcomayo River Basin, funded by AMTSK to support their remediation initiative. This river has experienced extensive pollution due to historic and ongoing industrial mining activities, significantly impacting water quality, local ecosystems, and downstream communities. The site visit played a critical role in shaping the analysis by providing opportunities to connect with the Bolivian federal government, the government of Potosí, and partners from Tomás Frías University as well as incorporate local knowledge from impacted communities and conduct field observations of pollution extent and conditions.

During the visit to Potosí, clear signs of contamination were observed firsthand, including rust-colored water and a strong sulfuric odor. The city's history and economy remains deeply tied to Cerro Rico, the "rich mountain" looming above the city, composed largely of silver ore. While the region had benefited from this resource abundance, it had also contributed to widespread degradation. Piles of mine tailings stretched throughout the city, and its close proximity to mining operations had severely impacted both river water and air quality. Some newer neighborhoods have even been constructed on top of capped tailings piles. Compounding these challenges was the absence of a municipal wastewater treatment system, despite the city's large and growing population. Although environmental standards and regulations existed at the national level, enforcement mechanisms were reported to be limited. Previous efforts led by foreign actors to clean up the river had reportedly resulted in exploitation rather than meaningful remediation.

Engagement with indigenous community members in Cantumarca, a neighborhood in Potosí located near the convergence of wastewater and acid mine drainage, provided insight into local experiences and historical land use. Community members described how, in past decades, their people farmed the land and relied on nearby wells for drinking water. Over time, they began experiencing serious health issues, including yellowing and loss of teeth and severe abdominal pain, later identified as cases of stomach cancer linked to lead-contaminated wells and groundwater. As a result, the community was forced to shift away from agriculture, with many members migrating elsewhere. In what was described as an unfair exchange, their ancestors ceded much of their land to the government in return for access to potable water infrastructure.

Today, residents continue to face ongoing exposure to contamination during both the rainy and dry seasons, primarily through mine tailings runoff and airborne dust particles. Community leaders reported high rates of respiratory illness, particularly in children, who also experienced elevated blood lead levels and increased incidence of leukemia. Residents observed trucks illegally dumping waste directly into river streams after dark, contributing to a pervasive and persistent odor that often prevented outdoor activity at night. Despite repeated attempts to report these activities, no government action was taken to regulate or prevent further pollution in the area.

Based on engagement with local Bolivian leaders and partners, adjustments were made to the scope and focus of the project. Initially, the project was solely focused on water pollution. However, after interviewing the community in Cantumarca, air pollution was identified as a major issue in the region. Air pollution analyses and remediation strategies were therefore included in this report. Due to a lack of knowledge regarding the locations of air pollution sources across the basin, air-related analyses focused on the city of Potosí. Additionally, communication with the community in Cantumarca and Ingeniero Edgar Bohrt, a professor at Tomas Frias University, helped identify specific locations as priorities for remediation recommendations: Devil's Gate and the San Miguel Dike.

1.3 Significance

AMTSK provides consulting services, technical assistance, water treatment, and field support to the mining industry, and has recently established a foundation to fund environmental remediation of the Pilcomayo Basin and medical treatment of affected populations.

The health effects of long-term heavy metal exposure are well-documented and include kidney and liver damage, cardiovascular disorders, cognitive impairment, and increased cancer risk (Rehman et al., 2018). These risks are particularly severe for pregnant women and children, with documented associations with miscarriage, stillbirth, birth defects, and impaired neurological development (CDC, 2026). Comprehensive estimates of health burden in the Pilcomayo region remain limited, and given the basin's centuries-long history of mining contamination, the consequences likely extend across multiple generations.

The history of Potosí and the Pilcomayo is also a history of structural inequity. Mining in the region began in 1545 under Spanish colonial rule, with indigenous populations coerced into hazardous labor. Though Bolivia gained independence, the legacy of exploitation persists. As of 2002, 60% of people living in the river basin were below the poverty line (Laboranti, 2011), and indigenous and low-income communities continue to bear a disproportionate share of the contamination burden while domestic and international corporations extract value from the region's mineral wealth. Indigenous communities, who constitute a substantial portion of the Bolivian Pilcomayo Basin's population and depend most directly on the river for water and food, face the greatest exposure risk.

As one of the first organizations to attempt large-scale cleanup of the Pilcomayo Basin, AMTSK faces significant challenges in scoping the problem. This project seeks to expand AMTSK's understanding of the geographic distribution and health implications of contamination, and to help guide targeted remediation strategies in coordination with the Bolivian government, giving greater visibility to communities that have long raised concerns about the river's safety.

1.4 Objectives

While contamination of the Pilcomayo river basin has been researched and documented in the past, there remains a gap in understanding between the contamination of the river and its health effects on the people living in the region. This project will assist AMTSK in its environmental remediation efforts by:

I. Identifying the most heavily polluted areas of the river system and characterizing the types and concentrations of pollutants present.
II. Assessing potential exposure pathways and health impacts of local pollutants.
III. Developing and applying a criteria-based ranking method to identify the towns, villages, and communities most at risk from contamination.
IV. Providing practical water, sediment, and air pollution remediation strategies for implementation by AMTSK or local governments to reduce environmental and public health impacts.
V. Presenting results in a comprehensive and accessible format so that they may be used by AMTSK, policymakers, NGOs, and local communities. This report will be developed in English and translated into Spanish for local stakeholders to access and utilize this information.

2. Background

2.1 Geography and Indigenous Peoples

2.1.1 Pilcomayo River Hydrology & Geography

The Pilcomayo River has long been a vital waterway for the region, sustaining ecosystems and communities across Bolivia, Paraguay, and Argentina. The Pilcomayo River Basin encompasses an area of ~272,000 km^2 (~98,100 km^2 in Bolivia) and was home to an estimated 1.5 million people as of 2002 (Laboranti, 2011). The Pilcomayo River originates in the Andes mountains in southern Bolivia, at roughly five kilometers above sea level, and eventually joins with the Paraguay River. The river is primarily fed by rainfall in the Andes, leading to high seasonal variability since around 90% of rainfall occurs in the wet season between mid-November and mid-July (Testa Tachinno, 2015). Flow rates can exceed 1,500 m^3/s during the wet season, and are around 5 m^3/s during the dry season (Martin-Vide et al., 2019). Soils in the upper Andean section are highly susceptible to erosion due to the small particle size of local geological formations (Smolders et al., 2002). Concentrations of suspended sediments can range from as low as 0.01 g/L in the dry season to as high as 60 g/L in the rainy season, with a mean annual sediment load of 140 million tons (Martín-Vide et al., 2014).

The Pilcomayo river flows southeast for 1,100 km (Figure 2.1.1a), initially descending through narrow canyons (Martín-Vide et al., 2014). Upon reaching the relatively flat Chaco Plains near Villa Montes, accumulation of silt due to high sediment loads causes the river to spread out across a 210,000 km^2 fan-shaped body of sediments (Iriondo, 1993). This lower section of the river accounts for roughly half of the deposition of sediments on the river's banks (Balaban et al., 2015). In their 2014 study, Martín-Vide et al. describe how this lower section starts as a braided river, characterized by numerous abandoned channels, and gradually turns into a single thread meandering channel. They go on to describe how downstream flow is suddenly obstructed by fine sediment and woody debris, resulting in extreme fluvial discontinuity, where the river channel disappears and water continues flowing as overland flow. Smolders, et al. (2002) describe how the point at which the channel 'disappears' has been gradually moving upstream since the mid-1940s. They also point out two critical points (Figure 2.1.1a) where, if this blockage point reaches one of the points, the course of the river could be dramatically altered. For the remainder of its course, the water and suspended sediments move as surface and groundwater down a mild slope (Martín-Vide et al., 2019), feeding a large marsh area before draining into the Paraguay river near Asunción, Paraguay (Orfeo, 1999). The lower Pilcomayo defines a small section of the border between Bolivia and Argentina, and a longer section of the border between Paraguay and Argentina.

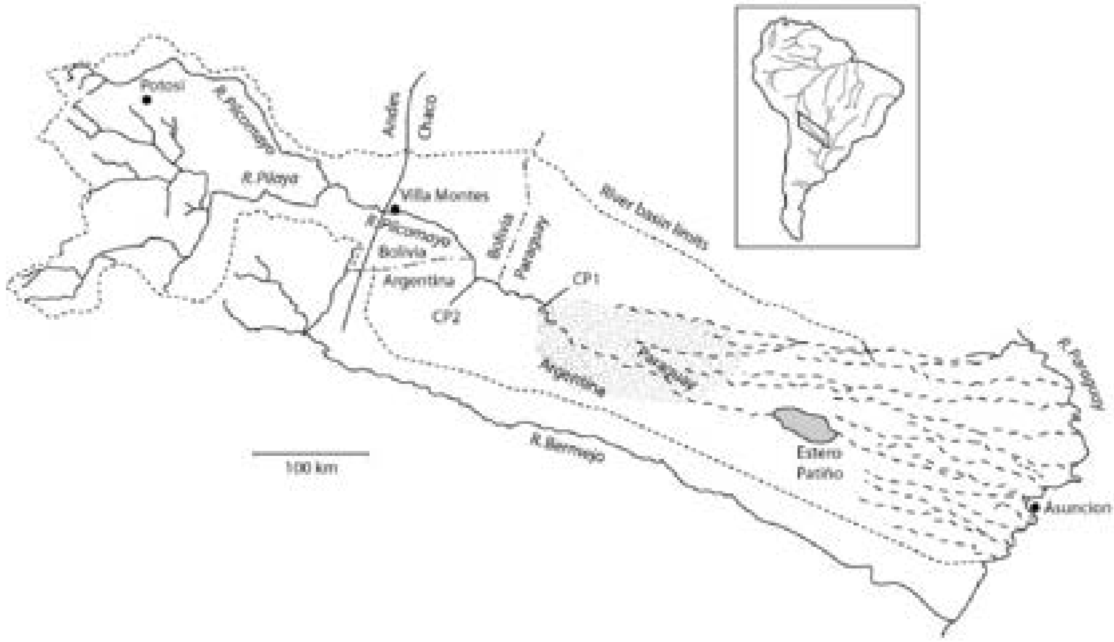

Figure 2.1.1a Map of the catchment of the Pilcomayo river. CP1 and CP2 indicate the two critical points. The dashed lines are (ancient) river beds that remain dry for large parts of the year. The shaded area marks the actual floodplain of the river. From Smolders et al., 2002.

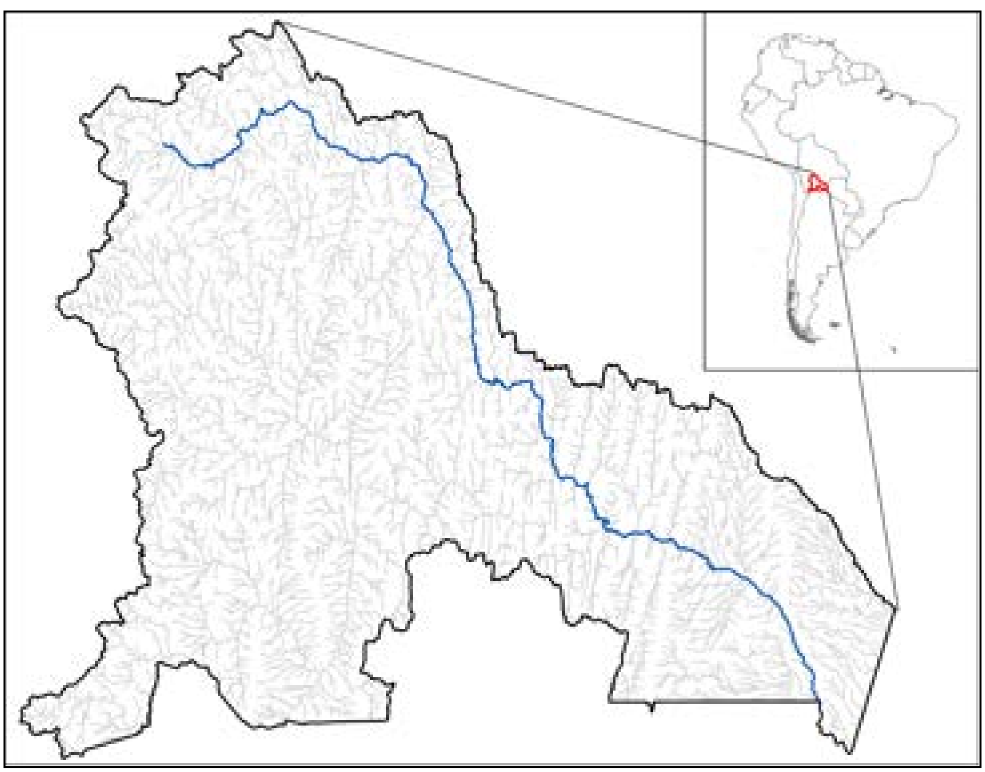

Figure 2.1.1b Map of Bolivian section of Pilcomayo River Basin. Basin border shown in black. Pilcomayo River shown in blue. Tributaries shown in light grey.

2.1.2 The Indigenous Peoples of Bolivia

Based on the sources we reviewed, between 44 and 60% of Bolivia's population are indigenous , comprising 36 recognized indigenous communities (Borgen Project, 2024; World Population Review, 2025). According to Indigenous Navigator (2025), the majority of indigenous peoples in Bolivia live in the Andes in the East, mainly Quechua-speakers (49.5%) and Aymara/Aymara-speakers (40.6%). They state that the remaining indigenous peoples live in the lowlands of the Gran Chaco – mainly Chiquitano (3.6%), Guaraní (2.5%), and Mojeño (1.4%). According to the United Nations Development Program (2021), indigenous people represent 75% of multidimensionally poor people (a definition of poverty accounting for factors such as access to education and basic infrastructure) despite making up 44% of Bolivia's population. In 2009, the redrafted Constitution of Bolivia declared the nation to be plurinational: made up of indigenous nations each with rights to autonomy, respect for cultural identity and language, and protection of its territories (Crabtree, 2017).

The Indigenous Relationship to Mining

The indigenous peoples of Bolivia have been intertwined with mining in Bolivia since silver was first discovered in Cerro Rico. During the Spanish colonial period, indigenous peoples were forced to work in silver mines. According to Mariscal et al. (2011), during the 1900s, many indigenous peoples were able to participate in the tin mining industry as part of the working class, which was a notable avenue of social and economic mobility. However, the establishment of mining enclaves during this period led to the fracturing and displacement of indigenous communities as well as widespread deforestation, large-scale use and pollution of water, and increased erosion. Lastly, in 1986, an economic crisis forced the closure of multiple mines, which led to the mass layoffs of indigenous miners (Mariscal et al., 2011).

2.1.3 Indigenous Groups in the Pilcomayo Basin

According to Castellón (2018), there are at least 20 indigenous groups living in the river basin. The Pilcomayo is featured in the founding myths and traditions of many peoples in the Great American Chaco, with the name "Pilcomayo" being derived from the Quechua words *phisqu* (bird) and *mayu* (river) (Castellón, 2018). According to Stassen et al (2012), indigenous peoples may constitute a specific-risk population from Pilcomayo river contamination; they are more likely to consume large amounts of fish and may even use the Pilcomayo as a source of drinking water;

Quechua-speakers

Quechua-speaking peoples are a large and highly diverse group, with multiple independent nations falling under the umbrella of "Quechua-speaking peoples". As of 2025, they number 2.5 million people, constituting roughly one-third of the total population (World Population Review, 2025; VisitBolivia, 2014). According to Minority Rights Group (2023), the Quechua live in both the Andean highlands and the lowlands. Many Quechua live in rural areas and subsist as small-scale *campesinos* (small-scale or subsistence farmers who own their own land), and many others have migrated into the cities and work as laborers (Minority Rights Group, 2023).

Aymara/Aymara-speakers

The Aymara live primarily in the altiplano of the central Andes, which spans across both Bolivia and Peru, and many live along Lake Titicaca (Countries and their Cultures, 2025). There are multiple other peoples with Aymara roots living in the Bolivian Andes which are commonly generalized as “Aymara-speaking peoples”. These peoples are estimated to number ~2 million (World Population Review, 2025). According to Countries and their Cultures (2025), the Aymara practice a traditional method of agriculture where only women are allowed to plant crops. This raises the possibility that Aymara women experience a disproportionate burden from soil and water contamination. Many Aymara are subsistence farmers and herders, some work in the mining industry, and others have migrated into cities for work (Countries and their Cultures, 2025; Minority Rights Group, 2023).

Weenhayek

According to Stassen et al. (2021), the Weenhayek live off the banks of the Pilcomayo River near the border of Bolivia and Argentina. Fishing is a vital part of their identity and critical to their economic livelihood, with Sábalo being the most culturally valued fish. In addition, the Weenhayek consume a notable portion of their fish harvest (Stassen et al., 2021). According to Baldivieso et al. (2023), the fish from the Pilcomayo makes its way all across Bolivia. They state that Weenhayek are trained to fish from a young age, with their primary method of fishing being diving into the river with nets. According to Baldivieso et al. (2023), the Weenhayek are concerned about the future of the Pilcomayo due to multiple environmental issues. The Pilcomayo is a naturally ecologically volatile system, and there are infrastructure projects undertaken by Argentina and Paraguay which impact water flow. The Weenhayek have also raised concerns about mining discharges from the upper basin and city sewage discharges contaminating the tributaries (Baldivieso et al., 2023). There is concern that the Weenhayek could be especially susceptible to pollution of the Pilcomayo River since they are frequently submerged in the river’s water and eat a large quantity of fish caught from the river. These concerns are supported by a study conducted by Stassen et al. in 2012, which found that the Weenhayek experienced elevated lead intake and health impacts compared to other indigenous communities (See *2.3.2 Documented Health Outcomes in the Pilcomayo Region*).

2.2 Historical and Contemporary Mining Impacts

Pilcomayo headwaters in the Bolivian highlands have hosted increased mining over the past decade; acid mine drainage, waste rock erosion, and tailing dumps discharge metals into tributaries, with dozens of mills dumping untreated waste as of 2018 (Stassen et al., 2012; Castelón, 2018). Few operations have proper tailings impoundments or safe dikes, risking dam failures; a 2014 break spilled thousands of cubic meters into a tributary (avoiding the main river only due to low dry season flows). Both the upper and middle Pilcomayo have been found to exceed WHO standards for heavy metals and arsenic (Castelón, 2018), threatening aquatic life and human health in river-dependent communities. Despite remediation efforts, legacy and ongoing contamination persist (Stassen et al., 2012).

2.2.1 History of Bolivian Mining

Large-scale mining began in 1545 with the discovery of Cerro Rico's silver by indigenous Peruvian Diego Huallpa, which fuelled Potosí's rise. From the 16th–18th centuries, Potosí supplied ~80% of global silver, minting "P"-marked piece-of-eight coins which circulated worldwide (Lane, 2015; Mann, 2011; Maxwell, 2020). Mining relied on forced labor, maintaining slavery-like abuse until the end of colonial rule (Abad & Maurer, 2025; Cole, 1985; Lane, 2015). Early smelting gave way to the mercury-based patio process, using ~1.5 kg mercury per kg silver and contributing to >25% of Latin American mercury emissions between 1500 and 1800. An estimated 8 million miners have died from poisoning and unsafe conditions, dubbing Cerro Rico "the mountain that eats men" (Nriagu, 1993; Robins & Hagan, 2012; Watson, 2018; Rehren, 2011).

Post-1825 independence, mining activity declined due to silver depletion. With the introduction of new technology allowing for the extraction of tin, mining was revitalized post-1900. The industry was dominated by exploitative "tin barons" who controlled politics and workers (Lane, 2015; Céspedes, 2018). The 1952 National Revolution overthrew oligarchic rule, enacting suffrage and nationalizing major tin mines under COMIBOL (Arnade & McFarren, 2025; Hudson & Hanratty, 1991). The revolution's momentum faded, opening the door to military dictatorships, labor suppression, and mineral decline. The 1980s tin price crash shrank mining to 4% of GDP by 1987, prompting neoliberal privatization and a period of inactivity for COMIBOL (Hudson & Hanratty, 1991). Cooperatives and small/medium mines dominated in this period, and by 2005 state mining revenue was $0. In 2005, Evo Morales revived COMIBOL, increasing state control/restrictions. Though the 2009 Constitution declared mineral resources to be state property, most mines remain private/coop (Benson, 2007; Hudson & Hanratty, 1991).

2.2.2 Modern Bolivian Mining (2000–Present)

There has been a resurgence of mining, particularly lithium extraction, in recent years. According to Céspedes (2025), the 2014 Mining Law, along with a 2023 resolution passed by AJAM (Mining Administrative Judicial Authority), significantly restricted mining-data access, hindering indigenous consultation regarding projects in their territories. Pending lithium contracts include Uranium One (Russia) and Hong Kong CBC (China). Many indigenous communities fear displacement/degradation due to mining (Céspedes, 2025). Approximately 80% of operators lacked environmental permits as of 2018 and state oversight of mining activities is minimal (Castelón, 2018; Céspedes, 2025).

2.2.3 Mining Pollution in Bolivia

Water and Sediment Pollution

Suspended sediments in the Pilcomayo contain toxic metals released from mining activity in the Potosí area. In 2001, Hudson-Edwards et al. found that many metal concentrations were significantly elevated above background levels in the river's upper reaches. High dissolved metal concentrations were found over 500 km downstream from Cerro Rico and are dependent on flow conditions (Smolders et al., 2002).

In 2002, Miller et al. utilized an isotopic fingerprinting approach that identified mining and milling operations as a dominant source of lead to channel bed sediments of the upper reaches of the Pilcomayo. They found that lead concentrations generally decreased with distance from mining activity at Potosí. The most significant decreases occurred between sampling sites R0 and R2, and between sites R3 and RSS (Figure 2.2.4a). It should be noted that, along with mining pollution, the region has experienced pollution from farmland expansion, grazing pressure, road construction, and urbanization (Brandt & Townsend, 2006).

Smolders et al. (2002) found that during the wet season, high total suspended sediment loads dilute the concentration of toxic metals at Villa Montes, while during the dry season metal concentrations are much higher (see Table 2.2.4a). However, they noted that high pH (between 7 and 9) led to low bioavailability and low overall concentrations of dissolved metals in lower reaches of the river, and only an estimated 0.3% of sediment is contaminated by the time it reaches Villa Montes. Balaban et al. (2015) noted that, despite these low concentrations in the lower Pilcomayo, there could still be risk from sediment deposition during annual floods.

In 2015, Balaban et al. identified a 100 km stretch near and upstream of Puente Sucre as the most contaminated part of the basin, where agricultural activity makes use of contaminated soil on the floodplain (see Figure 2.2.4b). They found no indication of agricultural activities utilizing floodplain sediments along the lower Bolivian Pilcomayo.

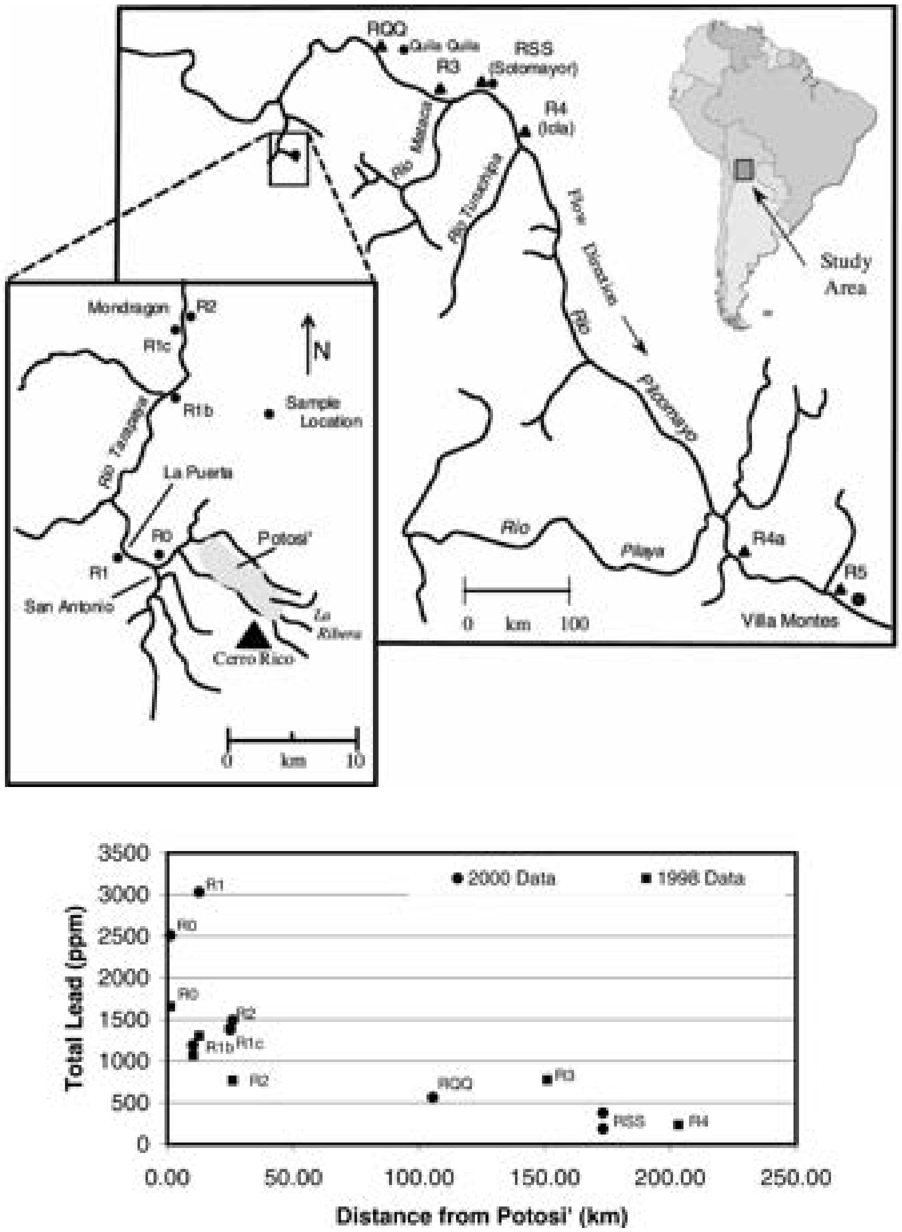

Figure 2.2.4a [Top] Location map showing the position of sampling sites along the Rio Pilcomayo, southern Bolivia. [Bottom] Total Pb concentrations in low-water channel deposits between R0 and R4. From Miller et al. (2002).

Table 2.2.4a Concentration of suspended solids, and Zn, Cu, Pb and Cd contents of suspended solids (mg kg^{-1} DW) during the rainy (January - April) and the dry (May - July) seasons at Villa Montese. Concentrations are means of six sampling dates. Concentration ranges are given in parentheses. Table and caption from Smolders et al., 2002.

	Solids ($g\ l^{-1}$)	Zn (ppm)	Cu (ppm)	Pb (ppm)	Cd (ppm)
January–April ($n = 6$)	23.6 (10.7–56.0)	139 (106–170)	23.9 (19.5–27.2)	35.3 (26.8–44.8)	0.55 (0.16–2.1)
May–July ($n = 6$)	0.011 (0.003–0.021)	19 327 (4386–32 635)	1107 (214–2514)	1495 (352–3345)	12.4 (6.5–15.9)

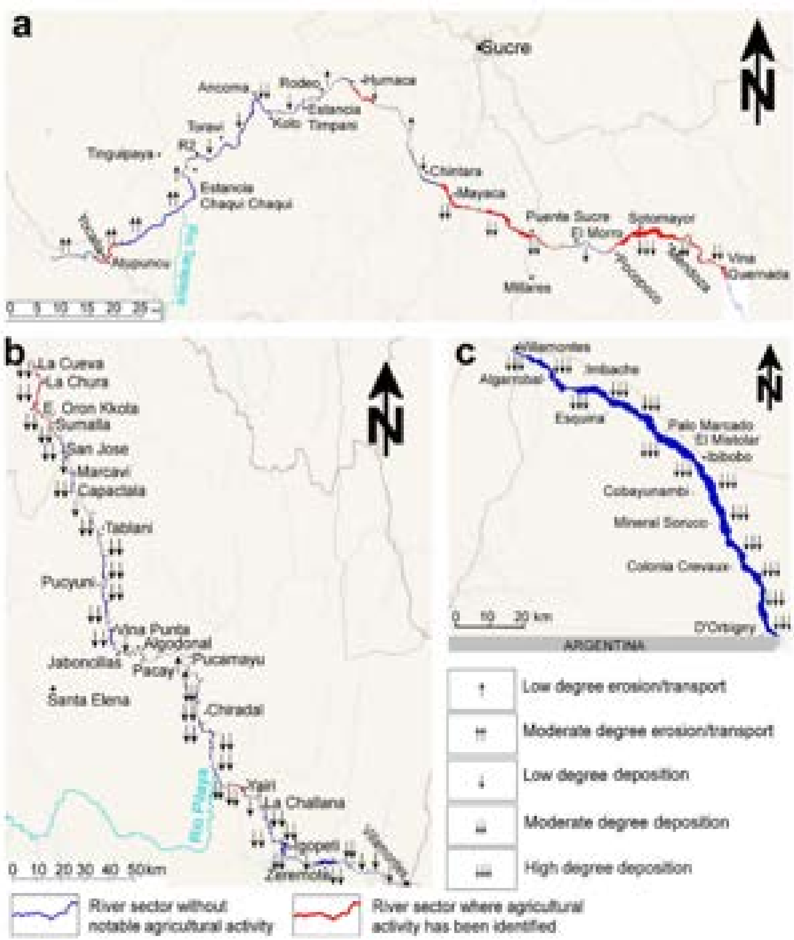

Figure 2.2.4b The most notable riverine communities, sectors with visible agricultural activities on the banks and dominant erosion/deposition processes for the a) Upper b) Mid and c) Lower Bolivian Pilcomayo. From Balaban et al., 2015.

Sources of Air Pollution

Mining processes contribute significantly to air pollution. In 2024, Kasongo et al. performed a literature review of major scientific databases focused on sources and concentrations of atmospheric dust from mining processes as well as environmental and human health risks from metal(loid)s (Figure 2.2.4c). They found that multiple mineral extraction and refining processes release both coarse and fine particulates into the atmosphere, which can contain minerals and metals. Rock blasting produces large clouds of atmospheric dust containing a significant portion of coarse (PM_{10}) particles; coarse dust is also released into the atmosphere through refining processes (crushing, grinding, and screening), tailing ponds, and waste rock piles. Most notably for our project, windblown dust from tailings contains both coarse and fine ($PM_{2.5}$) particles. Additionally, particles which have deposited onto surface soils are resuspended by the frequent vehicle traffic on mineral transport routes (Kasongo et al., 2024). While in Potosí , the team witnessed firsthand that trucks carrying mineral products in Potosí were often uncovered, meaning that they likely contributed windblown dust particles directly. Kasongo et al. (2024) found that smelting of minerals releases fine and ultra-fine (PM_1) particles, which may contain high concentrations of toxic heavy metals. They also found that smelting produces secondary inorganic aerosols such as sulfate, nitrate, and ammonium.

Figure 2.2.4c Diagram summarizing sources and transport of metal-bearing airborne particles from mining activities. From Kasongo et al., 2024.

Kasongo et al (2024) found that emission of particulates from tailings varies significantly with season and moisture. A study of tailings dams in Namibia revealed that the concentration of total suspended particles (TSP) near the tailings was 3 times greater when the surface of the tailings dam was dry compared to when two-thirds of the surfaces were covered with water. In the study, TSP concentrations were lowest in the rainy season and highest in the dry season (Kasongo et al., 2024). This corroborates what local leaders from Cantumarca indicated when interviewed. Kasongo et al (2024) found that the concentration of metals in PM samples varies widely depending on the materials extracted at a site, location, and other factors. They reviewed multiple studies focused on the size of metal particulates. Notably, they found that lead and cadmium were more common in fine particles, whereas iron and zinc were more common in coarse particles, and arsenic was highly variable.

2.2.4 Ongoing EJ Conflict: San Miguel Dike

One of the most striking landmarks from the team's 2025 trip to Bolivia was the San Miguel Tailings Dike (Figure 2.2.5a). The stories-tall grey tailing falls squarely in the middle of the community of Cantumarca in Potosí , which, according to Huerrera (2021), was home to 6,300 people as of 2021. The tailings dam had accumulated 4.3 million tons of sulfurous and oxide tailings as of 2021, spanning 18 hectares – an area so large it can be seen clearly from satellite imagery. Metals and minerals stored in the San Miguel Dike include: quartz, clay, pyrite, sphalerite, galena, chalcopyrite, silver, lead, tin, and zinc. Harmful material from the San Miguel Dike is carried directly into the homes of people in Cantumarca. During the dry season, wind carries metal-laced dust through windows, and during the wet season, rain washes metal-laced slurries into the community (Huerrera, 2021). During the team's trip to Bolivia, community leaders from Cantumarca referenced these transport mechanisms, specifically. According to Huerrera (2021), the tailing dam also poses serious environmental risks such as: Acid Mine Drainage, biodiversity loss, surface and ground water pollution, deforestation, and large-scale disturbance of hydro-geological systems. They state that the Cantumarca community is also highly concerned with the possibility of the dam breaking, which could have devastating effects.

Figure 2.2.5a Image of San Miguel tailing dike. From (Bolivian Thoughts, 2026)

According to Huerrera (2021), the San Miguel dike has been under the control of the Bolivian Mining Company (COMIBOL) since 1985. Since it came under COMIBOLs' control, residents of Cantumarca have attempted to negotiate with COMIBOL for its removal. In addition, the community in Cantumarca seeks reparations for the damages to human health and the environment caused by the dike over its 30+ year lifespan. The community in Cantumarca have organized protests, blockades, petitions, and independent scientific reports in an effort to persuade COMIBOL and the federal government of Bolivia to take action on the removal of the San Miguel Dike. In response, COMIBOL began an effort to move material from the San Miguel Dike to Agua Dulce for reprocessing in 2016 (Huerrera, 2021). However, when witnessed firsthand by our team in 2025, it is unclear how much progress has been made since 2016 or if the relocation is still ongoing.

2.3 Assessing Human Health Risks & Implications

Centuries of silver, lead, and tin mining have contaminated the Pilcomayo watershed with heavy metals, polluting water, soil, and food; ongoing unregulated operations worsen ecosystem degradation and health risks, disproportionately affecting rural and indigenous communities reliant on the river for drinking water, fish, and irrigation (Stassen et al., 2012). Understanding exposure pathways and health effects is critical for public health, environmental management, and remediation; however, due to long extraction history and poor data collection, the full generational impacts remain largely unknown.

2.3.1 Exposure Pathways in the Pilcomayo River Basin

Drinking Water
Direct exposure occurs via drinking water contaminated by upstream mining effluents; while major cities avoid it, indigenous groups and villages face high risks. In Quila Quila (75 km downstream from Potosí mines), infiltrated Chayanta River water showed cadmium at 23-63 µg/l (vs. WHO limit 3 µg/l); 80% of surveyed residents (mostly children) reported headaches, stomachaches, vomiting, diarrhea, and skin issues, attributed to water quality (Rojas et al., 2007). All pollutant regulatory limits for drinking water can be found in the *Appendix* (*A.2.2 Reference Standards*).

Inhalation and Ingestion
Beyond drinking water, heavy metals enter via food from irrigated crops or contaminated soils; potatoes, lettuces, onions, and carrots show elevated cadmium/lead, with soil exceeding background levels and are higher in lower terraces near flooding river (Garrido et al., 2009, 2017; Oporto et al., 2007; Miller et al., 2004). In 2004, Miller et al. examined heavy metal concentrations in soil, crops, and drinking water at four sites in the upper Pilcomayo basin, concluding that "the most significant exposure pathway appears to be the ingestion of contaminated soil particles attached to and consumed with vegetables," and that "drinking water does not appear to be a major exposure pathway, nor do vegetables." This is supported by Kasongo et al (2024),who state that ingestion is the main pathway by which the general public is exposed to metal contaminants. They state that ingestion also often occurs through dust or dirt which people unknowingly ingest (i.e. getting dust on your hands and putting your hand to your mouth). The authors also note inhalation as a significant exposure pathway; some elements have a low gastric bioaccessibility, but a high lung bioaccessibility (such as tin). They found that both gastric and lung bioaccessibility increase as particles become finer. Other exposure routes include dermal contact and bioaccumulated fish/aquatics.

Health Impacts of Metal Exposure
The health effects of chronic exposure to heavy metals found in the Pilcomayo River Basin, such as arsenic, cadmium, and lead, are well-documented. These metals are known to bioaccumulate in the body, affecting multiple organ systems and posing the greatest risk to vulnerable populations, particularly children and pregnant women. Globally, utilization of heavy metal-contaminated water is resulting in high morbidity and mortality rates, and may lead to health issues including cardiovascular disorders, neuronal damage, renal injuries, and increased risk of cancer and diabetes (Rehman et al., 2018). Table 2.3.1a summarizes potential health impacts associated with exposure to metals relevant to the basin, considering both ingestion and inhalation pathways.

Table 2.3.1a Target organs and acute and chronic effects of selected key metals. Information is summarized from ATSDR (n.d.) for copper and arsenic, WHO (n.d.) for arsenic and lead, and Cleveland Clinic (n.d.) for lead.

Metal	Target Organs/ Systems	Acute Effects	Chronic Effects
Arsenic (As)	Skin, cardiovascular system, nervous system, respiratory system	Nausea, vomiting, abdominal pain, diarrhea; numbness or tingling in the hands and feet	Skin lesions and hyperkeratosis; skin, lung, and bladder cancers; cardiovascular disease; diabetes; neurodevelopmental impairment
Cadmium (Cd)	Kidneys, bones, lungs, cardiovascular system	Nausea, vomiting, abdominal pain, diarrhea; bronchitis; chemical pneumonitis; pulmonary edema	Chronic kidney damage, possible renal failure; bone lesions; Chronic Obstructive Pulmonary Disease; lung cancer risk; possible mild anemia
Copper (Cu)	Gastrointestinal system, liver, kidneys, respiratory system	Nausea, vomiting, abdominal pain, diarrhea; inhalation may cause nose and throat irritation, dizziness, or headaches	Liver and kidney damage
Lead (Pb)	Brain/Central Nervous System, kidneys, cardiovascular system, liver, bones, reproductive system	Abdominal pain, cramps, vomiting, constipation; headache; fatigue; irritability; anemia; severe cases: seizures or coma	Infants/children: low birth weight; growth and developmental delays; reduced IQ; behavioral changes; attention deficit or learning difficulty Adults: hypertension; cardiovascular disease; kidney damage; reproductive toxicity (infertility, miscarriages); linked to bladder, lung, and stomach cancers

While acute poisoning can occur at high concentrations, many health effects result from chronic exposure to lower levels over time, as some metals accumulate in body tissues and disrupt normal biological processes. The severity of impacts depends on the concentration, duration, and route of exposure (Jaishankar et al., 2014). Several metals, including arsenic and cadmium, are classified as human carcinogens, with chronic exposure linked to increased cancer risk (Rehman et al., 2018), and for some contaminants, such as lead, there is no known safe level of exposure, with even very low blood lead concentrations linked to cognitive and behavioral effects in children (WHO, 2024).

2.3.2 Health Outcomes in Mining Regions

Documented Health Outcomes in the Pilcomayo Region
A study involving physical examinations and urinalyses of adults living in both mining and non-mining villages in Bolivia found significantly higher rates of hypertension, hematuria, and ketonuria among participants residing in mining regions (Farag et al., 2015). Individuals who reported consuming grains grown locally or watering their livestock at a water source

downstream from mine drainage sites were more likely to present with hematuria. Notably, participants with hematuria also exhibited significantly higher blood lead levels, suggesting a potential link between environmental exposure and renal effects.

On average, cadmium and lead concentrations were approximately three-fold higher among nonsmoking women from mining areas around Potosí compared to those from non-mining regions. For context, the average blood lead concentration in women in the United States is 0.4 µg/L, while women in these Bolivian mining regions had levels averaging 8.81 µg/L (ranging from 4.16 to 18.1 µg/L) (Farag et al., 2015). Blood lead levels exceeding 20 µg/L have been associated with somnolence, mood changes, renal dysfunction, decreases in bone density, hypertension, and cardiovascular disease. Hypertension was also found to be significantly more likely among mining region participants who reported eating poultry, legumes, and potatoes grown on their own farms, suggesting locally produced food as a potential exposure pathway.

While elevated heavy metal exposure has been documented in the general population, studies focusing specifically on indigenous communities and children reveal heightened exposure among vulnerable populations. Stassen et al. (2012) examined the Weenhayek, an Indigenous group living along the lower banks of the Pilcomayo River, and found that median hair lead concentrations were two to five times higher than those of a reference population. The study also reported that relative lead and cadmium intake was consistently higher in Weenhayek children compared to adults, with children exceeding the tolerable daily intake for all estimated percentiles of lead exposure.

Among the Weenhayek children studied, lactants exhibited higher levels of lead and cadmium in their hair than non-lactants, suggesting possible exposure during fetal development or through breast milk.In addition to elevated metal concentrations, the study identified potential health impacts within Weenhayek communities, including small family size, benign vascular tumors (hemangiomas and lymphangiomas), congenital anomalies, and delayed onset of walking (Stassen et al., 2012).

Beyond measurable health outcomes, it is equally important to consider the socio-economic and cultural dimensions that perpetuate health vulnerabilities in the Pilcomayo region. Today, mining remains one of the primary sources of employment in Potosí, making up nearly 40% of the local workforce (de Souza Santos et al., 2019). Despite the well-known occupational hazards of mining, including frequent accidents, chronic silicosis, and a life expectancy of just ~40 years, many local residents continue to support the industry. Driven by a lack of economically viable alternatives, local support reflects a long-standing dependency on the industry. These structural legacies persist and even today, residents fear that after the exhaustion of natural resources, the city will be forgotten and abandoned, highlighting the extent to which the mining industry remains a core part of regional identity (de Souza Santos et al., 2019).

Comparative Health Outcomes in Other Mining Regions

Global patterns from mining-affected regions corroborate findings from the Pilcomayo Basin. Water, soil, and dust contamination from mining operations has been linked to elevated rates of waterborne illness, miscarriage, birth defects, and respiratory and skin diseases in Burkina Faso, Mozambique, and Tanzania (Leuenberger et al., 2021). In Ghana, studies report 18% skin disease prevalence and 27% respiratory infection rates near active mines, with elevated blood concentrations of arsenic, cadmium, and lead associated with kidney disease, malaria, and diarrhea (Emmanuel et al., 2018).

Experiments on mice conducted by Witten et al. (2019) found that exposure to tailing dust in early development (in utero and childhood) negatively impacts lung development. Impacts include increased airway sensitivity, structural damage, and increased inflammation. Kasongo et al (2024) found that mine workers are at significant risk of respiratory illness. Ailments experienced by mine workers in studies they reviewed included: silicosis, pneumoconiosis, and pulmonary tuberculosis. Ingestion and inhalation of metals from mining operations can affect the nervous system, damage vital organs, and cause lung inflammation. In addition, inhalation of toxic metal particulates from smelting can cause bloodstream contamination. Communities living near mining areas have experienced silicosis and childhood lead poisoning, which causes behavioral issues as well as decreased intelligence quotients (Kasongo et al., 2024). Notably, when interviewed, local leaders from Cantumarca described data they had collected from ~100 children in their community exposing high levels of lead in blood. The local leaders also indicated that they do not source water from the Pilcomayo River and that they were more concerned with air quality than water pollution, citing issues with respiratory illness. Taking these statements in combination with the findings of Kasongo et al (2024) implies that airborne toxic metal particulates emitted during mining processes could be having a major impact on the community of Cantumarca and the wider Potosí area.

2.4 Remediation Strategies for the Pilcomayo Basin

2.4.1 Acid Mine Drainage

Johnson & Hallberg (2005) details the occurrence and implications of acid mine drainage. Acid mine drainage (AMD) may occur in active or abandoned mines, once pyrite (FeS_2) and other sulfide-bearing ores are exposed to oxygen and water. Oxygen is freely available in mining systems unless specifically capped, and groundwater, which is typically pumped to artificially low levels during the extraction process, returns to flood the mine and start a four-step reaction. This set of reactions lead to reduced water quality: reduced pH, increased heavy metals, and the visually characteristic ‘yellow-boy’ (a phenomenon caused by “the precipitation of complex iron hydroxy-sulphates” on the streambed) (Ravengai et al., 2007). Streams in the Pilcomayo exposed to AMD were determined to have elevated metal concentrations compared to non-impacted segments (Strosnider et al., 2011). Figure 2.4.1a shows AMD's various impacts on the local environment, which include elevated acidity and metal concentrations.

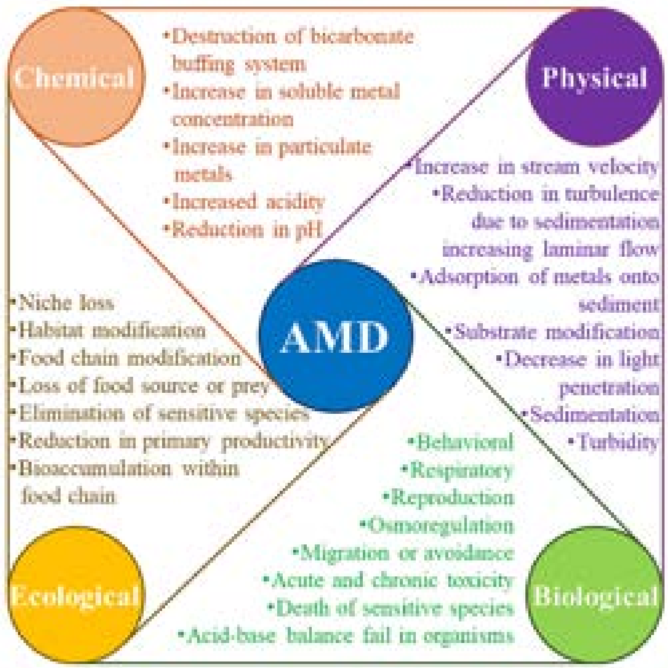

Figure 2.4.1a The impact of AMD on the environment. From Yang et al., 2023.

It's important to note that AMD reactions, once begun, follow a positive feedback loop as long as there are available reservoirs of pyrite or ferrous iron in the AMD stream (Strosnider et al., 2011). As pyrite and iron are readily available in the environment and can continue production of AMD, prevention is a cost-effective approach over remediation. Methods to prevent AMD during and after mining mainly focus on reducing available oxygen and water in the exposed region, as can be seen in Figure 2.4.1b.

Figure 2.4.1b Various approaches that have been evaluated to prevent or minimize the generation of mine drainage waters. From Johnson & Hallberg, 2005.

Efforts to Remediate AMD in the Pilcomayo Basin

AMD, among other pollution issues that have arisen over time due to mining activity, require remediation to prevent further pollution to the local communities utilizing the Pilcomayo River Basin (Strosnider et al., 2014). AMD has been a long-term concern on the Pilcomayo and remediation has been recommended from various perspectives, yet few regional remediation attempts have been discussed in academic literature (Sun et al., 2019). A plethora of studies have focused on the contamination of the Pilcomayo Basin by acid mine drainage, yet few have researched methodologies into regional remediation.

Bolivia expressed their interest in conducting remediation of specific regions along the Pilcomayo Basin by developing the *National Commission for the Pilcomayo and Bermejo Rivers* in 1989 (*Strategic Action Program for the Binational Basin of the Bermejo River*, 2010). Regulations from the formation of the commission established the *National Technical Office for the Pilcomayo and Bermejo Rivers (OTNPB)* in 1992. It's unclear the impacts or actions taken from these new agencies (*Technical Cooperation Project for the Sustainable Development of the Pilcomayo River Basin*, 1997). Local governments, such as the Municipality of Tarija (downstream of La Paz), have considered implementing localized solutions to end-result problems such as drinking water contamination, rather than contributing to source solutions (Villena-Martínez et al., 2023). This decentralized governmental approach to handling AMD-related concerns has reduced local capacity to take action at the contaminating sources.

While operational owners stated their intent to rehabilitate the Pilcomayo Basin in 1997, it took until 2004 for construction of tailings dams and canals to redirect approximately 360,000 tonnes/year of ore-processing sludge effluent away from the Pilcomayo (Smolders et al., 2002; Strosnider et al., 2011). Rain events until approximately 2010 continued to overflow the tailings canals and flotation ponds (Strosnider et al., 2011). Contamination continues to occur from past and present mining operations; remediating acid mine drainage will be an important aspect toward restoring the ecosystem services of the region.

2.4.2 Acid Mine Drainage Remediation Strategies

Designing AMD remediation in Bolivia requires balancing technical performance with constraints such as limited grid electricity in remote areas, high transport costs from major cities, limited local availability of reagents, and variable long-term operation and maintenance (O&M) capacity. Operationally simple, low-cost technologies are therefore preferred. Mining companies can reduce implementation barriers by leveraging their own supply chains, equipment, and staff to support local remediation (see *A.1.2 Remediation Strategy* for further detail on individual strategies). Table 2.4.2a links each broad treatment strategy to how it may be utilized in the Bolivian context and whether it is recommended as a primary option.

Table 2.4.2a Strategies for remediating acid mine drainage hazards

Strategy	Biological	Abiotic
Active	Oxidative selective metal precipitation Electrochemical (microbial)	Electrochemical (air-cathode) Membrane (thermal)
Passive	Wetland (oxic) Compost Wetland (anoxic)	Limestone adsorption Alkaline leach beds
Hybrid	Phytoremediation	–

Active & Biological
Active biological systems can achieve high removal efficiencies for metals and acidity but require continuous operation, trained staff, and reliable supplies of organic substrates and alkalinity (Rambabu et al., 2020). In Bolivia, these systems are most realistic at large, centralized mining facilities that already have on-site laboratories, process control, and access to power and reagents. They are poorly matched to remote legacy sites and small, informal operations where long-term O&M cannot be guaranteed.

Active & Abiotic
Membrane technologies (reverse osmosis, nanofiltration) and electrochemical systems can produce high-quality effluent and even recover saleable products (e.g., iron oxides), but they are capital-intensive, energy-intensive, and technically complex (Rambabu et al., 2020). They also generate secondary waste streams (e.g., brines, spent electrodes) that need careful management (Sun et al., 2015). In Bolivia, these methods may be feasible only in a few locations: near grid-connected industrial facilities or where there is a clear economic driver such as water reuse for processing plants. At most AMD sources, transport, energy, and maintenance demands will be prohibitive.

Passive & Biological
Passive biological systems align well with low-cost, low-energy settings. Constructed wetlands, compost-amended anoxic ponds, and subsurface reactive barriers rely on natural microbial processes and require relatively simple infrastructure. They are especially suitable where flows are moderate, land area is available, and local communities or mine operators can perform basic maintenance (e.g., clearing vegetation, inspecting flow paths). In Bolivia, these systems are promising for legacy sites near communities (e.g. Cantumarca), provided land rights are resolved and designs consider local climate, elevation, and extreme-event hydrology. Risks include metal accumulation in biota and the need to manage contaminated sediments or plant material over time.

Passive & Abiotic
Passive AMD remediation relies on geochemical processes. Limestone-based neutralization is widely used to raise pH and precipitate metals (Aubé & Zinck, 1999), and steel slag–an industrial byproduct–can also be utilized. Limestone leach beds can be installed at AMD seeps, as pre-treatment units, or in high concentration regions within the river basin, where AMD flows through coarse alkaline media, dissolving it and generating alkalinity (Jeff Skousen et al., 2017). These systems are simple and robust but require sufficient hydraulic retention time, which limits the maximum flow they can effectively treat. Diversion wells can offer additional retention time, as discussed in the Appendix (*A.1.2 Remediation Strategy*).

3. Data & Methods

3.1 Data Acquisition

The primary water and sediment quality data used in this project were obtained from the Comisión Trinacional para el Desarrollo de la Cuenca del Río Pilcomayo. This data includes various physical and chemical parameters relating to water or sediment quality, including metal concentrations, pH, temperature, salinity, dissolved oxygen, and other chemical concentrations. Measurements span 850 km of the Pilcomayo River and its tributaries. Our analyses focused on data collected between 2016 and 2024. Data from the year 2024 included extremely high values that exceeded known solubility limits, and was excluded from some of our analyses. Of the 46 locations sampled in this timeframe, 21 were located in Bolivia. Across all sampling events in this timeframe, 1.2% measured only sediment, 35.6% measured only water, and 63.1% measured both. The data also includes the date, time, and coordinates where each sample was taken. Each location was sampled no more than one time per sampling campaign, and there were typically between two and four sampling campaigns per year.

3.2 Hazard Scoring

3.2.1 Hazard Quotient and Standard Limits

This analysis is based on data from the Trinational Commission. To evaluate risk across parameters whose effects on human health differ in orders of magnitude, Hazard Quotients (HQ) were utilized, which compare a sampled pollutant concentration (C_{sample}) to a known acceptable level, such as a standard or reference limit from environmental guidelines or regulations ($C_{standard}$). Various case studies have utilized HQ in the past to consider hazards to humans caused by environmental contamination (Xu et al., 2025; Raj et al., 2022; Li et al., 2018). The HQ is a ratio ($HQ = C_{sample} / C_{standard}$); HQ values less than one indicate that a pollutant exists in concentrations within regulatory levels, while HQ values greater than one indicate that it exceeds regulatory levels. While using the HQ has literature support in leading remediation projects, HQ should be considered a recommendation of how to focus resources, and not an accurate assessment of current risk experienced by the population (Tannenbaum et al., 2003). Standard limits and reference values for pollutants considered in this analysis were compiled from various sources including the EPA, WHO, CDC, Bolivian Law 1333, USEPA, USGS, CDC, and FAO as they relate to drinking water and sediments (Table 3.2.1a). Bolivian Law 1333 sets classification levels for allowable usages at each quality level (Table 3.2.1b). If multiple standard/reference values were found for a single pollutant, the strictest standard/reference was used for HQ calculations. Hazard quotients were not calculated for contaminants without reference standards. In situations where sampled data is named as a group and has no clear reference standard (e.g., “hydrocarbons”), reference standards were not selected to avoid mischaracterization of sampled risk.

Table 3.2.1a Regulatory standard sources utilized for sample comparison and hazard quotient calculations. All utilized regulatory values are listed in the Appendix (*A.2.2 Reference Standards*).

Regulator	Media	Source
Bolivian Law 1333	Drinking Water	(Ministerio de Desarrollo Sostenible y Medio Ambiente, 1992)
USEPA	Drinking Water	(US EPA, 2025)
USGS	Sediment	(Zimmerman & Breault, 2003)
WHO	Drinking Water	(World Health Organization, n.d.)

Table 3.2.1b Bolivian water standards under Law No. 1333.

Class	Definition
Class A	Water of maximum quality; potable with little to no treatment.
Class B	General-use water, but to be suitable for human consumption requires physical treatment and bacteriological disinfection.
Class C	General-use water which, to be used for human consumption, requires full physical-chemical treatment + bacteriological disinfection.
Class D	Minimum quality water; in extreme cases of public necessity for human consumption would require sedimentation, full treatment, and special disinfection for parasites.

Samples which surpass the water quality threshold set by Class D were segmented into a fifth category, called “Beyond D.” While this is not an official designation, it is not recommended that water quality within this category be used for any purpose. The USEPA and WHO are utilized for calculating HQ when regulatory standards are stricter than the Bolivian Law 1333 (see Table 4.2.2a for a comparison for water quality standards). Drinking water regulations from Bolivian Law 1333 and other regulatory bodies (USEPA, 2025; WHO, 2022) were compared, finding that the Bolivian standards for drinking water are not specifically more or less lenient across all parameters (Table 3.2.1c).

Table 3.2.1c Comparison of Bolivian water quality standards with standards used in the US and abroad for pollutants where both limits exist. Bolivian standards were utilized as the stricter standard in approximately half of contaminants.

Parameter	BOL: Bolivian Class A Standard (mg/L)	INTL: Strict WHO/EPA Standard (mg/L)	Strictest
Antimony	0.01	0.006	INTL (x1.7)
Arsenic	0.05	0.01	INTL (x5.0)
Boron	1	2.4	BOL (x2.4)
Cadmium	0.005	0.003	INTL (x1.7)
Chromium	0.05	0.05	Equal
Copper	0.05	1.3	BOL (x26.0)
Iron	0.3	2	BOL (x6.7)
Lead	0.05	0.01	INTL (x5.0)
Manganese	0.5	0.08	INTL (x6.2)
Mercury	0.001	0.002	BOL (x2.0)
Nickel	0.05	0.07	BOL (x1.4)
Nitrate	2	10	INTL (x2.0)
Selenium	0.01	0.04	BOL (x4.0)
Silver	0.05	0.1	BOL (x2.0)

Bolivian Law has no standards for contaminant concentrations in soils. Reference standards used by many countries are the US Geological Survey's Sediment Quality Guidelines (Zimmerman & Breault, 2003). The SQGs are divided into three categories:

1. Above Probable Effect Level (PEL): the concentration above which adverse biological effects are expected to occur frequently (in more than 50% of cases).
2. Above Threshold Effect Level (TEL): the concentration of a contaminant above which adverse biological effects are expected to occur in approximately 15-50% of cases.
3. Below Threshold Effect Level: the concentration of a contaminant below which adverse biological effects are expected to occur rarely (i.e., in less than 10–15% of cases).

3.2.2 Cancer Risk

Cancer risk was evaluated as part of the chronic health assessment but was not included in final results because available data could not reliably link measured environmental concentrations of the parameters to individual cancer risk. Multiple sources were reviewed to identify oral cancer slope factors (for ingestion of food and water) and inhalation unit risks (for exposure via breathing), including USEPA's IRIS (US EPA, n.d.) and California OEHHA's Cancer Potency Values (OEHHA, 2025). However, few parameters in this study had matching cancer potency values, and uncertainties in sampling methods across monitoring campaigns prevented meaningful comparison with existing Pilcomayo data. For example, cobalt has separate oral cancer slope factors for ingestion of water-insoluble compounds and water-soluble compounds;

although cobalt in sediments may be predominantly water-insoluble, other environmental processes could alter its speciation or sorption behavior, and assigning a specific cancer risk without this information would be misleading.

3.3 Combined Risk Assessment

3.3.1 Water Quality

Hazard Quotient Aggregation
To generate station-level scores from multiple parameters measured over time, two aggregation steps were required. First, temporal aggregation determined how to handle multiple measurements of the same parameter at different time points at a given station. Options included using only the most recent measurement, filtering to a specific time period, or applying a time-weighted average that emphasizes recent data. Second, parameter aggregation combined HQ values across different pollutants measured at each station. Available methods included mean, median, maximum, sum, 95th percentile, or the Nemerow index (which gathers a geometric mean from the average and maximum values; Nemerow, 1974).

Sensitivity analysis was conducted to explore the uncertainty introduced due to the variety of calculation methodologies and, along with expert solicitation, a model was selected (see Section 3.9 and 4.9 *Sensitivity Analysis*). Aggregated hazard quotients utilized data from the most recent 5-years of data. Though the last reported sample was from June 2024, data from 2024 was excluded due to extremely high values that may be a result of data entry errors (see Section 4.*1 Data Limitations*). After removing 2024 data, the most recent sample was from December 2023. Therefore, the data was filtered to include all samples between December 2018 and December 2023, resulting in a five-year window. Each station was given a score using the 95th percentile of HQ values across all recent (within the last five years) observations and parameters. Each parameter was given a score using data from all stations within the last five years. This is a conservative approach that highlights the worst contamination experienced within the last five years while accounting for extreme outliers.

Interpolation Across River Network
Rivers were delineated in ArcGIS Pro using a digital elevation model (see Section *3.8.1 River Network Delineation*). Station locations were snapped to the nearest river segment within 2 kilometers to ensure spatial accuracy along the network. R scripts were written to interpolate these scores along the river network. First, network topology was constructed, allowing for tracing along the network and identifying adjacent nodes, segments and stations. Adjacent station pairs were identified, where a station pair is two stations that are positioned sequentially along the network. For each station pair, segments of the river network that connect the two stations to

one another were identified. Along each of these segments, uniformly spaced points were generated every 50 meters. These points were assigned interpolated HQ values using the formula:

$$HQ_{point} = (1 - \frac{d_{point}}{d_{total}}) \times HQ_{upstream} + (\frac{d_{point}}{d_{total}}) \times HQ_{downstream}$$

where d_{point} is the distance along the network from the upstream station to the point, and d_{total} is the total distance between stations. These interpolated point values were then rasterized onto a regular grid, where each raster cell was assigned the mean HQ value of all points falling within that cell. To extend risk values beyond the immediate river corridor, inverse distance weighted interpolation was applied to propagate values 2 km outward from the river. This resulted in a buffered gradient formed between each pair of adjacent stations. Where these gradients overlapped (i.e. downstream of confluences), the mean of overlapping cells was used. The final result was a raster whose cells represent interpolated water HQ values based on the HQ value of the closest upstream and downstream water quality station(s) and confluence(s).

Limitations

The purpose of this analysis was to highlight locations with the highest level of contamination, and to identify upstream and downstream areas that may be at risk. The analysis was not intended to be a highly accurate model for predicting water quality at any given point, but rather to reveal large-scale patterns that would justify further investigation into potentially high-risk areas. Due to the large spatial scale and lack of data, a more in-depth analysis was not feasible for this project. With that said, this method has several limitations worth highlighting.

Firstly, it attempts to aggregate hazard quotients across several parameters to describe each location with a single score using the 95th percentile across all parameters. In doing so, locations with extremely high values for a small number of parameters may be scored higher than locations with moderately high values for a large number of parameters. Performing this aggregation may also ignore differences in health impacts and exposure pathways of different contaminants, since all parameters are weighted equally. Additionally, though all locations were scored using the past five years of data, not all stations were sampled uniformly through time. As a result, seasonality may play a role that is not reflected in the result.

Secondly, the interpolation method assumes that water and sediment quality will vary linearly with distance between stations. This method does not account for more complex factors that may influence the transport and behavior of contaminants such as partitioning, sediment deposition and erosion, or flow rates.

Lastly, this method handles confluences by taking the mean value of overlapping segments. This does not account for differences in flow between the two converging tributaries, and instead weighs them equally. Additionally, this analysis only included tributaries for which water quality data exists. Uncontaminated tributaries may dilute contamination after confluences, a consideration that was not included in this analysis.

3.3.2 Sediment Quality

The same exact processes outlined in Section 3.3.1 were completed using the sediment quality data. First, sediment quality measurements were scored using hazard quotients, and aggregated to the station level (see *3.3.1 Hazard Quotient Aggregation*). The same aggregation methods (5-year window, 95th percentile across all observations) were used for sediment as water. These aggregated sediment hazard scores were interpolated along the river network (see *3.3.1 Interpolation Across River Network*), resulting in an interpolated sediment risk raster. The same limitations described when discussing water (see *3.3.1 Limitations*) apply to sediments.

3.3.3. Modeling Air Pollution from Wind-Blown Tailings Dust

Study Area, Model Selection, and Data

During the team's interview with the local leaders of Cantumarca, it was indicated that air pollution from windblown tailing dust is a major concern during Bolivia's dry season. Review of relevant literature also indicated that air pollution from mining activities can have major adverse impacts on the health of local populations (See *2.3.2 Comparative Health Outcomes in Other Mining Regions*). In order to integrate these concerns into our project, the team decided to analyze air pollution. Since the primary focus of our project is water and sediment contamination, the scope of air pollution analysis was limited. This analysis is targeted at air pollution from windblown tailing particles in the city of Potosí.

Based on the team's research, Potosí does not have any dedicated air quality monitoring stations. Publicly accessible records of air quality data in Potosí could not be sourced. Therefore, risk from air pollution was assessed by modeling the atmospheric transport and fate of particulate matter containing heavy metals from tailings piles throughout the city of Potosí. The purpose of this modeling was to estimate the magnitude and spatial distribution of risk from tailing-related air pollution within the city of Potosí.

The model selected for air pollution analysis was the HYSPLIT model (ARL, 2026) developed by the National Oceanic and Atmospheric Administration (NOAA) Air Resources Laboratory. HYSPLIT uses meteorological data to predict how pollutants from a point source travel through air and how they accumulate over time. The model's calculations use a hybrid of the Langrangian approach, which uses a moving frame of reference to calculate advection and dispersion, and the Eulerian method, which uses a fixed 3D grid as a frame of reference to calculate concentrations (ARL, 2026). For this analysis, particle concentrations were modeled using the particle dispersion model. In the particle dispersion model, a fixed number of particles are advected within the study grid based on the mean wind field and are dispersed using a turbulence component (ARL, 2026).

The meteorological data inputted into HYSPLIT was NOAA's 2024 Global Data Assimilation System (GDAS) data (*Global Data Assimilation System*, 2020). GDAS interpolates meteorological data recorded from observation stations around the globe into a 3-dimensional grid. This data is provided in a 1 degree by 1 degree resolution grid and is reported in 3 hr time intervals.

Estimating Emission Rates

A key input for HYSPLIT modeling is the hourly emission rate (units/hr). Based on the team's research, windblown particle emissions from tailings are not measured in Potosí. Therefore, emission rates were estimated using the wind speed data provided in the GDAS data set. To estimate emission rates, Stovern et al's (2014) models of windblown tailing dust from Arizona tailings were used as a basis. The formulas used were derived based on the EPA AP 42 report (Stovern et al, 2024). The first formula estimates the friction velocity of the near-surface wind. u^* is friction velocity (m/s), u_z is the wind speed (m/s) at height z (m), z_0 is roughness height (m), and K is the Von Karman constant.

$$u^* = (u_z/K) * ln(z/z_0)$$

u_z was calculated using NOAA's GDAS data. In the GDAS data, wind speed (m/s) is given in an eastward (u) and a northward (v) component. Absolute wind speed |V| (m/s) is given by combining the two components using the following equation (Weather Classes, n.d.):

$$|V| = \sqrt{u^2 + v^2}$$

In the GDAS data, the surface wind is measured at 10 m above ground level, so z = 10 m. According to Stovern et al (2014), roughness height of tailing walls is approximately 0.0014 m, so z_0 was set to 0.0014 m. Sensitivity analysis revealed that the value of z_0 had a significant impact on model outputs, therefore, the decision was deferred to Stovern et al's (2014) expertise.

Friction velocity determines the erosion potential of a tailing. In order for particles to erode from tailings, the friction velocity must exceed a critical threshold (u^*_t). According to Stovern et al (2014), experiments on a copper tailing found a critical threshold of 0.172 m/s. If the friction velocity is below 0.172 m/s, the erosion potential is zero. If the friction velocity is above 0.172 m/s, erosion potential is given by the following formula:

$$P = 58 * (u^* - u^*_t)^2 + 25 * (u^* - u^*_t)$$

Finally, the emission factor is calculated using the erosion potential and particle diameter. Emission factor is expressed as g/m^2.

$$E = k * \sum_{i=1}^{N} P$$

k is the particle size multiplier (0.5 for PM_{10} and 0.075 for PM2.5). Erosion potential (P) is aggregated based on the number of disturbances (N) in a given time period. For this analysis, the number of disturbances was set to 1. A disturbance is an instance when the frictional wind speed is greater than the threshold (u^*_t). Determining a number of disturbances within each time step of the modeling would require meteorological data at a higher temporal resolution than the GDAS data. Therefore, the number of disturbances was set to 1.

For HYSPLIT modeling, emission rate is given as "units" per hour, where "units" can be interpreted based on the value of emission rate input. For this analysis, emission rates were calculated in grams. In addition, emission rates were calculated using 3-hr resolution meteorological GDAS data, then averaged per day. For the purpose of modeling, each day was assumed to have a constant hourly emission rate equal to its daily average. The equation used to calculate emission rates uses units of g/m^2 per event and HYSPIT's input for emission rate uses units of units per hour. To align these units, each tailing was assumed to experience 1 emission event per hour.

Based on this assumption, the emission factor (E) was fed into HYSPLIT as the hourly emission rate. However, the emission factor is given in g/m^2. To get the true estimation of emission rate, the emission factor was multiplied by the surface area of each tailing location (discrete tailings or groups of tailings). The surface area of each tailing location was estimated using the Measure Area tool in ArcGIS Pro. For the estimates of surface area produced for each tailing location, see Table 3.3.3a. Sensitivity analysis revealed that spatial patterns of concentrations in model outputs stayed similar, even when estimates of surface area differed by a factor of 100,000 m^2. Magnitudes of concentrations differed by a factor of 10, however.

Spatial and Temporal Inputs

Key inputs for HYSPLIT concentration modeling include: meteorological data, emission rate (units per hour), point source location, time period, and physical characteristics of the particle species.

A subset of tailings in the city of Potosí were selected as the point sources for modeling. In order to determine the location of these point sources, a GIS layer was created to map the location of tailings and overburden heaps in the city of Potosí metropolitan area. This layer was created by combining two layers: 1) an existing shapefile uploaded to ArcGIS Online by an online user titled “INGENIOS MINEROS POTOSI ” which detailed the locations, names, and classifications of mining facilities and tailings piles within the city of Potosí (kevinvalle.1999, 2023), and 2) a layer created by our team by tracing polygons over tailings, overburdened heaps, and mineral refineries using Google Earth Pro satellite imagery from March 2025. The second layer was added to supplement the original data given that there was uncertainty surrounding the data from kevinvalle.1999 (2023). It appeared that the data was several years old. Notably, the size of certain tailings identified by kevinvalle.1999 had increased considerably. These two layers were combined using ArcGIS Pro, with the INGENIOS MINEROS POTOSI layer being given priority. Where the two layers overlapped, the classification of the INGENIOS MINEROS POTOSI layer was used. For tailings which grew in size between the INGENIOS MINEROS POTOSI layer and the team’s Google Earth Pro layer, the layers were merged to reflect the geometry of the Google Earth Pro layer. Using the combined layer of tailing and mine facility locations, four tailing locations in and around the city of Potosí were selected as point sources for particulate modeling. Each tailing location included multiple tailings, with the exception of the San Miguel tailing dike. These four locations were chosen based on their proximity to the city of Potosí. The locations can be seen in Figure 3.3.3a.

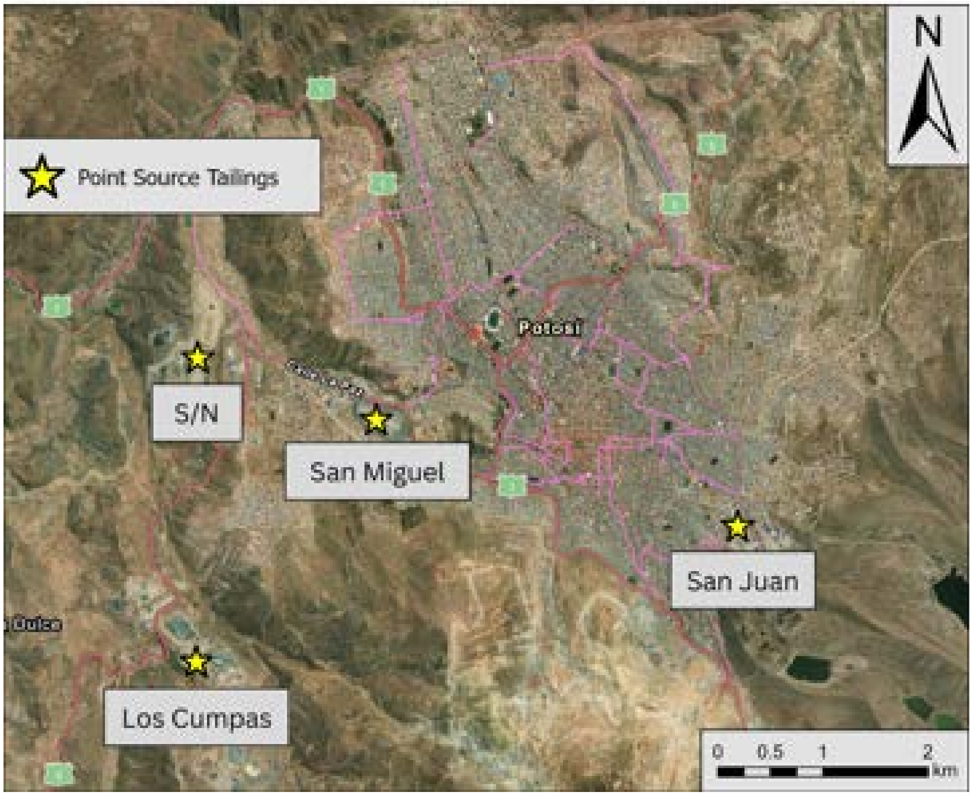

Figure 3.3.3a Location of tailings used as point sources in HYSPLIT concentration modeling.

Table 3.3.3a. Point source locations and starting heights. Names were assigned based on names listed in Google Earth Pro. "San Juan" and "Los Cumpas" refer to specific mining companies within a cluster of mining facilities. "San Miguel" refers to the name of the tailing dam.

Name	**Starting Height (meters above ground level)**	**Surface Area Estimate (m^2)**
San Miguel	8	212,000
S/N	4	276,000
San Juan	0	10,000
Los Cumpas	4	176,000

In addition to the location of the point sources, HYSPLIT also requires the height (meters above ground level) of the point sources. For this analysis, height was estimated based on Google Earth imagery and by viewing images of the selected tailings online (Table 3.3.3a). Sensitivity analysis revealed that the starting height did not significantly affect model outputs.

HYSPLIT models the spread of pollutants within a “meteorological grid” set by the user. For this analysis, a 0.2 x 0.2 decimal degree grid was set as the study area at a resolution of 0.02 x 0.02 decimal degrees. 0.01 x 0.01 resolution was tested as well, but was not significantly different. HYSPLIT also models at distinct “vertical levels” set by the user. For this analysis, 2m above ground level was set as the level height. This was selected as it is roughly the height of an adult human, therefore particulate concentrations at this height would most directly impact human health.

Six days from Bolivia’s dry season in 2024 were used to represent a range of possible dry season conditions. To estimate the range of emission conditions in the city of Potosí, summary statistics of the emission rates were calculated based on the GDAS data. The GDAS data was processed in Python using *ARLreader*. Emission rates were calculated for each 3 hour interval in the GDAS data, then averaged within each day (24 hrs). Then, the mean and standard deviation of all the daily average emission rates between May 1st, 2024 and October 31st, 2024 were calculated. See Table 3.3.3b for the summary statistics of the emission rates and Figure 3.3.3b for a histogram of the distribution of PM_{10} emission rates. After examining the distribution, two days with emission rates close to the mean, two days with emission rates above the mean, and two days with emission rates below the mean were selected for modeling. In addition, days were chosen across a range of months to account for temporal variation in wind directions. The days selected for modeling were: May 5th, June 3rd, July 4th, July 5th, August 10th, and September 5th. See Table 3.3.3c for the selected days and their emission rates for both PM_{10} and $PM_{2.5}$.

Table 3.3.3b Summary statistics of estimated daily average PM_{10} emission rates (g/m^2-hr) in the city of Potosí between May 1st and October 31st, 2024.

Minimum	Maximum	Median	Interquartile Range	Mean	Standard Deviation
0	7.69	0.809	[0.287, 1.83]	1.27	1.31

Table 3.3.3c Modeled days and their 24-hr emission rates.

Date	PM_{10} Emission Rate (g/m^2-hr)	$PM_{2.5}$ Emission Rate (g/m^2-hr)
May 5, 2024	1.940	0.291
Jun 3, 2024	0.043	0.0065
July 4, 2024	4.941	0.741
July 5, 2024	2.869	0.122
August 10, 2024	0.0162	0.0024
September 5, 2024	1.23	0.185

Pollutant Characteristic Inputs

The molecular weight, density, and diameter of a pollutant particle are required for HYSPLIT modeling. For our analysis, we decided to use one pollutant species as a representative for heavy metal-laden particulate matter from tailings, generally. To decide on a representative pollutant, we reviewed the water and sediment sample data from the Trinational Commission as well as literature regarding the composition and characteristics of Bolivian tailings to determine which metals were common in Bolivian tailings. Sources reviewed can be seen in Table 3.3.3d.

Based on this review, cadmium, lead, zinc, arsenic, iron, and copper were identified as some of the most common metals in Bolivian tailings. Of these, lead was chosen as a focus because it is an element of high interest to our project given its documented health impacts. Additionally, according to Huerrera (2024), the San Miguel tailing dike contains lead in the form of galena (PbS). Galena often oxidizes to anglesite ($PbSO_4$) as it is exposed to oxygen in the atmosphere, which is why anglesite was chosen as the representative pollutant. This was further supported by Hayes et al. (2012), who found that anglesite was enriched in the surface level of similar tailings in Arizona. The physical properties of anglesite used as HYSPLIT inputs were: density = 6.3 g/cm^3, molecular weight = 303.3 g/mol. We performed sensitivity analyses using different particles as the basis for modeling and found that the effect of molecular weight and density on the model output were marginal.

Table 3.3.3d Sources used to inform the decision of pollutant particle species for modeling.

Source	Information Utilized
(Huerrera, 2021)	Composition of San Miguel tailings dam in Potosi
(Alfonso et al., 2022)	Composition of Llallagua tailings
(Romero et al., 2014)	Composition of Llallagua tailings
(Sun et al., 2019)	Composition of Tarapaya tailings
(Hudson-Edwards et al., 2001)	Composition of Pilaya tailings
(Tapia et al., 2012)	Composition of Oruro tailings
(Comisión Trinacional para el Desarrollo de la Cuenca del Río Pilcomayo, n.d.)	Elements present in water and sediment samples taken from the Pilcomayo River
(Hayes et al., 2012)	Effect of weathering on lead in arid mine tailings (Arizona)

HYPLIT modeling was run using both 10 microns and 2.5 microns for the particle diameter input. This was based on Kasongo et al's (2024) finding that windblown dust from mine tailings can contain both PM_{10} and $PM_{2.5}$ particulates. In addition, selecting PM_{10} and $PM_{2.5}$ allowed us to keep our modeling simple while retaining important insights. In reality, there are many more particle sizes that could be considered for modeling, especially those between PM_{10} to $PM_{2.5}$ and those less than $PM_{2.5}$. Based on sensitivity analysis, modeling particle diameters in-between PM_{10} and $PM_{2.5}$ had a significant impact on model outputs and warrants further consideration. Modeling particle diameters greater than PM_{10} had a marginal impact, however.

The last physical property of pollutant particles used for HYSPIT's models is "shape," with a shape = 1 indicating that pollutant particles are perfect spheres and a shape = 0.5 indicating that 50% of the "sphere" is void space. The shape was set to 1, as our team could not source a method of estimating the geometry of pollutant particles. In addition, sensitivity analysis revealed the effect of changing the shape parameter to be marginal.

Model Outputs

HYSPLIT was run through Python rather than using HYSPLIT's built-in graphical user interface. This allowed a larger number of models to be run, easier modification of model, and enhanced reproducibility. For modeling at scale, this approach is recommended.

Each HYSPLIT model was run at a 1-hr timestep, which produced a shapefile of anglesite concentration in air for each hour. The models account for how pollutants accumulate over the course of a day (24 hrs). Concentrations are reported in g/m^3. One set of hourly HYSPLIT models was produced for each of the six days and each of the four tailing point sources. These models were run using PM_{10} as the particle diameter, and repeated using $PM_{2.5}$, for a total of 48 sets of hourly models. HYSPLIT concentration models output their results in a unique data format, which the program can then convert into a .kml shapefile compatible with Google Earth Pro. Each .kml shapefile stores 4 polygon layers, with each layer denoting an order of magnitude of pollutant concentrations.

Once all the models were produced, the hourly concentration shapefiles were rasterized. The rasters from each of the four tailing point sources were aggregated within each hour via addition. Then, the hourly aggregations were grouped by date and the average concentration of each day was evaluated within each raster cell. This was done for PM_{10} concentrations and $PM_{2.5}$ concentrations separately, resulting in 12 rasters of daily average concentration.

To compare the relative level of risk from windblown air pollution from tailings between different areas of the city of Potosí, the modeled average PM_{10} and $PM_{2.5}$ concentrations of the six days were compared to reference WHO standards. According to the WHO (World Health Organization, 2021), the limit for human health concern for PM_{10} and $PM_{2.5}$ concentrations in air are as follows (See Table 3.3.3e).

Table 3.3.3e WHO standards for PM_{10} and $PM_{2.5}$ concentrations in air. According to the WHO, the 24-hr average standard should not be exceeded more than 3 to 4 times per year.

Particle Diameter	24-hr average (3–4x per year) ($\mu g/m^3$)	Annual average ($\mu g/m^3$)
$PM_{2.5}$	15	5
PM_{10}	45	15

To standardize the concentration values from PM_{10} and PM2.5, the hazard quotient (HQ) for each particle diameter was calculated. To calculate the HQ, the daily average PM_{10} concentration value of each cell was divided by 45 $\mu g/m^3$ and the daily average $PM_{2.5}$ concentration values were divided by 15 $\mu g/m^3$. This resulted in 12 rasters of daily HQ values. Then, the 80th percentile of HQ values across the six days was calculated within each raster cell for each particle diameter, resulting in 2 rasters (one for each particle diameter). Processes involving concentration rasters were completed in Python (version 3.13.12) using the *rasterio*, *shapely*, and *scipy* packages.

Any areas where the 80th percentile of concentrations exceed the WHO's 24-hour standards are projected to be in excess of the 3-4 days per year threshold set by the WHO. Based on the distribution of emission rate values (See Figure 3.3.3b), Potosí would likely experience 80th percentile emission conditions up to 10 or 20 days during the dry season alone. It is possible that PM_{10} or $PM_{2.5}$ concentrations are in exceedance of the WHO's 24-hr average far more than 3-4 times a year. An important caveat is that the WHO's PM_{10} or $PM_{2.5}$ standards do not account for the chemical composition of the particles. Since heavy metal particulates are being modeled, it is likely that a “safe” level of metal-laden PM_{10} or $PM_{2.5}$ in air is lower than the general WHO standard for those particle sizes.

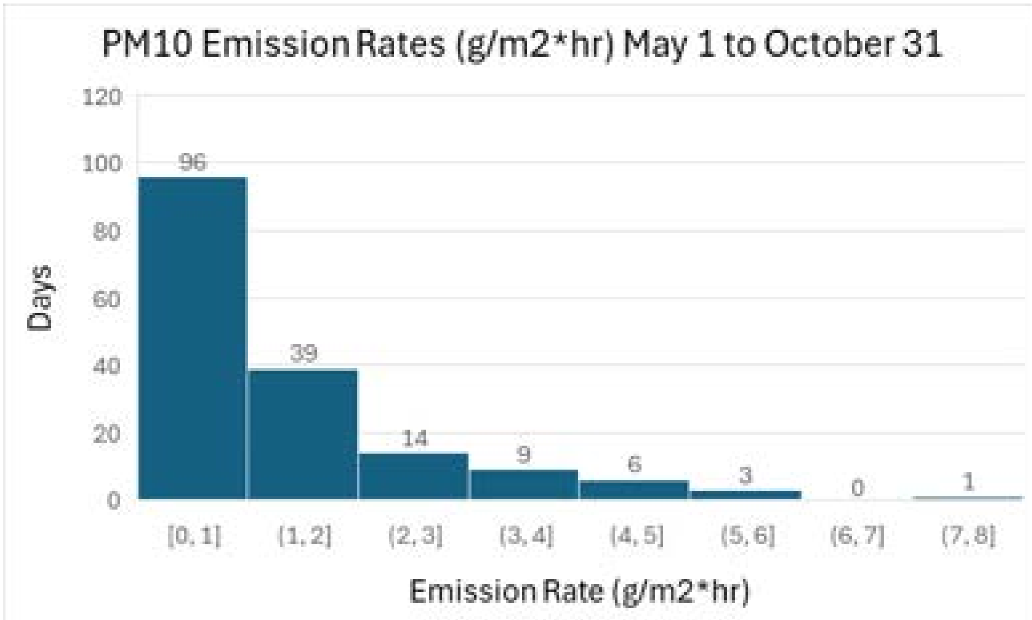

Figure 3.3.3b Histogram of estimated daily average PM_{10} emission rates (g/m^2-hr) in the city of Potosí between May 1st and October 31st, 2024.

In order to provide a consolidated assessment of human health risks due to windblown tailing particles, the 80th percentile hazard quotient layers for PM_{10} and $PM_{2.5}$ were aggregated. The HQ values of each layer were added together within each cell, producing an aggregated hazard quotient score. This was done to provide an indication of where hazards are high for both PM_{10} and $PM_{2.5}$. Lastly, the aggregated hazard quotients were divided into multiple classes of relative risk using 6 quantiles (See Table 3.3.3f). The lowest quantile (0) was excluded, leaving 5 classes of relative risk. The same class definitions were used to categorize risk from PM_{10} and $PM_{2.5}$ individually. These classes assign a level of priority for action (preventative or treatment) to different regions based on projected concentrations of air particulates. They do not define an absolute level of health risk. This was done to facilitate communicating our results and making spatial comparisons.

Table 3.3.3f Classes of priority for remediation based on modeled air particle concentrations

Classification	HQ Range
Lowest Priority	0 – 0.74
Low Priority	0.74 – 2.97
Moderate Priority	2.97 – 5.94
High Priority	5.94 – 17.46
Highest Priority	> 17.46

Limitations

There were some computational, data, and human resource limitations present in the generation of the HYSPLIT models. The approach initially considered for generating these models was to produce one model for every day of Bolivia's dry season (May through October). However the amount of processing power and time required to accomplish this was beyond the scope of this project. Therefore, the team decided to generate models based on six days representative of Bolivia's dry season. Additionally, the team could not acquire data on the specific composition of tailings within Potosí. Therefore, anglesite was used as a stand-in for heavy metal particulates in general. In reality, particulate matter from tailings would likely be a complex mixture of clays, metals, minerals, etc.

It is important to note that these predictions are made with key modeling assumptions. Sensitivity analysis was performed to see the relative effect of different modeling assumptions (See 4.3.6 *Sensitivity Analysis*). The most significant assumption made for modeling was the estimation of emission rate. To gain a true understanding of emission rates, real-time in-field measurements would be necessary. Additionally, the timestep of meteorological data used and the timestep at which HYSPLIT was set to increment its models have a significant effect on the predictions of the model. Lastly, the time period which models were run over and the locations designated as point sources have a significant effect. Installing air quality stations would provide more accurate data to base future models. Given more time and greater access to on-the-ground data, models of Potosí's air quality can be made with higher temporal and spatial resolution.

The purpose of these models is to predict the magnitude of air pollution and regions experiencing the largest impacts from windblown tailing in order to inform further research. It should be noted that the hazard classes are defined relative to each other. Areas of "Low" or "Lowest" priority should not be considered "safe", as there is not an appropriate standard relating to heavy metal laden particulate matter by which projected concentration values could be compared.

3.3.4 Environmental Justice Index (EJI)

Study Area & Spatial Unit of Analysis
This analysis examines populations located within the Pilcomayo River Basin, including the Bolivian departments of Chuquisaca, Oruro, Potosí, and Tarija. All indicators were aggregated to the municipal level. This spatial unit was selected because it's the smallest available resolution that would ensure consistency across population, housing, and mortality data, as well as facilitate integration with air and water quality indicators, which are available at finer spatial scales. Environmental justice analyses often benefit from high-resolution spatial data; therefore, this index should be interpreted as a preliminary screening tool to identify municipalities of elevated concern rather than a precise measure of localized risk.

Data Sources
Data was sourced from the 2024 Bolivian Population and Housing Census (INE, 2024). This includes individual-level demographic and socioeconomic records. It also covers household-level housing and infrastructure characteristics, as well as registered mortality rates. From these characteristics, a subset was selected and expressed as proportions of each municipality's total population. Any missing or invalid values were omitted from calculations for corresponding indicators.

The indicator selection and ranking methodology followed the U.S. Centers for Disease Control and Prevention's (CDC) 2024 Environmental Justice Index (EJI) technical documentation. This approach measures cumulative impacts of environmental burden. It does so through the intersection of environmental burden, social disadvantage, and health outcomes. The emphasis is on relative, rather than absolute, vulnerability across geographic units. All data processing, analysis, and visualization for the EJI were conducted in R (version 4.5.2) using packages including *tidyverse*, *sf*, and related supporting packages.

Environmental Justice Framework
The EPA defines environmental justice (EJ) as "the fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies." Under this definition, "fair treatment" ensures that environmental harms and risks are distributed equitably and should not disproportionately impact any group of people (EPA, 2013). The framework addresses cumulative impacts which describe the combined human health effects that arise from the interaction of environmental stressors, pre-existing health conditions, and social influences. Cumulative impact assessment approaches build on conventional risk assessment techniques by incorporating both quantitative and semi-quantitative data to evaluate the joint effects of social, environmental, and health-related factors on community well-being (CDC, 2024).

Semi-quantitative methods use numerical or ranked scores to compare relative conditions when precise metrics are unavailable or impractical. The EJI is designed to be flexible, allowing indicators and modules to be added or removed, and rankings re-calculated, with minimal effort to support communities in customizing the index to reflect local needs and priorities (CDC, 2024). By applying these principles and a holistic analysis, this study examines how multiple dimensions of vulnerability, beyond proximity to mining-related contamination, shape communities' susceptibility to harm and their capacity to recover from hazard exposure.

Indicator Modules

A total of 20 indicators were used in the analysis, categorized into three modules: health vulnerability (5 indicators), environmental burden (10 indicators), and socioeconomic disadvantage (5 indicators).

Unlike the CDC EJI, which incorporates detailed chronic health conditions, comparable measures were not available in the Bolivian census data used for this study. Health vulnerability was therefore characterized using demographic and population-based proxies, including the presence of vulnerable age groups, access to formal healthcare and insurance, and premature mortality rates.

Environmental burden indicators capture conditions associated with elevated risk, including employment in occupations that may involve direct contact with mining contaminants or air pollutants, structural vulnerability of housing based on construction materials, and household access to safe water, sanitation, and waste disposal infrastructure.

Socioeconomic indicators reflect structural disadvantage and reduced adaptive capacity, including indigenous identity, disability status, education level, and employment conditions.

Indicator Definitions

All indicators were constructed as municipal-level proportions designed to capture the share of a population's vulnerable individuals or households. Indicators were calculated such that higher values consistently represent higher levels of vulnerability, thereby eliminating any need for inversion. As a result, percentile transformation and subsequent aggregation reflect relative vulnerability across municipalities rather than relative advantage. Age-based filtering was applied to education and labor indicators to ensure conceptual and statistical accuracy, with individuals under the age of 15 excluded from relevant denominators. A detailed table describing individual indicators and their definitions is provided in the Appendix (*A.2.3 EJI Methods*). Figure 3.3.4a provides a schematic overview of the EJI framework, including its three modules and a subset of selected indicators.

Figure 3.3.4a Workflow for the construction of the EJI and overview of module indicators.

Percentile Transformation
All indicators were transformed into percentile ranks prior to module aggregation, following the methodology of the CDC EJI. For each indicator, each municipality was assigned a rank (r) that was converted to a percentile using the formula $(r - 1) / (n - 1)$, where n represents the total number of municipalities included in the ranking for that indicator. If two or more municipalities had the same value for an indicator, they were assigned the average of the ranks they would occupy so that municipalities with identical values would receive the same percentile. Missing values were excluded from the calculation of ranks and percentiles.

Percentile ranking standardizes indicators measured on different scales and enables direct comparison across municipalities by expressing each value relative to the distribution of all municipalities. Percentiles range from 0 to 1, where 0 represents the lowest observed vulnerability and 1 represents the highest.

Composite EJI Scoring
Within each module, percentile-transformed indicators were aggregated using an unweighted arithmetic mean to produce module-specific vulnerability scores for health, environmental burden, and socioeconomic status (CDC, 2024). The overall EJI score was then calculated as the mean of the three module scores, assigning equal eights to each module. Municipalities were subsequently ranked based on EJI scores, with higher scores indicating greater relative vulnerability, consistent with standard EJ screening approaches (CDC, 2024; EPA, 2024).

Index Validation: Correlation & Leave-One-Out Analyses
Spearman rank correlations were calculated among indicators within each module to assess potential collinearity among measures of vulnerability (see Appendix *Figure A.2.3a*). When two indicators were highly correlated ($p > 0.85$), one was excluded from the calculation of the overall module score and was therefore excluded from the composite EJI.

To evaluate the relative contribution of individual indicators to overall vulnerability patterns, we calculated Spearman rank correlations between each percentile-transformed indicator and the overall EJI score. All indicators included in the construction of the EJI were analyzed, and results were visualized to assess the relative strength of association across all three modules (Figure 3.3.4b).

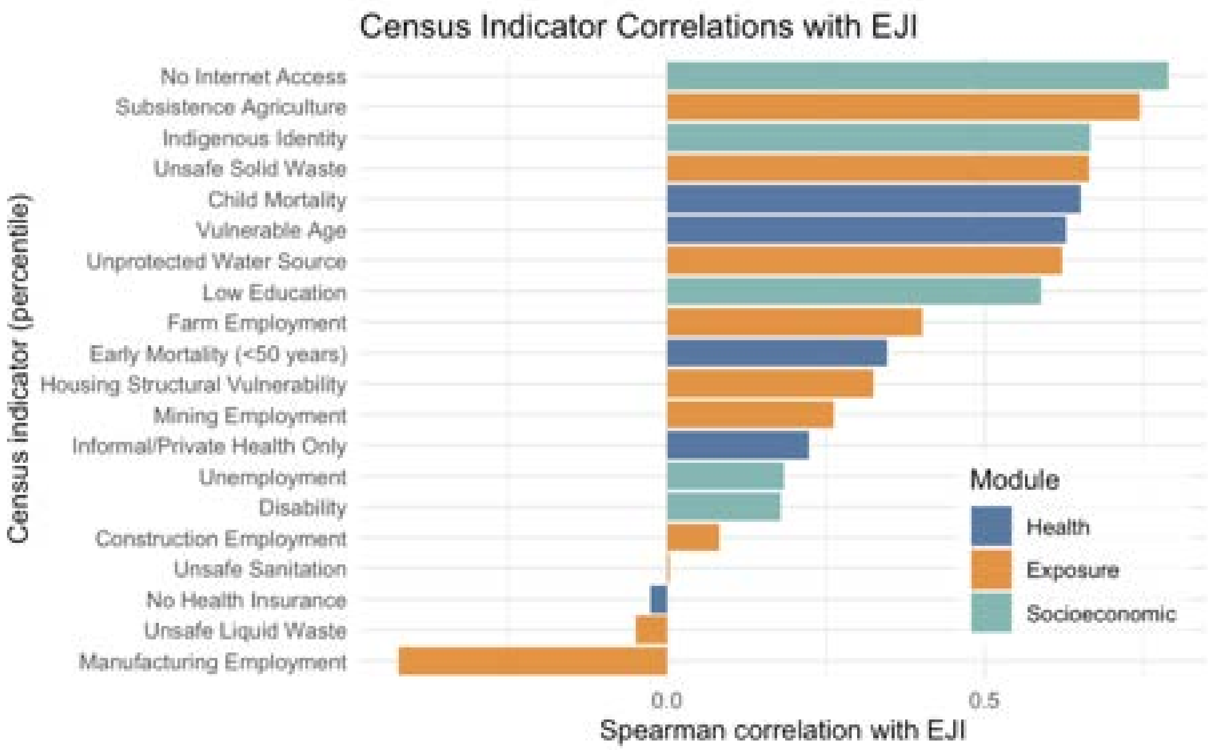

Figure 3.3.4b Spearman correlations between individual census indicators and the composite EJI, where each bar represents the strength of association between a percentile-transformed indicator and the overall score, grouped by module. Higher correlations indicate that the indicator closely aligns with patterns of overall vulnerability.

Finally, correlations between the overall EJI score and modular scores showed strong positive relationships with the health, environmental burden, and socioeconomic modules correlating at 0.72, 0.72, and 0.85 respectively, suggesting that each module contributes substantially to the composite vulnerability.

A leave-one-out analysis was also conducted to assess the influence of individual indicators on the overall EJI. Spearman correlations between the original EJI and recalculated EJI scores leaving out one indicator each time were all very high (>0.95), indicating that no single indicator disproportionately drives the overall score. Because the composite EJI was robust to the removal of individual indicators, we did not perform further sensitivity or influence analyses.

Limitations
Because the EJI was developed for application in the U.S., there are assumptions embedded in the underlying framework that may not be transferable to Bolivia's social, environmental, and institutional context. Adaptation of the CDC framework is further constrained by data availability and the relatively coarse spatial resolution of Bolivian census indicators. This data may miss or not accurately reflect the experiences of less accessible populations, including rural, remote, Indigenous, and migratory communities. As a result, this index should be interpreted as a high-level, preliminary screening tool rather than a comprehensive representation of lived experiences or observed health outcomes.

While indicators were selected as objectively as possible to reflect multiple dimensions of vulnerability, they cannot capture the full range of environmental risks and burdens local populations face. This analysis was designed with a specific focus on mining-related water and air pollution impacts, and does not target broad stressors such as climate change, which may further exacerbate individual- and community-level risk.

3.3.5 Composite Hazard Ranking

In order to understand where populations potentially experience the greatest impact to human health from water contamination in the Pilcomayo River, we created a composite hazard ranking which accounts for water and sediment contamination, population vulnerability (evaluating using EJI framework), and population size. This composite ranking used the following data layers:

1. Interpolated Water Quality Aggregated Hazard Quotient Raster
2. Interpolated Sediment Quality Aggregated Hazard Quotient Raster
3. EJI score by municipality
4. Raster of population density (World Population Review, 2024)

It is worth noting that the air quality analysis was excluded from the final combination because it was only conducted for the city of Potosí. Including this metric risked implying that air pollution was limited to Potosí, when similar hazards likely occur in other unassessed areas. In future iterations, air quality is a dimension warranting consideration. There were also significant gaps in water and sediment data, but they provided sufficient enough coverage of the basin that their inclusion provided meaningful results. Both EJI vulnerability and population density covered the entirety of the basin.

The Water and Sediment Quality Hazard Quotient layers (Layers 1 & 2) summarize the level of risk to human health from water and sediment contamination. The EJI Score (Layer 3) summarizes the vulnerability of populations to impacts from contamination, including concentration of high-risk age groups, access to healthcare facilities, and other factors. The population raster (Layer 4) denotes the concentration of individuals who may be impacted by local contamination. Thus, a location with a high number of highly-vulnerable people who live near a section of the river with high levels of contamination would be considered the highest priority for remediation efforts.

Scores for each layer were separated into five risk categories, where higher scores indicate higher risk (Table 3.3.5a). Interpolated water and sediment HQs were binned using station scores (see Section 3.3.1). Population vulnerability was binned using quantile binning. The population size raster was square-root transformed due to high variance, then classified utilizing Jenks natural breaks where higher population density equates to higher risk (G. F. Jenks, 1977).

Table 3.3.5a Classification Ranges for each input layer.

Classification	EJI Score Range	Sqrt Population Range	Water HQ Range	Sediment HQ Range
Lowest Priority	0–0.4	0–3.29	0–12.5	0–15.44
Low Priority	0.4–0.45	3.29–9.21	12.5–18.37	15.44–15.73
Moderate Priority	0.45–0.5	9.21–29.11	18.37–28.85	15.73–21.11
High Priority	0.5–0.6	29.11–75.57	28.85–50.5	21.11–43.83
Highest Priority	> 0.6	> 75.57	> 50.5	> 43.83

Each data layer exists at a different spatial scale and format. The EJI layer is made up of polygons and the population size and Interpolated Water Quality and Sediment Quality Hazard Quotient layers are rasters. To integrate these, the EJI polygon layer was converted to a raster matching the cell size of the Water and Sediment Quality layers. The population raster was then resampled to the same resolution. With all layers aligned, the four binned layers were summed to produce a Composite Score ranging from 4 to 20.

Table 3.3.5b Classification of risk based on aggregated qualitative risk assessments

Final Classification	Combined Score
Lowest Priority	< 4
Low Priority	4–8
Moderate Priority	8–12
High Priority	12–16
Highest Priority	16–20

It is important to note that these classes are defined qualitatively and not quantitatively. The aim of assigning numerical values and adding those values together was to gain an understanding of where stressors from multiple sources overlap and provide a numeric method for prioritizing regions for remedial action. It should not be interpreted that an area of "high priority" is twice as hazardous as an area of "low priority."

These scores were extracted to settlement locations. This analysis was limited to settlements within 2 km of rivers to focus on communities with a reasonable likelihood of exposure to contaminated water or sediment. The 2km buffer zone is also consistent with the buffer used in the interpolated water and sediment risk layers. The composite score and individual layer scores were extracted to settlements, then classified into bins of final qualitative risk (see table 3.3.5b). Settlements near priority areas were displayed in Figures 5.1a-e, and the ten settlements with the highest combined score were chosen for display in Figure 5.1f (see Conclusions & Recommendations).

3.4 Contaminant Migration Analysis

3.4.1 Three-Dimensional Contamination Surface Models

Interactive three-dimensional contamination surface models were generated for priority metals (As, Cd, Pb, Zn) using water column concentration data to visualize the evolution of pollution pulses across approximately 820 kilometers of the basin over the eight-year monitoring period. River distances were assigned to each monitoring station based on manually classified hydrological position within the basin, correcting for the complex tributary structure of the upper Pilcomayo. Each model plots elapsed monitoring time along the x-axis, downstream distance along the y-axis, and metal concentration along the z-axis. Concentrations were interpolated across a regular 50×50 spatiotemporal grid using linear interpolation, with nearest-neighbor fill applied at grid edges. These models represent concurrent spatial measurements across the monitoring network rather than tracking of individual water parcels downstream.

3.4.2 Seasonal Variation and Migration Patterns

Seasonal metal dynamics were analyzed across both water and sediment matrices for As, Cd, Pb, and Zn to characterize the cyclical transfer of metals between aqueous and solid phases. Monitoring stations were classified into three regions: Potosí Mining Zone, Bolivia Non-Mining, and Argentina/Paraguay. Wet and dry seasons were defined empirically from precipitation records, with wet season months identified as the four months of highest mean precipitation and dry season months as the four lowest, corresponding to November through April and May through October respectively. Median metal concentrations were calculated separately for each season, region, and matrix combination, and compared using grouped bar plots to quantify the inverse relationship between water column and sediment concentrations across the seasonal cycle.

3.4.3 Empirical Sediment-Water Partitioning

To quantify metal distribution between solid and aqueous phases, sediment and water samples collected from identical monitoring stations during the same sampling campaigns were matched for pairwise analysis across 27 monitoring stations. Where multiple grain size fractions were analyzed from the same station and campaign, the median metal concentration across fractions was calculated to produce a single representative value per sampling event, preventing stations with greater sieve fraction coverage from disproportionately influencing results. This matching protocol ensured temporal and spatial consistency, yielding 265 matched pairs for Pb, As, and Zn and 264 pairs for Cd. Two dissolved oxygen measurements were identified as erroneous via cross-check against concurrent saturation percentage and water temperature data and excluded from dissolved oxygen correlation analysis. The site-specific partitioning coefficient (K_d, L/kg) was calculated by dividing total extractable metal in the sediment fraction (mg/kg) by total metal in the aqueous phase (mg/L), with water concentrations converted from micrograms per liter to milligrams per liter to ensure dimensional consistency:

$$K_d = \text{Partitioning Ratio (L/kg)} = C_{sediment}\ \text{(mg/kg)} \div C_{water}\ \text{(mg/L)}$$

Due to the non-normal distribution of environmental data spanning multiple orders of magnitude, Spearman rank correlation was used to evaluate the monotonic relationship between K_d and four physicochemical parameters: pH, clay content, organic matter, and dissolved oxygen. A similar approach was taken to compare pollutant concentrations in deposited sediments against dissolved concentrations.

3.4.4 Mechanistic Hypothesis Testing

To identify the geochemical phases controlling metal retention, four distinct retention mechanisms were tested against the empirical field data rather than assuming standard theoretical behavior. The Oxide Surface Hypothesis tested whether iron and manganese oxides act as the primary sorption substrates for metals as predicted by surface complexation models. The Aluminum Control Hypothesis evaluated aluminum content as a distinct sorption substrate, particularly for arsenate. The Mineral Co-Precipitation Hypothesis tested whether calcium, phosphorus, and sulfur content drive retention for cadmium, which may precipitate as carbonates or phosphates rather than adsorbing to oxides. The Organic Complexation Hypothesis assessed whether organic matter acts as a significant metal sink or plays a negligible role in this mineral-dominated system.

Each mechanism was evaluated using Spearman rank correlation between the calculated K_d and the concentration of the corresponding mineral phase proxy in co-located sediment samples. Mineral phase proxies included total iron (% Fe, n = 846), total manganese (mg/kg Mn, n = 846), total aluminum (% Al, n = 560), total calcium (% Ca, n = 560), total phosphorus (% P, n = 560), organic matter content (% OM), and clay fraction (% clay). Iron and manganese served as proxies for Fe(III) and Mn(IV) oxide abundance under the predominantly oxic conditions observed across the basin. All geochemical variables were log-transformed prior to correlation analysis to account for multi-order-of-magnitude ranges typical of mining-impacted sediments. A mechanism was considered verified where Spearman correlation with K_d was statistically significant and positive ($p < 0.05$), while non-significant or negative correlations were interpreted as falsification of that retention pathway for that metal.

3.4.5 Sediment Texture Analysis

To evaluate physical controls on metal accumulation in the Potosí mining zone, 417 sediment samples from 12 monitoring stations were categorized into five textural classes: sand, sandy loam, sandy clay loam, loam, and silty loam. Priority metals included As, Cd, Pb, and Zn, with mercury (Hg) included as an additional analyte given emerging concentration trends observed in the later years of the monitoring record. Median metal concentrations were calculated per textural class and Spearman rank correlation was used to evaluate the relationship between clay content and metal concentration across all classes. Stations were ordered by descending arsenic median to visualize the spatial decay gradient from source-proximal to distal stations.

Seasonal patterns were examined using the same wet and dry season definition applied in Section 3.4.2 (November through April and May through October respectively), comparing median metal concentrations by textural class across both periods. Temporal trends were assessed across 16 sampling campaigns spanning 2016 to 2024, with basin-wide median concentrations and interquartile ranges plotted per campaign to identify long-term trajectories. Sediment textural composition was also tracked across campaigns to evaluate whether shifts in the dominant texture class over the monitoring period could independently influence metal retention patterns.

3.4.6 Hypoxic Events

A time-series analysis was conducted to evaluate the impact of hypoxic events, defined as dissolved oxygen concentrations below 4 mg/L, on water column metal concentrations at Potosí mining zone stations. Three hypoxic episodes were identified between 2016 and 2024: Tarapaya in April 2016 (DO = 3.36 mg/L), Potosí – Naciente río La Ribera in October 2017 (DO = 2.84 mg/L), and La Quiaca in February 2024 (DO = 2.61 mg/L). Each event was analyzed independently using the full monitoring record to contextualize metal behavior relative to the event date. Given the complex tributary structure of the upper Pilcomayo, before-and-after enrichment analysis was restricted to stations with confirmed hydrological connectivity to the event station to avoid unfounded assumptions about downstream transport across separate sub-tributaries. For Event 2 at Naciente río La Ribera, three confirmed connected stations were included Tarapaya, Pilcomayo – agua arriba confluencia, and San Antonio – Potosí, in addition to the event station itself. Events 1 and 3 are presented as time series only because no pre-event baseline data exist for Tarapaya, where systematic monitoring commenced concurrently with the 2016 event, and no post-event samples are available for La Quiaca, which coincides with the final sampling campaign in the dataset.

Where both pre- and post-event samples were available within a 180-day window on either side of the event date, enrichment factors were calculated as the median post-event concentration divided by the median pre-event concentration for Pb, As, Cd, Zn, Fe, and Mn in the water column. Differences in pre- and post-event concentrations were evaluated using the Mann-Whitney U test. Statistical power limitations arising from small sample sizes ($n = 1–2$ per window at most stations) are explicitly acknowledged, and enrichment factors are reported as descriptive effect sizes rather than statistically confirmed results where sample sizes preclude formal inference.

3.5 Cost Analysis of Remediation Technology Options

Case studies were reviewed for data availability of implementation costs for different remediation technologies. Cost ranges and averages were provided when sources provided a cost range. Costs were described in cost per ton of material remediated per year when possible; otherwise, the units given in the literature were kept. Several remediation technologies did not have cost estimates readily available to describe in this study, but were included due to their promise for remediation effectiveness and feasibility. USD values were converted to Bolivian Boliviano (BOB) at an exchange rate of 1 USD = 6.92 BOB (as of February 9, 2026). It is important to note that cost estimates are relative to the value of USD in the year in which each source was published. Not all sources had an associated date, however.

3.6 Wastewater Fingerprinting

To identify stations most impacted by wastewater contamination in the river network, concentration data for three wastewater indicators—ammonia, BOD, and fecal coliforms—were analyzed. Since these parameters are measured in different units (ammonia and BOD in mg/L; fecal coliforms in CFU/100mL or MPN/100mL), samples were normalized using parameter-specific percentile rankings to enable comparison across indicators. An 85th percentile threshold was established for each parameter to identify atypically elevated concentrations for a given station. Samples were then grouped by station and date to identify concurrent contamination events where all three parameters were measured and all exceeded their respective

85th percentile thresholds. These dates represent acute wastewater contamination incidents. Stations with frequent concurrent exceedances were prioritized as candidates for wastewater treatment intervention.

To assess regulatory compliance, all sampling dates where ammonia, BOD, and fecal coliforms were measured concurrently were classified according to Bolivia's Law 1333 (Ministerio de Desarrollo Sostenible y Medio Ambiente, 1992). This regulation establishes four water quality classes (A through D) based on maximum permissible concentrations for each parameter. A sampling event was classified into a given class only when all three parameters met that class's limits; samples exceeding even the minimum quality standard (Class D) were classified as 'Beyond D.' The frequency distribution of water quality classes was calculated for each station to characterize regulatory compliance patterns. International standards for all three contaminants were also reviewed and regulatory compliance was similarly determined.

3.7 Hardness and Metal Toxicity

Hardness can impact the toxicity of various heavy metals. Reference metal toxicity levels utilize an assumed hardness level (the USEPA has utilized both 50 or 100 mg/L $CaCO_3$), but having information on water hardness can lead to more detailed hazard quotients (EPA, 1996). Increased hardness generally reduces uptake of metals and therefore reduces toxicity from the same contaminant concentration. Hardness data from the Trinational Commission was categorized using the US Geological Survey Classification scheme to determine how frequently stations experienced samples from different hardness levels.

Bolivian Law 1333 does not include information on hardness when considering contaminant limits, and the US EPA provides formulas to calculate Criterion Maximum Concentrations (CMCs) and Criterion Continuous Concentrations (CCCs).

3.8 Hydrologic GIS Analyses

3.8.1 River Network Delineation

In ArcGIS Pro, tools from the Spatial Analyst Hydrology toolset were used to delineate the river network in the Pilcomayo Basin. First, Google Earth Engine was used to obtain the ALOS DSM: Global 30m v3.2 digital elevation model (DEM) for our study (JAXA Earth Observation Research Center, 2021). Then, each cell in the DEM was assigned a flow direction value based on the gain or loss in elevation when compared to adjacent cells. Next, each cell was assigned a flow accumulation value, calculated as the number of cells that flow into that cell. A conceptual diagram for this process, provided in the ArcGIS Desktop documentation (Desktop Help 10.0 - How Flow Accumulation Works, n.d.), is shown below (Figure 3.8.1a).

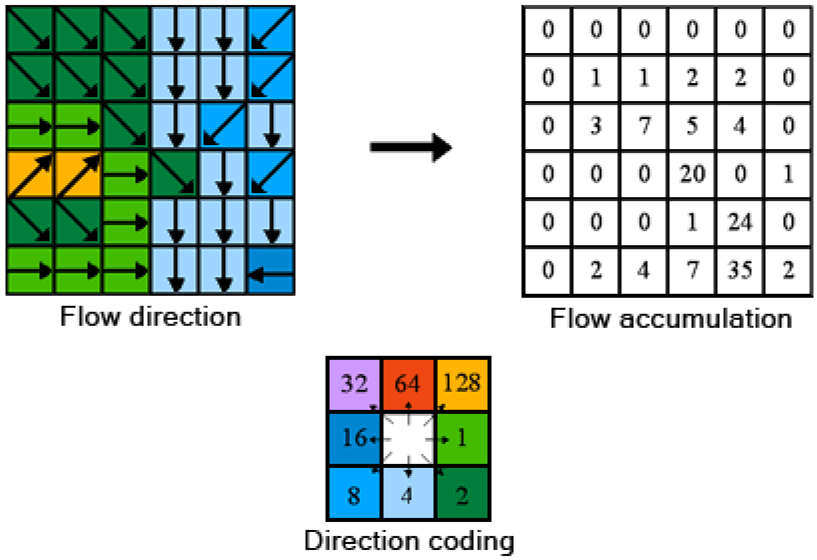

Figure 3.8.1a Conceptual model of flow direction and accumulation workflow in ArcGIS Pro.

Cells with high flow accumulation represent areas of concentrated flow, and represent stream channels. Using a cutoff value of 10,000, each cell was classified as either being part of a stream channel or not part of a stream channel. The value of 10,000 was chosen by comparing results to satellite imagery, and choosing the result that matched the best with the locations of rivers. The resulting raster was converted to a line feature class, wherein stream channel cells were converted into line features. Finally, the flow direction raster was used in combination with the river network line feature class to create a trace network. Trace networks are similar to line feature classes, but they include directionality. In other words, each segment of the network has a “from” node and a “to” node. This allows for tracing along the network upstream or downstream from a given point.

3.8.2 Floodplain Delineation

In ArcGIS Pro, floodplains were delineated using the Cost-Distance tool. The aforementioned stream channel raster was used as the source raster, and the DEM was used to create a slope layer, which was used as the cost raster. The Cost-Distance tool calculates the accumulated effort (or cost) required for water to move from a source location to outward locations. The cost raster determines the effort required to move through each location. As a result, areas with gentle slopes near source locations will have low accumulated cost, and areas with steep slopes, or areas that are far away from source locations will have high accumulated cost. Reclassifying the resulting raster to only include cells with low accumulated cost isolated floodplain areas.

3.8.3 Upstream Mining Exposure

Locations of settlements and known mining activity were compiled (see Table 3.8.3a) and used to create mining exposure risk maps (Figure 3.8.3a). Settlements and mining locations were snapped to the closest point on the river network. For each settlement, an upstream trace was performed using the river trace network (see Section 3.8.1). For each mine or mining facility identified in this trace, the upstream distance from the settlement to that location was recorded. The settlement was then assigned a mining exposure score, defined as $\sum \frac{1}{1+d_n}$, where d_n is the upstream distance to the nth mine or mining facility. This calculation was performed for thousands of settlements in the Pilcomayo Basin using an R script for workflow automation.

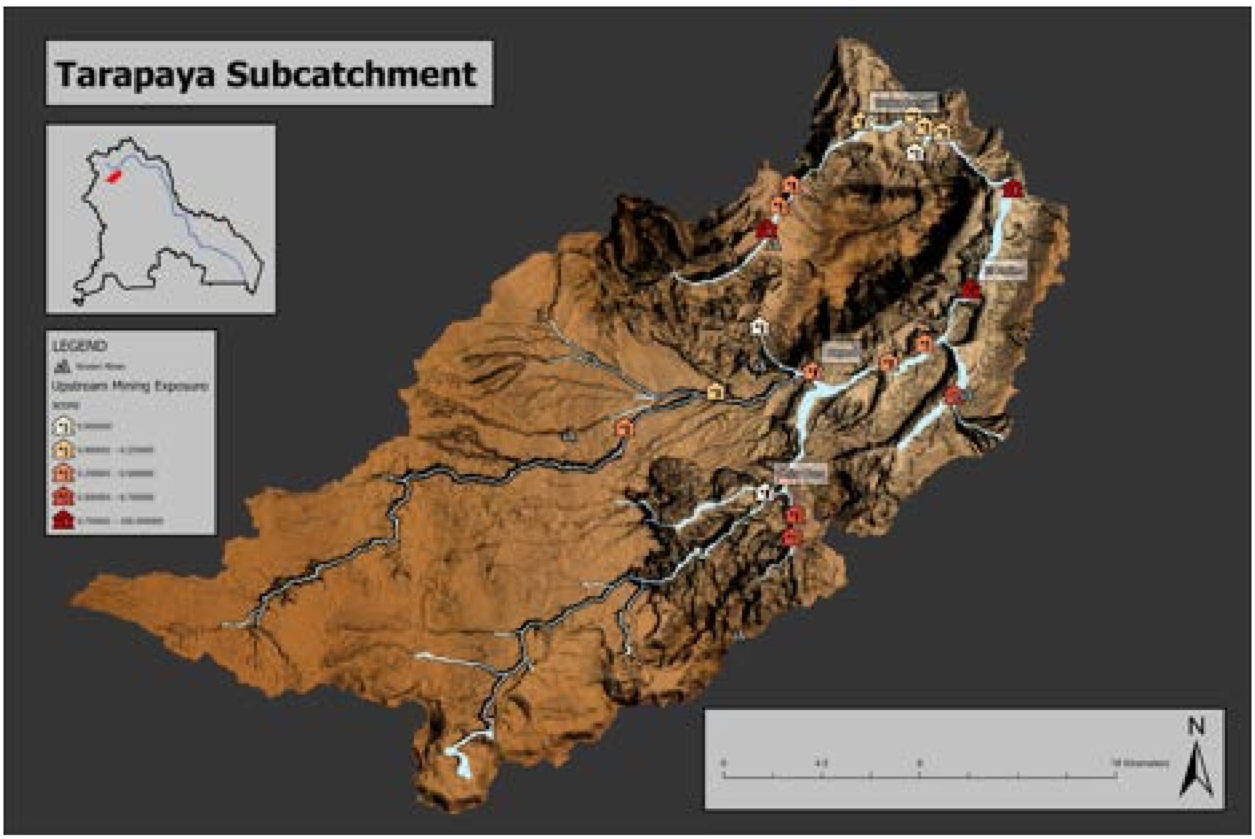

Figure 3.8.3a Upstream mining exposure score for settlements in the Tarapaya subcatchment. Darker red indicates higher mining exposure (high upstream mining activity). Three highest population settlements labeled. River floodplains shown in light blue.

Table 3.8.3a Data sources for upstream mining exposure analysis

Type	Name	Data Format	Source
Settlements	Poblaciones	Shapefile (.shp)	GeoBolivia
Mining	Mineral Facilities of Latin America and the Caribbean	Table (.csv)	U.S. Geological Survey
Mining	Mining Activity in the Pilcomayo Basin	Table (compiled manually)	Mindat.org
Mining	Mineras	Table (.csv)	GeoBolivia

3.9 Sensitivity Analysis

3.9.1 HQ Calculation for Water and Sediment Quality

Algorithms developed for assessing hazard scores for parameters and stations on a basin-wide scale can utilize various methods to aggregate individual samples. Two key methodological decisions affect final rankings: (1) how to aggregate hazard quotients across multiple measurements, and (2) how to combine data across multiple years of sampling. As method selection can impact the final prioritization list provided to the client, a series of sensitivity analyses were conducted to assess the impacts of these decisions.

HQ aggregation summarizes hazard quotients across multiple values. Four methods were tested: 95th percentile (emphasizes high-end exposures), median (reduces influence of outliers), mean (balances all values equally), and maximum (captures worst-case scenarios). For station rankings, HQ aggregation combines values within a station across multiple samples; for parameter rankings, it combines values across all stations where that parameter was measured.

Temporal aggregation combines hazard quotient data across multiple years of sampling. Four methods were tested:

1. Weighted: Applies exponential decay weights to prioritize recent data over older measurements.
2. Recent: Uses only data from a specified window prior to the most recent sampling date at each location (e.g., recency = 5 includes data from the five years preceding each station's last sampling date, so a station last sampled in 2020 would include data from 2015-2020, while a station last sampled in 2024 would include data from 2019-2024).
3. Mean: Averages all available years equally.
4. Max: Selects the maximum value across all years.

A full factorial sensitivity analysis tested all combinations of temporal and HQ aggregation methods (16 combinations), with additional variations in the recency window parameter for the recent temporal method. Each method combination produced rankings (rank 1 = highest concern) for both stations and parameters. Spearman rank correlations quantified agreement between all method pairs to assess ranking stability. For each method combination, the five highest-ranked entities were identified and compared across methods to determine how method selection would impact final recommendations to the client.

3.9.2 Air Particulate Concentration Modeling

To test the impact of the multiple assumptions and inputs used for modeling concentrations of airborne particulate matter, multiple sensitivity analyses were performed. First, a base model was produced, then variant models were produced for comparison. With some exceptions, the base model was compared to all variant models. The base model was produced using the inputs listed in Table 3.3.6a. For each variant model produced, the concentration output shapefile produced was visually compared to the output shapefile of the base model.

Table 3.9.2a HYPSLIT model parameters used as the basis for comparison in sensitivity analysis

HYSPLIT Input	Value
Date	July 24, 2024
Layer Height	2 m
Point Source Location	Los Cumpas
Emission Rate	1.27 g/hr
Concentration Multiplier (surface area)	167,000 m^2
Particle Diameter	2.5 μg
Density	6.3 g/cm^3
Molecular Weight	303.3 g/mol
Shape	1

For each HYSPLIT input being tested, only the value of the input under review was modified; all other input values were kept the same as the base model. The only exceptions were particle diameter and date. For testing the date input, both the date and the emission rate inputs were changed, as emission rates vary depending on daily wind speeds. In addition, model outputs were compared using 10 µg rather than 2.5 µg. For testing particle diameter, both the emission rate and the particle diameter inputs were modified. Particle diameter is a critical assumption used in the equations from Stovern et al. (2014) used to estimate emission rates, therefore, the emission rates were recalculated using the appropriate multiplier as described by Stovern et al. (2014). The effects of two other key assumptions used to calculate the emission rate were tested as well. Emission rates were recalculated using a different number of disturbances and using different values for surface roughness height (See 3.3.4 *Estimating Emission Rates*).

4. Results & Discussion

Water-based analysis focused on the locations impacted and the pollutants of greatest magnitude compared to their reference standard. Location-based analysis examined each location for all parameter concentrations during the period of 2018-2023. Parameter-based analysis examined the magnitude of pollutant concentrations at each location during 2018-2023. A spatiotemporal analysis depicted pollutant migratory patterns for a subset of high-priority parameters across all years of data (2016-2024). These analyses led into the development of a priority map of areas which would benefit from remediation. Additional analysis on drinking water standards, concerns of wastewater exposure in the basin, and a review of reported water hardness were included.

4.1 Data Limitations

Uncertainty due to the reported speciation of metals makes determining whether waters are considered saturated or unsaturated difficult, as the contaminant speciation impacts toxicity. A metal's speciation also impacts its solubility; metallic silver is considered insoluble in water (Agency for Toxic Substances and Disease Registry (US), 1990), while $AgNO_3$ (silver nitrate) is highly soluble (reported between 1g/L to 2.4kg/L; NOAA, n.d.; National Institute of Health, n.d.). Environmental conditions impact solubility, with influences due to the presence of organic compounds, changes in pH (where increased acidity can cause leaching in metals), the mineralogical composition of rocks, and other factors that may compete in sorption and desorption activity (Lewińska & Karczewska, 2025; Angel et al., 2020).

Pollutant concentrations reported in environmental samples in 2024 were found at highly elevated concentrations beyond typical solubility limits; for example, the highest sample of silver in water was reported by the Trinational Commission at 1,643,160 µg/L (1.6g/L) in 2024 at Cotagaita. Dissolved silver concentrations averaged 162 mg/L in 2024, whereas reported silver concentrations in all other years averaged 0.08 mg/L with only four instances where silver surpassed the WHO's 0.1mg/L recommended limit (Figure 4.1a).

Given silver's low solubility, it was unlikely that this reported silver concentration was possible. Available literature and news were considered to determine potential causes of the spike, and the Trinational Commission was contacted regarding the drastic spike. As there is no evidence of an event that would cause silver (among other pollutants) to spike by a 3000x magnitude, and beyond known solubility limits in the instance of silver, the most likely cause was determined to be a data input error. All 2024 data were removed from final analysis (except in limited cases when noted); the most recent date included in this analysis is December 12, 2023.

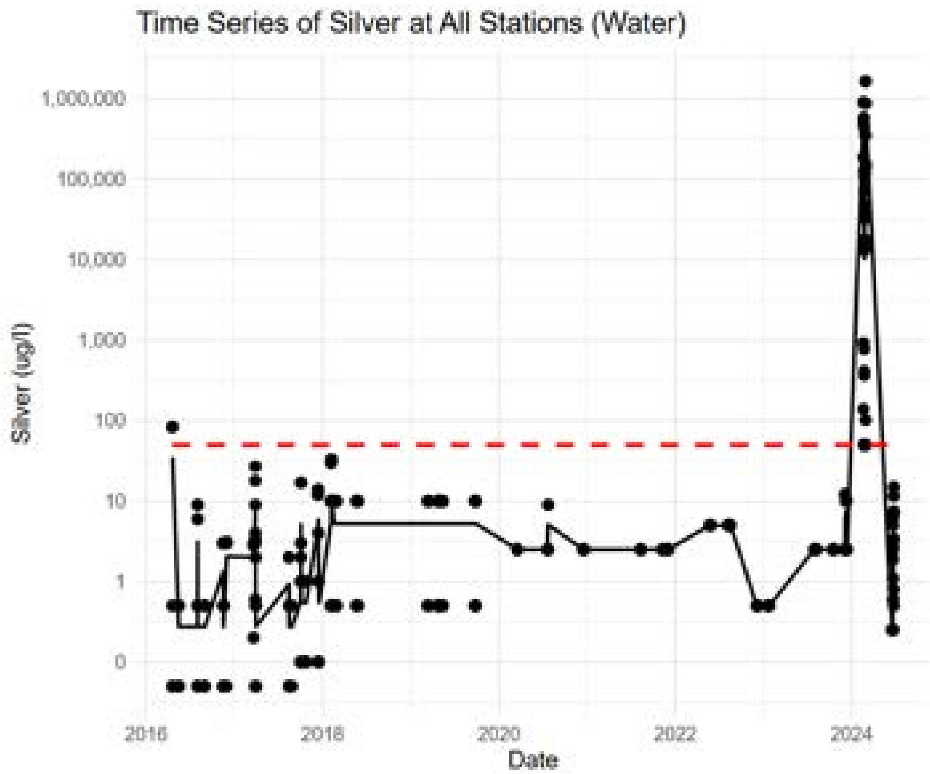

Figure 4.1a Time series of silver concentrations plotted over time. Y-axis (concentration) scaled to show order of magnitude differences. Concentrations stay near regulatory levels (using Bolivian Law 1333's 0.05mg/L limit) until 2024, when some concentrations spike beyond 30,000x the limit. Dots indicate specific samples; the red dotted line signifies the regulatory limit.

4.2 Parameter Analysis

Hazard quotients (HQ) provided a standardized method of considering the severity of each contaminant at detected concentrations. The 95th percentile HQ were determined across all collected samples for each analyte for the last five sampling years and across all sampling stations (Figures 4.2a & 4.2b). Zinc, lead, and arsenic were identified within the top 10 pollutants for both water and sediment media. Only eight parameters in the sediments database had available USGS regulatory sediment limits. Chromium has a reported 95th percentile HQ<1, signifying that it is within the USGS regulatory limit for sediments (37.3 mg/kg) for the majority of samples collected.

Concentration distributions of arsenic and lead in water and sediment media are described in detail; a non-exhaustive list of additional contaminants with high HQ values are discussed further in the Appendix (*A.3.1 Priority Contaminants*).

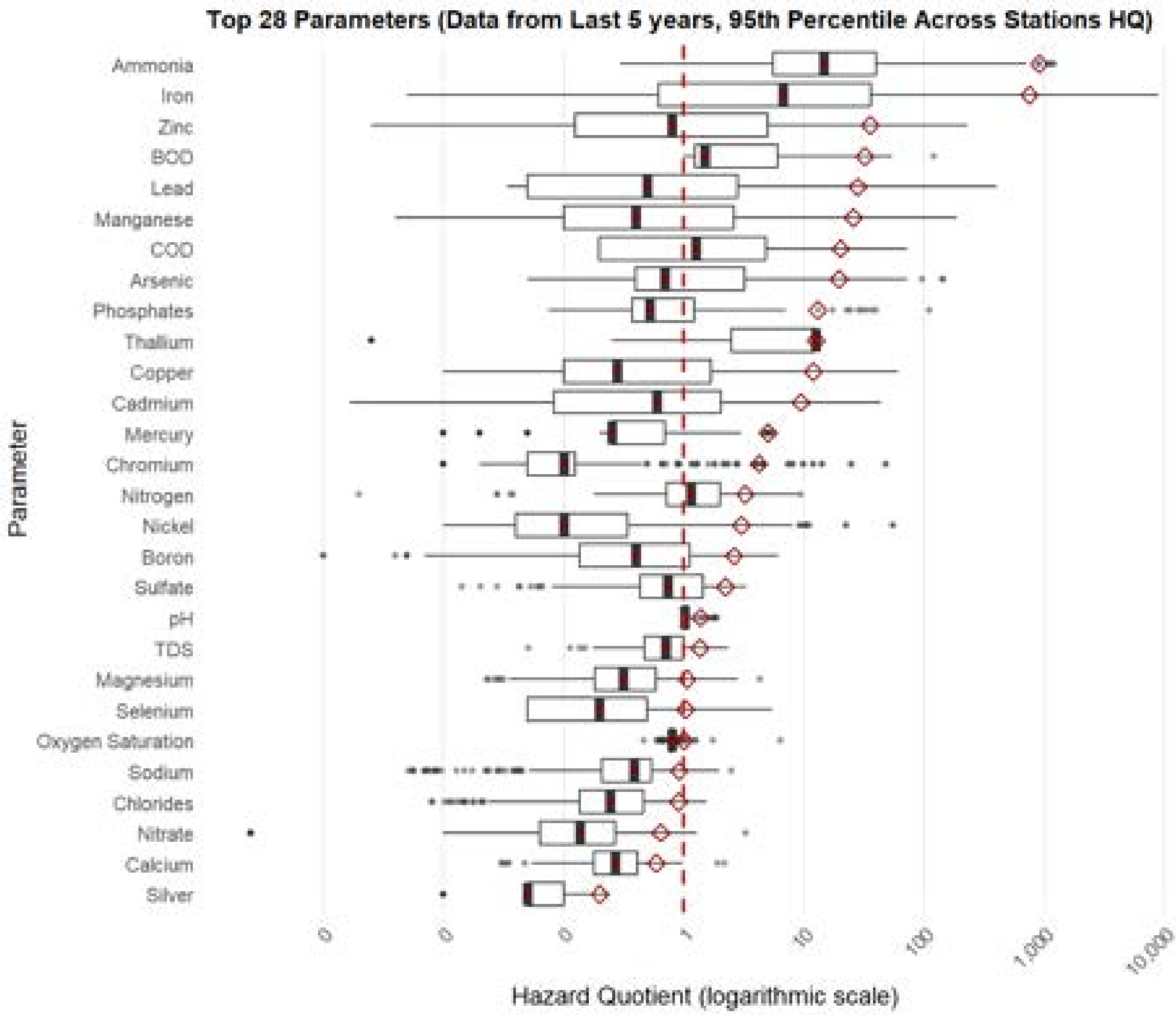

Figure 4.2a List of the 10 "most concerning" parameters, defined as those with the highest 95th percentile HQ scores for water quality samples collected across the Bolivian portion of the basin during 2018-2023. The red diamond signifies 95th percentile value. The red line signifies HQ=1, where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic.

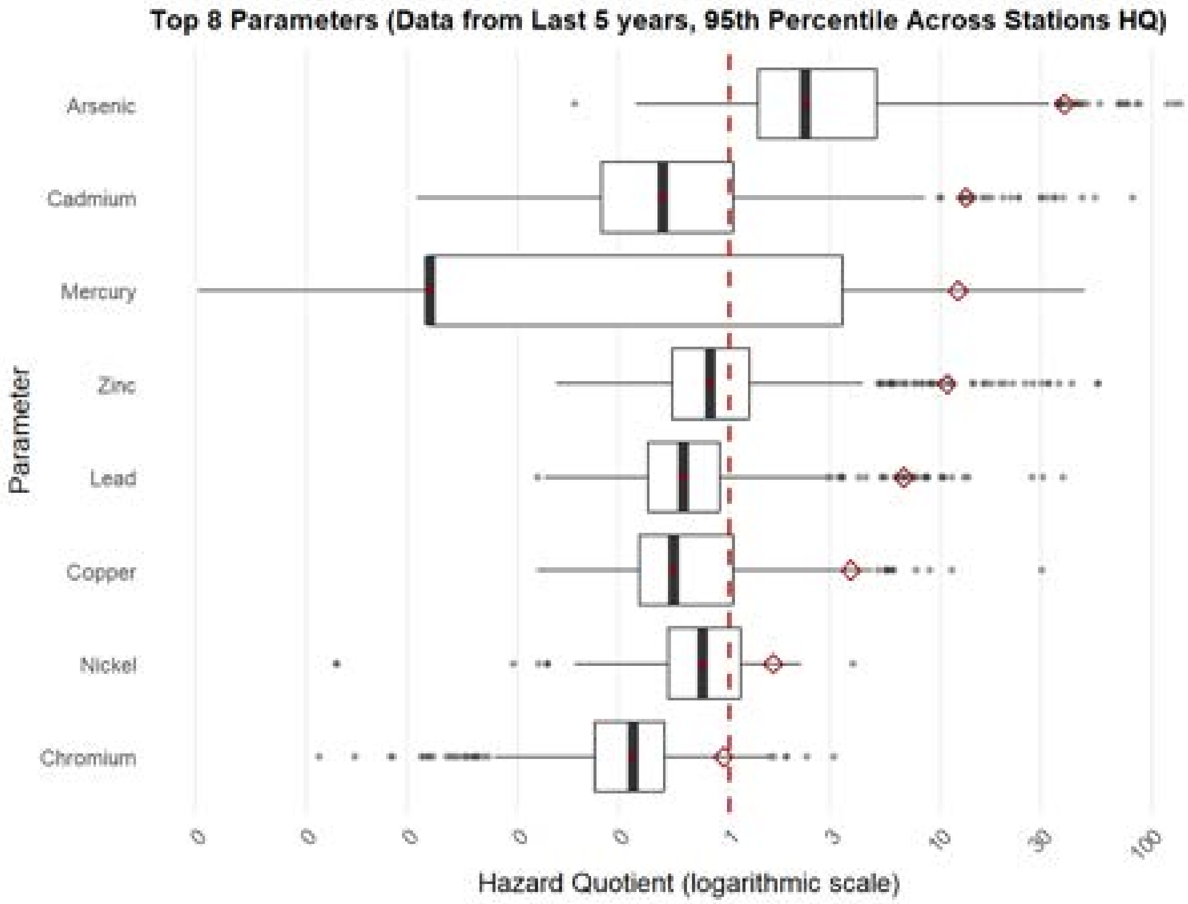

Figure 4.2b List of the "most concerning" parameters for sediment quality across the Bolivian portion of the basin. The red diamond signifies 95th percentile value. The red line signifies HQ=1, where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic. X-axis values are non-transformed but the scale is logarithmic.

4.2.1 Arsenic

Arsenic (As) was the most concerning parameter in sediments and a parameter of concern in water. Hazard quotients were calculated using the strictest regulatory limit, which is given by the USEPA at 10 µg/L and is stricter than the Bolivian Law No. 1333 standard (50 µg/L for Classes A-C). Water-based arsenic concentrations exceeded the USEPA limit in 44% of all samples (Table 4.2.1a). The median As concentration in water averaged 10 µg/L for all years, which means that 50% of samples are out-of-compliance with USEPA reference standards. The same analyses were conducted on As concentrations in sediments, finding that the strictest regulatory standard (USGS: 5.9mg/kg) was surpassed in 85% of sampled sediments with the maximum reported concentration 379x above the regulatory standard.

Each station was reviewed to understand general loading patterns of each parameter of interest stated above. Using the 95th percentile method, stations were characterized by their aggregate hazard quotient and plotted (Figures 4.2.1a and 4.2.1b). The majority of 95th percentile hazard quotients in both media were above HQ=1; most sediment median values were HQ>1, while many water quality medians were reported at HQ<1, signifying elevated arsenic hazard risks from sediments. In sediment samples, arsenic was distributed across a variety of particle sizes, with the highest concentrations in large particulates (>2mm) followed by fine sands and fine silts (Figure 4.2.1c).

Table 4.2.1a Mean and maximum arsenic concentrations and percent of samples exceeding health standards in water and sediment samples across all stations by year.

	Water Samples (μg/L)				Sediment Samples (mg/kg)			
Year	n	Exc. (%)	Med.	Max (Station)	n	Exc. (%)	Med.	Max (Station)
2016	83	37%	6.0	2805 (Tarapaya)	109	83%	12.8	856 (Pilcomayo arriba Tacobamba)
2017	97	52%	11.0	405 (Canutillos 3)	120	98%	15.1	2240 (Palca Grande)
2018	62	55%	22.5	2030 (Cotagaita)	83	92%	15.2	187 (Colavi - Canutillos)
2019	71	37%	10.0	268 (Tarapaya)	118	94%	13.6	515 (Tarapaya)
2020	37	52%	13.0	397 (Tarapaya)	19	100%	25.4	271 (Tarapaya)
2021	70	29%	3.5	214 (San Antonio - Potosi)	63	95%	13.0	498 (Tarapaya)
2022	85	44%	5.0	960 (Colavi - Cantillos)	74	66%	10.68	758 (Villamontes)
2023	166	48%	7.5	1438 (El Puente)	100	69%	12.35	802 (El Puente)
All Years	671	44%	10.0	2805 (Tarapaya)	686	85%	14.0	2240 (Palca Grande)

Notes: n = number of samples; Exc. = Exceedance rate; Med. = Median. Sediment exceedances utilize a USGS standard of 5.9 mg/kg.

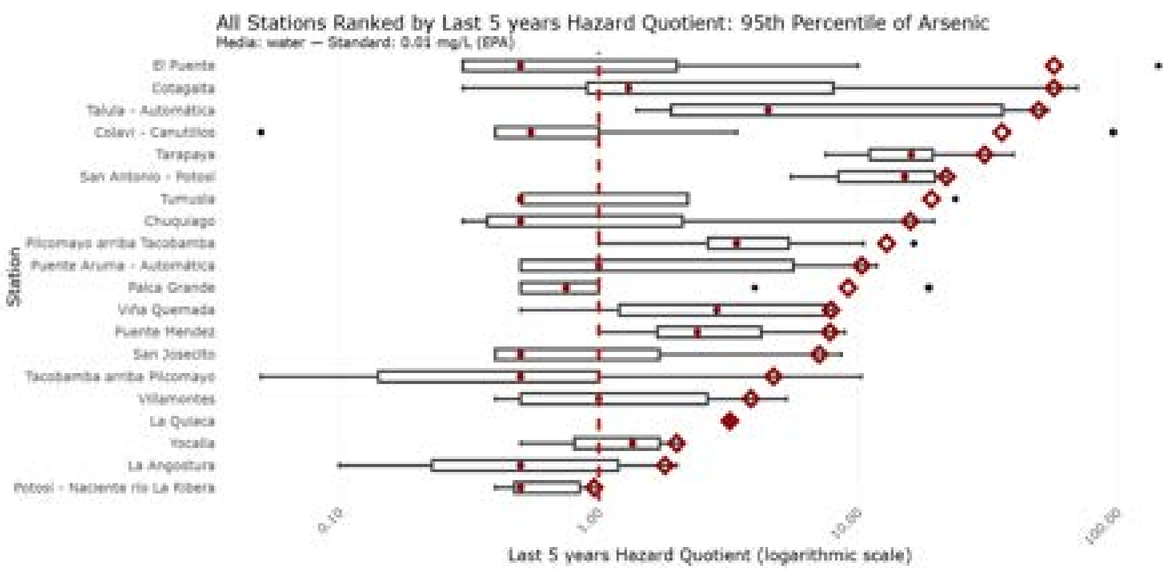

Figure 4.2.1a Top ten water-based arsenic concentrations aggregated by station using the 95th percentile for the most recent 5 years of data. Red line, dots and diamonds signify HQ = 1, median, and 95th percentile respectively. X-axis uses a log scale for display.

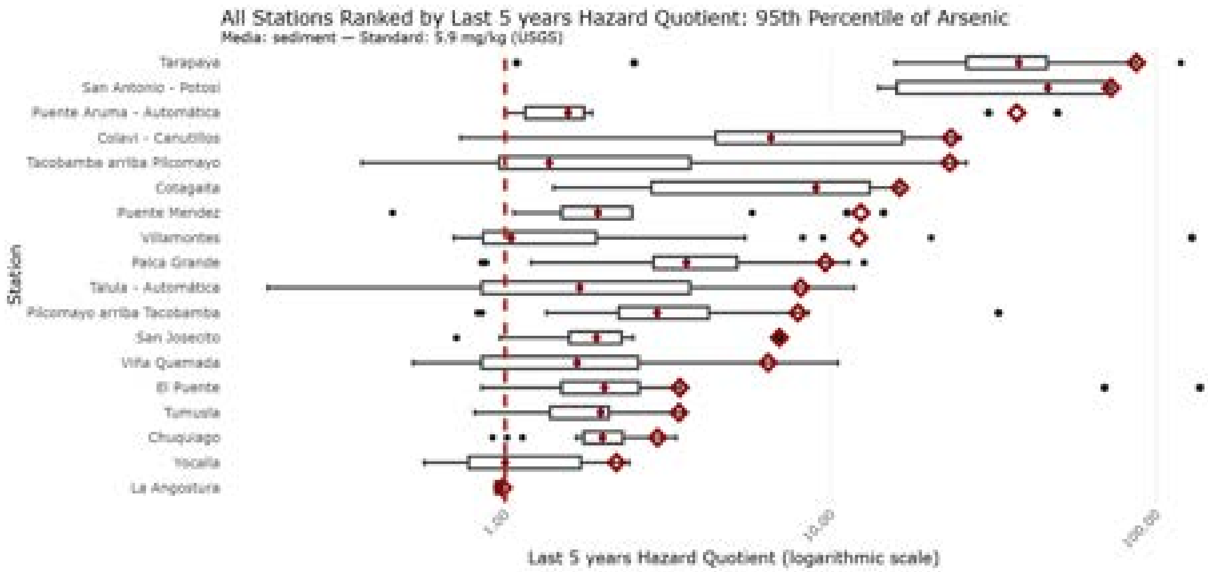

Figure 4.2.1b Top ten sediment-based arsenic concentrations aggregated by station using the 95th percentile for the most recent 5 years of data. Red line, dots and diamonds signify HQ = 1, median, and 95th percentile respectively. X-axis uses a log scale for display.

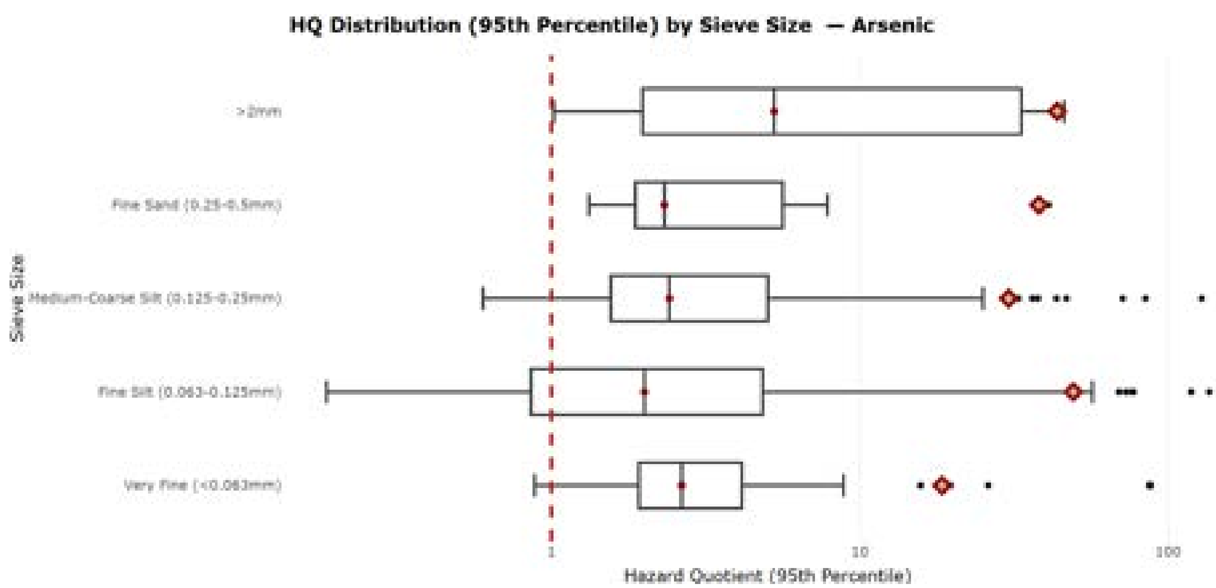

Figure 4.2.1c 95th percentile arsenic hazard quotients reported within sediments at each station for each sieve size. The red dot signifies median value, red diamond signifies 95th percentile value. The red dotted line signifies where arsenic levels surpass regulatory standards (HQ=1). X-axis values are untransformed but the scale is logarithmic. All grain sizes were found to be above safe limits, but all reported concentrations contained by the largest sediments (>2mm) were beyond regulatory limits (HQ>1).

4.2.2 Lead

Lead (Pb) was the 5th most concerning parameter in both water and sediment samples, and local communities have utilized lead chelation and other treatments to combat lead poisoning. Hazard quotients were calculated using the strictest regulatory limit, which is given by the WHO at 10 µg/L for water matrices. Water-based lead concentrations exceeded this limit in 35% of all samples, while sediments exceeded the strictest regulatory standard in 19% of samples (Table 4.2.2a). Compared to water samples, sediment samples have a lower exceedance rate, yet sediment concentrations may act as a continued loading source of lead in the environment and degrade water quality. Maximum concentrations were consistently found at Tarapaya station between 2018-2023, while Pilcomayo arriba Tacobamba had the highest detected concentration.

Table 4.2.2a Mean and maximum lead concentrations and percent of samples exceeding health standards in water and sediment samples across all stations by year.

	Water Samples (µg/L)				Sediment Samples (mg/kg)			
Year	***n***	**Exc. (%)**	**Med.**	**Max (Station)**	***n***	**Exc. (%)**	**Med.**	**Max (Station)**
2016	74	28%	21	2868 (Tarapaya)	109	17%	21.6	7500 (Pilcomayo arriba Tacobamba)
2017	97	39%	1.0	331 (Tarapaya)	120	18%	20.1	419 (San Antonio - Potosi)
2018	62	52%	25.6	2015 (Cotagaita)	83	12%	20.3	114 (Tarapaya)
2019	79	46%	10.0	168 (Palca Grande)	118	23%	22.9	359 (Tarapaya)
2020	32	34%	1.6	632 (Tarapaya)	19	21%	21.2	236 (Tarapaya)
2021	59	27%	0.5	203 (Puente Mendez)	63	16%	23.3	304 (Tarapaya)
2022	81	26%	5.0	572 (Colavi - Canutillos)	74	23%	14.1	1326 (Tarapaya)
2023	141	30%	1.0	4050 (El Puente)	100	24%	19.7	948 (Tarapaya)
All Years	625	35%	4.65	4050 (El Puente)	686	19%	20.9	7500 (Pilcomayo arriba Tacobamba)

Notes: n = number of samples; Exc. = Exceedance rate; Med. = Median. Sediment exceedances utilize a USGS standard of 35 mg/kg.

Pb concentrations in sediments exceeded the strictest regulatory standard in 19% of samples (Table 4.2.2b). Compared to water samples, sediment samples have a lower exceedance rate, yet sediment concentrations may act as a continued loading source of lead in the environment and degrade water quality. Maximum concentrations were consistently found at Tarapaya station between 2018-2023, while Pilcomayo arriba Tacobamba had the highest detected concentration.

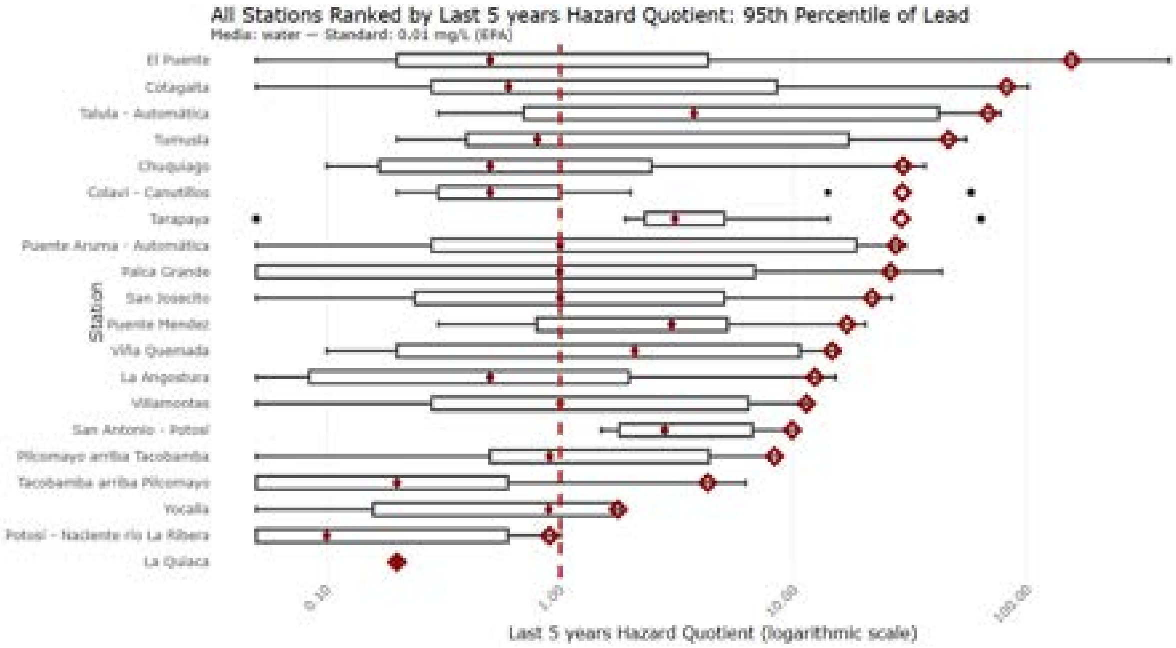

Figure 4.2.2a Top ten lead concentrations in water aggregated by station using the 95th percentile for the most recent 5 years of data. Red line, dots and diamonds signify HQ=1, median, and 95th percentile respectively. X-axis uses a log scale for display.

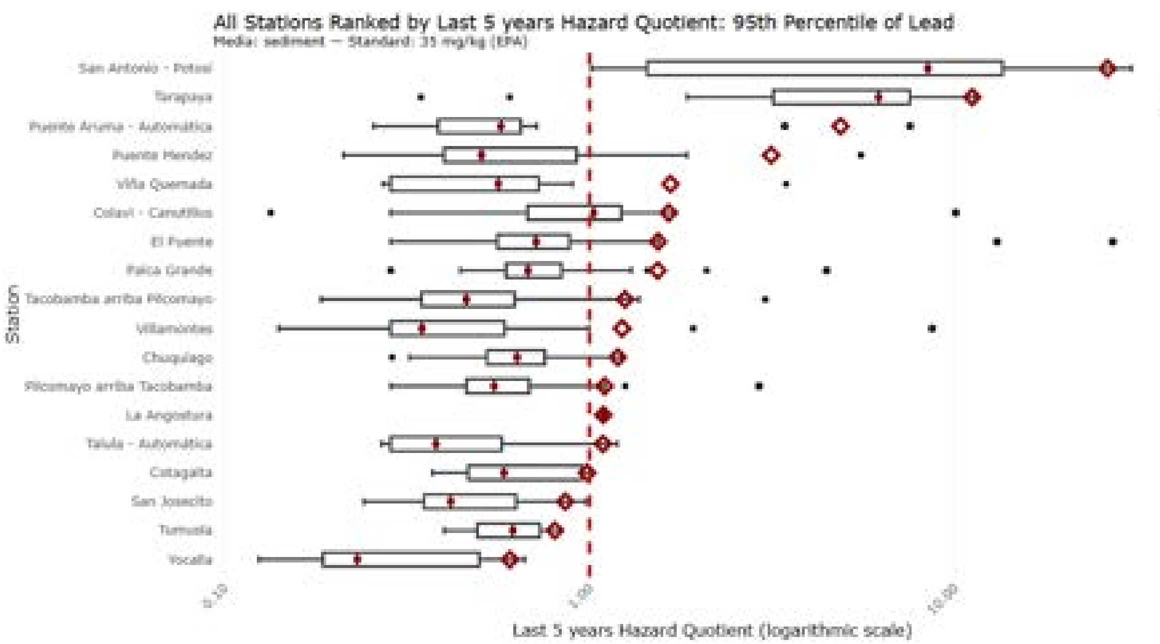

Figure 4.2.2b Top ten lead concentrations in sediments aggregated by station using the 95th percentile for the most recent 5 years of data. Red line, dots and diamonds signify HQ=1, median, and 95th percentile respectively. X-axis uses a log scale for display.

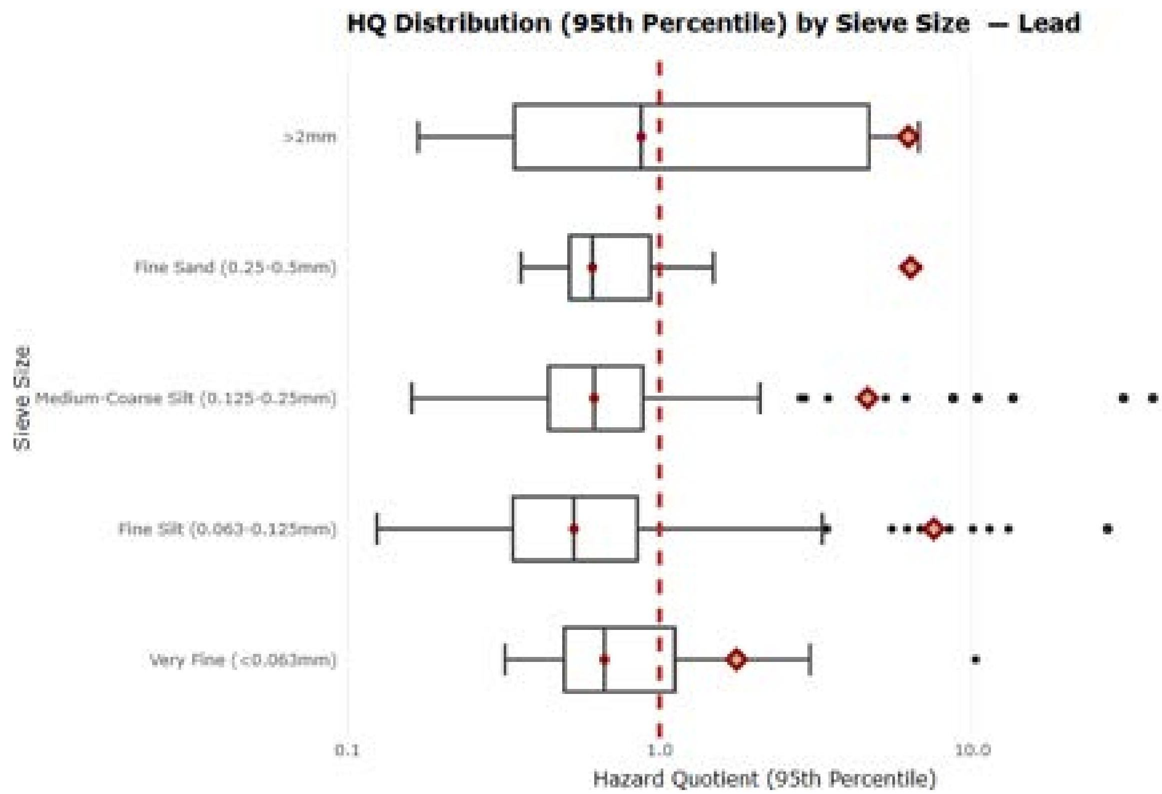

Figure 4.2.2c 95th percentile lead hazard quotients reported within sediments at each station for each sieve size. The red dot signifies median value, red diamond signifies 95th percentile value. The red dotted line signifies where lead levels surpass regulatory standards (HQ=1). X-axis values are untransformed but the scale is logarithmic. Very fine grain sizes were found to have the lowest 95th percentile values, and all medians were found below the HQ=1 limit.

Each station was reviewed to understand general loading patterns of each parameter of interest stated above. Using the 95th percentile method, stations were characterized by their aggregate hazard quotient and plotted (Figures 4.2.2a & 4.2.2b). El Puente had the most extreme reported lead concentration in water, and water-based samples generally experienced HQ values a magnitude higher than for sediment. Only two samples for dissolved lead were reported at La Quiaca, both collected in 2023. San Antonio - Potosí had 100% of lead samples exceed regulatory standards set by the USEPA (35 mg/kg). Lead was mainly sorbed to fine sands (Figure 4.2.2c), yet all grain sizes were found to have unsafe levels of lead, posing a risk mainly to animals and children.

4.2.3 Other Contaminants of Concern

Additional reported contaminants were compared in water and sediments to the strictest regulatory limits found, highlighting several of the top contaminants reported between 2018-2023 (Table 4.2.3a & Table 4.2.3b). All contaminants with determined regulatory limits were found to have concentrations above regulatory limits at one or more stations except for silver. Calcium was least often reported above regulatory limits (n=3, 1.0% of samples) while ammonia was most often above standard limits (n=149, 100%), indicating low dissolved oxygen levels and potential wastewater discharges.

Table 4.2.3a Total percent of water samples exceeding health-based reference standards for parameters with top 15 exceedance rates between 2018-2023.

Parameters	Standard reference limit (μg/L)	Exceedance Rate Across 2018–2023 (%) (n = total)
BOD_5	Bolivian Law: 2,000	100% ($n = 149$)
Ammonia	Bolivian Law: 50	98% ($n = 93$)
Thallium (Tl)	USEPA: 2	85% ($n = 305$)
Iron (Fe)	Bolivian Law: 300	69% ($n = 468$)
Nitrogen	Bolivian Law: 5,000	57% ($n = 91$)
COD	Bolivian Law: 5,000	54% ($n = 156$)
Zinc (Zn)	Bolivian Law: 200	47% ($n = 430$)
Arsenic (As)	USEPA: 10	42% ($n = 429$)
Sulfate (SO_4^{2-})	Bolivian Law: 30,000	39% ($n = 157$)
Manganese (Mn)	WHO: 80	37% ($n = 453$)
Cadmium (Cd)	WHO: 3	33% ($n = 413$)
Copper (Cu)	Bolivian Law: 50	33% ($n = 392$)
Lead (Pb)	USEPA: 10	33% ($n = 378$)
Boron (B)	Bolivian Law: 1,000	31% ($n = 449$)
pH	Bolivian Law: 6–8.5	31% ($n = 164$)

Table 4.2.3b Total percent of sediment samples exceeding health-based reference standards for all parameters with standard references.

Parameters	Standard reference limit (mg/kg)	Exceedance Rate Across 2018–2023 (%) (n = total)
Arsenic (As)	USGS: 5.9	83 ($n = 374$)
Zinc (Zn)	USGS: 123.0	38 ($n = 374$)
Nickel (Ni)	USGS: 18.0	33 ($n = 373$)
Mercury (Hg)	US EPA: 0.13	32 ($n = 341$)
Copper (Cu)	USGS: 35.7	27 ($n = 374$)
Cadmium (Cd)	US EPA: 0.596	26 ($n = 374$)
Lead (Pb)	USEPA: 35.0	22 ($n = 374$)
Chromium (Cr)	USGS: 37.3	4 ($n = 374$)

4.3 Combined Hazard Risk

4.3.1 Interpolated Water Hazard

Hazard scores were aggregated and calculated the 95th percentile value across all water quality parameters across a five-year window (2018-2023; see Section 4.1). The highest risk levels were located at San Antonio - Potosí, Talula - Automatica, and Tarapaya (Figure 4.3.1a & 4.3.1b).

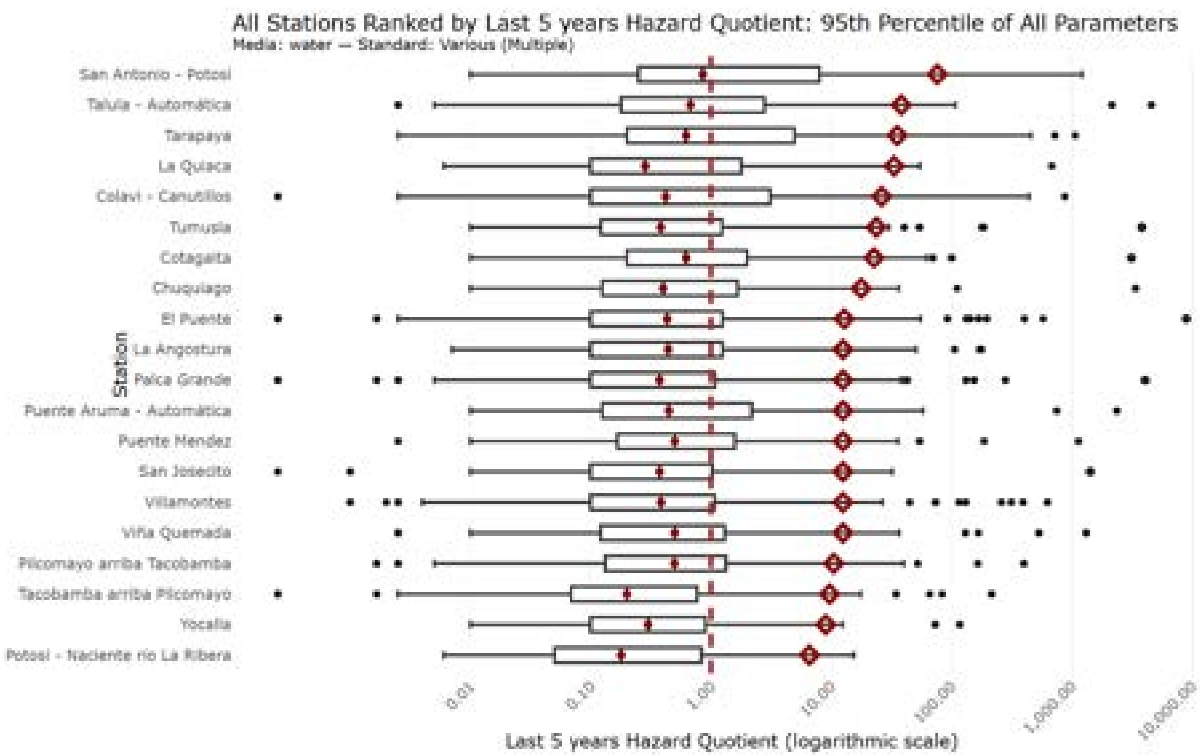

Figure 4.3.1a. All Bolivian stations scored using water quality data for all parameters. Stations are ranked using the 95th percentile for the most recent 5 years of data (2019-2024). Red dot signifies median value, red diamond signifies 95th percentile value. The red line signifies HQ=1, where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic.

After each water station was scored, these station scores were interpolated across the river network, creating a water risk raster, highlighting areas that are more likely to experience high water contamination. Quantiles generated from the scored stations were used to bin the risk raster, categorizing areas into risk classifications from 1 (lower risk) to 5 (higher risk).

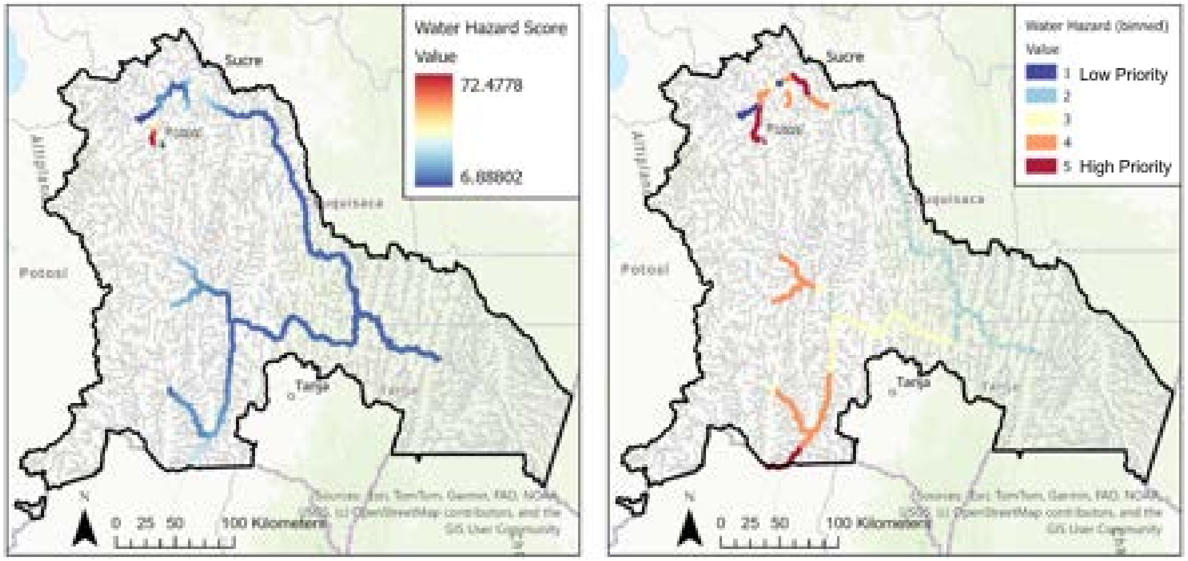

Figure 4.3.1b Water hazard quotients aggregated to the station level and interpolated across the river network (see section 3.3.1). Red indicates high risk of water contamination. The map on the right shows these values binned into 5 categories using quantiles.

4.3.2 Interpolated Sediment Hazard

San Antonio - Potosí and Tarapaya exhibited the most elevated risk scores, which aligns with the water quality data (Figure 4.3.2a). A remediation which allows the local population to utilize the Pilcomayo Basin with limited exposure risk will require the reduction of all contaminants within all media to below regulatory limits.

After each sediment station was scored, these station scores were interpolated across the river network, creating a sediment risk raster, highlighting areas that are more likely to experience high sediment contamination. Quantiles generated from the scored stations were used to bin the risk raster, categorizing areas into risk classifications from 1 (lower risk) to 5 (higher risk).

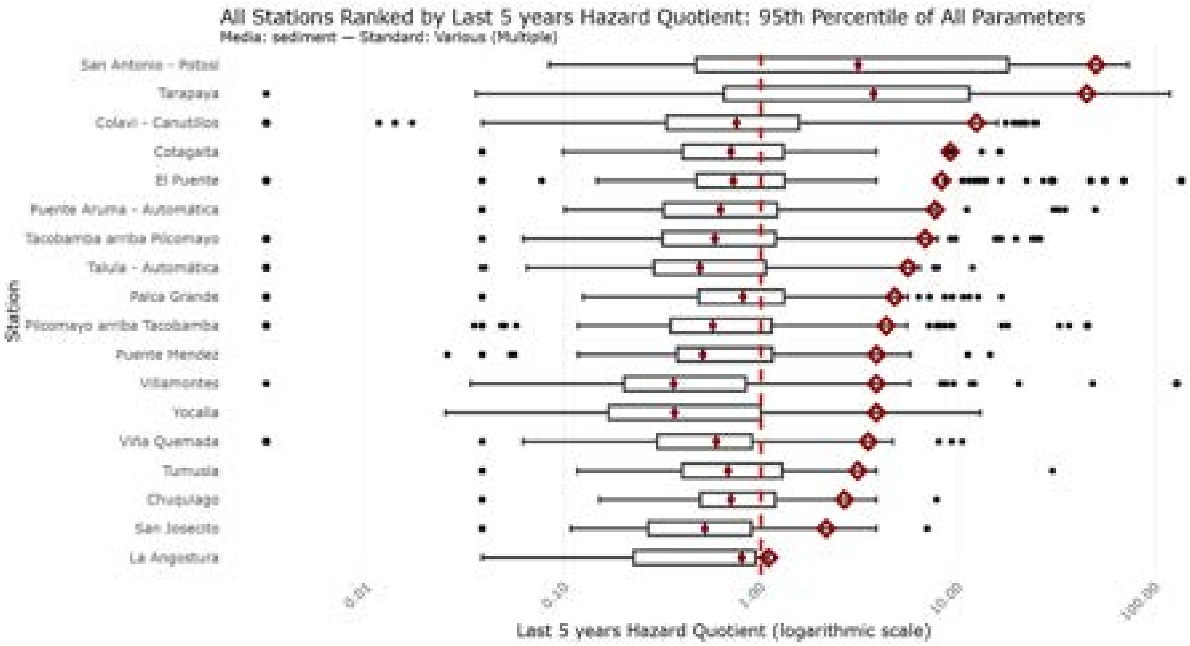

Figure 4.3.2a All Bolivian stations scored using sediment quality data for all parameters. Stations are ranked using the 95th percentile for the most recent 5 years of data (2018-2023). The red dot signifies median value, the red diamond signifies 95th percentile value, and the red line signifies HQ=1 where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic.

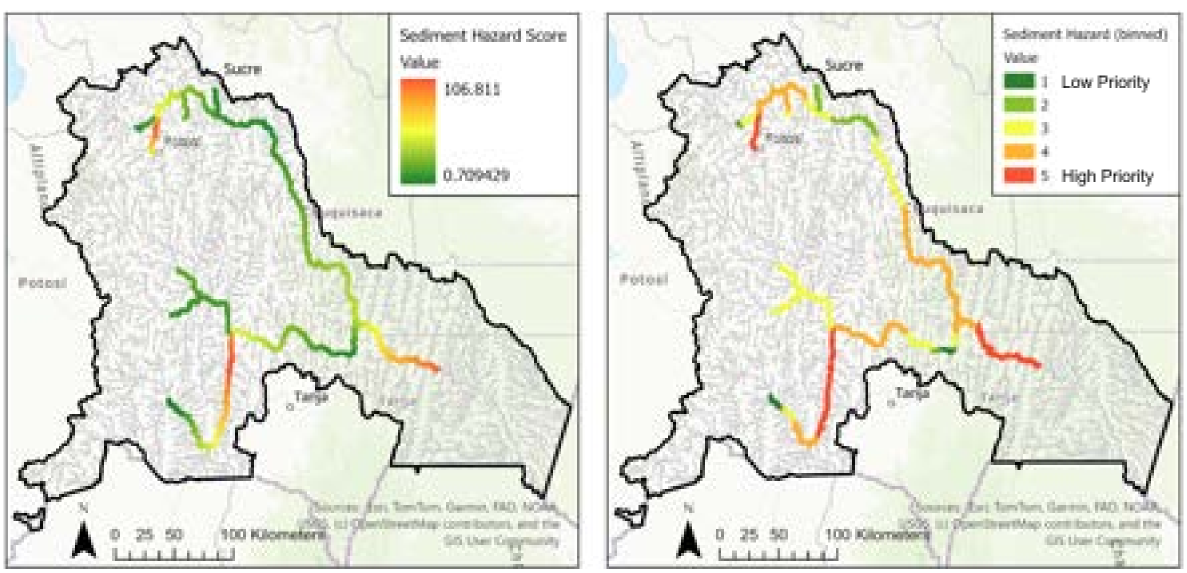

Figure 4.3.2b Sediment hazard quotients aggregated to the station level and interpolated across the river network (see section 3.3.2). Red indicates high risk of sediment contamination. The map on the right shows these values binned into 5 categories using quantiles.

4.3.3 Modeled Air Particulate Hazards

The models of air particulate concentrations predicted the combined impacts of $PM_{2.5}$ and PM_{10} to be most severe in the southwest areas of the city of Potosí and near the eastern boundary of the municipality of Potosí (Figure 4.3.3a). The models predict that the majority of the city of Potosí falls in the range of "Moderate Priority" (aggregated HQ = 2.97 – 5.94). to "Highest Priority" (aggregated HQ > 17.46) (Figure 4.3.3a). When examined separately, PM_{10} concentrations were predicted to be most extreme in the southwest area of Potosí. However, PM_{10} concentrations were predicted to be above WHO standards across the majority of the city of Potosí (See Figure 4.3.3c). $PM_{2.5}$ concentrations in air were predicted to be most extreme along the eastern boundary of the municipality of Potosí. They were predicted to be multiple times above WHO standards in the southwest edge of the city of Potosí, specifically the community of Cantumarca (See 4.3.3c & Figure 4.3.3b). When comparing the risk assessment of each particle diameter to the aggregated risk assessment, our models predict that PM_{10} concentrations contribute the majority of air particle hazards within the city of Potosí. $PM_{2.5}$ concentrations were predicted to be highest outside of the city, on the other hand.

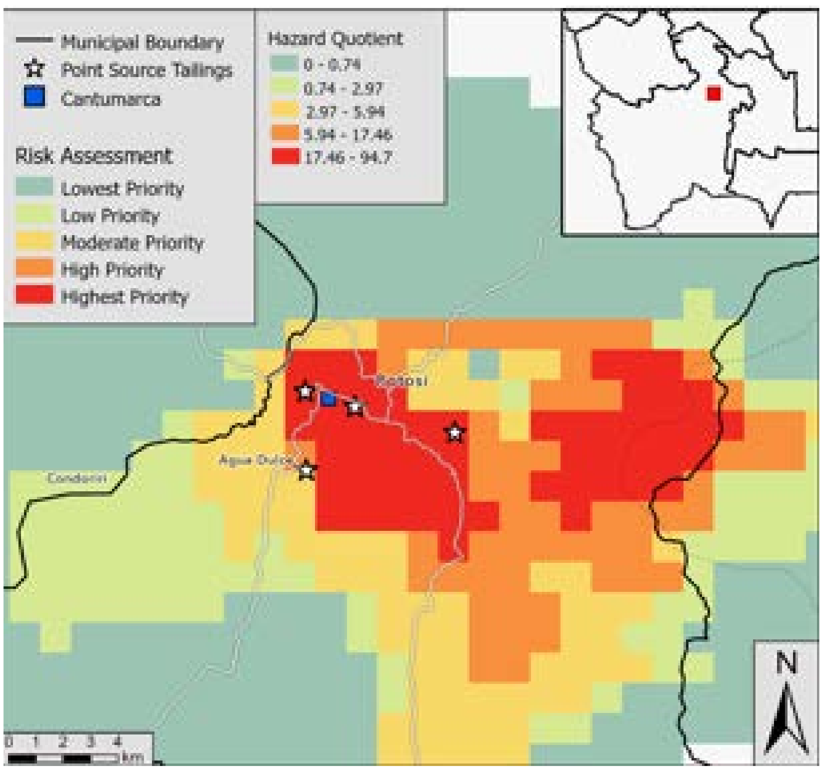

Figure 4.3.3a Risk assessment based on 80th percentile hazard quotients of PM_{10} and $PM_{2.5}$ concentrations in air originating from windblown tailing dust from several tailings in the city of Potosí. The blue square marks the community of Cantumarca. The white stars mark the tailings used as point sources in the model.

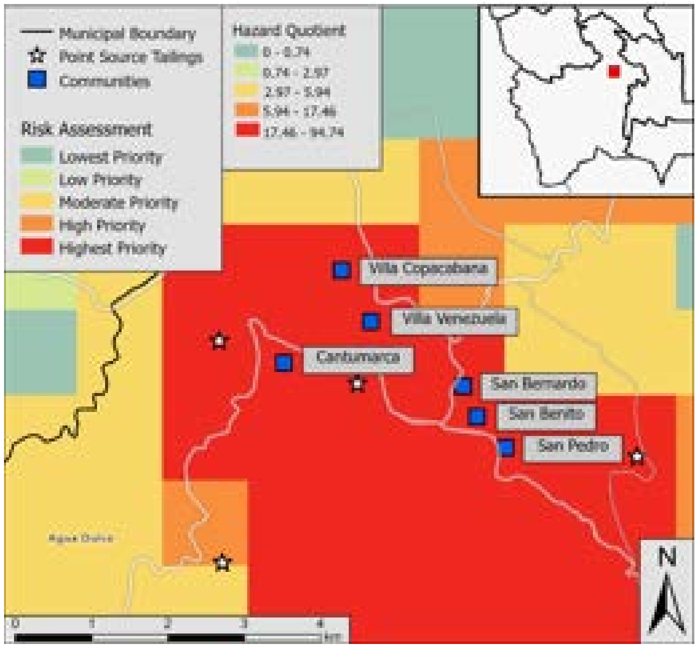

Figure 4.3.3b Risk assessment based on 80th percentile hazard quotients of PM_{10} and $PM_{2.5}$ concentrations in air zoomed into highest risk areas with impacted communities highlighted. Blue squares represent communities within the city of Potosí. White starts mark the tailings used as point sources in models

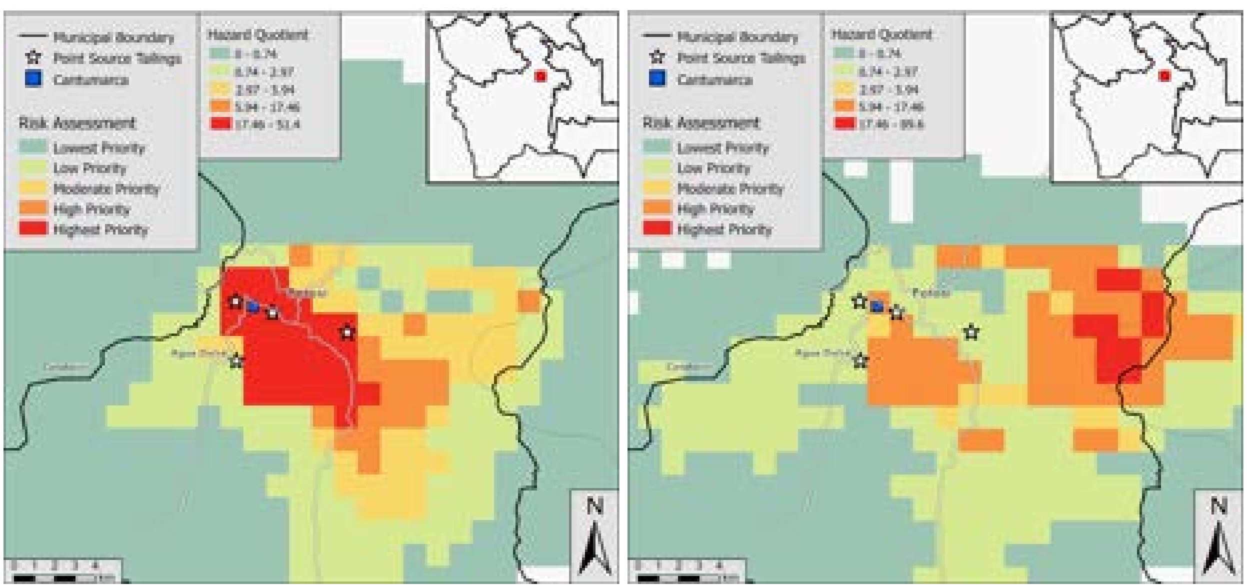

Figure 4.3.3c Risk Assessment based on 80th percentile of PM_{10} [left] and $PM_{2.5}$ [right] concentrations in air originating from windblown tailing dust from several tailings within the city of Potosí. The blue square marks the community of Cantumarca. The white stars mark the tailings used as point sources.

Based on these models, the community of Cantumarca is one of the most impacted from particulates matter in air – experiencing both PM_{10} and $PM_{2.5}$ impacts (Figure 4.3.3c). Other heavily impacted communities include: Villa Copacabana, Villa Venezuela, San Bernardo, San Benito, and San Pedro (Figure 4.3.3b). Our models predict that PM_{10} concentrations are most impactful in these communities.

These results are concerning given the conservative approach considered during modeling. Based on the distribution of emission rates across Bolivia's dry season (See Figure Figure 3.3.3b), these models raise the possibility that particulate matter concentrations exceed 24-hr WHO standards far more than 3 or 4 days a year. Overall, PM_{10} particles remain near the tailing sources, whereas $PM_{2.5}$ particles travel outside of the city. Based on sensitivity analysis testing the HYSPLIT model using other particle diameters, it is likely that concentrations of particles between PM_{10} and $PM_{2.5}$ are highest at a distance in-between the concentration plumes of the two modeled diameters.

It should be noted that wind direction can be highly variable across time. Sensitivity analysis revealed that wind direction in Potosí can even be reversed depending on the date. However, the highest concentration plumes seen when testing multiple dates travelled east. This indicates that the models' estimations may still capture significant trends in the direction and severity of air particulate plumes in the city of Potosí. Overall, the magnitude of concentrations and distance of plumes from tailing sources are the most significant takeaways from these models, with the direction of travel of particles being a secondary takeaway.

Air pollution modeling was performed using four tailings within the city of Potosí as point sources. This gives an idea of how the exposed tailings affect the communities of Potosí. However, this does not account for other sources of particulate matter which pose a significant risk from metal-bearing fine particles, such as: waste rock piles, emissions from uncovered truck beds transporting ores, and mining activities like rock blasting, crushing, and smelting (Figure 4.3.3d; Kasongo et al., 2024).

It is highly likely that the models produced are an underestimation of the true concentration of mining-related particulates containing heavy metals and minerals experienced by the communities of Potosí. This is highly concerning considering that the models predict both $PM_{2.5}$ and PM_{10} concentrations to be in exceedance of WHO standards. An additional cause for concern is that the WHO standards for $PM_{2.5}$ and PM_{10} apply to those particulate size classes, generally, not accounting for the chemical composition of particulate matter. Given that the literature suggests mining-related particulates often contain significant amounts of heavy metals and minerals, the true acceptable level of concentration of these materials in air could be far lower than these WHO standards suggest. Taken together, heavy metal and mineral concentrations in the air of Potosí may be orders of magnitude above what would be considered an acceptable level for human health.

Figure 4.3.3d [Top Left] Example of an uncovered haul truck transporting ore from Zanko et al., 2019. [Top Right] Example of a smelting facility from Bruce McAllister, 1972. [Bottom Left] Example of rock blasting from Tri-State Drilling & Blasting, n.d. [Bottom Right] Uncovered waste rock pile in the city of Potosí captured by our team in September, 2025.

During our analysis, other potential sources of air pollution from mining activities were identified. Figure 4.3.3e identifies mining facilities, including refineries, tailings, and other storage. This map can be a useful starting point for identifying sources of metal and mineral air pollution in the city of Potosí, along with identifying regions at risk of PM_{10} and coarser particle pollution, as they tend to travel between 1 and 10 kilometers from their source (Zeb et al., 2022). Therefore, any areas within a few kilometers of a mining facility, particularly storage and rock blasting, are at a probable risk of impacts from coarse particles. Lung inflammation, asthma, and other respiratory illnesses are typical of coarse particle exposure. These mining facilities may also be sources of pollution to surface waters, groundwater, and other resources (Kasongo et al., 2024). Of particular note, the community of Cantumarca lies between two large tailings and lies adjacent to a cluster of refineries and storage facilities. It is likely this community faces threats from air pollution not captured by the models, particularly $PM_{2.5}$ pollution from smelting activities.

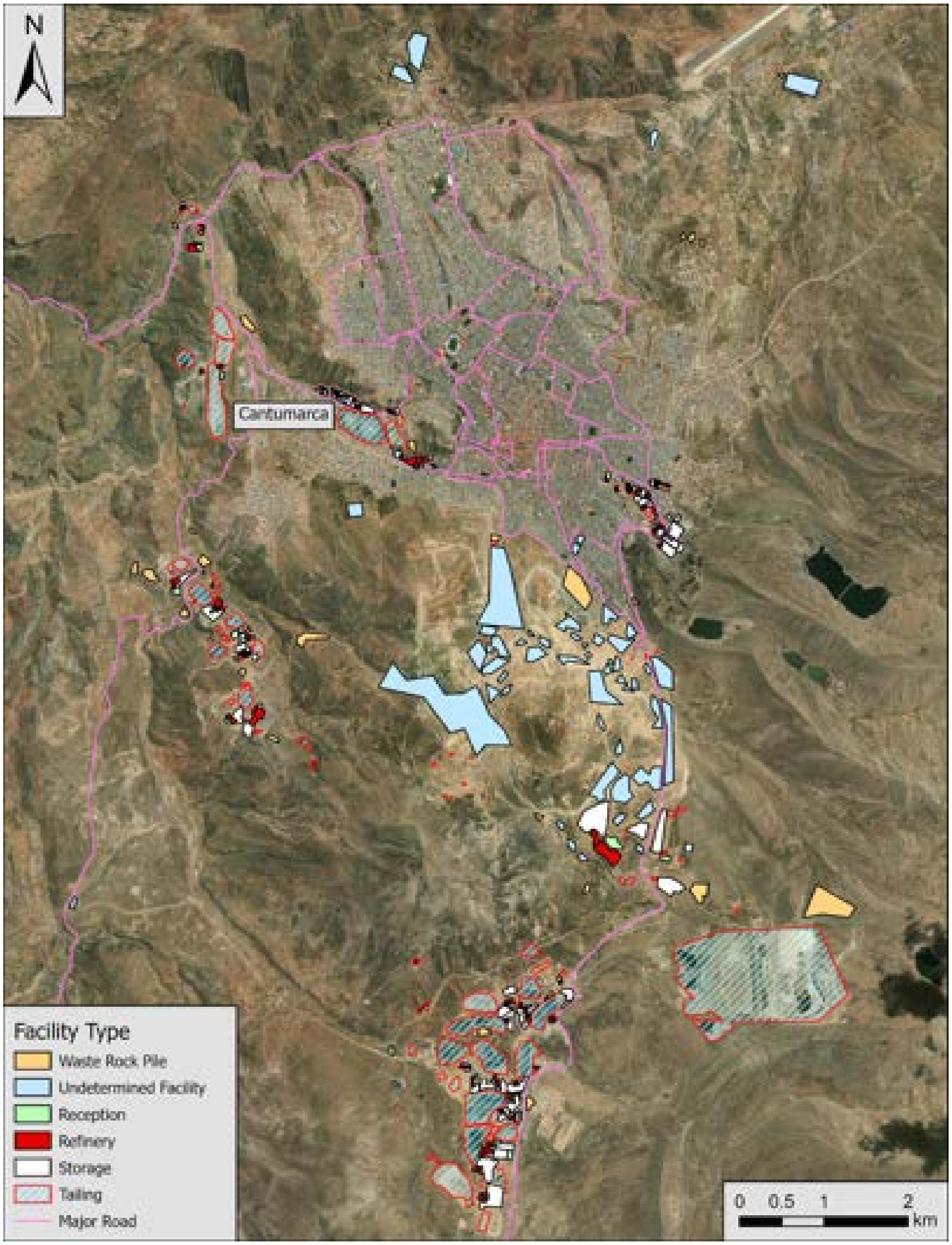

Figure 4.3.3e Map of mining facilities, tailings, and waste rock piles in the city of Potosí.

4.3.4 EJI Results

Results show that Potosí accounts for 87.5% of municipalities in the top 20% of EJI scores, indicating a strong geographic concentration of environmental vulnerability. This pattern is illustrated in Figure 4.3.4a, which highlights a clustering of high-vulnerability municipalities in the northern region of Potosí. The top five ranked municipalities, from highest to lowest EJI score, are:

1. Tacobamba, Cornelio Saavedra (Potosí)
2. Arampampa, General Bernardino Bilbao (Potosí)
3. Tinguipaya, Tomás Frías (Potosí)
4. San Pedro de Macha, Chayanta (Potosí)
5. San Pedro, Charcas (Potosí)

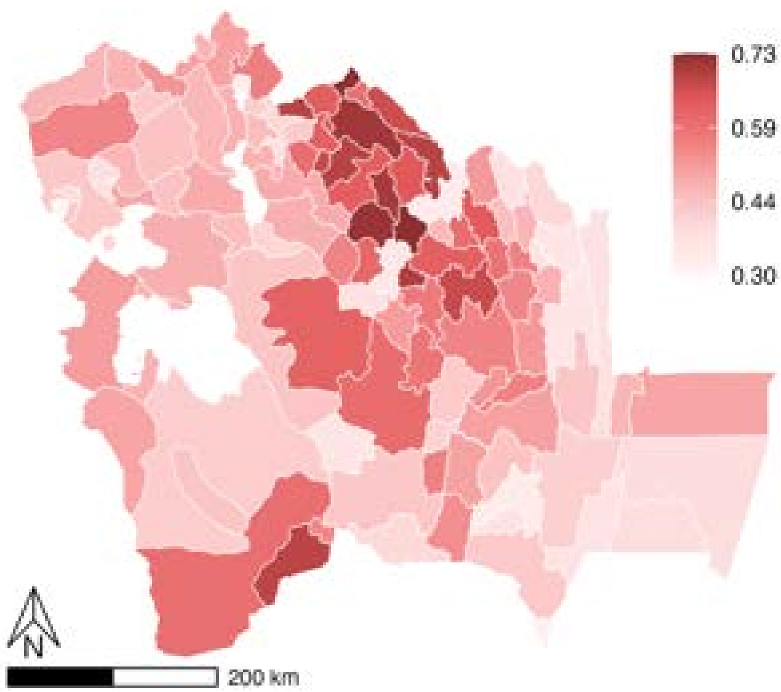

Figure 4.3.4a Environmental Justice Index (EJI) scores by municipality for river basin departments: Chuquisaca, Oruro, Potosí, and Tarija. The index is a composite score of health, environmental, and socioeconomic indicators with higher values and darker shades representing greater vulnerability. White areas indicate municipalities with missing or unavailable data.

Given this concentration, targeted interventions and further study in the department of Potosí may help better address and support the municipalities most at risk. To better understand underlying drivers, composite EJI scores can be further contextualized using module-specific scores, which help identify how different dimensions of vulnerability vary spatially.

Figure 4.3.4bA presents health vulnerability scores across the basin. Moderate to high health scores are observed throughout the central basin, spanning both the northern and southern regions of Potosí. Analysis of specific health indicators reveals that five municipalities in Potosí exhibit high proportions of deaths under 50 years. These municipalities are Chuquihuta Ayllu Jucumani (49%), Colquechaca (45%), Tinguipaya (43%), Atocha (40%), and San Pedro de Macha (40%). Conversely, several municipalities in Oruro exhibit relatively high proportions of residents without health insurance, including Esmerelda (45%), La Rivera (41%), Sabaya (39%), Todos Santos (38%), Yunguyo del Litoral (38%).

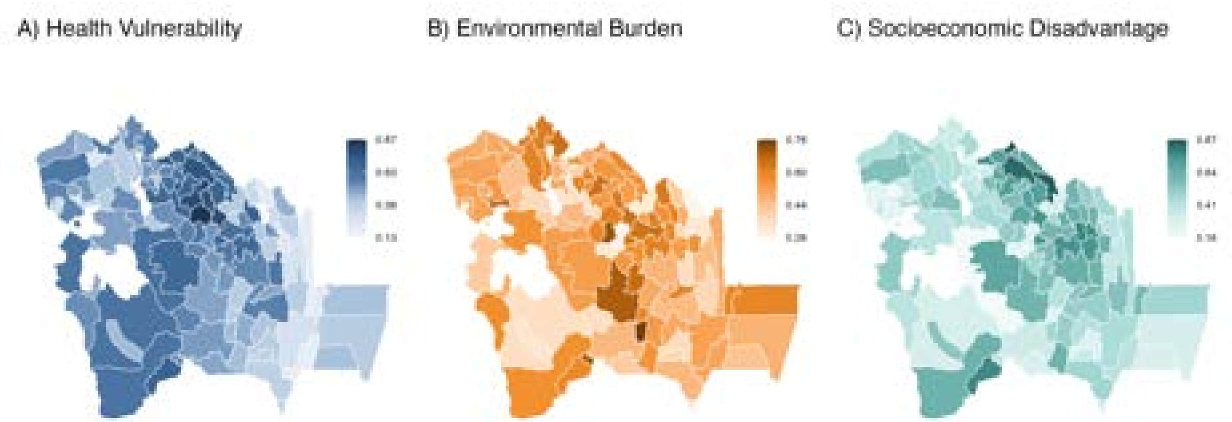

Figure 4.3.4b Component indices contributing to the composite EJI. (A) Health Vulnerability Index; (B) Environmental Burden Index; (C) Socioeconomic Disadvantage Index. Darker shades indicate higher values, corresponding to greater vulnerability. White areas indicate municipalities with missing or unavailable data.

Figure 4.3.4bB reflects environmental burden, with higher scores clustering in the central basin spanning Chuquisaca, Potosí, and Oruro. Analysis of specific indicators highlights localized exposures. Porco and Atocha (Potosí) exhibit the highest proportions of residents employed in mining, while Arampampa and San Pedro report high participation in agricultural production (67% and 64%, respectively), with roughly 45% of participants producing at least some crops for household consumption. More than half of households in El Choro and Tacobamba rely on unprotected water sources (e.g., wells, rivers, ditches, or rainfall), affecting approximately 74% of households in El Choro and 51% in Tacobamba.

Figure 4.3.4bC reports the distribution of socioeconomic disadvantage across the basin, with higher scores concentrated primarily in municipalities in the northern regions of Potosí and Chuquisaca. Notably, Arampampa, San Pedro, and San Pedro de Macha — all ranked among the top five municipalities in the composite EJI — also rank among the five highest in socioeconomic vulnerability, indicating that socioeconomic conditions are a key driver of overall environmental justice risk in these areas. Across municipalities with the highest EJI scores, more than 80% of residents identify as Indigenous. Although not included in the index, adult illiteracy exceeds 18% in four of the five municipalities (excluding Arampampa), with the highest rates observed in Ravelo (28%) and Ocurí (25%). In addition, more than 37% of households in these municipalities lack internet access, including San Pedro (46%), Arampampa (42%), and Tacobamba and Tinguipaya (both 41%).

These results illustrate the cumulative impacts of EJ vulnerability across the Pilcomayo River Basin. High EJI municipalities generally experience elevated vulnerability across multiple domains, though no single threshold defines what constitutes "high" vulnerability. The composite EJI reflects relative patterns across municipalities and may not fully capture specific facets of local context or lived experience. Nevertheless, these findings identify priority areas of concern and provide a framework for more detailed, targeted analyses in the future.

In particular, additional analysis of health vulnerability is needed. The current assessment relies on self-reported data and does not include information on chronic health conditions or the locations and capacities of healthcare facilities. Future work should explore where healthcare services are located, what resources they provide, and whether they adequately serve the most vulnerable municipalities.

4.3.5 Composite Hazard Ranking

To identify regions of high risk to human health from mining-related contamination in the Pilcomayo River, a qualitative risk categorization of areas along affected rivers was developed. Areas which had multiple overlapping risk categories were considered priority regions for immediate remediation, and are recommended for priority resource allocation to initially focus on these areas (Figure 4.3.5a).

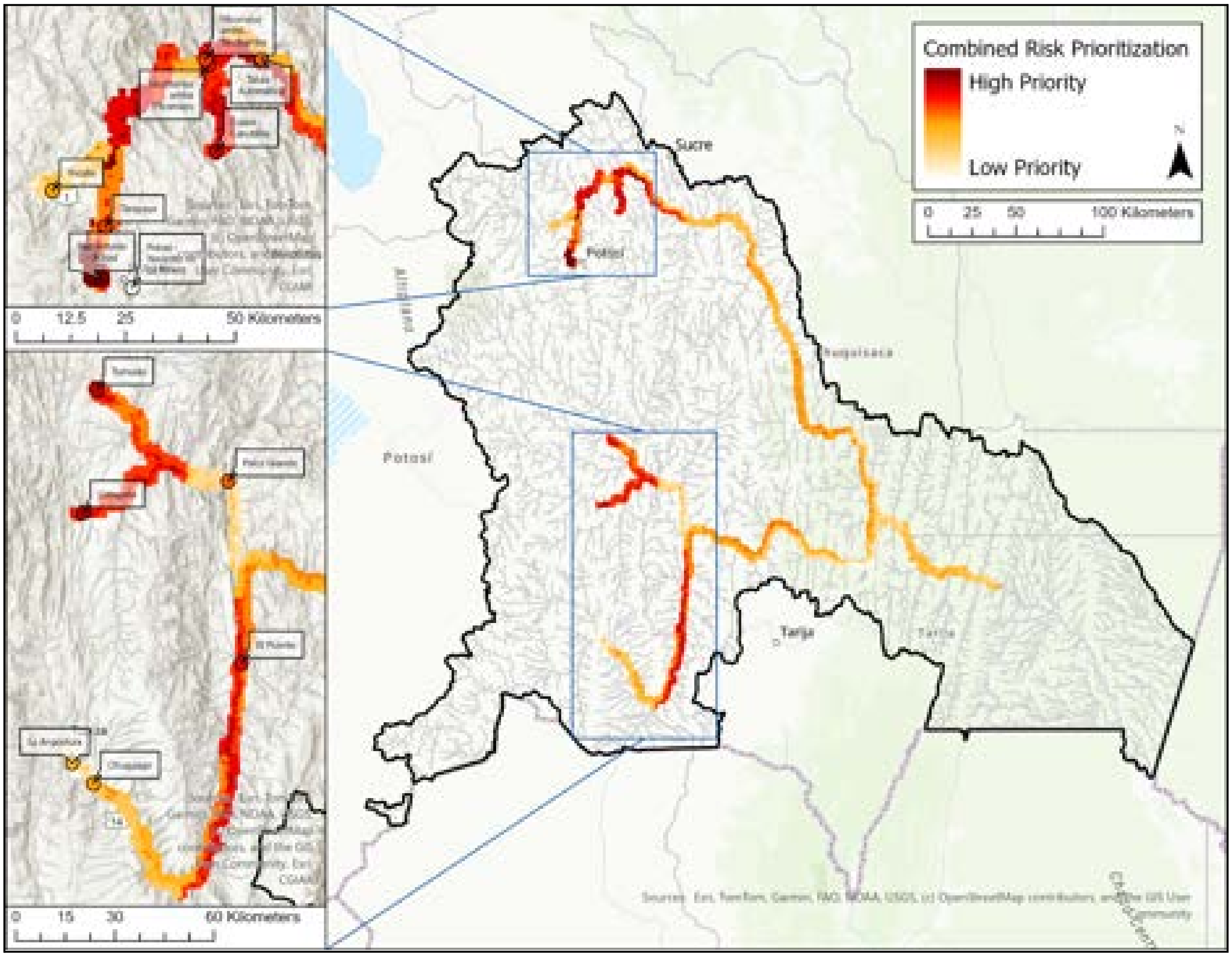

Figure 4.3.5a Combined hazard prioritization map. Created by summing four binned input layers: water risk, sediment risk, EJI vulnerability, and population density. Red indicates areas where these layers overlap, resulting in a high priority for remediation.

Areas in the Potosí-Tarapaya region ranked in the highest priority category for both sediment and water contamination. Areas along the Rio Blanca near Cotagaita, the Rio Tumusla near Tumusla, and the Rio Pilcomayo near Talula ranked highly for water-based risk. Areas near the Rio San Juan del Oro by El Puente and the Pilcomayo River by Villamontes ranked highly for sediment contamination. When considering the combined impact of water, sediment, vulnerability, and population, the areas of greatest concern are those in the Upper Pilcomayo region, the Potosí area, along the rivers near Cotagaita and Tumusla, and along the Rio San Juan del Oro near El Puente.

4.4 Contaminant Migration Analysis

4.4.1 Seasonality Exchange Cycle

As flow velocity decreases in the dry season, dissolved metals adsorb to fine particles and settle into the riverbed, creating persistent contamination reservoirs while water column concentrations decline. This inverse relationship between aqueous and solid phases is visible across all three basin regions in Figure 4.4.1a. In the water column, wet-season median concentrations are substantially higher than dry-season values for most metals and regions, for example, median arsenic at Potosí drops from 47.5 µg/L in the wet season to 15.0 µg/L in the dry season, and cadmium follows a similar pattern (9.25 to 2.5 µg/L). Lead shows the most pronounced seasonal contrast basin-wide, falling from 292.0 µg/L in the wet season to 10.0 µg/L in the dry season at Argentina/Paraguay stations, consistent with its high sediment affinity and strong partitioning ratio. Wet-to-dry season concentration ratios in the water column range from approximately 3× for arsenic at Potosí to nearly 30× for lead and cadmium at Argentina/Paraguay stations, reflecting both the strength of wet-season mobilization and the pronounced dry-season settling of more strongly partitioning metals at distal locations.

The sediment phase shows a more muted and metal-specific seasonal response. The clearest dry-season enrichment is in zinc at Potosí, where median sediment concentrations nearly double from 88.8 mg/kg in the wet season to 179.0 mg/kg in the dry season. Arsenic and lead show modest dry-season increases at Potosí (16.5 to 18.4 mg/kg and 19.1 to 21.5 mg/kg respectively), while cadmium sediment concentrations remain essentially stable across seasons basin-wide. This metal-specific pattern is consistent with the partitioning analysis: zinc's promiscuous multi-mechanism binding means it accumulates readily in sediment under low-flow conditions, whereas cadmium's reliance on acid-soluble calcium-phosphate minerals limits dry-season accumulation. These dynamics are further examined by texture class in section 4.4.2.

As shown in Figure 4.4.1b, arsenic concentrations decline sharply with distance from the Potosí headwaters, with source-proximal stations recording median values more than an order of magnitude higher than distal stations at the Argentina/Paraguay border. This gradient confirms that dilution and sedimentation attenuate peak contamination loads along the 800 km transport corridor, while the persistent non-zero concentrations at downstream stations confirm that contamination reaches transboundary waters throughout the monitoring record.

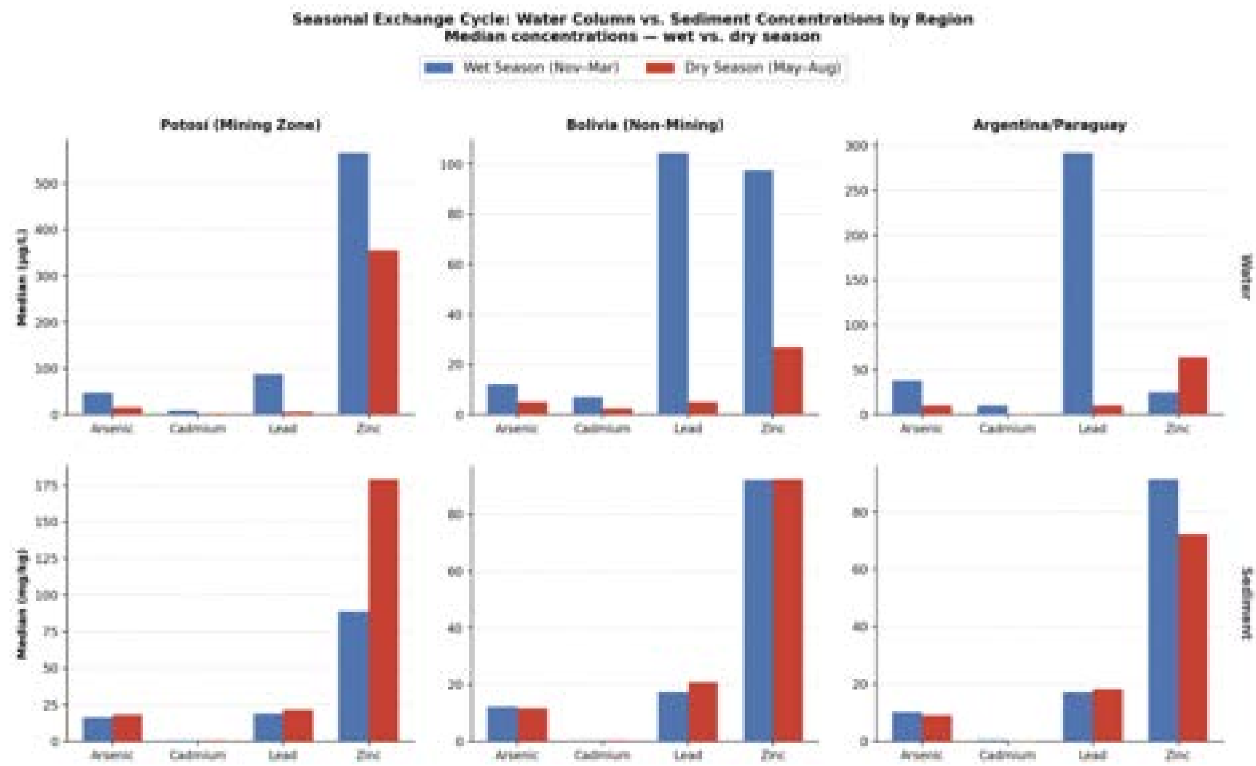

Figure 4.4.1a Median concentrations of arsenic, cadmium, lead, and zinc in the water column (µg/L, top row) and sediment (mg/kg, bottom row) during wet and dry seasons across three basin regions. The inverse pattern between rows, higher water concentrations in the wet season, higher sediment concentrations in the dry season, illustrates the seasonal mobilization and settling exchange cycle. Sample sizes range from n = 21–76 per water group and n = 40–175 per sediment group.

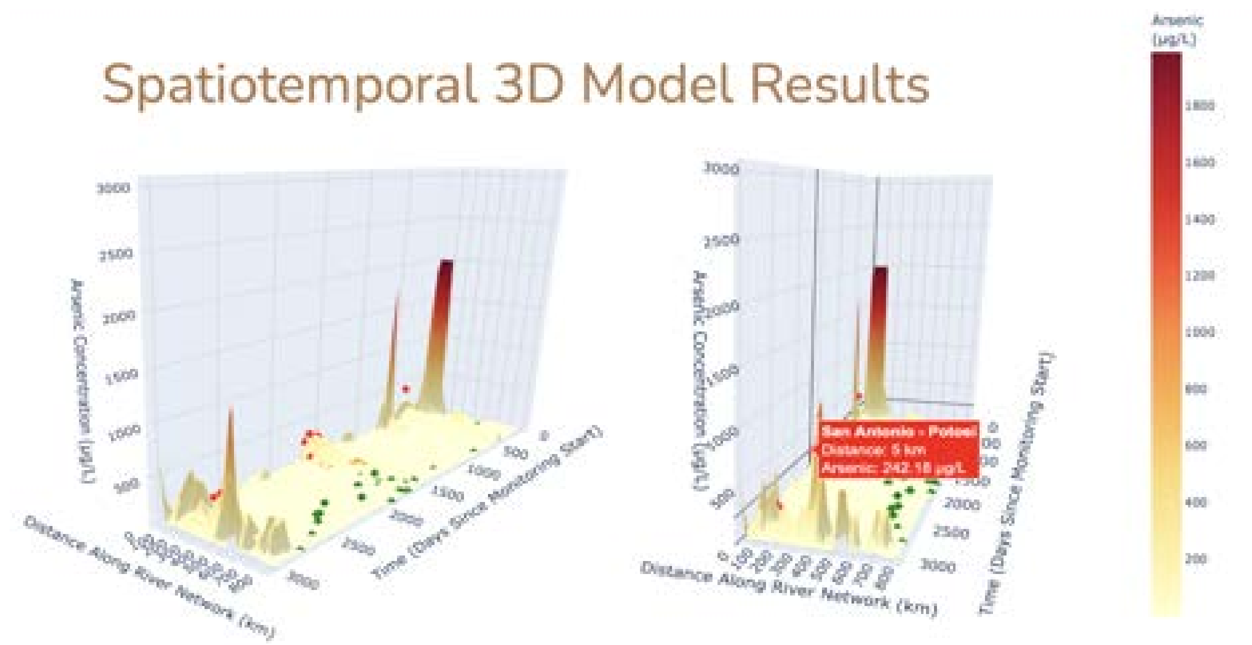

Figure 4.4.1b Three-dimensional surface of arsenic concentrations (µg/L) across the 800 km longitudinal gradient from Potosí headwaters to the Argentina/Paraguay border over the eight-year monitoring period. Diamonds indicate sampling locations colored by region (red = Potosí mining zone, orange = Bolivia non-mining, green = Argentina/Paraguay). Note: the surface represents concurrent spatial measurements, not downstream parcel tracking.

4.4.2 Potosí-zone Sediment Characteristics

Detailed characterization of sediment texture in the Potosí mining zone reveals that mining-derived loading is pervasive enough to override natural textural sorting effects. In typical fluvial systems, finer-textured sediments accumulate significantly more metals due to their greater surface area and cation exchange capacity; however, arsenic medians in this basin remained relatively consistent across all five texture classes, ranging only from 15.0 to 20.7 mg/kg. This narrow range indicates that the sheer magnitude of mining discharge saturates all available sorption sites regardless of grain size, from coarse sand to silty loam.

The dry season further concentrates metals in sediment due to reduced stream discharge and lower dilution of mining effluent, which is the inverse of water-phase behavior, in which high wet-season flow mobilizes metals into the water column. As shown in Table 4.4.2a, this seasonal mechanism produces substantial dry-season enrichment across texture classes, most notably a

157% increase in cadmium in sandy loam and a 531% spike in arsenic in silty loam, though the latter is based on n = 14 dry-season observations and should be interpreted with caution.

Table 4.4.2a Wet and dry season median sediment concentrations (mg/kg) by texture class for priority metals in the Potosí mining zone, with percent change from wet to dry season.

Metal	Texture Class	Wet Median (mg/kg)	Dry Median (mg/kg)	Change
Arsenic	Sand	17.3	22.1	+28%
	Sandy Loam	14.2	17.3	+22%
	Silty Loam	14.0	88.4	+531% [a]
Cadmium	Sand	0.22	0.42	+91%
	Sandy Loam	0.21	0.54	+157%
Lead	Sand	20.6	22.3	+8%
	Sandy Loam	20.7	18.8	−9%
	Silty Loam	11.0	38.5	+250%
Mercury	Sand	0.11	0.21	+91%
	Sandy Loam	0.00 [b]	0.29	—

[a] $n = 14$ dry-season observations; interpret with caution.
[b] Below detection limit; value represents $\frac{1}{2}\times$ detection limit substitution.

The eight-year monitoring record reveals a shift in sediment texture composition across the study period, as shown in Figure 4.4.2a. Sand dominated early campaigns from 2016 to 2018, while sandy loam became the modal texture class from 2019 onward. The drivers of this compositional shift are not fully resolved; it may reflect changes in sampling coverage, interannual hydrological variability affecting sediment transport, or a combination of both. Regardless of cause, a move toward finer sediment textures in later campaigns could independently influence metal retention capacity and should be considered when interpreting long-term concentration trends.

Despite this textural shift, the most striking pattern in the long-term record is the emergence of mercury and cadmium contamination that cannot be explained by geomorphic change alone (Figure 4.4.2b). Mercury concentrations, which remained near zero through 2019, escalated rapidly to 1.3 mg/kg during the 2021 dry season before reaching a record high of 2.3 mg/kg in the 2024 wet season, more than thirteen times the USEPA freshwater sediment guideline of 0.174 mg/kg. Simultaneously, cadmium spiked to 5.0 mg/kg in 2024, the highest value in the eight-year record. The synchronous nature of these two spikes, occurring in metals with distinct geochemical behaviors, suggests a discrete upstream source change rather than gradual accumulation, potentially tied to shifts in ore processing methods or the introduction of new discharge points in the headwaters.

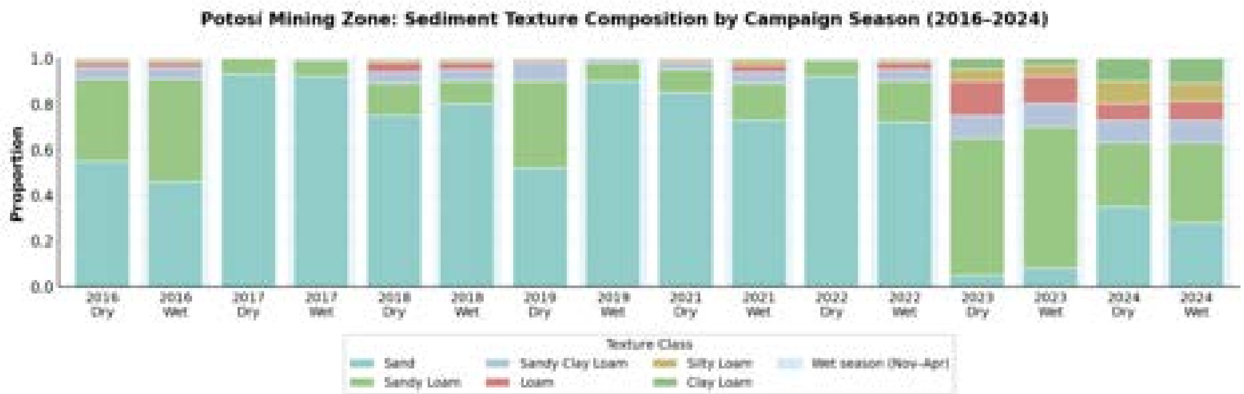

Figure 4.4.2a Stacked bars show the proportional abundance of five texture classes across 16 sampling campaigns. Blue shading indicates wet season campaigns (November–April).

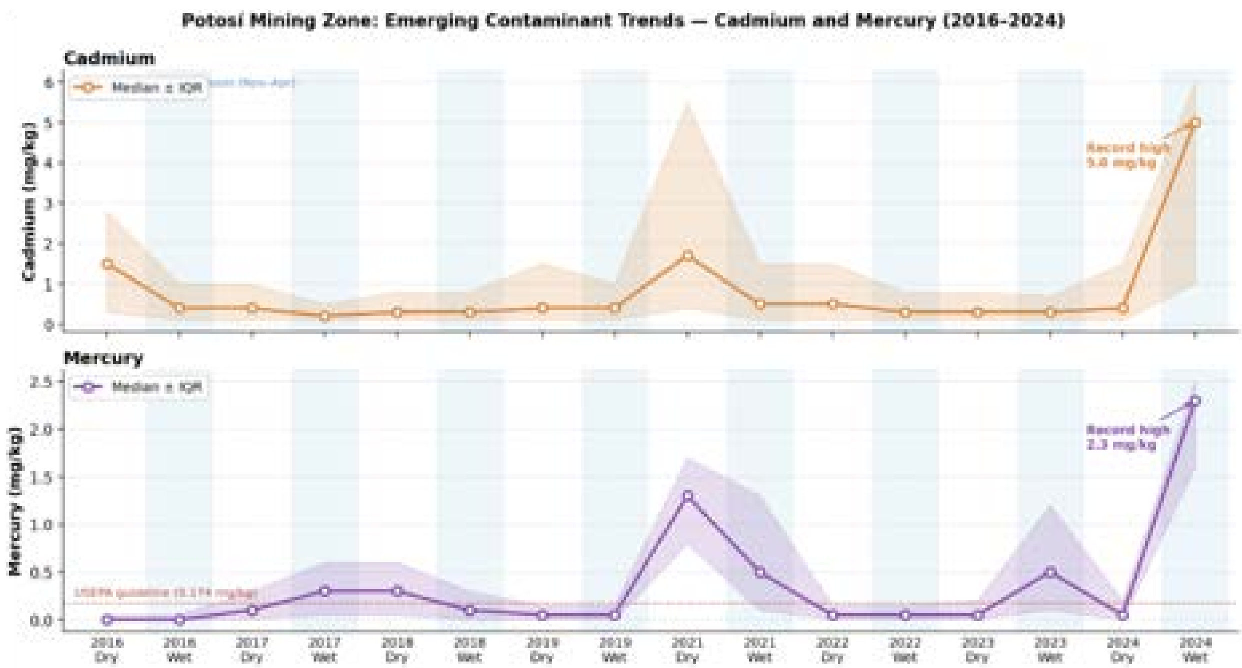

Figure 4.4.2b Basin-wide median concentrations (mg/kg) with interquartile range (shaded band) across all Potosí stations per campaign. The dotted red line in the mercury panel indicates the USEPA freshwater sediment quality guideline (0.174 mg/kg). Blue shading indicates wet season campaigns.

4.4.3 Distribution and Retention Hierarchy

The magnitude of sediment-water partitioning across the Pilcomayo Basin reflects a distinct hierarchy of metal mobility, with median K_d values spanning nearly an order of magnitude between the most and least sequestered elements. Lead (Pb) exhibits the highest affinity for the solid phase (median K_d: 1,667 L/kg), suggesting robust immobilization under prevailing basin conditions. Arsenic (As) and Zinc (Zn) show moderate-to-high retention (K_d: 908 L/kg and 664 L/kg, respectively), while Cadmium (Cd) remains significantly more labile (median K_d: 200 L/kg), representing the greatest risk for long-distance downstream transport. As shown in Table 4.4.3a, the wide ranges and high standard deviations for these ratios—particularly for lead and zinc—are indicative of the basin's dramatic environmental gradients, where partitioning efficiency fluctuates based on local water chemistry.

Table 4.4.3a Site-specific sediment-water distribution coefficients derived from matched field samples (2016–2024).

Metal	n	Median K_d (L/kg)	Mean K_d (L/kg)	Std. Dev.
Lead (Pb)	265	1,667	11,294	47,612
Arsenic (As)	265	908	4,244	15,423
Zinc (Zn)	265	664	12,260	84,645
Cadmium (Cd)	264	200	3,179	15,117

Note: The large discrepancy between median and mean K_d values reflects the highly right-skewed distribution of field-derived partitioning ratios, driven by extreme values at source-proximal mining stations. The median is the preferred measure of central tendency for this dataset.

K_d values were calculated across all contaminants within the Trinational Commission monitoring dataset using both suspended and deposited sediment fractions (Figure 4.4.3a). Median K_d values exceeded 1 L/kg for the priority metals analyzed here, confirming that the majority of contaminant mass partitions out of the dissolved water column. Notably, K_d medians derived from deposited bed sediments were substantially higher than those from suspended sediments. This difference reflects the reduced metal-water exchange capacity of settled sediments: once deposited, metals bound to bed sediments are largely immobile and do not readily exchange with the water column flowing above, whereas metals associated with suspended particles remain in active contact with the dissolved phase and are more susceptible to desorption and continued downstream transport. This distinction has direct implications for interpreting long-distance transport risk, particularly for cadmium, whose lower overall K_d suggests greater potential for mobilization relative to lead and arsenic under changing hydrological conditions.

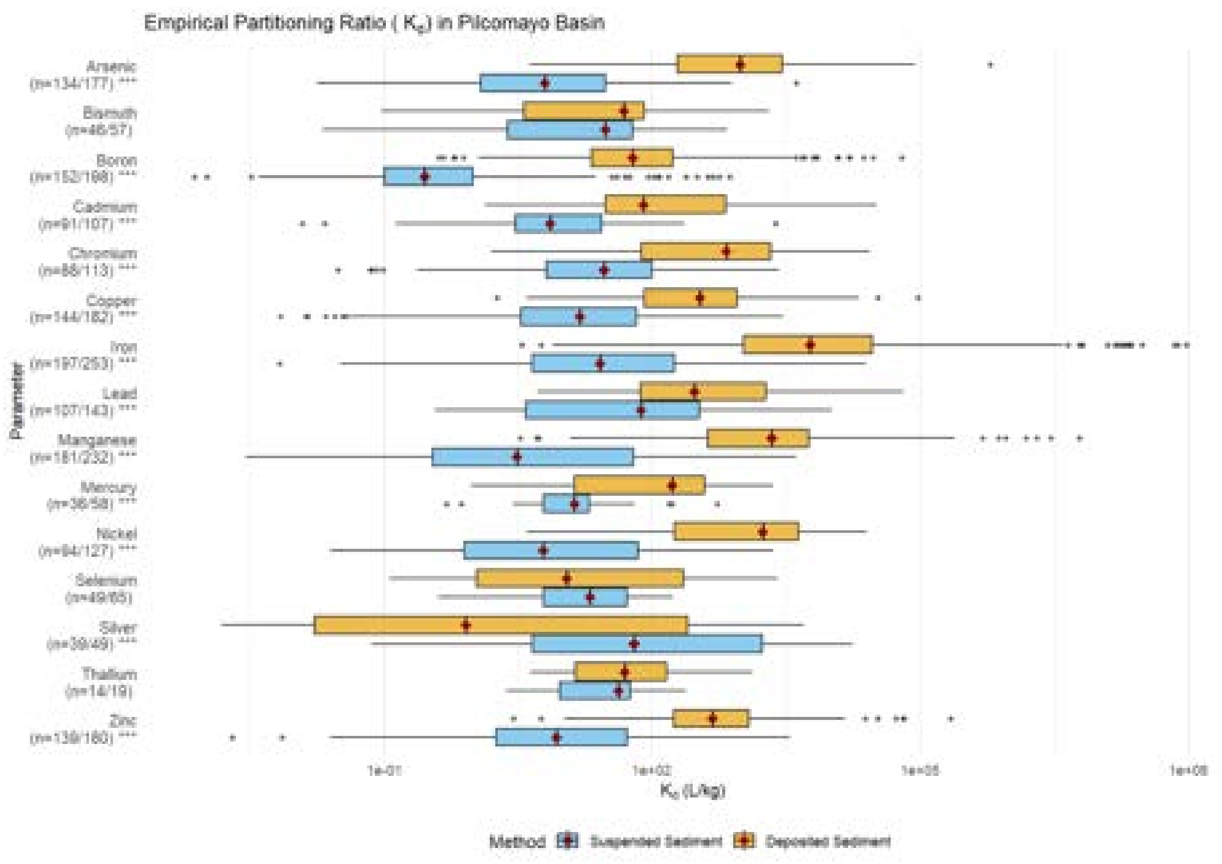

Figure 4.4.3a Distributions of empirical partitioning ratios for each pollutant based on sediments suspended in the water column versus sediments deposited in the river basin. Sample sizes describe the total pairs utilized (format: n=[deposited pairs / suspended pairs]). Medians signified using a red dot. X-axis values are untransformed but the scale is logarithmic. Asterisks represent statistical significance in difference between method values (***: p<0.001).

4.4.4 Mechanistic Verification of Sediment-Water Partitioning

To identify the specific chemical processes driving these partitioning patterns, a mechanistic verification was conducted using Spearman rank correlation between partitioning ratios and sediment mineral composition (Fe, Mn, Al, Ca, P, S) and water quality parameters (pH, DO), distinguishing theoretical geochemical models from actual field behavior.

The impacts of pH, clay content, and organic matter on metal partitioning were first evaluated using Spearman rank correlation (Figure 4.4.4a), before dissolved oxygen was incorporated into a comprehensive assessment of all physicochemical controls (Figure 4.4.4b). The full heatmap reveals that pH is the dominant control, with three of the four metals showing significant negative correlations reflecting the mobilizing effect of acidic mining-zone conditions on sediment-bound metals. Dissolved oxygen, which governs the redox state of the system, operates as a consistent secondary driver across all four metals, with significant positive correlations confirming that the basin's predominantly oxic conditions (98.5% of sampling events with DO >4 mg/L) are essential for maintaining the stability of the oxide minerals that sequester these contaminants. Clay and organic matter play secondary, metal-specific roles that are discussed in the mineral-phase analysis below.

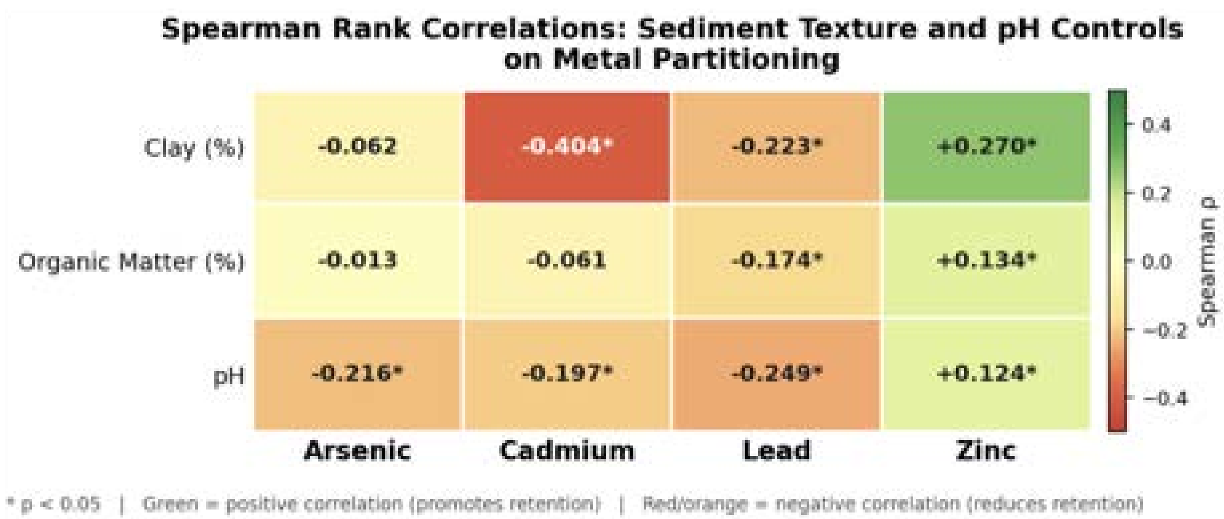

Figure 4.4.4a Spearman ρ values for clay content, organic matter, and pH across all four metals (n = 255–265 per metal). Asterisks indicate $p < 0.05$. Green = positive correlation (higher parameter associated with better sediment retention); red/orange = negative correlation (higher parameter associated with worse retention).

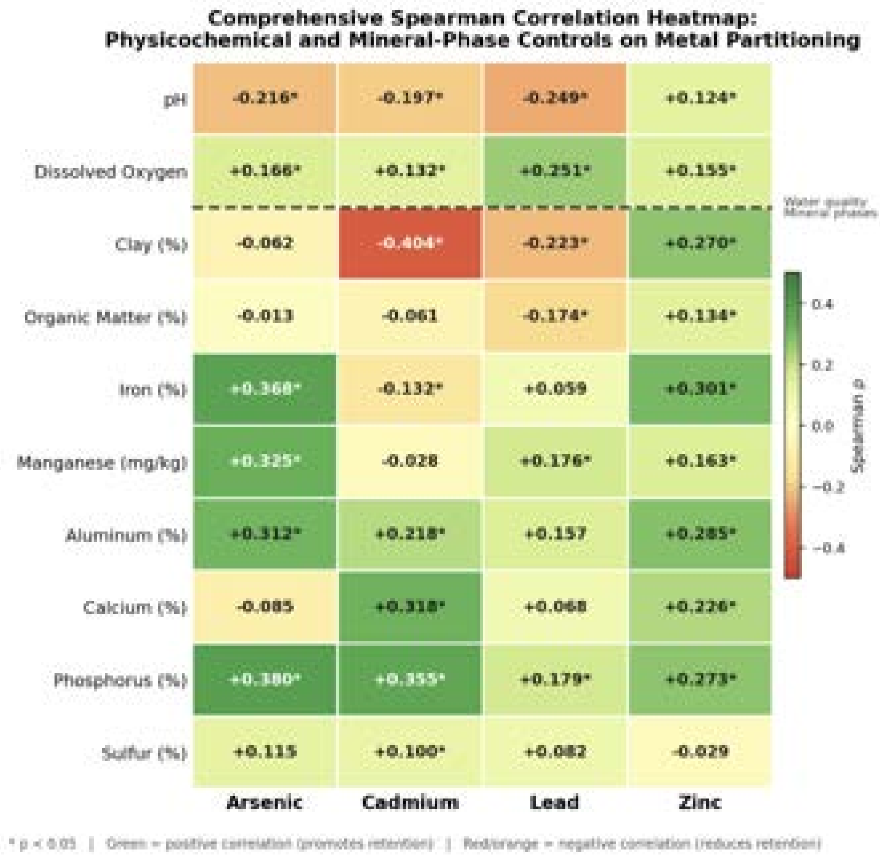

Figure 4.4.4b Spearman ρ values quantifying the monotonic relationships between metal partitioning (K_d, L/kg) and physicochemical parameters (pH, dissolved oxygen, clay content, organic matter) across all four metals (n = 255–265 per metal). Color scale reflects the direction and magnitude of the correlation: green indicates positive ρ (higher parameter value associated with greater sediment retention); red/orange indicates negative ρ (higher parameter value associated with reduced retention and increased aqueous mobility); yellow/neutral indicates no meaningful correlation ($\rho \approx 0$). Color intensity scales with the absolute magnitude of ρ. Asterisks (*) denote statistical significance at $p < 0.05$.

4.4.5 Metal-Specific Drivers of Retention

Whereas the heatmap (Figure 4.4.4b) identifies the environmental conditions governing partitioning, Figure 4.4.5a resolves the underlying mineral-phase controls for each metal individually, revealing which specific sediment constituents (iron oxides, aluminum hydroxides, calcium minerals, or phosphate phases) drive retention. This distinction is important because two metals may respond similarly to pH or DO yet rely on fundamentally different binding substrates, with direct implications for how contamination responds to geochemical disturbance. While pH and DO were examined as universal environmental controls in the preceding section, they are retained in Figure 4.4.5a to contextualize their relative effect sizes against the mineral-phase variables, allowing direct comparison of geochemical and environmental drivers for each metal in a single view. The following subsections examine each metal in turn, moving from the most oxide-dependent to the most geochemically anomalous retention pathway identified in this dataset.

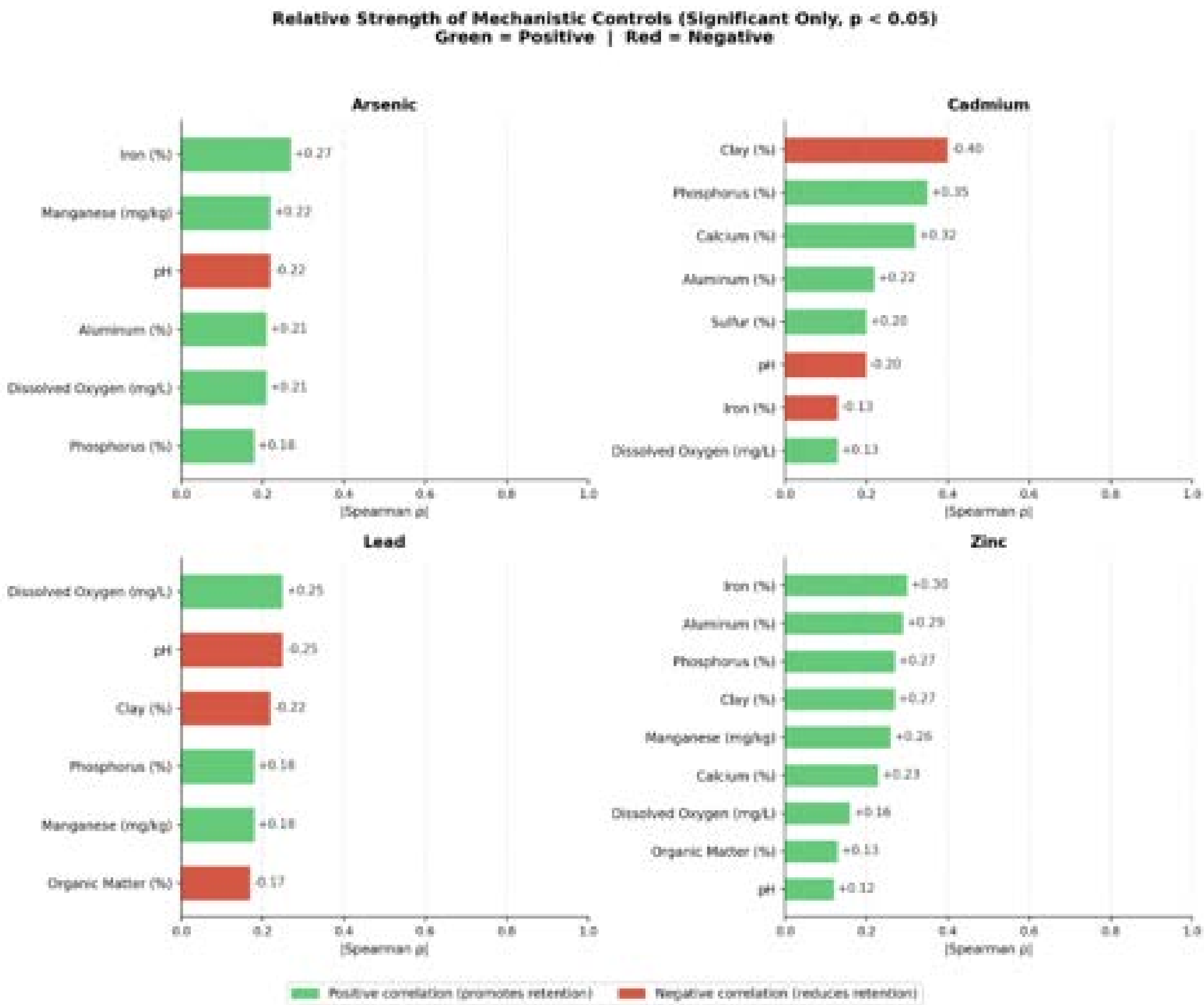

Figure 4.4.5a Each panel shows one metal. Bars represent the absolute value of Spearman ρ for statistically significant parameters only ($p < 0.05$), sorted by effect size. Green bars indicate positive correlations (parameter promotes sediment retention); red bars indicate negative correlations (parameter reduces retention and increases aqueous mobility). Parameters include sediment mineral phase proxies (Fe, Mn, Al, Ca, P, S) and water quality variables (pH, DO).

Arsenic: Oxide-Dominated Retention

For arsenic, the mechanistic analysis confirms that iron and aluminum oxides are the co-dominant controls on sediment retention, with manganese oxides providing additional secondary support. Sediment iron content shows the strongest mechanistic correlation in the entire dataset (Spearman $\rho = +0.268$, $p < 0.0001$, $n = 265$, Figure 4.4.5b), validating the hypothesis that arsenic exists primarily as arsenate [As(V)] under the predominantly oxic conditions of the basin (98.5% of samples with DO > 4 mg/L). Arsenate forms stable inner-sphere complexes with Fe-OH surface groups on goethite, ferrihydrite, and hematite phases through direct As-O-Fe bonding (Fendorf et al. 1997), explaining why iron content is such a reliable predictor of retention across the basin. Manganese oxides provide an additional significant pathway ($\rho = +0.225$, $p = 0.0002$, $n = 265$, Figure 4.4.5g), and aluminum hydroxide phases contribute a moderate but significant secondary mechanism ($\rho = +0.212$, $p = 0.014$, $n = 135$).

The dissolved oxygen correlation for arsenic ($\rho = +0.206$, $p = 0.0008$) reflects a dual mechanism not present for the other metals. High DO not only maintains the stability of Fe and Al oxide mineral phases but also promotes the As(V) oxidation state over As(III). Under low-DO conditions, arsenic is reduced to arsenite [As(III)], which exists as a neutral molecule (H_3AsO_3) with far weaker affinity for oxide surfaces than anionic arsenate. This compound redox effect means arsenic is more acutely sensitive to oxygen depletion than its K_d alone would suggest. Arsenic retention is also significantly reduced by acidity ($\rho = -0.216$, $p = 0.0004$), as shown in Table 4.4.5e, though it remains less pH-sensitive than lead.

A critical mechanistic interaction between aluminum content and pH is revealed in Figure 4.4.5c, which plots pH against aluminum content with arsenic retention shown as a color gradient. The highest arsenic retention values are concentrated at pH 3-5 rather than at neutral pH, which is counterintuitive but geochemically coherent: Al-OH surface groups become protonated under acidic conditions, generating a net positive surface charge that strongly attracts anionic arsenate through electrostatic attraction. At pH 7-8, these surface sites deprotonate and become negatively charged, sharply reducing arsenate affinity regardless of aluminum abundance. This interaction carries a critical remediation implication: raising pH to neutrality in the mining zone, while beneficial for cadmium and lead mobility, would paradoxically reduce arsenic retention on aluminum oxide surfaces and risk mobilizing arsenic currently sequestered under acidic conditions. Organic matter plays no role in arsenic retention ($\rho = -0.013$, $p = 0.833$, Figure 4.4.5h), consistent with electrostatic repulsion between anionic arsenate and negatively charged organic surfaces.

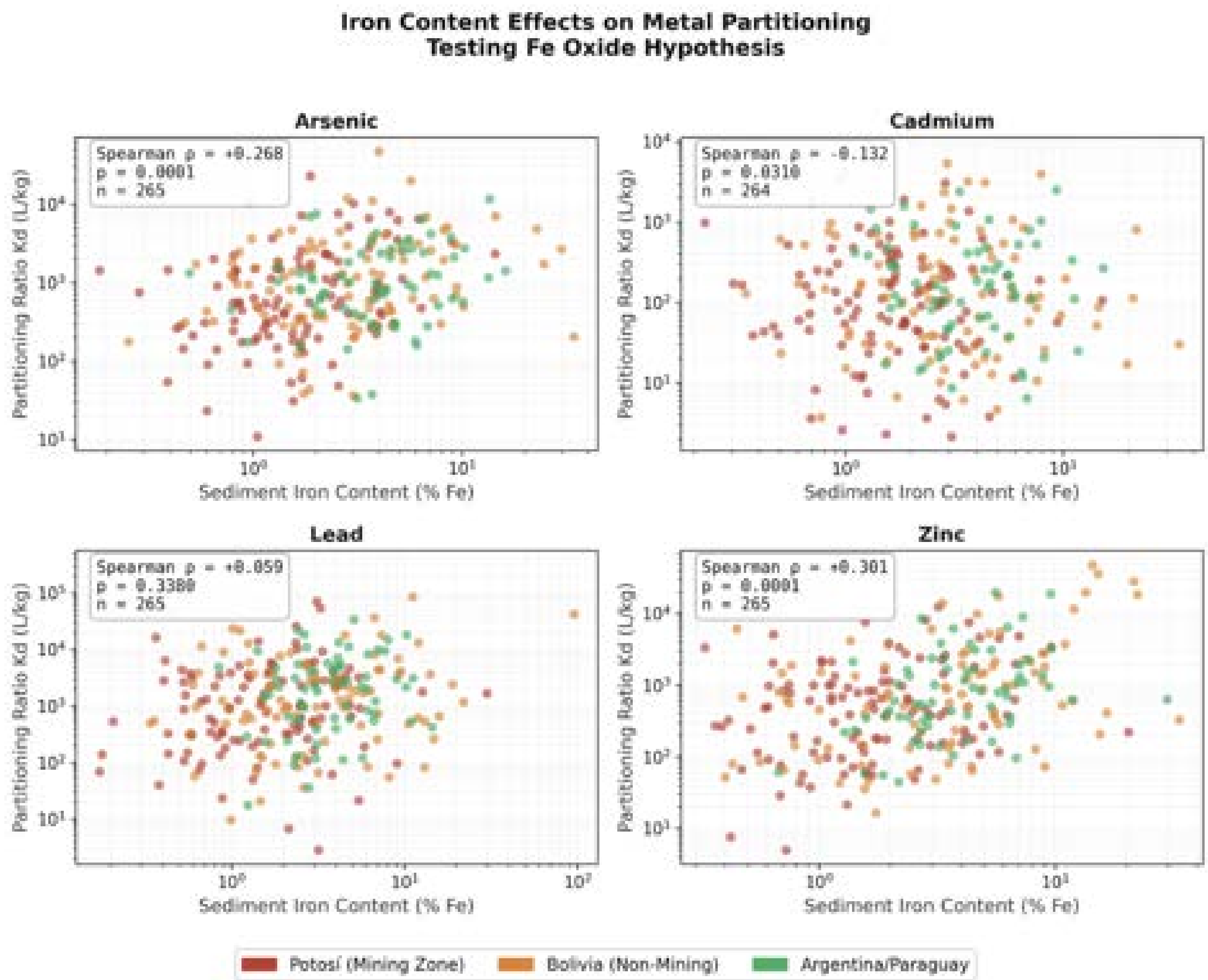

Figure 4.4.5b Each panel shows one metal. X-axis: sediment iron content (% Fe, log scale). Y-axis: partitioning ratio (K_d, L/kg, log scale). Points are colored by basin region: red = Potosí mining zone, orange = Bolivia non-mining, green = Argentina/Paraguay. Spearman ρ, p-value, and sample size are reported in the upper left of each panel. Arsenic and zinc show clear positive trends confirming Fe oxide sorption. Cadmium shows a significant negative trend, falsifying the Fe oxide hypothesis for this metal. Lead shows no systematic relationship with iron content.

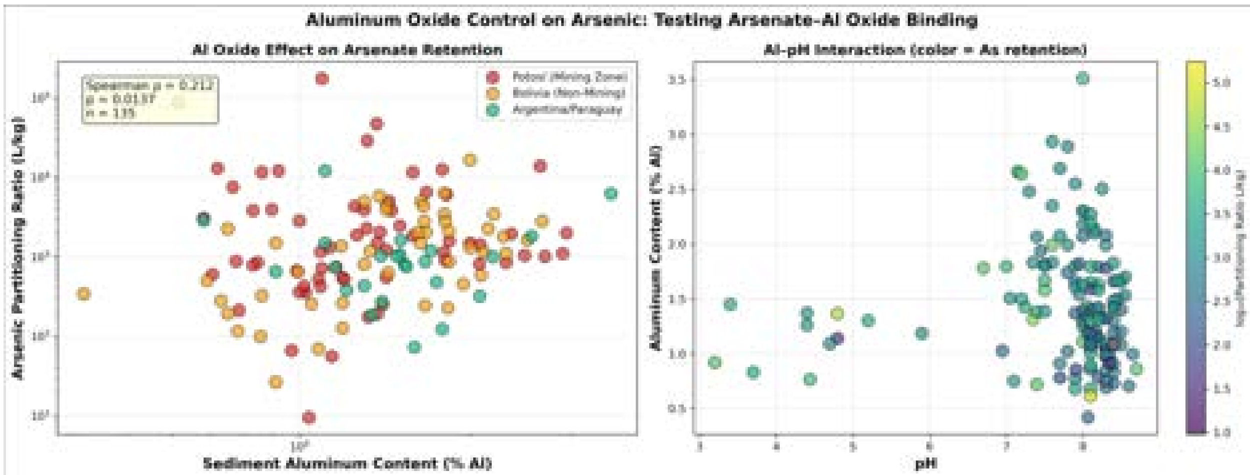

Figure 4.4.5c Left panel: X-axis = sediment aluminum content (% Al, log scale); Y-axis = arsenic partitioning ratio (K_d, L/kg, log scale); points colored by basin region. Spearman ρ = +0.212 (p = 0.014, n = 135) confirms aluminum oxides as a significant secondary sorption substrate for arsenate. Right panel: X-axis = pH; Y-axis = sediment aluminum content (% Al); color indicates arsenic retention on a log scale, where yellow/bright represents highest retention and purple/dark represents lowest.

High arsenic retention is concentrated at pH 3-5, confirming that protonated Al-OH surfaces under acidic conditions strongly bind arsenate. Retention collapses at pH 7-8 regardless of aluminum content (Table 4.4.5e).

Table 4.4.5e pH Effects on Metal Partitioning (Spearman Rank Correlation).

Metal	Spearman ρ	p-value	Interpretation
Lead	−0.249	< 0.0001	Strongest pH sensitivity; low pH severely reduces Pb binding
Arsenic	−0.216	0.0004	Retention decreases under acidic conditions
Cadmium	−0.197	0.0013	Acidic water mobilizes Cd from Ca/P minerals
Zinc	+0.124	0.044	Weak positive; higher pH slightly improves Zn retention

Cadmium: Calcium-Phosphate Co-Precipitation

The mechanistic analysis reveals a retention pathway for cadmium that fundamentally contradicts standard geochemical expectations. Rather than binding to iron or manganese oxides as textbook models predict (Spadini et al., 1999, Yin et al., 2011), cadmium correlates most strongly with phosphate content (ρ = +0.355, p < 0.0001, n = 135) and calcium content (ρ = +0.318, p = 0.0002, n = 135), identifying co-precipitation with calcium-phosphate minerals as the primary retention mechanism. Cadmium ions (Cd^{2+}) substitute for calcium (Ca^{2+}) in calcite and apatite crystal lattices due to their similar ionic radii, forming solid solutions and cadmium carbonate phases ($CdCO_3$, otavite) that sequester cadmium from the dissolved phase.

Critically, the iron oxide hypothesis is definitively falsified for cadmium: Fe content shows a significant negative correlation with cadmium partitioning ($\rho = -0.132$, $p = 0.031$, $n = 264$, Figure 4.4.5b), meaning cadmium retention actually decreases in iron-rich sediments despite high Fe availability throughout the basin. Manganese oxides show no significant relationship ($\rho = -0.028$, $p = 0.646$, Figure 4.4.5g). The reliance on calcium-phosphate minerals rather than oxide phases explains cadmium's uniquely low basin-wide retention (median K_d = 200 L/kg) and its disproportionate mobility risk: calcium carbonate and phosphate minerals are acid-soluble, dissolving readily at the pH 3-5 conditions prevalent in the Potosí mining zone and releasing co-precipitated cadmium into the water column. This acid sensitivity is confirmed by cadmium's significant negative pH correlation ($\rho = -0.197$, $p = 0.0013$), and its weak dissolved oxygen response ($\rho = +0.132$, $p = 0.033$) reflects the redox-insensitive nature of Ca/P mineral phases, which neither dissolve nor precipitate in response to changes in oxygen availability.

Lead: Maximum Retention and Redox Sensitivity
Lead exhibits the highest median retention of the four priority metals (median K_d = 1,667 L/kg), yet the mechanistic analysis reveals that this retention is sustained by a narrower and more redox-sensitive set of binding mechanisms than the high K_d value alone would suggest. The iron oxide hypothesis is not verified for lead: the Fe correlation is positive but statistically non-significant ($\rho = +0.059$, $p = 0.338$), and aluminum oxides similarly show no significant relationship ($\rho = +0.157$, $p = 0.069$). Instead, manganese oxides represent the only confirmed oxide retention pathway for lead in this system ($\rho = +0.176$, $p = 0.004$, $n = 265$, Figure 4.4.5g), alongside phosphate minerals ($\rho = +0.179$, $p = 0.038$, $n = 135$).

This dependence on manganese oxides directly explains why lead exhibits the strongest dissolved oxygen correlation of all four metals ($\rho = +0.251$, $p < 0.0001$), as shown in Table 4.4.5f. Mn(IV) oxide minerals are stable only under oxidizing conditions; when dissolved oxygen drops, microbial reduction converts Mn(IV) to soluble Mn(II), dissolving the primary lead binding substrate and releasing adsorbed lead into the water column (Lovley & Phillips 1988; Nealson & Saffarini 1994). The basin's predominantly oxic conditions (98.5% of samples with DO > 4 mg/L) therefore serve as the principal mechanism sustaining lead's high sediment retention under current monitoring conditions. Lead is simultaneously the most pH-sensitive metal in the study ($\rho = -0.249$, $p < 0.0001$), meaning that the combination of low pH and low DO represents the worst-case scenario for lead mobilization, a condition documented during the October 2017 hypoxic event at Naciente río La Ribera. Organic matter shows a significant negative correlation with lead retention ($\rho = -0.174$, $p = 0.005$, $n = 258$, Figure 4.4.5h), most likely because organic-matter-rich sediments in the basin coincide with low-pH mining zone conditions where H^+ competition overwhelms all retention mechanisms rather than reflecting a direct mobilizing effect of organic matter on lead.

Table 4.4.5f. Dissolved Oxygen Effects on Metal Partitioning (Spearman Rank Correlation).

Metal	Spearman ρ	p-value	Interpretation
Lead	+0.251	< 0.0001	Strongest DO sensitivity; high DO maintains Mn oxides that bind Pb
Arsenic	+0.206	0.0008	Direct redox control; high DO promotes As(V) and Fe/Al oxide stability
Zinc	+0.155	0.011	Higher oxygen preserves Fe/Al/Mn oxides across all Zn binding pathways
Cadmium	+0.132	0.033	Weakest DO response; Ca/P minerals are redox-insensitive

Arsenic and zinc show positive trends consistent with manganese oxide sorption as a secondary mechanism alongside iron oxides. Lead shows a weak but statistically significant positive trend ($\rho = +0.176$, $p = 0.004$), identifying manganese oxides as the only verified oxide retention pathway for lead in this system. Cadmium shows no relationship with manganese content, consistent with its independence from oxide-based sorption.

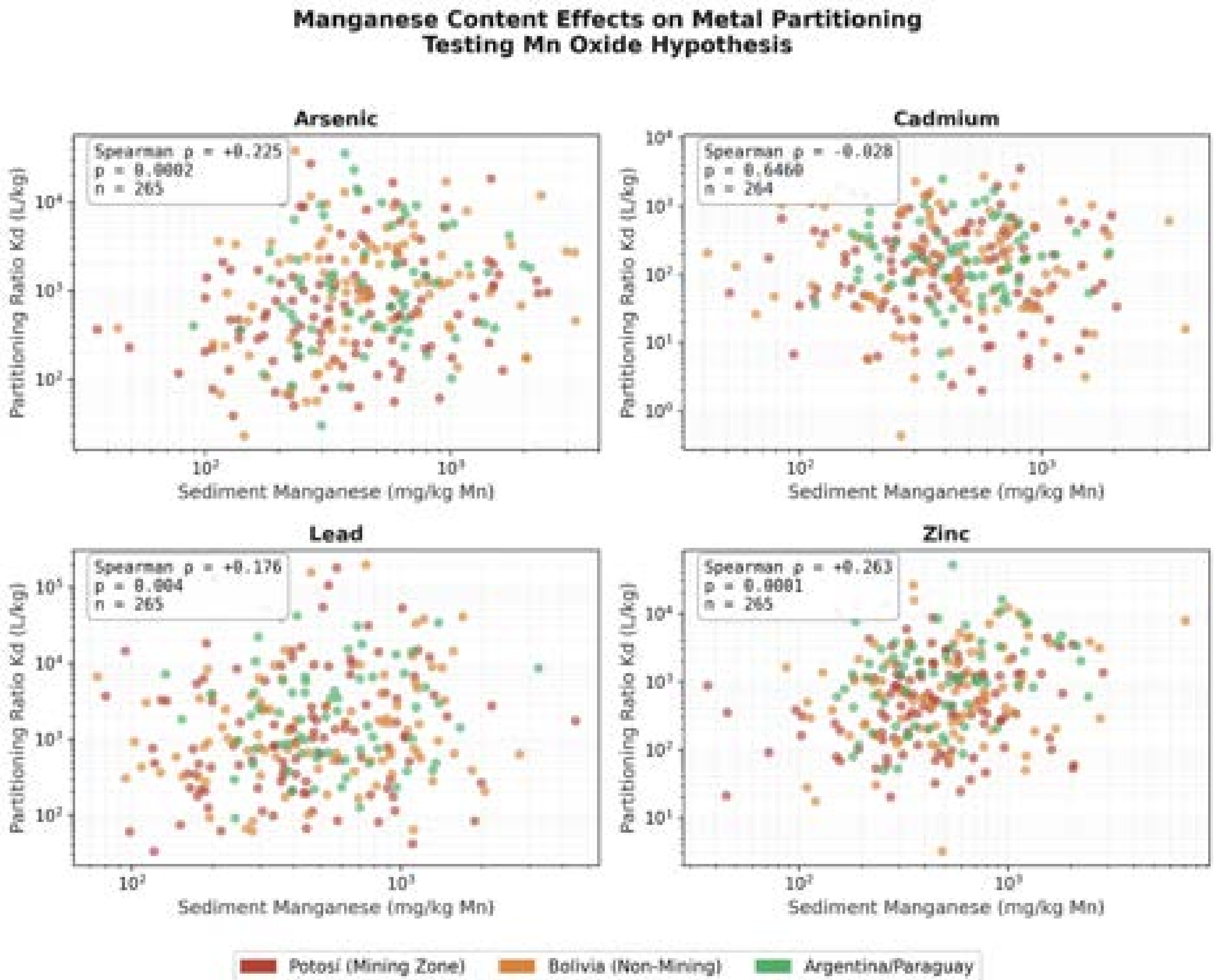

Figure 4.4.5g Each panel shows one metal. X-axis: sediment manganese content (mg/kg Mn, log scale). Y-axis: partitioning ratio (K_d, L/kg, log scale). Points are colored by basin region: red = Potosí mining zone, orange = Bolivia non-mining, green = Argentina/Paraguay.

Zinc: Diversified and Resilient Binding

Zinc is distinctive in its broad, multi-mechanism binding behavior, showing statistically significant correlations with every mineral phase tested in this analysis. Its primary drivers are iron oxides ($\rho = +0.301$, $p < 0.0001$, $n = 265$) and aluminum oxides ($\rho = +0.285$, $p = 0.0008$, $n = 135$), with additional significant contributions from phosphate minerals ($\rho = +0.273$, $p = 0.0013$, $n = 135$), clay exchange sites ($\rho = +0.270$, $p < 0.0001$, $n = 256$), manganese oxides ($\rho = +0.263$, $p < 0.0001$, $n = 265$), calcium minerals ($\rho = +0.226$, $p = 0.009$, $n = 135$), and organic matter ($\rho = +0.134$, $p = 0.032$, $n = 258$). Zinc is the only metal among the four studied to show a significant positive correlation with organic matter (Figure 4.4.5h), reflecting its capacity for complexation with humic substances in addition to mineral sorption. However, given the low median organic matter content of Pilcomayo sediments (0.90%), this pathway contributes minimally to overall retention relative to oxide and clay mechanisms.

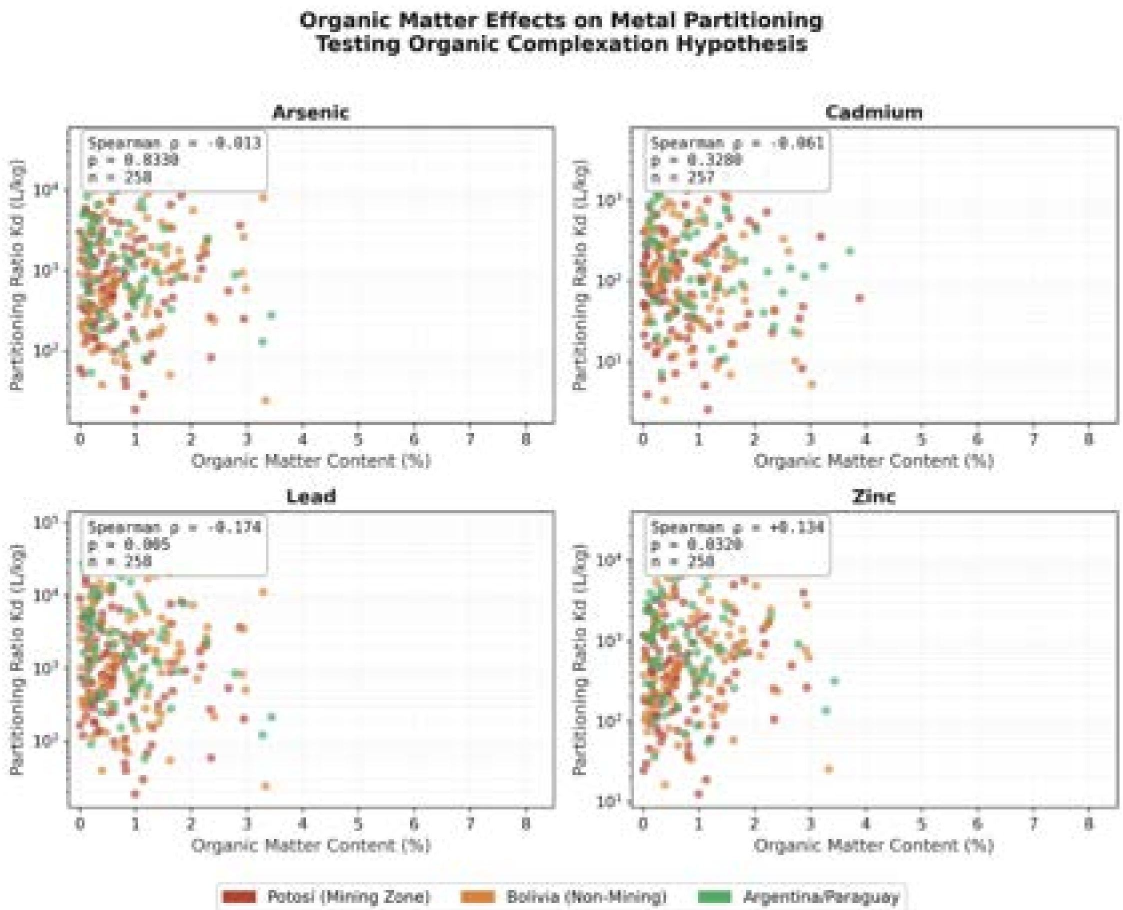

Figure 4.4.5h X-axis = organic matter content (%). Y-axis = partitioning ratio (K_d, L/kg, log scale). Organic matter is consistently low across Pilcomayo sediments (median 0.90%). Arsenic and cadmium show flat, scattered distributions confirming that organic matter is not a meaningful retention control for these metals. Lead shows a weak negative trend (ρ = -0.174), likely reflecting the co-occurrence of organic-matter-rich sediments with low-pH mining zone conditions rather than a direct mobilizing effect. Zinc is the only metal with a significant positive organic matter correlation (ρ = +0.134), confirming it as the sole metal with any organic complexation signal in this mineral-dominated system.

This diversity of binding mechanisms provides zinc with meaningful geochemical resilience: when one retention pathway is disrupted, for example, Fe oxide dissolution during a low-DO event, other mechanisms including clay exchange and aluminum oxide binding remain active, buffering against catastrophic release. This distributed binding explains zinc's moderate but stable basin-wide retention (median K_d = 664 L/kg) and its absence of a clear enrichment signal during the documented hypoxic events, in contrast to the more pronounced responses observed for lead and cadmium. Zinc is also the only metal to show a positive pH correlation ($\rho = +0.124$, $p = 0.044$), reflecting its cationic behavior across a broader pH range and the contribution of clay exchange mechanisms that function across the full basin pH gradient of 3-8.5.

4.4.8 Hypoxic Events: Water Column Metal Mobilization

Research identified three critical hypoxic episodes, defined as dissolved oxygen (DO) falling below 4 mg/L, at Potosí mining zone stations between 2016 and 2024. These events represent discrete departures from the basin's predominantly oxic baseline, where 98.5% of samples maintain DO levels above 4 mg/L, and act as chemical triggers that disrupt the stable sediment-water equilibrium. While flow-driven mobilization occurs seasonally, these hypoxic events trigger the reductive dissolution of iron (Fe) and manganese (Mn) oxide minerals, co-mobilizing associated toxic metals from the sediment reservoir into the water column. Events 1 and 3 are excluded from the enrichment analysis. No pre-event baseline data exist for Tarapaya, where systematic monitoring commenced concurrently with the 2016 event, and no post-event samples are available for La Quiaca, which coincides with the final sampling campaign in the dataset. Event 2 therefore provides the only opportunity for paired before-and-after comparison and is the focus of the quantitative analysis below.

Event 2: Potosí – Naciente río La Ribera (October 2017)
Event 2, occurring on October 4, 2017, at the Potosí - Naciente río La Ribera station with a recorded dissolved oxygen level of 2.84 mg/L, provides the most complete dataset for before-and-after enrichment analysis within a 180-day window. This event empirically demonstrates the relative vulnerability of metals predicted by the site-specific distribution coefficients (K_d) established under baseline conditions.

Lead exhibited the most pronounced post-event increase at the hydrologically connected Tarapaya station, with median water column concentrations rising from 40.0 µg/L to 119.5 µg/L, representing an enrichment factor of approximately 3.0×. This surge reflects the disruption of lead's large sediment reservoir (median K_d = 1,667 L/kg) and its high sensitivity to oxygen loss (Spearman $\rho = 0.251$). Cadmium also showed a modest increase from 11.0 to 18.0 µg/L. Given that cadmium relies on acid-soluble calcium minerals rather than iron oxides, its release is likely attributable to the pH depression that frequently accompanies hypoxia. Arsenic and iron concentrations remained effectively stable during this period, potentially due to buffering by the abundant iron oxide substrate found in Pilcomayo sediments. While Tarapaya showed clear enrichment, other connected stations lacked sampling data within the specific 180-day windows required for comparison. As visualized in Figure 4.4.8a, these contrasts provide empirical evidence of oxygen depletion triggering metal release from the sediment "source reservoir."

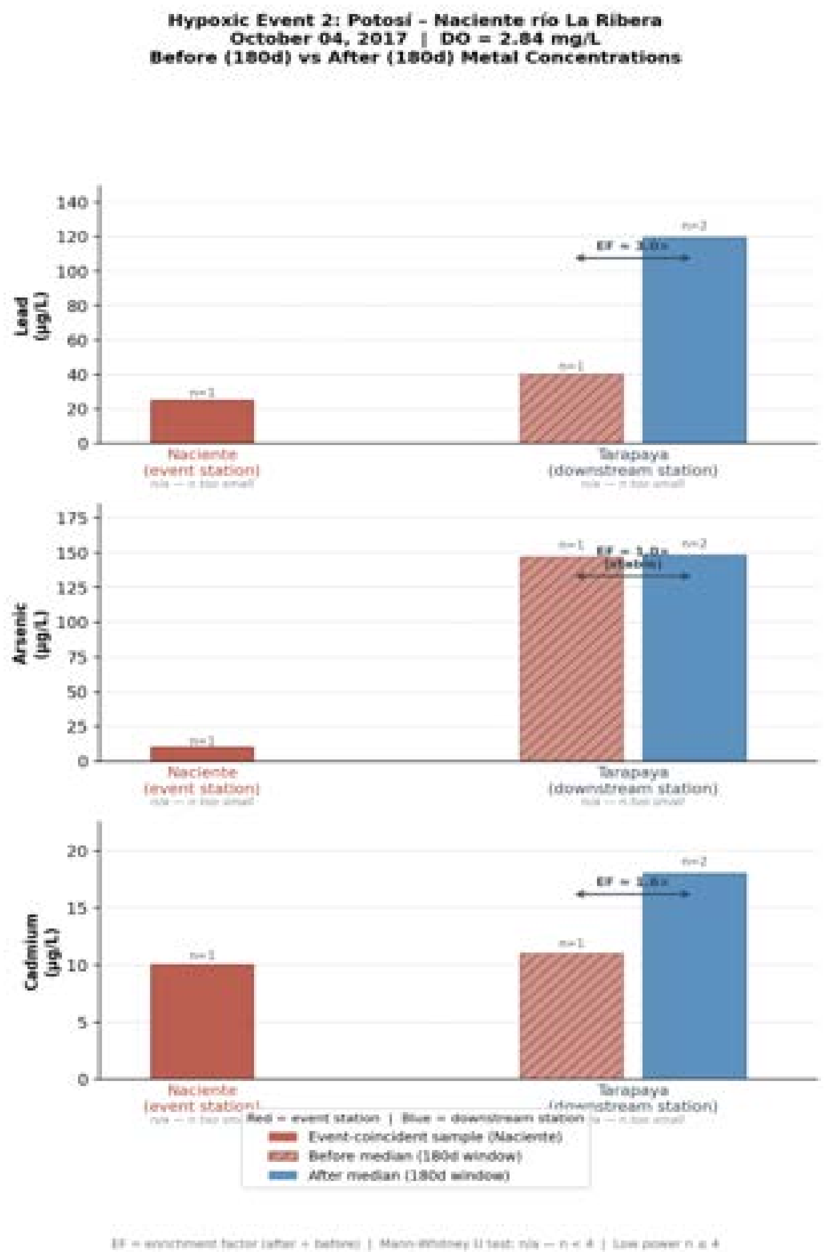

Figure 4.4.8a Median metal concentrations (µg/L) in the water column within 180-day pre-event (hatched bars) and post-event (solid bars) windows surrounding the October 4, 2017 hypoxic episode (DO = 2.84 mg/L) at Potosí – Naciente río La Ribera. The Naciente bar represents a single event-coincident sample; no pre-event baseline exists for this station, precluding enrichment factor calculation. Tarapaya is a hydrologically confirmed downstream station (n = 1 before / n = 2 after). Enrichment factors (EF) are reported as descriptive effect sizes only; sample sizes preclude formal significance testing.

Temporal Dynamics and Long-Term Trends
The full eight-year monitoring record at the Naciente station contextualizes the 2017 event within longer-term contamination trends. The October 2017 event marked a significant period of elevated concentrations for several priority metals, with lead showing an immediate spike followed by a sustained multi-year decline through 2021. Arsenic and cadmium followed a similar trajectory, peaking during the 2017-2018 post-event period before declining through 2022.

Iron and manganese concentrations exhibited variations of one to two orders of magnitude across the record, with the post-event window capturing elevations consistent with the reductive dissolution of oxide minerals during low-oxygen conditions. This mechanism compounds existing wet-season loading and overrides the typical "stable state" of the sediment reservoir. The record also highlights that multiple metals reached their highest recorded values during the 2024 wet season, aligning with the emerging cadmium crisis documented in the basin's sediment record (Figure 4.4.8b).

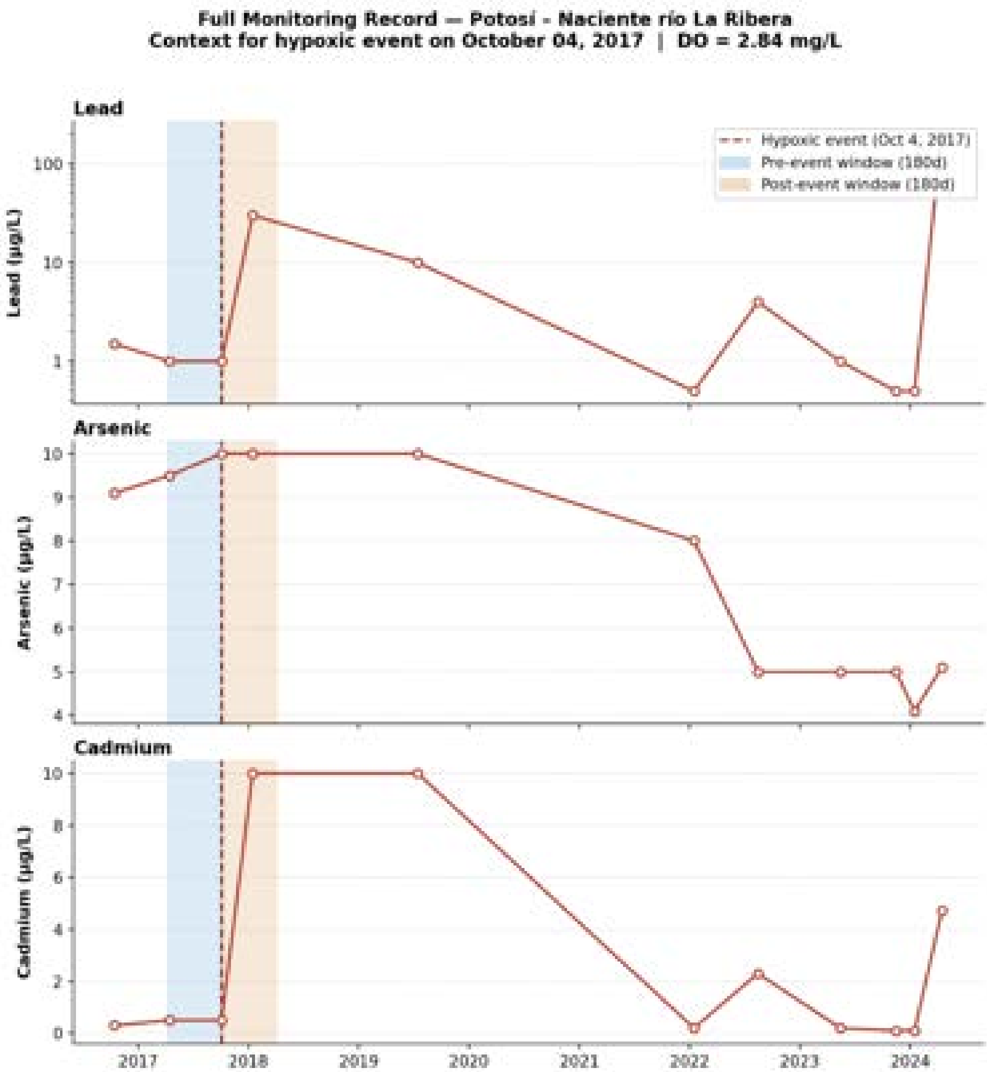

Figure 4.4.8b Time-series of water column concentrations (µg/L) for lead, arsenic, and cadmium across the eight-year monitoring record. The dashed line marks the October 4, 2017 hypoxic event (DO = 2.84 mg/L). Blue shading indicates the 180-day pre-event window; red shading indicates the 180-day post-event window. Lead is plotted on a log scale to capture the multi-order-of-magnitude range across the record.

4.5 Cost Estimates of Recommended Remediation Technologies

This section provides decisionmakers and policymakers a high level summary of the remediation technology options recommended thus far as well as estimates of the cost of each technology option (Table 4.5a).

Table 4.5a Cost estimates of remediation technologies compiled from different sources. Cost estimates are relative to the year in which the source was published, though not all sources have an associated date. Units in $/ton/yr reference the cost of the system per ton of material remediated per year.

Technology	Currency	Range of Cost Estimates	Mean Estimate	Source
Limestone Leach Beds	USD BOB	$31 - $408 /ton/yr 215 - 2,823 Bs/ton/yr	$51 /ton/yr 353 Bs/ton/yr	Ziemkiewicz et al.
Phytoremediation	USD BOB	$20 - $40 per tonne 138 - 277 Bs per tonne	$30 per tonne 208 Bs per tonne	Islam et al., 2024
Optical Particle Counter (OPC)	USD BOB	$20,000 - $25,000 per unit 138,400 – 173,000 Bs per unit	N/A N/A	Dupont et al., 2025
OPC: Low cost variant	USD BOB	$50 - $1,000 per unit 346 – 6,920 Bs per unit	N/A N/A	Dupont et al., 2025
Engineered Tailings Covers	USD BOB	$100,000 - $1,575,000 /ha 692,000 – 10,899,000 Bs/ha	$730,000 /ha 5,051,600 Bs/ha	ATA, 2013
Chemical Dust Suppressant	USD BOB	$23,900 - $29,900 per hectare 165,388 – 206,908 Bs/ha	N/A N/A	Arizona Dept. of Env. Quality, 2004

Lotken (n.d.) projects the installed cost of a High Density Sludge (HDS) system at $3,431,250 [Bs 23,744,250] and the operating cost (excluding labor) at $368,484 [Bs 2,549,909]. Skousen & Ziemkiewiez (2005) provide an aggregation of costs and efficiency for passive remediation systems of AMD which were employed across the United States before 2005 (Table 4.5b). Their findings can be used as a basis for comparing the cost of different passive remediation options, though it is important to note that these technologies may have improved efficiency or cost-effectiveness since 2005.

Table 4.5b Costs and efficiency metrics of passive AMD treatment systems. AeW = Aerobic Wetlands, ALD = Anoxic Limestone Drain, OLC = Open Limestone Channel, SLB = Slag Leach Bed, LSB = Limestone Leach Bed, VFW = Vertical Flow Wetland, AnW = Anaerobic Wetland. Note that the conversion rate is $1 per 6.92 BOB (February 9, 2026) and that values are relative to the value of USD in 2005. From Skousen & Ziekiewiez, 2005.

System Type	Number of Units	Average Acid Treated (t/yr)	Average Total Cost ($)	Average Removal (g/day/t, g/m^2/day)	Average Cost ($/t/yr)	% of Systems That Removed Acid
AeW	6	9.1	10,565	16.3 g/m^2/day	59	100
ALD	37	15.4	29,726	86.1 g/day/t	96	92
OLC	10	10.6	30,813	30.2 g/day/t	145	90
SLB	2	76.3	54,604	2,334 g/day/t	36	100
LSB	17	17.6	72,835	18.1 g/day/t	207	100
VFW	16	11.6	51,151	87.5 g/m^2/day	220	94
AnW	17	8.0	94,515	16.4 g/m^2/day	591	71
Ponds	11	0.8	10,035	1.7 g/m^2/day	1,468	82

4.6 Wastewater Fingerprinting

Bolivia's Regulation on Water Contamination (Bolivia Ministerio de Desarrollo Sostenible y Medio Ambiente, 1995) provides limits for each contaminant considered for determining the presence of wastewater (Table 4.6a). Class A supports drinking water (using disinfection treatment), recreation, and industrial water supply; Class D supports vehicular travel, industrial supply, and drinking water in extreme conditions (using pre-sedimentation and complete treatment).

A total of eight sampled dates across three stations reported wastewater indicator contaminants (ammonia, BOD, fecal coliforms) above the 85th percentile of all sampled concentrations (Table 4.6b). Most concentrations found at these points were above all regulatory limits set by Bolivian Law 1333. San Antonio - Potosí was found to have parameters above the 85th percentile in five sampled instances (out of 12 total dates sampled, 42%), indicating a potentially ongoing concern with contamination from waste.

Table 4.6a Bolivian Law 1333's Regulation on Water Contamination regulatory limits.

Chemical	Class A	Class B	Class C	Class D
BOD_5 (mg/L)	≤ 2	≤ 5	≤ 20	≤ 30
Ammonia (mg/L)	≤ 0.05	≤ 1	≤ 2	≤ 4
Fecal Coliform (CFU or MPN/100 mL)	≤ 200	$\leq 1,000$	$\leq 5,000$	$\leq 10,000$

Table 4.6b Stations with high concentrations for all wastewater indicators, including date, concentrations reported, and the minimum percentile for the analyte concentrations compared to all reported concentrations of that analyte. Concentrations are augmented to include the Bolivian Law 1333 classification supported for the given concentrations (Classes A-D).

			Reported Concentration		
Station	Date	Minimum Parameter Percentile	Ammonia (mg/L)	BOD (mg/L)	Fecal Coliform (CFU/100 mL)
La Quiaca	2023-12-13	0.92	34.04*	109*	1.5×10^5*
San Antonio - Potosi	2023-12-07	0.93	55.70*	73*	4.7×10^4*
San Antonio - Potosi	2023-08-03	0.95	61.10*	240*	1.5×10^5*
San Antonio - Potosi	2020-07-20	0.96	45.70*	65*	4.6×10^7*
Tarapaya	2020-07-20	0.89	20.00*	86*	4.3×10^6*
San Antonio	2017-10-04	0.91	57.47*	94*	7.2×10^3 (C)
Tarapaya	2017-10-04	0.89	37.02*	49*	4.5×10^3 (D)
San Antonio - Potosi	2016-08-02	0.91	63.52*	235*	7.2×10^3 (D)

* Concentration exceeds all regulatory standards set by Law 1333.

All stations with reported concentrations for ammonia, BOD, and fecal coliform on a single date were reviewed to determine how the water quality would score when considering all wastewater parameters (Figure 4.6a). No sampled stations received a Class A score, and over 25% of all sampled dates did not meet Class D limits. Tarapaya, San Antonio - Potosí, and La Quiaca never met Class D limits and should be prioritized for wastewater treatment. Potosí - Naciente rio La Ribera had the highest percentage of samples in Class B; this station is near a drinking water reservoir upstream of the city.

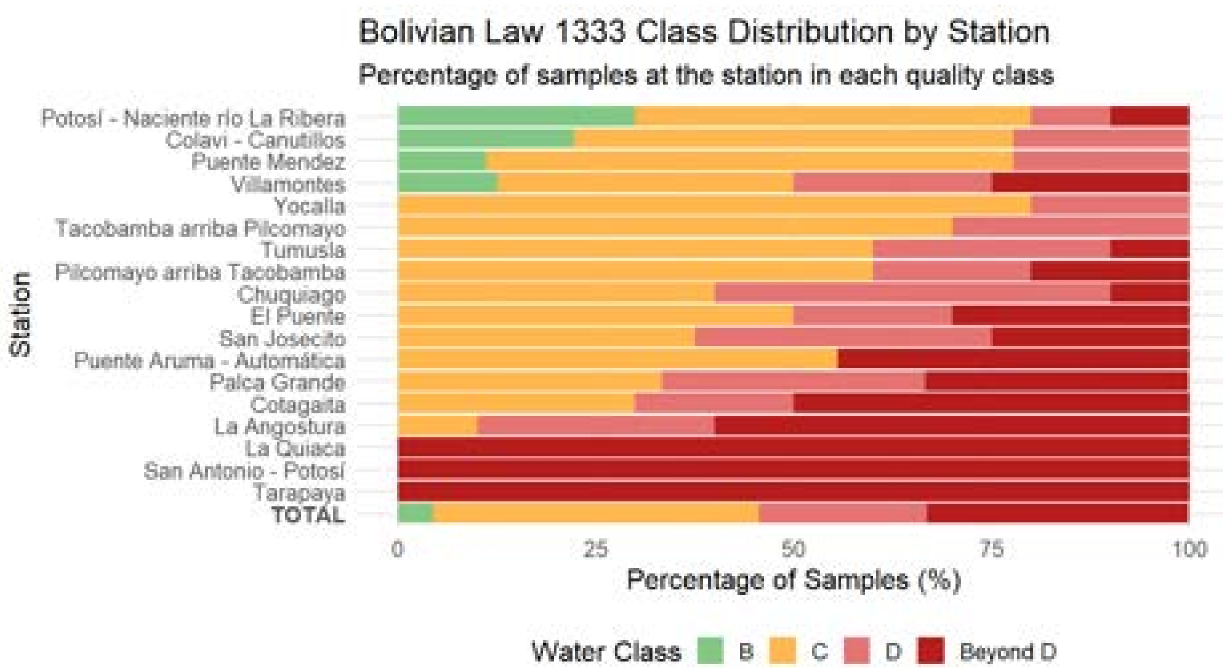

Figure 4.6a Station water quality classifications for wastewater indicators are shown as a percent of the total number of samples collected at that location. No stations reported water quality within Class A for any parameter, and few reported concentrations within Class B limits for all three parameters.

The WHO and USEPA seek to have no detectable coliform within a standard 100mL sample (Table 4.6c). The WHO does not recommend a specific ammonia limit, citing that there are minimal health concerns at naturally occurring concentrations; the USEPA similarly sets no health related limits, but surface waters have an acute limit of 17mg/L (sampled as Total Ammonia Nitrogen). Neither agency has health recommendations for BOD, but surface water discharges are regulated under the Clean Water Act

Table 4.6c International standards for the selected wastewater indicators in drinking water or surface waters

Parameter	WHO	USEPA
Fecal Coliform	Drinking water: "Must not be detectable in any 100 mL sample" (World Health Organization, 2022)	Drinking water: Maximum Contaminant Level Goal of 0 (non-detectable), with an enforceable maximum contaminant limit of 5% of samples within a month (US EPA, 2025)
Ammonia	Drinking water: "No health-based guideline value is proposed" (Chemical Fact Sheet: Ammonia, 2022)	Drinking water: No health limit set. Surface water (Acute): 17 mg/L (as Total Ammonia Nitrogen)
BOD	No reference found	Drinking water: No reference found. Surface water discharges: 30 mg/L average across 30 days (40 CFR Part 133 – Secondary Treatment Regulation, 1984)

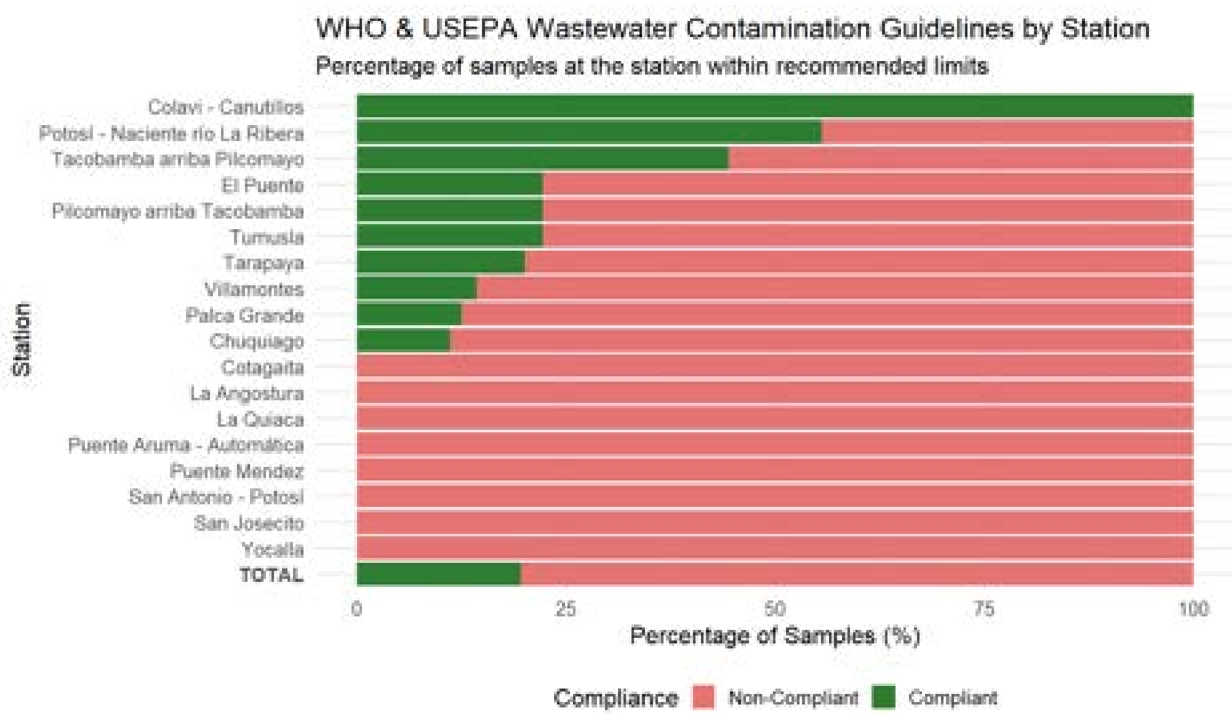

Figure 4.6b Percent of samples within USEPA & WHO regulatory recommendations for wastewater indicators. Samples counted include date-station pairs which had data for all three parameters. Total sample count described as the final bar; fewer than 25% of sampled date-stations were within regulatory limits for surface waters.

All date-station pairs with reported concentrations for all three wastewater indicators (n=143) were included in analysis against international standards (Figure 4.6b). Fewer than 25% of samples were within international regulatory recommendations for surface waters. Many stations only reported non-compliant results. Colavi - Canutillos met recommendations for all parameters in 100% of collected samples (n=8); in comparison with the station's Bolivian compliance score, this shows how rigorous the Bolivian standards are for some parameters.

4.7 Variation of Metal Toxicity due to Hardness

Hardness classification was found for each station as a frequency percentage (Figure 4.7a). Softer waters increase toxicity from metal contaminants, thus higher toxicity risks are expected at stations: Canutillos 3, Colavi - Canutillos, Potosí - Naciente rio La Ribera, and Yocalla. Further research should include hardness as a parameter for metal toxicity when developing a hazard quotient, along with finding drinking water-specific regulations that include toxicity calculations based on hardness.

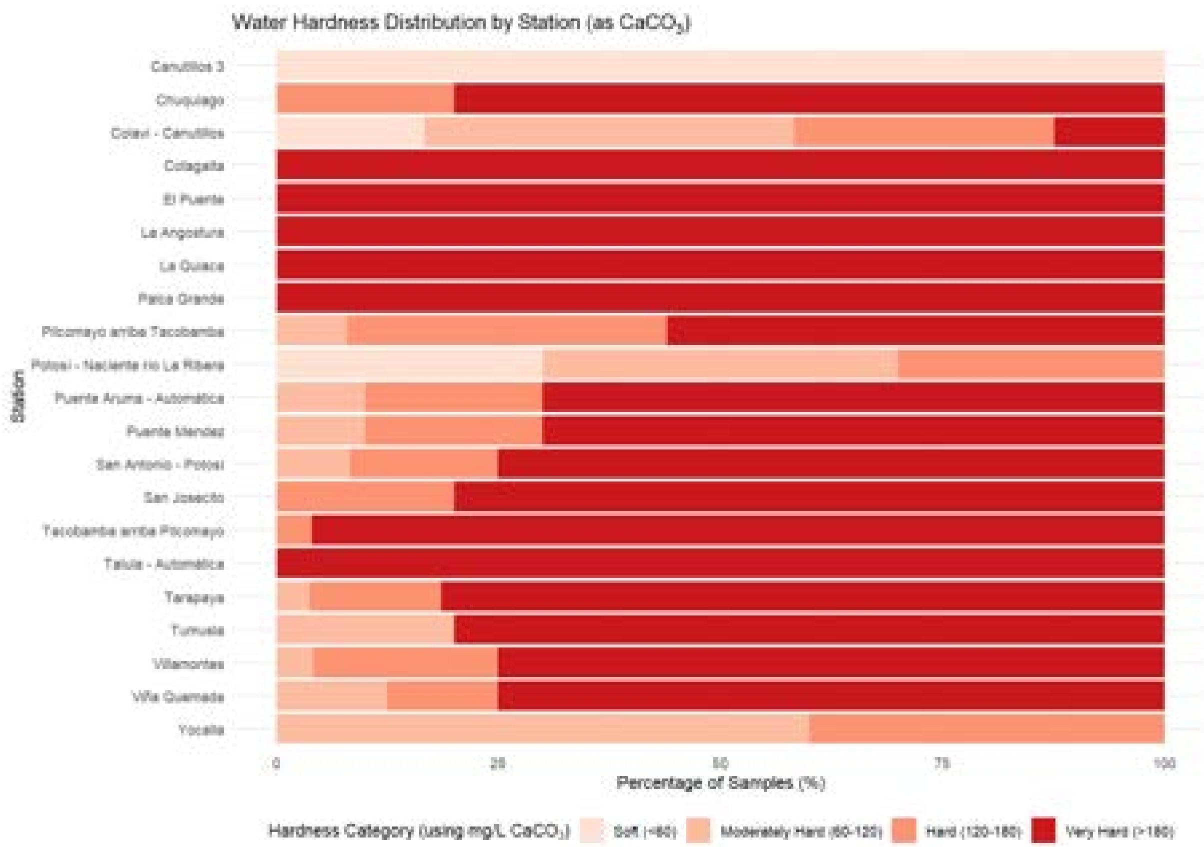

Figure 4.7a The percent of samples found at each station was grouped by hardness classification. Canutillos 3 was found to have only soft water (<60 mg/L $CaCO_3$), while many stations were found to have very hard water.

Recommended limits for aquatic life are calculated for various metals and hardness using USEPA formulas (Table 4.7a). Although the impacts of water hardness on aquatic life and their response to heavy metals contamination, these studies were not readily available for human drinking water standards (Ha et al., 2020; Mohammad Ebrahimpour et al., 2010). These values are not the same as drinking water standards, as utilized in other sections of this study, and should be cautiously compared with existing regulatory limits. Drinking water standards do not closely align with CMCs and drinking-quality standards for each metal should be determined using hardness before proceeding with a hardness-dependent model.

Table 4.7a Criterion Maximum Concentrations (CMCs) formulas developed by the US EPA for aquatic life are shown for each contaminant with an empirically determined relationship to hardness.

Contaminant	EPA Standard (μg/L)	CMC Formula	CMC (μg/L) 60 mg/L $CaCO_3$	CMC (μg/L) 180 mg/L $CaCO_3$
Cadmium	5	$e^{1.128 \cdot \ln(\text{hardness}) - 3.6867}$	2.53	8.77
Chromium (III)	100	$e^{0.819 \cdot \ln(\text{hardness}) + 3.7256}$	1182.41	2917.94
Copper	1300	$e^{0.9422 \cdot \ln(\text{hardness}) - 1.700}$	8.62	24.36
Nickel	70**	$e^{0.846 \cdot \ln(\text{hardness}) + 2.255}$	303.43	771.43
Zinc	5000*	$e^{0.8473 \cdot \ln(\text{hardness}) + 0.884}$	77.44	197.16

* National Secondary Drinking Water Regulations are non-enforceable standards that may impact water aesthetic or cause cosmetic changes (EPA, 2025).
** Nickel does not have a standard under the EPA, but the WHO recommends a guideline value (WHO, 2021).

4.8 Drinking Water Quality

Water quality data collected between December 12, 2018 and December 13, 2023 were compared with Bolivian Law 1333 to characterize the allowable uses within the river basin for the last 5 years (Figure 4.8a. See *4.1 Data Limitations* regarding 2024 data). Many contaminants were found to be mostly within Class A standards, which can be utilized for drinking water after filtration. Class D waters can be used for drinking water only in extreme situations and after extensive treatment, according to Law 1333; many (~15%) samples were found to be above those limits. Compared to international standards (where they were readily available from reputable sources), a similar percentage of samples were considered within regulatory limits (Figure 4.8b). Regulatory limits are listed in the Appendix (*A.2.2 Reference Standards*).

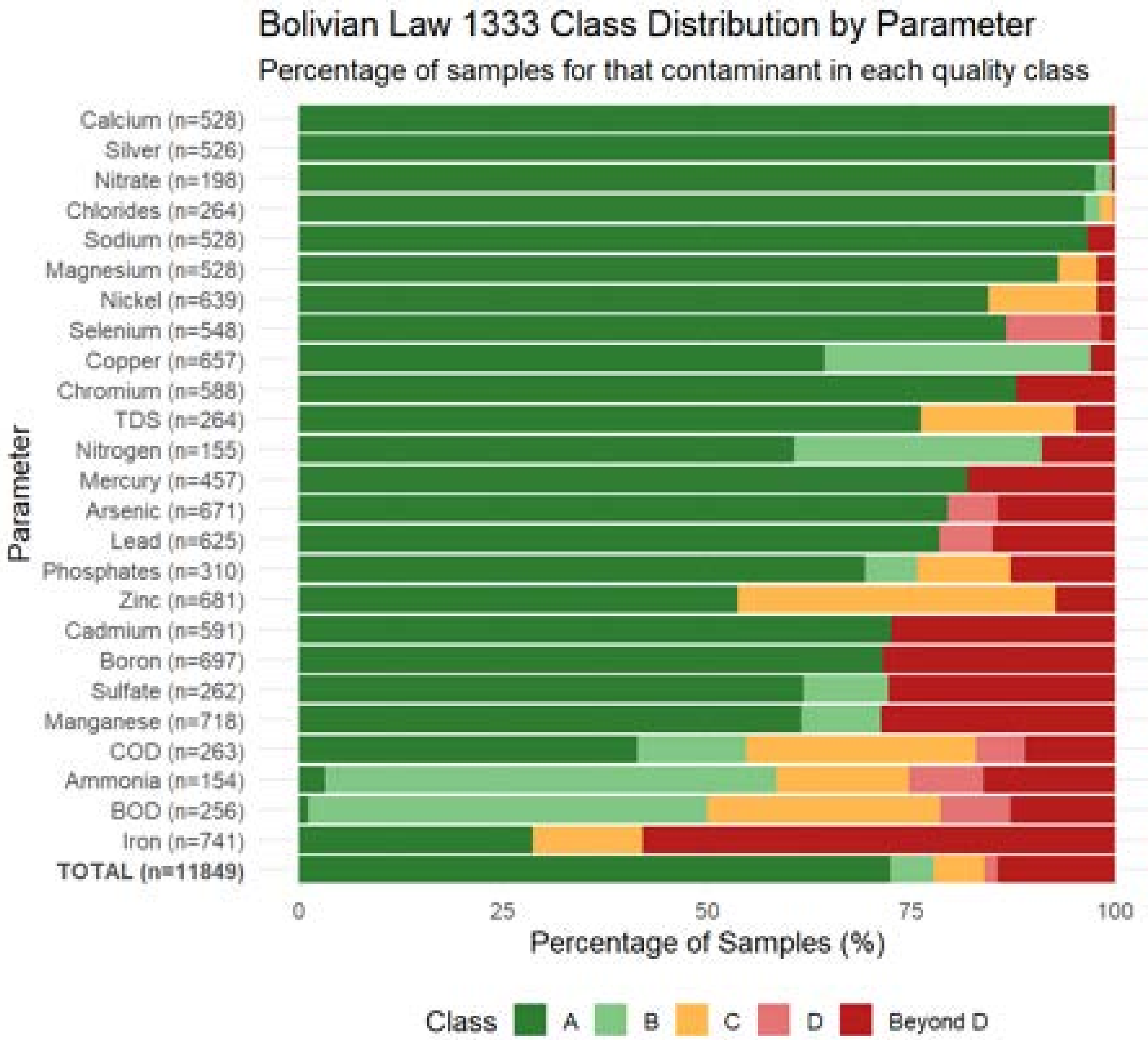

Figure 4.8a Water quality classification for each sample is shown as a percentage of all samples taken within the last 5 years of sampling. Water quality classification uses Bolivian Law 1333 standards. Colors represent the Bolivian Law 1333 classification for that analyte sample. Approximately 70% of sampled contaminants were within Class A limits for use as drinking water.

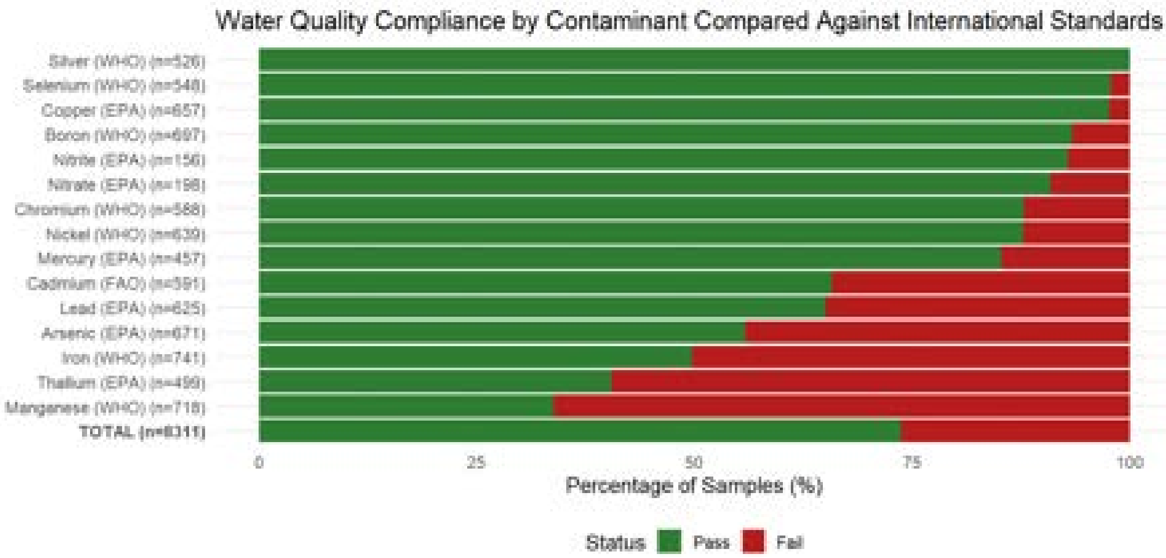

Figure 4.8b Water samples compared against strict global regulatory limits (US EPA, 2025; *Chemical Hazards in Drinking-Water*, n.d.).

4.9 Sensitivity Analysis

4.9.1 Water and Sediment Parameters

Sensitivity analysis on individual parameters showed small correlation differences and did not capture the full uncertainty in hazard calculations; see the Appendix (*A.3.5 Sensitivity Analysis*) for configuration details on specific model configurations. A comprehensive sensitivity analysis was then conducted across all model configurations, varying both HQ aggregation methods and temporal selection criteria, to identify patterns in station prioritization using water quality data. Note that this analysis was run utilizing all available years of data (2016-2024); future analysis may want to consider whether the variation in 2024 data would substantially impact the results of this sensitivity analysis.

Comparing models utilizing dissolved pollutant data demonstrated a clear impact when scoring stations or parameters using median concentration values rather than mean, max, or 95th percentile (Figure 4.3.6a). Expert solicitation and reviewed literature on utilizing hazard quotients led us to select a 95th percentile model (ATSDR, 2023; Kim et al., 2013; Waters et al., 2015).

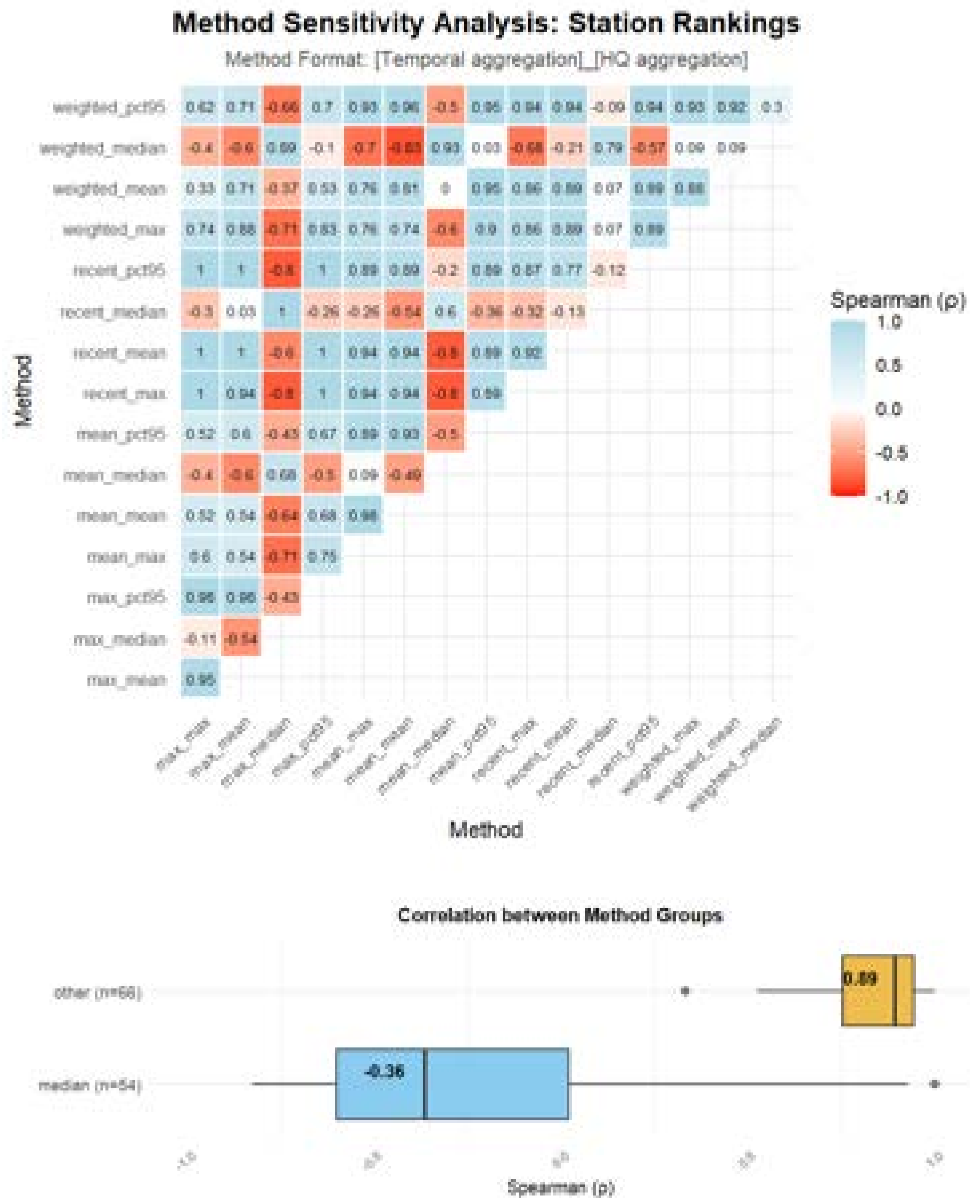

Figure 4.3.6a (Top) Spearman correlation plots for each model configuration calculated using water quality data. Models using median HQ aggregation deviate from other methods, and model selection can be generally reduced to selecting median HQs versus higher values. Recent models utilize the most recent date of sampling. (Bottom) Boxplot of Spearman correlations between models running the median HQ aggregation method and those using other scores. Median correlation value is shown within the boxplot. A significant difference can be seen between models running the median HQ aggregation method.

Selecting the most recent years of available data highlighted current conditions within the basin. To assess whether the recency window size impacts results, time windows from 0 years (only the most recent date sampled) to 5 years were compared using a correlation plot (Figure 4.3.6b) and by ranking the top 5 parameters (Figure 4.3.6c). All models agreed on the top four parameters, while median-HQ models typically differed on one parameter.

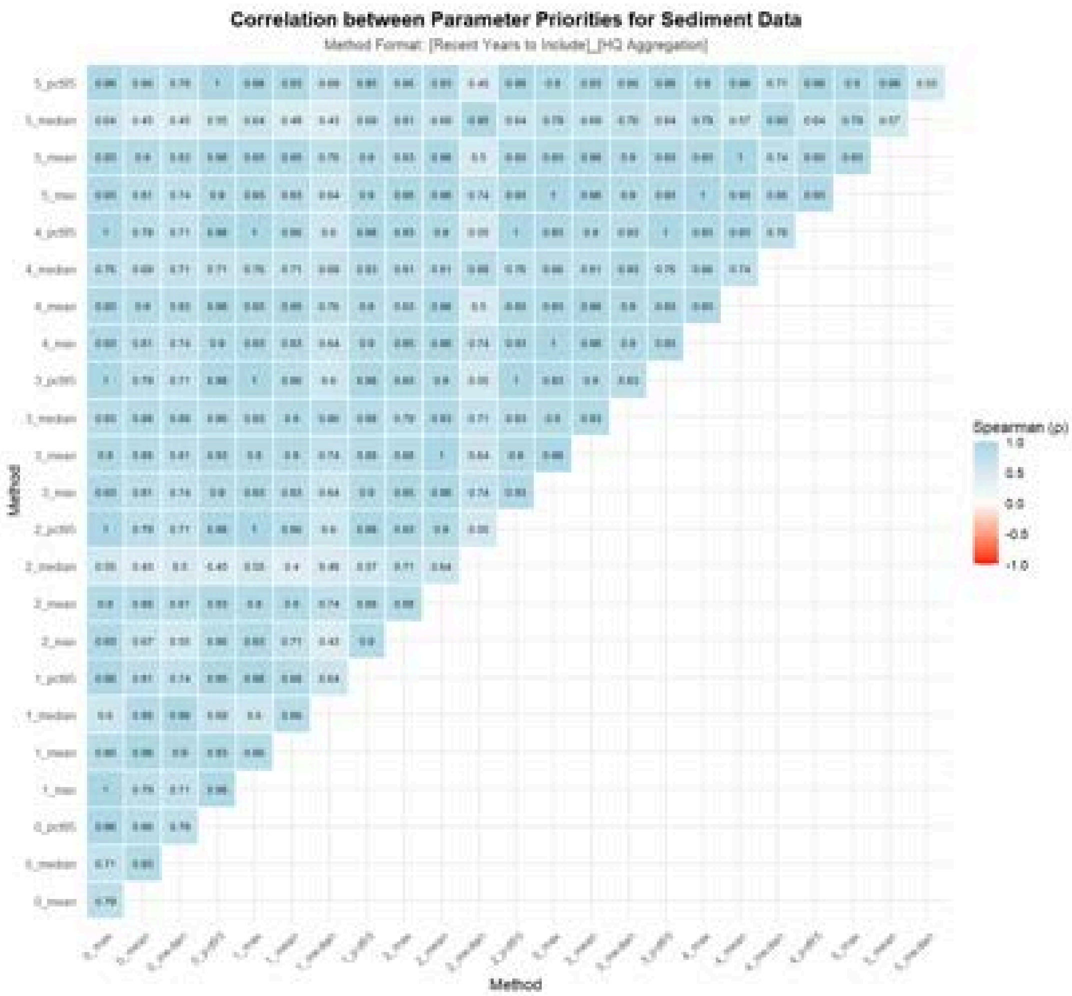

Figure 4.3.6b. Spearman correlations for output parameter priorities for sediment data are shown. Method configurations across HQ aggregation and years within the recency window (between 0 years, where only the most recent date was considered, and 5 years) were compared. Median HQ aggregation for years 2 and 5 were the least correlated with other model output, but all models generally agreed upon the same list of parameters.

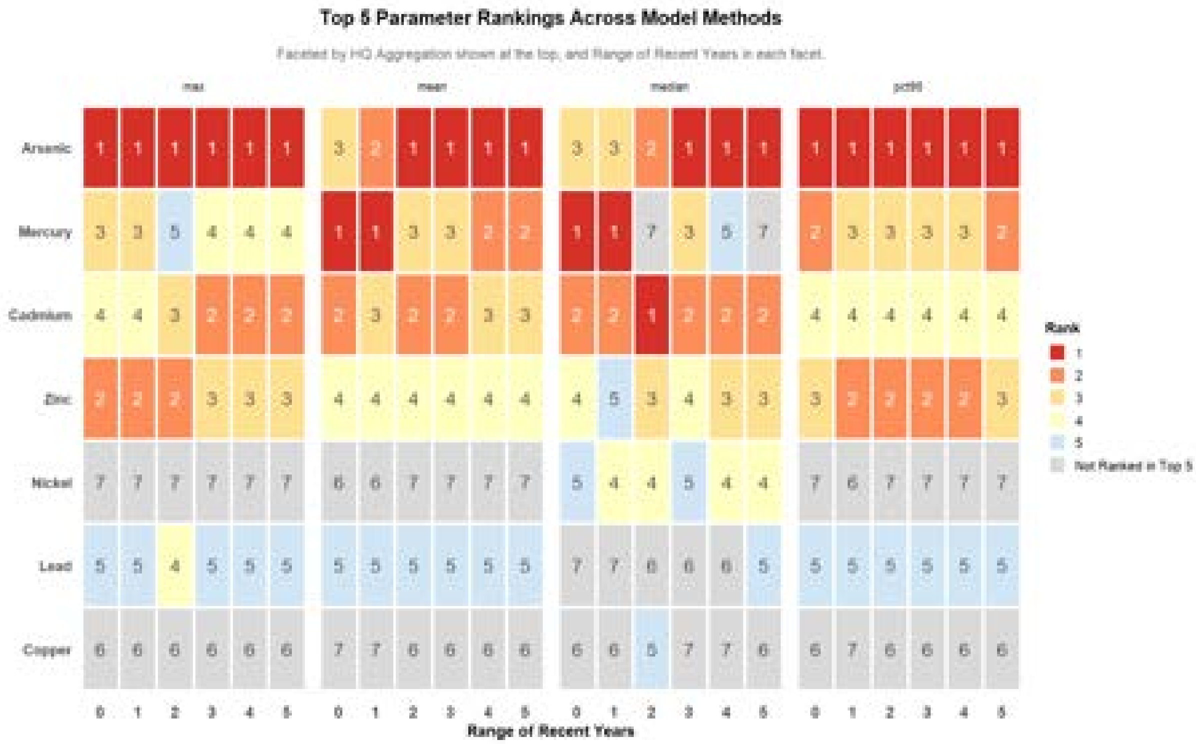

Figure 4.3.6c. Parameter priorities for sediment data across various model configurations. All models prioritized arsenic, mercury, cadmium, and zinc. Only one model (median HQ within the last 2 years) prioritized copper within the top 5 parameters of concern.

Three key modeling decisions emerged from this analysis:

1. **HQ Aggregation**: 95th percentile versus median.
2. **Temporal Filtering**: Recent data versus all historical samples.
3. **Recency Window**: Most recent date versus 5-year window

The 95th percentile was selected across the study to prioritize contaminants of highest concern and capture acute exposure potential, consistent with expert recommendations and management objectives focused on worst-case scenarios rather than chronic background conditions. Temporal recency was chosen as a time-based filter to prioritize stations with ongoing contamination, which supports AMTSK's focus on current conditions rather than allowing historical data (e.g., from 2016) to dominate prioritization. A 5-year window balances current conditions with evidence of ongoing contamination. Sensitivity analysis showed low output variation across window sizes of 2-5 years, with the 5-year window providing the greatest reporting size and a stronger sense of how slower-moving contaminants have migrated through the basin.

4.9.2 Air Particulate Concentration Modeling

Sensitivity analysis was performed to test the effect of assumptions used for the estimation of emission rates and for key HYSPLIT inputs.

Temporal Variation

The concentration model outputs produced were examined for each of the six days modeled to gain an understanding of the impact of temporal variation on our risk assessment layers (see Appendix *Figure A.3.5l*). The direction and magnitude of emission plumes varied significantly across time. In some cases, wind direction was even reversed between different days (See Figure 4.3.6d & Figure A.3.5l). Based on this analysis, it is likely that temporal variation is the largest contributor of uncertainty in the air particulate modeling performed.

July 4th, 2024 had the highest concentration plumes of the six days selected for the HYSPLIT analysis. This indicates that July 4th has the largest impact to the final air particulate hazard assessment layer of the six days modeled. Conversely, August 10th, 2024 had the lowest concentration plumes, likely contributing the least to the final risk assessment.

The outputs of two days which were not selected for the final air particulate modeling were also examined to gain a sense of the temporal variability not captured by our models. The alternate days tested can be seen in Table 4.3.6a.

Table 4.3.6a Alternate dates used for sensitivity analysis and their average 24-hr emission rates.

Date	PM_{10} Emission Rate (g/m^2-hr)	$PM_{2.5}$ Emission Rate (g/m^2-hr)
July 24, 2024	1.27	0.190
September 8, 2024	1.27	0.190

Between days with very similar emission rates, the size and concentration of plumes varied more than initially expected (See Figure 4.3.6d). The two main factors which may contribute to this trend are differences in the vertical motion of winds and differences in turbulence between days. Given that the HYSPLIT model output was generated solely for concentrations in ground-height air (2 m above ground level), vertical winds which transport pollutants higher into the air column are not accounted for. Some days may have more downward or upward winds, which would cause a difference in concentrations seen by our model outputs even between days with near identical emission rates. In addition, differences in turbulence cause differences in the area which pollutants spread over.

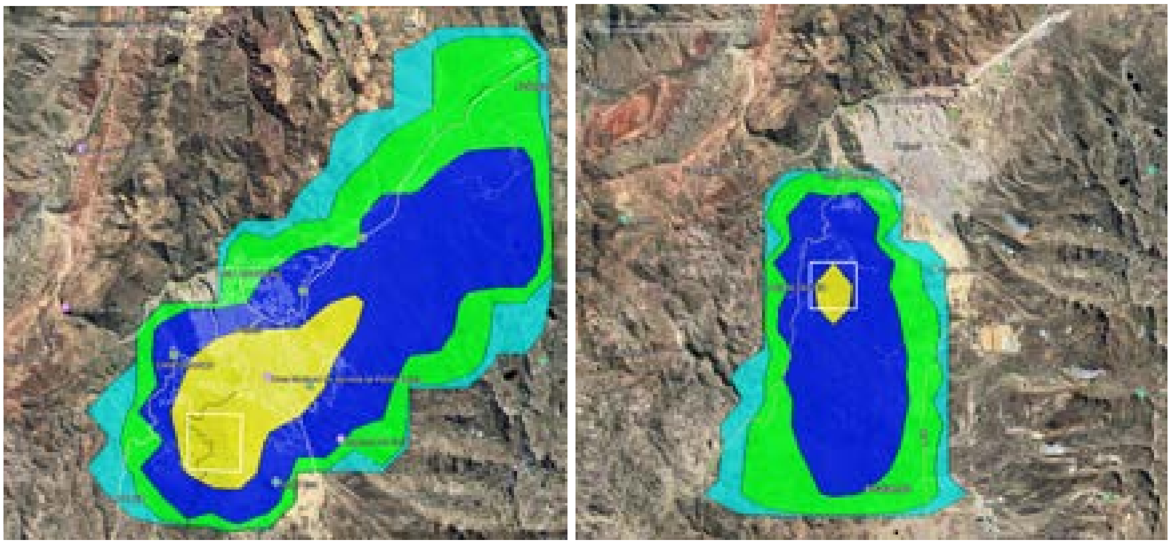

Figure 4.3.6d. HYSPLIT concentration outputs for alternate days tested for modeling. Both figures show the model produced at 6:00 AM using PM_{10}. [Left] July 24, 2024. [Right] September 8, 2024. Yellow = 0.001 g/m^3. Dark Blue = 1E-4 g/m^3. Green = 1E-5 g/m^3. Light Blue = 1E-6 g/m^3.

This analysis indicates that this modeling approach is worth further examination using a larger number of days for modeling. To gain a more complete understanding of windblown particulate dynamics, modeling using a higher resolution of temporal data is highly recommended. It is likely that this approach could produce significantly different results than the models used for this project's air particulate risk assessment.

Number of Disturbances

Sensitivity analysis was performed to see the effect of a higher number of disturbances per hour. Emission rates were calculated using 2 and 3 disturbances per hour. This had a significant effect on the concentration of pollutants. When using 2 disturbances, there was an increase in the spread of high concentration plumes compared to the 1-disturbance model. Additionally, some smaller plumes had concentrations 10 times the magnitude of the highest concentration plumes of the base model. When testing 3 disturbances, there was a significant spreading of highest concentration plumes, but no significant increase in concentrations when compared to the 2-disturbance model (Figure 4.3.6e and Figure 4.3.6f). In the models used for the air particulate hazard assessment, the number of disturbances was left at 1; there were no criteria or data to inform the true number of disturbances the region experiences. This is an important caveat, as concentrations of wind-blown pollutants from tailings experienced in the city of Potosí could in reality be much higher than predicted by the model. However, the spatial trends estimated by the models were unchanged.

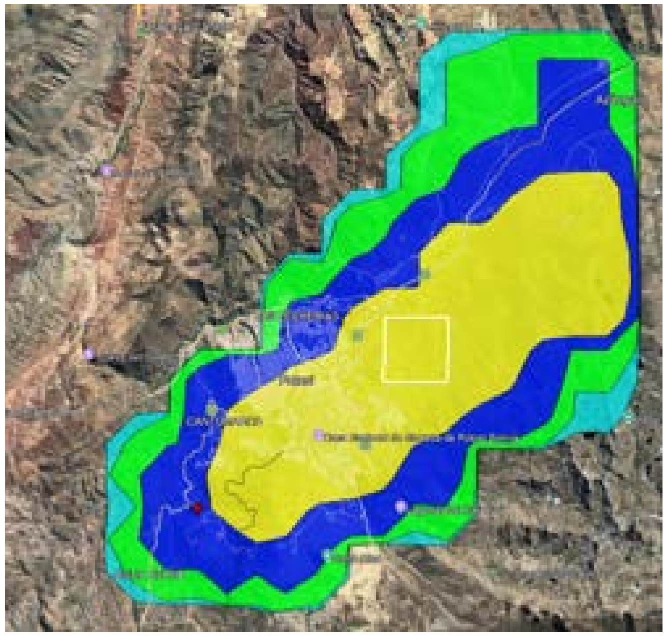

Figure 4.3.6e HYSPLIT model produced for the Los Cumpas point source at 6:00 AM July 24, 2024 using 1 disturbance to calculate the average emission rate. This is the base model by which many of the sensitivity analysis variants were compared against. Yellow = 0.001 g/m^3. Dark Blue = 1E-4 g/m^3. Green = 1E-5 g/m^3. Light Blue = 1E-6 g/m^3

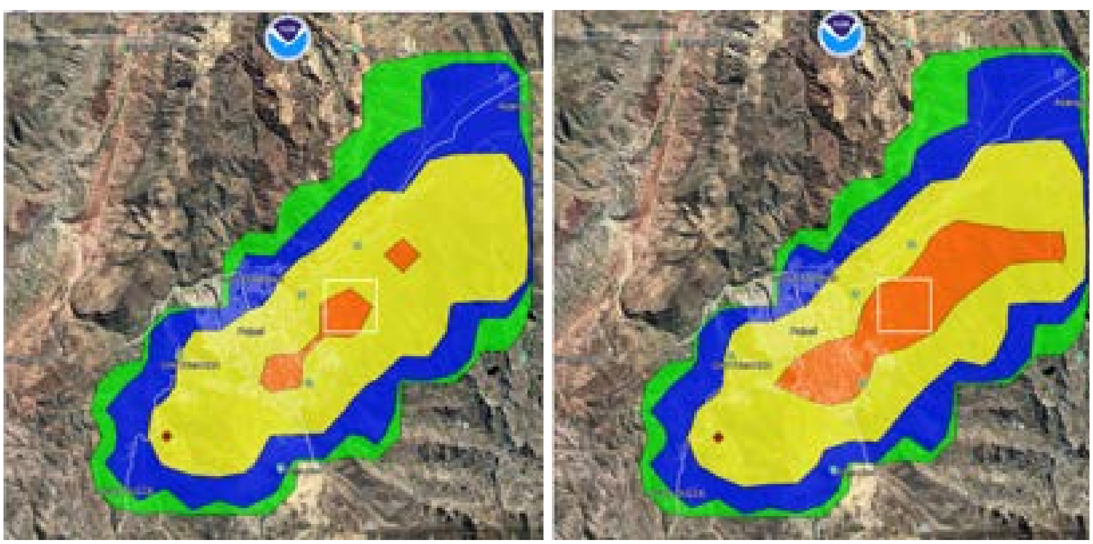

Figure 4.3.6f HYSPLIT models produced for the Los Cumpas point source at 6:00 AM July 24, 2024 using 2 disturbances [left] and 3 disturbances [right] to calculate the emission rate. Orange = 0.01 g/m^3. Yellow = 0.001 g/m^3. Dark Blue = 1E-4 g/m^3. Green = 1E-5 g/m^3. Light Blue = 1E-6 g/m^3

Roughness Height

Sensitivity analysis was performed to test the effect of roughness height (z_0). Roughness height is a factor used to estimate the roughness or smoothness of surfaces. It can vary by factors of 100 or more depending on the surface. The effect of using a roughness height of 0.000014 m and 0.14 m was tested. This had a significant effect on the concentration of plumes: when the roughness height varied by a factor of 100, the concentration of plumes differed by a factor of 10. This is expected given that $\ln(z/z_0)$ is a multiplier in one of the equations used to estimate emission rates, where z = 10 (See 3.3.3 *Estimating Emission Rates*). The shape and direction of the plumes were largely unchanged, however. The difference in concentrations is apparent when comparing Figure 4.3.6g to Figure 4.3.6e.

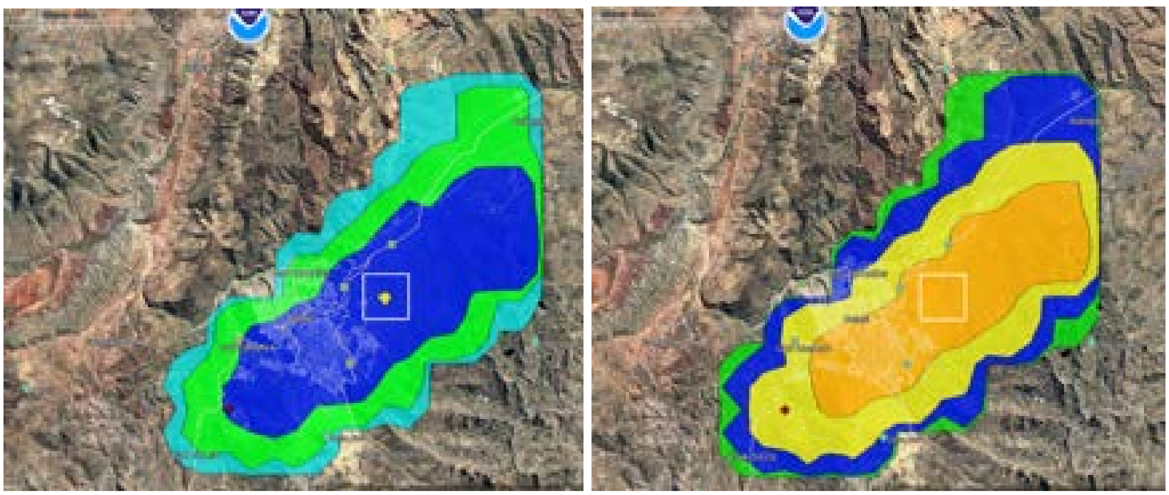

Figure 4.3.6g HYSPLIT models produced using different values for roughness height. Each model was produced on July 24 at 6:00 AM at the Los Cumpas point source. [Left] Roughness height = 0.000014. [Right] Roughness height = 0.14. Orange = 0.01 g/m^3. Yellow = 0.001 g/m^3. Dark Blue = 1E-4 g/m^3. Green = 1E-5 g/m^3. Light Blue = 1E-6 g/m^3.

Particle Diameter

The effect of changing the particle diameter used in the model was tested. Particle diameter is an input in both the estimation of the emission rate and in HYSPLIT. Based on initial models produced, it was known that the difference between using 2.5 µm and 10 µm was significant. In sensitivity analysis, 5 µm and 30 µm were tested as well. Particle size had a significant effect on the shape, size, and magnitude of concentration of particle plumes. In the equation for estimating emission rates, particle size is used as a multiplier on the final output. The multiplier values provided in the paper are as follows: $PM_{2.5}$ = 0.075, PM_{10} = 0.5, PM_{15} = 0.6, and PM_{30} = 1. For the sake of testing, 0.15 was estimated as the multiplier value for PM_5 (double the multiplier value for $PM_{2.5}$). PM_5 was tested because in order to see the effect of particle sizes between 2.5 and 10 microns, as they can have varying and significant health effects. Of the four particle sizes tested, PM_5 had the largest spread of the highest concentrations. Further examination into particle diameters between 2.5 and 10 microns may produce significant findings. The outputs of the model using $PM_{2.5}$, $PM_{5,}$ PM_{10}, and PM_{30} can be seen in Figure 4.3.6h.

An interesting result of this analysis is the magnitude of difference between the output using PM_5 and $PM_{2.5}$. There was a 2 times difference in emission rates between these 2 particle diameters, but a 10 times difference in the concentration plumes outputted by HYSPLIT. This is likely a result of the thresholding used by HYSPLIT's concentration functionality. Concentration plumes are produced in layers that differ in magnitudes of 10. It is likely that the difference in emission rates between $PM_{2.5}$ and PM_{10} caused the calculation of concentrations to cross a magnitude threshold.

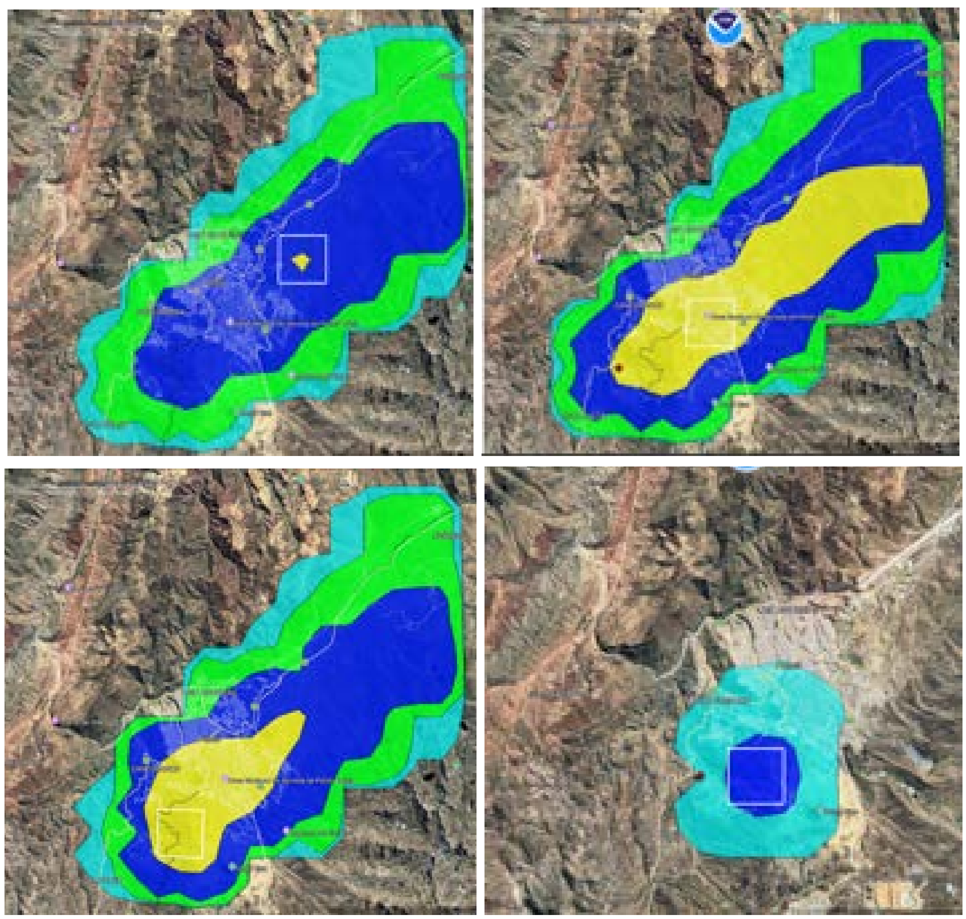

Figure 4.3.6h HYSPLIT models produced for the Los Cumpas point source at 6:00 AM July 24, 2024 using different particle diameters. Top left is $PM_{2.5}$. Top right is PM_5. Bottom left is PM_{10}. Bottom right is PM_{30}. Yellow = 0.001 g/m^3. Dark Blue = 1E-4 g/m^3. Green = 1E-5 g/m^3. Light Blue = 1E-6 g/m^3

Surface Area

The effect of deviating the estimate of tailing surface area by 30,000, 60,000, and 100,000 m^2 in both directions was tested. Predictably, surface area was positively correlated with higher plume concentrations. The magnitude of concentrations differed by a factor of 10 only when the tailing area surface differed by a factor of 100,000 m^2. The spatial trends in relative concentration were largely similar between different surface area estimations.

Other Variables

Different values for the following model inputs were tested as well: pollutant density, pollutant molecular mass, pollutant shape, layer height, starting height, and point source location. Varying the values of density, molecular mass, shape, starting height, and location produced marginal differences in model outputs. Varying the value of layer height produced significant differences, however, this was expected given that layer height was a model input we decided based on our interests rather than an assumption made for the sake of modeling. Therefore, none of these variables were determined to significantly alter the results of our HYPLIT model outputs. For a full breakdown of these analyses, see the Appendix (*A.3.5 Sensitivity Analysis*).

5. Conclusions & Recommendations

AMTSK's commitment to both environmental remediation and community health in the Pilcomayo Basin requires a strategy that is scientifically informed and feasibly implementable in Bolivia's political and economic context. AMTSK currently has an ambitious remediation plan, incorporating water remediation, tailing and sediment control, and air pollution prevention. In this section, recommendations are provided for targeted and feasible remediation strategies.

5.1 Recommended Location for AMTSK Pilot Treatment System

The locations facing the highest risk from water quality, sediment quality, and social vulnerability between 2018 and 2023, given the available reported data:

1. San Antonio - Potosí
2. Cotagaita
3. Colavi - Canutillos
4. Tumusla
5. El Puente

Of these, it is recommended that AMTSK install its pilot treatment system between San Antonio - Potosí and Potosí - Naciente rio La Ribera (Figure 5.1a). The stark contrast between water and sediment quality at these two stations support prior assumptions that the city of Potosí and heavy mining activity in surrounding areas are the most significant sources of contamination. Remediating upstream of the Pilcomayo will in principle have the effect of preventing accumulation of contaminants downstream. It should be noted that these named locations refer to the sampling stations used by the Trinational Commission for the Development of the Pilcomayo River basin.

The results of the included analyses support placing the pilot technology somewhere along the Huaynamallu River, but the data resolution is too coarse to make a more specific recommendation (given the distance between reporting stations). Water quality worsens significantly between Potosí - Naciente río La Ribera and San Antonio - Potosí. By sampling along this stretch of river, AMTSK can determine precisely where this change occurs, and place remediation machinery at that location. There may be significant mine discharge or other discrete contaminant flows going into this stretch of river which would serve as a meaningful target for AMTSK's pilot machine. Another suitable location for AMTSK's pilot treatment system is the tributary upstream of "Cueva del Diablo" (The Devil's Cave), which lies just downstream of San Antonio - Potosí (Figure 5.1a). At this location, a stream of AMD and wastewater converges. This location was recommended by a local expert (Ingeniero Bohrt) as contaminated water from Cerro Rico discharges here at a rate of 350 L/s (30.24 million L/day).

Figure 5.1a Potosí and surrounding areas with water sampling locations (circles) and settlements (house icons). Red icons indicate higher combined risk due to water contamination, sediment contamination, population vulnerability and population density. River segments connecting water quality sampling stations highlighted in teal. Cueva del Diablo shown in green.

Another highly suitable location for the pilot treatment systems is Cotagaita (Figure 5.1b). This station was identified as the highest priority in the composite analysis, though it is part of a different tributary branch than San Antonio - Potosí . Other potential locations for the initial deployment of treatment systems include Colavi - Canuntillos, Tumulsa, and El Puente (Figures 5.1c-e). As with the recommendation for San Antonio - Potosí, a high resolution sampling of these river stretches may help identify where discrete sources of contaminants are discharged, and can lead to determining suitable locations for treatment technology. In the future, AMTSK or the Bolivian government may be interested in expanding the remediation effort to a larger scale. The initial deployment of AMTSK's technology is projected to start with a pilot treatment system, followed by 6 or 7 additional installations. As the project continues, another batch of systems may be installed, or further remediation strategies may be employed, across the Pilcomayo basin. To assist long-term planning, the composite analysis supports expanding remediation to other parts of the basin. (Figure 5.1f).

Figure 5.1b Cotagaita and surrounding areas with water sampling locations (circles) and settlements (house icons). Red icons indicate higher combined risk due to water contamination, sediment contamination, population vulnerability and population density.

Figure 5.1c Colavi and surrounding areas with water sampling locations (circles) and settlements (house icons). Red icons indicate higher combined risk due to water contamination, sediment contamination, population vulnerability and population density.

Figure 5.1d Tumusla and surrounding areas with water sampling stations (circles) and settlements (house icons). Red icons indicate higher combined risk due to water contamination, sediment contamination, population vulnerability, and population density.

Figure 5.1e El Puente and surrounding areas with water sampling locations (circles) and settlements (house icons). Red icons indicate higher combined risk due to water contamination, sediment contamination, population vulnerability and population density.

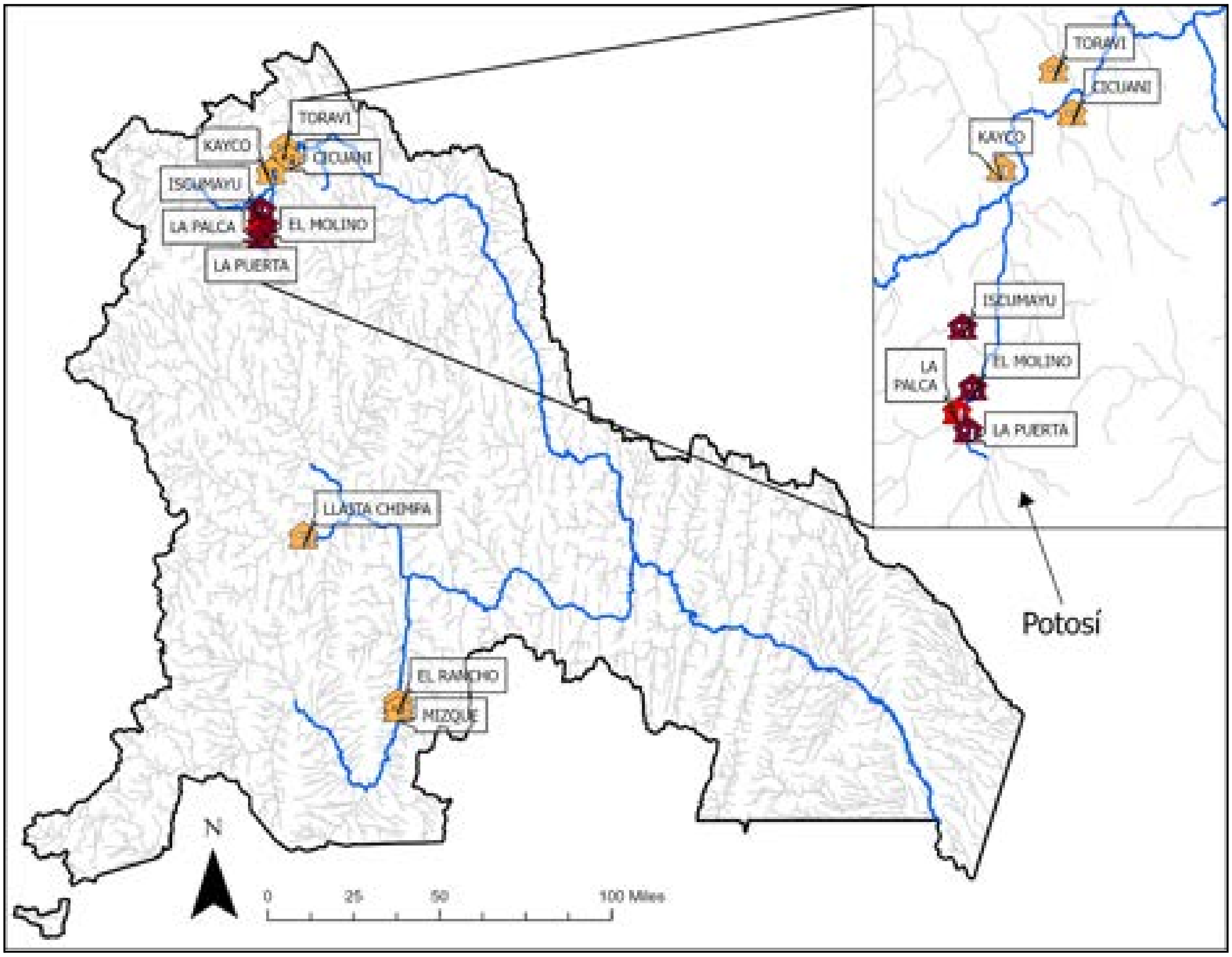

Figure 5.1f Top ten settlements with highest combined priority score due to water contamination, sediment contamination, population vulnerability, and population density. Red icons indicate higher combined priority. Rivers downstream of water quality stations shown in blue. Other rivers shown in light grey. Location of Potosí shown with an arrow.

5.2 Recommended Water Remediation Strategies

There are multiple remediation strategies which can be employed to support AMTSK's technology. While AMTSK deploys its technology in the Pilcomayo Basin, there are actions which can be taken by the Bolivian federal government, the government of Potosí, Tomas Frías University, or other organizations to assist the remediation effort.

Passive, Abiotic Remediation

Based on our analysis, we believe passive, abiotic remediation techniques will be most effective for the Pilcomayo River. A common approach is limestone-based neutralization, which is widely used to raise pH and precipitate metals (Aubé & Zinck, 1999). Our analysis revealed that toxic heavy metals concentrations are the primary issue impacting the Pilcomayo River. Limestone leach beds cause AMD and other mine waste streams to flow through coarse alkaline media, dissolving it and generating alkalinity (Jeff Skousen et al., 2017). They can be installed at AMD sites, as pre-treatment units, or in high concentration regions within the river basin. These systems are simple and robust but require sufficient hydraulic retention time, which limits the maximum flow they can effectively treat. This is especially a concern during Bolivia's rainy season, where flow rates often spike due to increased precipitation. Industrial byproducts such as steel slag can substitute for limestone in similar configurations. According to Mapukata et al. (2024), oxidative precipitation is another common approach, whereby oxidizing agents such as hydrogen peroxide (H_2O_2) convert metal ions to a less soluble, oxidized state. They state that the choice between limestone and oxidative precipitation depends on site-specific water chemistry.

It is recommended that limestone treatments be paired with diversion wells. Diversion wells can be dug to offer increased hydraulic retention time (Jeff Skousen et al., 2017). AMD is directed through a vertical tank filled with small-diameter alkaline material, where turbulence promotes dissolution and heavy metal precipitation. A downstream settling pond then captures the precipitated solids. While these systems require periodic replenishment of alkaline material, they have relatively low operational complexity and are well-suited to remote sites where power and technical capacity are limited.

Another passive, abiotic technique which can be employed at low cost is filtration. According to Mapukata et al. (2024) filtration can be performed using simple materials such as sand, diatomaceous earth, activated carbon, cellulose, and anthracite coal. There are two types of filtration, depth filtration, where contaminants penetrate into filter media, and surface filtration, where particles are retained on the top layer of the media. Depth filtration is recommended as it allows for higher contaminant-holding capacity than surface filtration (Mapukata et al., 2024).

Passive, Biological Remediation

Other remediation strategies which could be effective in the Pilcomayo Basin are passive, biological treatment options. Passive biological systems align well with low-cost, low-energy settings. Constructed wetlands, compost-amended anoxic ponds, and subsurface reactive barriers rely on natural microbial processes and require relatively simple infrastructure. They are especially suitable where flows are moderate, land area is available, and local communities or mine operators can perform basic maintenance (e.g., clearing vegetation, inspecting flow paths). As with limestone leach beds, spikes in flow rates due to precipitation may pose a challenge. In Bolivia, these systems are promising for legacy sites near communities (e.g. Cantumarca), provided land rights are resolved and designs consider local climate, elevation, and extreme-event hydrology. Risks include metal accumulation in biota and the need to manage contaminated sediments or plant material over time.

Active Remediation

For areas with access to trained personnel and higher monetary resources, the High Density Sludge (HDS) process is recommended. According to Mapukata et al. (2024), HDS has been the industry standard for treating AMD for decades. Compared to other treatment options, it produces low levels of metals in final effluents and a much smaller volume of waste sludge. In the HDS process, a neutralising reagent is added to AMD to precipitate and remove metal ions and a flocculant is added to aggregate fine particles. The mixture settles in a settling tank and forms a sludge layer concentrated with heavy metal contaminants. The sludge is then separated and the treated water can be discharged or treated further. Metals can be recovered or beneficiated from the sludge, however, sludge is most commonly landfilled, stabilized, or recycled. It is important to note that HDS is expensive due to the specialized equipment and chemicals used and the difficulty of disposing of the waste sludge (Mapukata et al., 2024).

In addition, it is highly recommended that the city of Potosí install a wastewater treatment plant. Currently, the city discharges its wastewater into various bodies of water. Based on wastewater fingerprinting analysis, parameters correlated with wastewater pollution are far above WHO standards for safe water in the Pilcomayo River. Treating wastewater before discharge into bodies of water would significantly decrease the hazard posed by wastewater contamination.

Preventing AMD From Tailings

In order to ensure long-term viability of remediation, further contamination of water must be prevented. Tailings are a major source of contaminants, with slurries of toxic material being carried by rains. Engineered covers prevent rain water from mixing with the tailings, thus preventing this pathway of pollution. Engineered covers also limit the exposure of tailing materials to oxygen, thus mitigating Acid Mine Drainage (AMD). Material from tailings can also be reprocessed to extract leftover valuable materials. This process reduces the overall volume of waste material, but does not eliminate the need for waste storage. To deal with existing AMD or other waste streams from tailings, channels can be built to divert harmful streams away from the Pilcomayo river. These streams can be collected in waste ponds, which can be remediated. Lastly, strict regulation should be applied to any new tailings which are developed in Bolivia. Namely, any new tailings should be required to have a lining in order to prevent groundwater contamination. Other regulations could include requiring engineered covers or secondary processing.

5.3 Recommended Sediment Remediation Strategies

The combined findings from spatiotemporal migration analysis, mechanistic partitioning verification, sediment texture characterization, and hypoxic event monitoring provide a geochemically grounded basis for targeting remediation interventions across the Pilcomayo River Basin. Effective treatment must account for the dual-phase nature of contamination, the metal-specific retention pathways identified through distribution coefficient (K_d) analysis, and the seasonal dynamics that govern when metals are most mobilized and most accessible for intervention.

Seasonal Intervention Priorities

The seasonal exchange cycle represents both the basin's primary contamination mechanism and its most tractable intervention point. Rainfall in the Potosímining headwaters triggers mobilization pulses that amplify water column concentrations by 3 to 30 times, with wet-to-dry season ratios reaching nearly 30 times for lead and cadmium at Argentina/Paraguay stations. Wet season interventions should therefore prioritize reducing the flux of dissolved and suspended metals entering the river system at the source. This can be accomplished through mechanical means such as diversion channels, berms, or drainage ditches that intercept mine drainage before it reaches the river channel.

During the dry season, metals settle and accumulate in sediment reservoirs, with the most pronounced enrichment occurring for cadmium in sandy loam and arsenic in silty loam. Because mining-derived loading in the Potosí zone saturates all available sorption sites regardless of grain size, texture-targeted dredging is not warranted; the dry-season focus should instead shift to in situ stabilization of contaminated bed material.

In Situ Stabilization

In situ stabilization uses chemical or microbial amendments to transform heavy metals into stable mineral forms that resist migration into the water column and uptake by biota. Amendment selection should follow the metal-specific retention hierarchy established by sediment-water distribution coefficients: Pb (1,667 L/kg) > As (908 L/kg) > Zn (664 L/kg) > Cd (200 L/kg).

For arsenic, mechanistic verification confirms that iron oxide adsorption is the dominant retention control; amendments consisting of iron, manganese, and aluminum (hydro)oxides that expand available sorption surface area are therefore recommended. For lead and cadmium, iron oxide-rich red mud and limestone amendments are effective at reducing extractable concentrations. Cadmium warrants particular attention: its low K_d reflects dependence on acid-soluble calcium carbonate and phosphate minerals rather than iron oxides, making it highly sensitive to the pH 3 to 5 conditions prevalent in the Potosí mining zone. Targeted alkalinity and calcium phosphate amendments are required rather than standard oxide-based treatments.

A critical caveat applies to pH amendment for arsenic: raising pH toward neutrality, while beneficial for cadmium and lead mobility, would reduce arsenic retention on aluminum oxide surfaces, whose protonated Al-OH groups provide strongest arsenate binding under acidic conditions. Any pH amendment program must be implemented gradually and monitored for arsenic release.

Maintaining dissolved oxygen above the 4 mg/L threshold that currently characterizes 98.5% of basin samples should be an explicit management target. The October 2017 Naciente hypoxic event demonstrated that a drop in dissolved oxygen can trigger a nearly threefold enrichment of lead in the water column through reductive dissolution of manganese oxides. Installing wastewater treatment infrastructure in the Potosí mining zone would substantially support this objective by reducing organic loading and microbial oxygen demand.

Mid-Basin Passive Treatment

In the mid-basin, remediation should focus on passive interception at key depositional zones, capitalizing on the longer hydraulic residence times during the dry season. Constructed wetlands and phytoremediation buffers are well-suited to this reach. Phytoremediation involves the planting of hyperaccumulator species to extract heavy metals through root uptake. Harvested plant material must be processed for metal recovery or disposed of through controlled burial to prevent secondary contamination exposure. Zinc's resilience across eight binding mechanisms — including clay, organic matter, iron, aluminum, manganese, and calcium mineral phases — suggests it will respond particularly well to passive approaches that preserve existing sorption capacity.

Priority Hotspots and Ex Situ Treatment

Targeted ex situ treatment through dredging and off-site stabilization should be reserved for the most severely contaminated hotspots identified by spatiotemporal modeling. Priority sites are San Antonio-Potosí and Tarapaya, where arsenic exceeds 200 mg/kg and lead exceeds 180 mg/kg, and locations where hypoxic events have historically driven acute downstream metal loading.

Emerging Contaminants

An emerging priority not captured by the four-year K_d baseline is the rapid escalation of mercury and cadmium contamination since 2019. Mercury concentrations reached 2.3 mg/kg in the 2024 wet season, more than thirteen times the USEPA freshwater sediment guideline of 0.174 mg/kg, and cadmium simultaneously spiked to 5.0 mg/kg, the highest value in the eight-year record. The synchronous nature of these increases suggests a discrete upstream source change, potentially tied to shifts in ore processing methods or the introduction of artisanal gold amalgamation. Urgent source identification is required before remediation planning for these metals can proceed.

5.4 Mitigating Impacts of Air Particulates from Tailings and Other Sources

Based on modeling and literature, the concentration of metal and mineral-laced particulates in the air of Potosí is above safe levels during Bolivia's dry season. The sources of this contamination include: smelting, rock blasting, particle size reduction processes (crushing, grinding, etc.), transportation (both resuspension from vehicle traffic and windblown emissions from uncovered vehicles), waste rock piles, and tailings.

Monitoring
To address the impacts of air particulates, we recommend enhanced data collection and monitoring. Air quality data can be used to issue public warnings when air quality poses a health hazard. It can also be used to inform decisions on where and how to implement solutions. We also recommend making the data publicly available in some form so that it can be utilized in further research and solution development.

The most comprehensive way to collect data on air particulates is through an air quality monitoring station. These provide real-time, robust data on air quality, including particulate matter and other forms of air pollution. However, air quality monitoring stations are expensive and technologically complex. There are a number of cheaper alternatives to collect air quality data. According to Kasongo et al. (2024), a cost-effective and reliable method of monitoring air pollutants is to analyze the composition of dust deposited on terrestrial biomass or on rooftops; this method gives an estimation of the chemical composition of atmospheric dust. The mass concentration of aerosols must also be measured, with gravimetric filtration being the reference method set by the European Directives. However, this method can be labor-intensive and difficult to apply at scale. A simpler alternative accepted by the European Directives is to use Optical Particle Counters to estimate mass concentration based on the number concentration of particles. This involves making some assumptions about particle characteristics, but is much easier to apply at scale (Kasongo et al. 2024).

Mitigating Windblown Particulate Emissions
There are a number of simple and effective methods to reduce the emission of mining-related particulate matter. Windblown particulates contribute a significant portion of mining-related air pollution. Covering sources of windblown dust can prevent the emission of pollutants. Sources that can be covered include: tailings, waste rock piles, and truckbeds carrying mineral products. Truckbeds can be covered with a simple tarp or a mechanical enclosure to prevent windblown

emissions. Additionally, reducing the speed of vehicles in mining areas can reduce resuspension of particles.

Tailings and waste rock piles can be covered with engineered covers for maximum reduction in windblown particles. However, this can be very expensive and takes a long time. According to Kasongo et al (2024), tailings and waste rock piles can also be regularly covered in water, which can reduce particulate emissions by a factor of three. They state that tailings and waste rock piles can be covered in layers of uncontaminated soil as well, however, the volume of soil and water needed may be too great for some situations. The cost of these techniques is dependent on the regional cost of soils and water, respectively. According to Kasongo et al (2024), there is a growing field of research on chemical dust suppressants. Most notably, food processing waste and byproducts can be utilized for dust suppression; these are of high interest because they are low cost and have a minimal environmental impact (Kasongo et al, 2024). Dust suppressants can also be applied to mining products during transportation.

Mitigating Particulate Emissions from Mining Processes

Dust emission from mining processes can be mitigated through simple methods. According to Kasongo et al. (2024), spraying water during rock blasting has been used. Additionally, the addition of water or exhaust ventilation to crushing, grinding, and screening has been employed. Remediation of contaminated soils can reduce the impact of windblown emission and resuspension. The direct recovery of contaminated topsoils and bioremediation have been employed in the past (Kasongo et al., 2024). Particulate emissions from smelting can be controlled by employing mitigation technology at smelting facilities. The specific equipment employed should be based on the size of particulates emitted at a particular site, and multiple technologies can be deployed to mitigate a range of size classes. Speight (2020) describes a number of filtering and scrubbing technologies which have been employed to control particulate matter, including: cyclone collectors, fabric filters, wet scrubbers, and electrostatic precipitators. It is recommended the federal government or government of Potosí issue increased regulations and standards for mitigating particulate emissions from mining activities. Some examples are mandating proper protective masks for miners, setting emissions limits for mining facilities, or setting technology standards for refineries. Inspiration for such policies can be taken from the United States Clean Air Act or similar European policies.

Reducing Population Exposure to Air Hazards

While less severe, waste rock piles also contribute to pollution in the Pilcomayo Basin. A potential strategy of reducing populations' exposure to airborne pollution from these piles is to consolidate them. Currently, waste rock is stored in an unorganized fashion in the Potosí area, with many separate small piles scattered haphazardly throughout the city. More effective consolidation of waste rock away from centers of population would reduce population's direct proximity to these sources of airborne contaminants.

A quickly implementable recommendation for the government of Potosí, the federal government of Bolivia, and other organizations is to assist communities in acquiring personal protective equipment. Protective face masks such as N95s, KN95s, and FFP2s are tested to block 95% of fine particles down to 0.3 microns. Distribution of these materials can be facilitated by the local government, especially during Bolivia's dry season, when windblown particulate emissions are most severe. These can help reduce the burden of respiratory illness experienced by communities in Potosí. These can be especially impactful for miners, who experience the most direct exposure to particulates. Other protective technologies include: indoor air purifiers, filters for HVAC systems, humidifiers, negative ion generators, or indoor air quality monitors. Wearing long clothing and protective eyewear can also reduce direct exposure to harmful particles. It is recommended that the government of Potosí issue advisories to communities for equipment and practices to prevent exposure from harmful particulates.

5.6 Policy and Enforcement Considerations

During outreach conducted in Bolivia, community members consistently identified inadequate enforcement of environmental regulations and a lack of transparency from mine operators as primary concerns. Enhanced monitoring and reporting requirements for mining companies, specifically relating to waste disposal and pollutant emissions, are recommended to strengthen the capacity of local and federal authorities to identify violations and hold responsible parties accountable under Bolivia's existing environmental regulatory framework.

A thorough examination of environmental statutes from the United States and European nations is advised as a reference point for potential reform. Relevant examples from the United States include the Resource Conservation and Recovery Act (RCRA), the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), the Clean Water Act, and the Clean Air Act. Of particular note are enforcement mechanisms for violating parties, including fines which act on a per-day-of-violation basis, injunctions, financial responsibility for remediation, and federal prosecution.

Of additional relevance are the procedural requirements these laws place on regulatory administrators. Under US law, EPA administrators are required to set environmental standards based on scientific evidence within defined timeframes. These provisions help ensure that standards are both scientifically grounded and updated at a pace commensurate with the urgency of environmental conditions. Enforcement mechanisms permit private actors to pursue legal action against federal agencies that fail to promulgate or enforce standards as required, allowing for local communities to contribute in ensuring accountability.

It should be noted that the US statutes referenced are of considerable age and carry well-documented limitations, including regulatory loopholes and implementation challenges. Bolivia is therefore positioned with a distinct advantage: the opportunity to draw selectively from multiple legal traditions, adapt those frameworks to the specific needs and priorities of the Bolivian people, and develop a more iterative and responsive regulatory approach informed by decades of international experience.

5.7 Future Use of the Hazard Ranking System

The composite hazard ranking system developed for this project is designed to compare areas of the river basin relative to one another in terms of short-term potential for heavy metal pollution. Its modular, multi-layer architecture allows individual layers to be updated, added, or removed as data becomes available or priorities change over time. Any new layer incorporated into the system must include a spatial component to maintain compatibility with the current design. Data layers of particular interest for future integration include local health outcomes such as blood lead test results or disease and chronic illness incidence rates, water sourcing information, and fish consumption data. As additional data is collected across the region, the modular design allows local stakeholders to evaluate relative risk under varying criteria.

The water hazard quotient layer is similarly designed for iterative refinement. New data from future sampling campaigns conducted by the Trinational Commission for the Development of the Pilcomayo River Basin can be incorporated as they become available, enabling stakeholders to assess whether the spatial distribution of water pollution risks has shifted over time. The methodology also accommodates testing of alternative aggregation approaches, including different temporal windows, alternative summary statistics such as the 90th percentile or mean, and the application of different health-based reference standards.

The current analysis is focused on acute human health risks; however, the same framework can be extended to assess additional dimensions of risk from mining pollution. Cancer risk assessments and ecological risk assessments for wildlife both have established concentration-based standards against which measured metal concentrations can be compared. Depending on community needs and stakeholder priorities, these dimensions can be evaluated independently or combined with the existing human health risk layers.

Finally, the methodology developed for this project is not specific to the Pilcomayo Basin. Where comparable data are available, the approach can be replicated and applied to other subcatchments in Bolivia or to mining-impacted river systems elsewhere. Potential applications include remediation planning, public health program development, and support for environmental justice advocacy efforts.

5.8 Web Applications

Two web-based applications were developed to provide interactive insights on environmental data to various stakeholders, including government officials, industry leaders, and remediation experts. These apps were developed in parallel to conduct different types of analyses and remain separate to better support different stakeholder groups. Both applications can be viewed in either English or Spanish.

5.8.1 Sediment and Water Quality Explorer

The Water and Sediment Quality Explorer application allows users to upload water and sediment quality data to generate dynamic data visualizations and maps of the Pilcomayo Basin (Mills et al., 2026).

Available Data
The application utilizes data from the Trinational Commission from 2016-2023, with the option to include data from 2024 (See *4.1 Data Limitations*). Users may filter the data by analyte, sampled station, and time ranges, and visualize results based on raw values, comparisons to standards, or risk-based factors such as hazard quotient. Users can upload new environmental data sampled in the region utilizing a provided formatting guide or the format utilized by the Trinational Commission, supporting the analysis of future environmental surveys conducted by AMTSK, local community organizations, government agencies, and other interested parties.

Ranking Plots
A series of plots rank the reported concentrations of water or sediment media by:

1) ranking individual reported samples by concentration or HQ,
2) ranking stations by HQ,
3) ranking hazardous parameters, and
4) ranking grain sizes by HQ

Each plot can be filtered to focus on specific parameters, media, fractions, or stations. Sediment parameters were grouped by sieve size (grain-size fraction) using the Wentworth grain size classification system (Wentworth, 1922). Additional information about these plots can be found in the Appendix (*A.2.1 Data Analysis*).

Time Series Plots
Time series plots were generated to inspect trends in water and sediment quality over time. Users may select a parameter and a location to generate a time series plot showing how measured values of that parameter have changed over time. Users may also display relevant reference limits to compare with measured values.

Combined Risk Maps
Layers that combine hazards from water and sediment quality data with municipality-level EJI scores and population density can be generated. Water and sediment risk layers are created by scoring stations (see *3.3.1 Hazard Quotient Aggregation*), and interpolating scores across the river network (see *3.3.1 Interpolation Across River Network*). Users may adjust the method for scoring stations. EJI scores and population density are included within the app.

Once layers are generated, and a score binning method is selected for each layer (e.g. Equal-Interval, Quantiles), layers can be included into a final prioritization map to provide priority scores along impacted areas along the basin. The locations of mines and settlements, rivers, tailings, ingenious and processing facilities, and subcatchments can also be shown. Subcatchments may be delineated using the water and sediment stations as outlet points, and are assigned the scores of their corresponding stations.

5.8.2 Contaminant Migration Analysis Site

An interactive web-based dashboard was developed to facilitate comprehensive visualization and analysis of heavy metal contamination across the Pilcomayo River Basin. The platform integrates water quality and sediment data from 46 monitoring stations spanning 2016 to 2024 and is organized into modular pages accessible through a sidebar navigation with three dropdown sections: Migration Cycle, Action Plan, and Project History. The main dashboard page provides a basin overview including an interactive station map, a summary of key findings across all analytical components, median partitioning coefficients by metal, and basin-wide statistics.

Migration Cycle
The Migration Cycle section contains six analytical pages. The Spatiotemporal Trends page presents water column and sediment time-series plots and seasonal comparison visualizations. The 3D Spatial Models page displays interactive three-dimensional contamination surface models for As, Cd, Pb, and Zn across the full downstream gradient. The Sediment Partitioning Chemistry Analysis page presents K_d distributions, Spearman correlation heatmaps, and mechanistic verification results. The Hypoxic Events page presents event-specific time-series and before-and-after enrichment analysis. The Potosí-Specific Sediment Contamination Analysis page presents texture class comparisons, seasonal decomposition, and temporal trend plots for the mining zone. The Sediment Texture Analysis page documents basin-wide textural classification distribution across all 959 sediment samples.

Action Plan
The Action Plan section presents the Remediation Strategy and Cost-Benefit Analysis pages, which provide intervention options and associated cost estimates for priority contamination zones.

Project History
The Project History page presents a chronological timeline of mining activity in the Potosí region from 1545 to the present alongside documentation of affected communities and associated environmental and health impacts.

5.9 Next Steps

AMTSK and Ingeniero Bohrt's lab at Tomas Frias University have plans in motion to monitor and remediate contamination of the Pilcomayo River. Following a review of their proposed methodology (Detailed Plan to Establish the Environmental Baseline of the Huaynamallu River), recommendations were developed by the team.

Parameter Selection
The current plan includes measurement of zinc, copper, iron, arsenic, and lead. Based on the results of the team's analysis, measurement of silver, cadmium, and selenium is also recommended, as these parameters produced some of the highest hazard quotient scores across all parameters assessed. The current plan also includes nitrates and phosphates to estimate nutrient concentrations. Measuring ammonia is also recommended, given that it yielded the second highest hazard quotient score of all parameters assessed. Measurement of specific fecal indicator bacteria, such as enterococcus, is also recommended to distinguish between nutrient contributions from agricultural runoff versus wastewater.

Sampling Design
The current plan focuses on sampling the Huaynamallu River, a decision well-supported by the findings of the team's analyses. The plan proposes sampling at 5-10 sites; if feasible, sampling at 10 sites is strongly recommended. A higher site density will allow for more precise identification of discrete contaminant inputs. Should sampling effort expand in the future, the Alja Mayu River is recommended for priority consideration, as it flows directly downstream of Huaynamallu and feeds into San Antonio-Potosí, the highest-risk sampling station identified through composite analysis. El Potrillo is another high-priority location for future sampling.

Risk Prioritization Framework
Bohrt's lab proposes an integrated risk matrix that incorporates human health risk (HQ), ecological risk (RQ), social vulnerability, and disaster risk, weighted according to the following formula:

$$\text{Priority} = [HQ \times 0.4] + [RQ \times 0.3] + [\text{Social Vulnerability} \times 0.2] + [\text{Disaster Risk} \times 0.1]$$

The inclusion of ecological risk is a notable strength of this framework, as it incorporates dimensions of risk which were of interest, but outside the scope of this project. However, the incorporation of disaster risk into this formulation warrants caution: actions taken to prevent catastrophic events such as tailings dam failures are fundamentally different in nature from those taken to remediate ongoing pollution, and conflating the two within a single priority score may obscure appropriate response actions. Evaluating disaster risk through a separate assessment process and pursuing preventative measures on an independent, expedited timeline is recommended.

Careful, stakeholder-informed selection of numerical weights is also advised. In the composite risk assessment conducted for this project, numerical weights were intentionally omitted, as the relative importance of different risk components depends on local priorities. These can be determined more appropriately by decisionmakers and affected communities. Weights can be a valuable tool when stakeholder priorities are clearly established through an inclusive engagement process.

Data Collection

Bohrt's plan addresses several data gaps encountered during this project, most notably through the incorporation of blood lead testing, fish tissue sampling, agricultural soil analysis, livestock milk sampling, and community health surveys covering water source, diet, and health symptoms. This methodology is well-aligned with approaches considered during the initial planning stages of this study. Expansion of this data collection effort to other critical areas of the basin is strongly encouraged.

Baseline Establishment and Adaptive Management

Overall, Bohrt's proposed plan is well-aligned with the analyses and methodology of this study and is expected to play a critical role in tracking upstream contamination sources. The Huaynamallu sampling effort will be essential for siting AMTSK's pilot remediation technology, and completion of this sampling as soon as practicable is recommended. The spatial resolution of data available from the Trinational Commission for the Development of the Pilcomayo River Basin was relatively coarse; the proposed sampling design will establish a substantially higher-resolution baseline for the area identified in this analysis as most critical.

Establishing an accurate baseline is a foundational step in the remediation process, as it enables rigorous evaluation of the effectiveness of interventions over time. Defining measurable, scientifically informed remediation goals at the outset is essential. Ongoing monitoring against the established baseline should inform adaptive management as needed. A Before-After Control-Impact Paired Series (BACIPS) framework is recommended for evaluating remediation outcomes, with stations upstream of any deployed remediation technology serving as controls.

The collaboration of Tomas Frias University, AMTSK, the Potosí regional government, and the Bolivian federal government will be central to the long-term success of future efforts. In addition, local participation and meaningful stakeholder engagement are critical factors in the viability and sustainability of remediation strategies in the basin. Ultimately, the purpose of this project is to support AMTSK's remediation efforts and address the environmental and health burdens facing communities in the Pilcomayo Basin, which are rooted in a long history of mining activity. It is the hope of our team that these findings contribute to planning and decisionmaking that leads to measurable improvements in human health across the region.

Data Availability Statement

To maximize accessibility, all tools and materials developed by the project are available in both English and Spanish.

- Interactive data visualizations of the Trinational Commission Data are hosted on Sediment and Water Quality Explorer: https://github.com/snakekn/pilcomayo_water_sediment_quality_app
- Comprehensive interactive visualizations supporting the contaminant migration analysis presented in Section 4.4, including sediment-water partitioning coefficients, physicochemical and mineral-phase correlation heatmaps, three-dimensional spatiotemporal contamination surfaces, hypoxic event enrichment analyses, and basin-wide sediment texture distributions, are hosted on the project's public-facing dashboard at https://katerinabischell.github.io/riverremedy/
- Annual water and sediment quality data are available upon request from the Trinational Commission for the Development of the Pilcomayo River basin, which collects and updates the repository annually.
- 2024 GDAS meteorological data is available at https://noaa-oar-arl-hysplit-pds.s3.amazonaws.com/index.html#gdas1/2024/
- 2024 Bolivian Census data is available at https://cpv2024.ine.gob.bo/index.php/principal/descargas/
- This report is available through the Bren School's Master's Projects repository.

Code availability: Analysis scripts are hosted through a GitHub repository to support reproducibility and independent verification of results.

Our team is open to inquiries and further discussions as the need may arise.

Appendix

A.1 Supplemental Background

A.1.1 Priority Region Selection Strategy

Remediation of past environmental damage has yet to occur due to a lack of resources committed to restore the river basin. A methodology to prioritize regions will be required to allocate finite resources in areas most affected by the history of mining. This methodology should incorporate the characterization of harms to local communities and the impacts on the environment posed by historical and ongoing mining.

Multi-index frameworks characterizing localized harms with socioeconomic demographics have been recommended from an environmental justice perspective for prioritizing regions for improvements (*EJSCREEN Fact Sheet*, 2016). Yang et al. (2021) developed a model to evaluate the health vulnerability of Chinese villages to heavy metals, which can be redeveloped for use in Bolivia. They determined population vulnerability by multiplying physical vulnerability and social vulnerability indicators into an environmental health vulnerability score. Figure A.1.1a indicates the various data considered within each index. Access to relevant data may be limited in our study region, thus a model specific to Potosí should be developed. Modifying Yang et al.'s methodology to incorporate health concerns from drinking water and other exposure routes can provide a meaningful quantitative approach to prioritize regions of greatest concern.

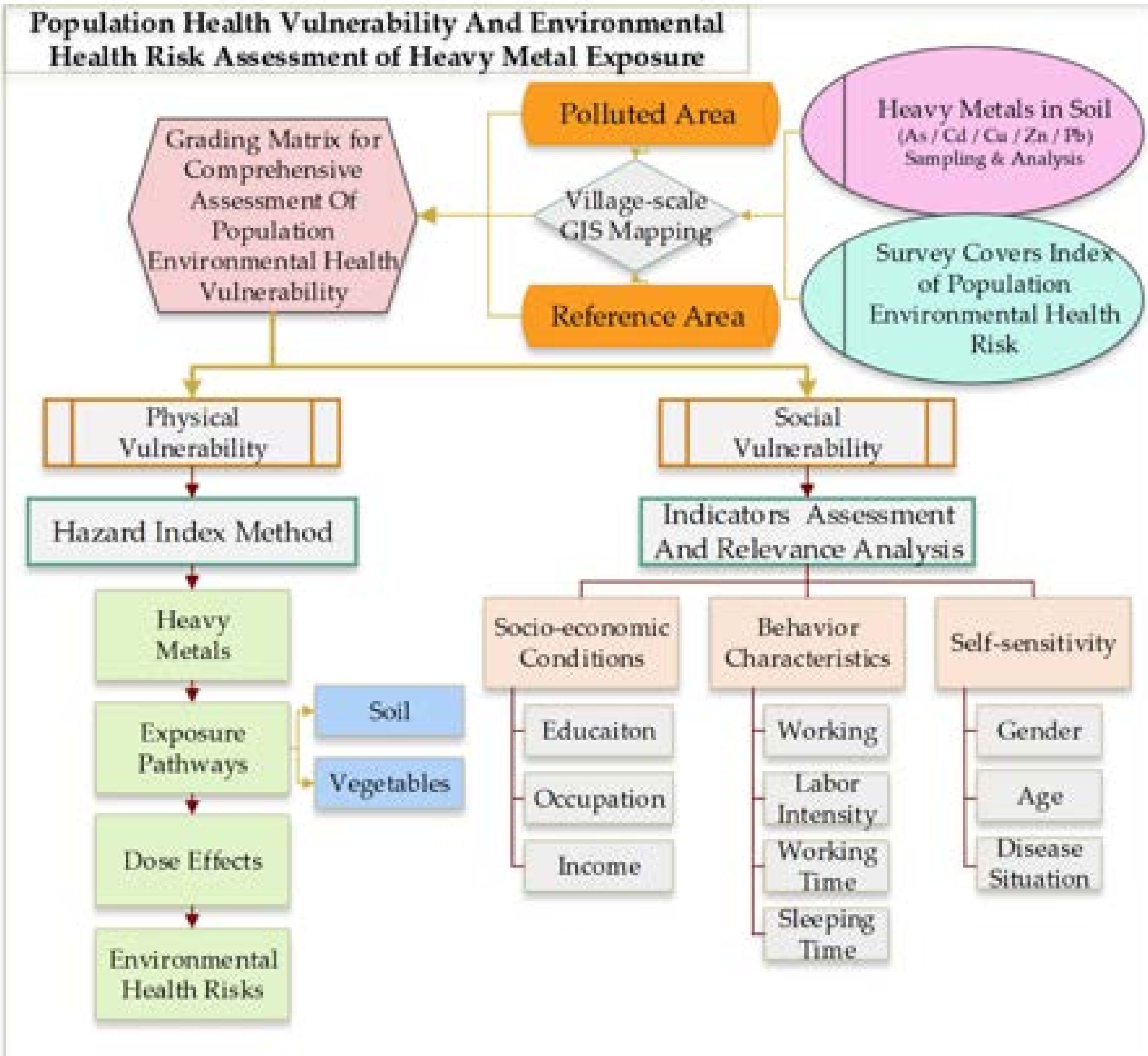

Figure A.1.1a Data considered within various indices to assess a population's environmental health vulnerability within a specified region using various sampled data. This method can be modified to provide a relevant assessment of impacted regions within the Pilcomayo river basin impacted by mining activities (Yang et al., 2021).

A.1.2 Remediation Strategy

AMD Remediation Decision Framework

Remediation treatments are segmented by abiotic versus biological processes, along with active or passive systems as shown in Figure A.1.2a and Table A.1.2a (Johnson & Hallberg, 2005; Rambabu et al., 2020). Selection criteria include metals removed, land area utilized, waste materials produced, cost of installation and maintenance, flow rates, and others. The

International Network for Acid Prevention has listed various treatment activities depending on AMD characteristics (*GARDGuide*, 2018).

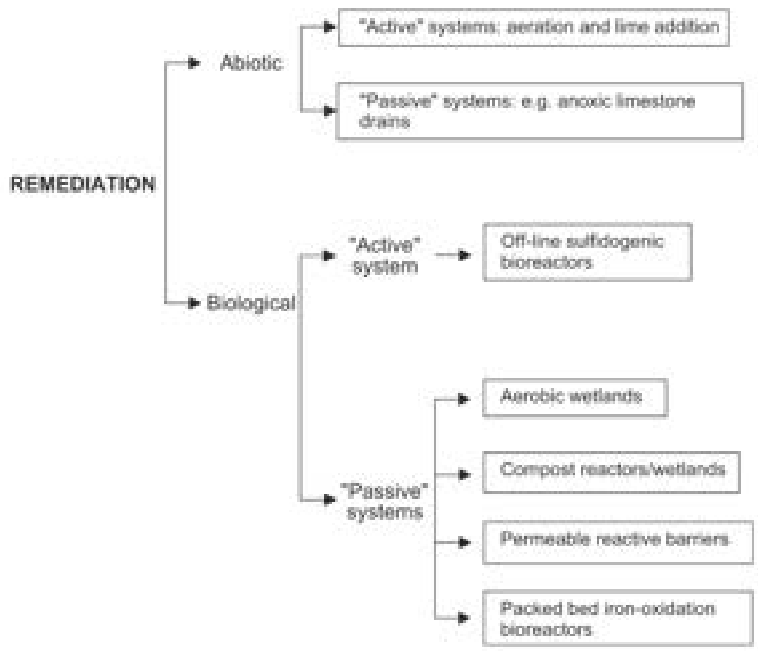

Figure A.1.2a Biological and abiotic strategies for remediating acid mine drainage waters. From Johnson & Hallberg, 2005.

Table A.1.2a. Categorical comparison of characteristics between passive and active biological treatments. From Rambabu et al., 2020.

Characteristics	Passive Treatment	Active Treatment
Cost of operation	Relatively Low ($0.3–$0.4/kg of metal removed)*	Relatively high (($0.7–$1/kg of metal removed)*
Labor requirement	Less	More
Area of Treatment	Large (0.1–2 m^2/kg of metal removed)†	Small (0.01–0.2 m^2/kg of metal removed)†
Recovery of metals	Difficult	Easy
System control	Poor	Good
Effluent predictability	Poor	Good

* cost typically accounts for capital and operational expenses. Maintenance cost for active treatment is 5–10 times higher than the passive treatment.
*,† Actual values can vary depending on the geography, AMD volume, specific composition of the AMD and its pollutant(s), climatic conditions and other influencing factors.

Various classification frameworks for AMD have been developed over the last century to support stakeholders engaging in AMD remediation (Thisani et al., 2020). The Global Acid Rock Drainage (GARD) framework is widely utilized, which considers pH, Total Dissolved Solids (TDS), and sulfate concentrations as indicators to classify mine water effluent (Joscha Opitz & Wendy Timms, 2016). Listed below are several other major classification systems that demonstrate the importance of characterizing waste as a means of beginning remediation (Table A.1.2b).

Table A.1.2b AMD Remediation Selection Frameworks. From Joscha Opitz & Wendy Timms (2016).

Framework	**Classification Goals**	**Water Quality Parameters**
Global Acid Mine Drainage	Open-source, simple determination, acid rock drainage (ARD) vs other drainage types	pH, SO_4^{2-}, Total Dissolved Solids (TDS), qualitative metal concentration ranges
Water Accounting Framework (Minerals Council of Australia)	Level of water treatment required to reach human consumption standards	pH, TDS, turbidity, coliforms
Hill (1968)	Distance to point-source pollutants	pH, acidity (equivalent $CaCO_3$), metal concentrations (Fe^{2+}, Fe^{3+}, Al), SO_4^{2-}
UK Classification System	Scope of impact on receiving waters	Area & length impacted, substrate quality for salmonid reproduction, iron deposition (qualitative) & concentration (quantitative), pH, DO, Al

AMD remediation is a long-term activity, so selecting designs that fit the needs of the local region is vital to crafting an effective remediation approach. A generalized context-focused selection strategy includes selecting high-potential treatments by comparing effectiveness of selected indicators in various treatment designs with current AMD characterization (Rojas-Marin et al., 2024). A context-driven strategy for net acidic waters was compiled using available data, where the process of determination identifies specific water quality parameters vital for effective treatment (Figure A.1.2b; Ali, 2011).

Selection strategies are not comprehensive, and novel methods have been developed and tested, yet only a few pilot projects exist or are well documented. Various pilot studies are underway to determine the feasibility of novel methods, such as the utilization of waste products (like fly ash from coal combustion) to increase alkalinity (Gitari et al., 2006). An organic substrate injection system utilizing llama droppings was co-designed by Bolivian and UK partners, yet was not widely utilized after the Bolivian financial crisis around 1998-2002 (Younger, 2007). Other theoretical models consider the dual benefits of utilizing waste destined for landfills to cover or introduce alkalinity into AMD reservoirs (Rezaie & Anderson, 2020).

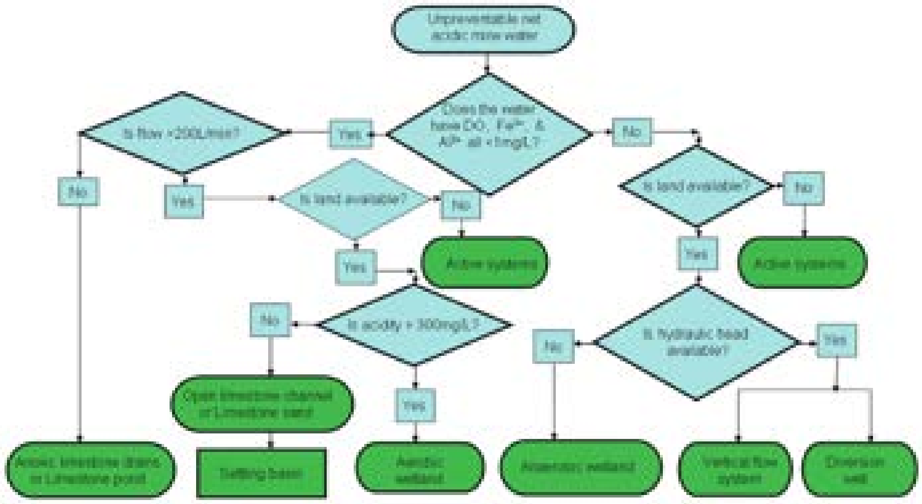

Figure A.1.2b Flow chart giving the decision logic for selecting remediation technologies for net acidic waters. From Ali, 2011.

AMD Remediation Strategies

<u>Active and Biological</u>

Biological and active treatment systems are often engineered continuous processes which require increased operational support (Rambabu et al., 2020). Biological sulfate reducers utilize sulfur as an electron acceptor to digest organic matter in a continuous flow, precipitating sulfate out of solution–a main characteristic of AMD–and into H_2S. Various mechanisms within these bioreactors, including carbonates, hydroxides, and phosphates, can help to precipitate metals from effluent. In high arsenic AMD flows, an anaerobic fixed-bed, sulfate-reducing bioreactor was found to efficiently remove arsenic, antimony, and other metals (Laroche et al., 2023). These processes are frequently consecutively paired with other systems or include alkaline-inducing materials. More recently, designers are considering integrating waste streams from other industrial processes to reduce costs and total waste volume (Rambabu et al., 2020).

Many processes utilize flocculated sludge to filter out precipitated metals at the risk of contaminating biological waste that can be used for its nutrient content in other contexts (Rambabu et al., 2020). The concentrations of sulfate and various metals pose toxicity concerns that would change the design of the reactor to cover various steps, which are widely discussed and many designs exist. Algal bioremediation has been seen as a novel, yet cost- and performance-effective, method for treatment (Rambabu et al., 2020). Various studied species of algae can bioaccumulate select metals and elements while producing high alkalinity. Engineering to maintain pH, dissolved oxygen, and temperature impact efficiency. Phytoremediation is considered a hybrid passive & active method to remove contaminants, and will be reviewed later on.

Active and Abiotic

Membrane technology has been adopted from wastewater treatment processes to treating AMD (Rambabu et al., 2020). Reverse osmosis and nano-filtration both require low chemical consumption yet require higher energy and maintenance costs to upkeep, and can achieve removal rates greater than 93%. Removal efficacy is dependent on pH, initial concentrations, allowable flux across the membrane, and temperature.

Electrochemical separation of ferrous iron from AMD has been found to recover goethite (FeOOH) while producing electricity, supporting cost recovery from AMD remediation projects (Sun et al., 2015). High operational complexity adds to the difficulty of implementing such a system.

Passive & Biological

Biological treatment methods are mainly focused on microbial treatments and phytoremediation methods.

Microbial treatment processes utilize naturally occurring geochemical activities under a wide range of environmental conditions (Rambabu et al., 2020). Selecting an effective treatment candidate requires knowledge of AMD parameters to ensure microbial activity and health, including pH, temperature, salinity, metal concentrations, and others. Organic substrate injection, permeable reactive barriers, infiltration beds, anoxic ponds, and wetlands are all highly researched and implemented methods, yet fluctuate in long-term performance and operational sustainability. Passive biological systems are more effective at lower flow rates, when microbial activity has a long enough retention time to interact with AMD.

Various concerns exist when selecting a passive biological method. Wildlife exposures to metal concentrations can increase contaminant concentrations in the local ecosystem, potentially leading to human exposure and consumption of contaminants. Passive methods require characterized environmental conditions to maintain optimal performance, so environmental fluctuations can greatly impact contamination reduction. Introduction of large-area passive treatment methods can interfere with localized hydrogeological processes, requiring additional monitoring to avoid other potential concerns.

Phytoremediation is a hybrid biological treatment option utilizing well-suited plants to extract, stabilize, or volatilize compounds in effluent (Hassan et al., 2024). Phytoextraction absorbs metal(loid)s via the plant's root system into the plant body, phytostabilization utilizes the plant's root system to reduces metal(loid) mobility in the soil and prevent further pollution, phytodegradation metabolizes organic contaminants into further degradates (which are often less harmful), and phytovolatilization allows concentrations to be absorbed then released into the atmosphere. As AMD is mainly the pollution of waterways by dissolved metals and acidity, phytovolatilization does not play a major role in AMD remediation.

A commonly utilized plant for AMD remediation is *Vetiver zizanioides*, which can tolerate acidic soil environments and conduct phytoextraction (Yang et al., 2023). The process is considered hybrid as the plant mass will bioaccumulate metals with minimal human support, but the mass must be manually removed from the environment, or else the accumulated metals will be redeposited into the environment. This allows for the potential for "plant mining," where metals are extracted from biomass and valorized. While this method is cheap (requires planting specific species and maintenance), plant growth and extraction is slow, and the plant biomass become hazardous for local fauna and communities who may be unaware of their high metal concentrations.

<u>Passive & Abiotic</u>

A method with fewer real-world examples are iron oxidation channels, which utilize low pH to oxidize iron (Ferrous iron to Ferric iron) and precipitate it as a method of lowering dissolved metal concentrations (Burgos & Macalady, 2014). As ferric iron is required to perpetuate the cycle of AMD, this method can reduce the long-term continuation of AMD impacts. Further remediation methods are required to reduce acidity and precipitate other metals.

Diversion wells can be dug to offer increased hydraulic retention time (Jeff Skousen et al., 2017). AMD is directed through a vertical tank filled with small-diameter alkaline material, where turbulence promotes dissolution and metal precipitation. A downstream settling pond captures precipitated solids. While these systems require periodic replenishment of alkaline material, they have relatively low operational complexity and are well-suited to remote sites where power and technical capacity are limited.

Additional Impacts of Treatment

While not the focus of our analysis, the development of a treatment system often does not consider the entire lifecycle of impacts. For example, transporting lime between its source and the remediation site consumes fossil fuels (Johnson & Hallberg, 2005). The United States Environmental Protection Agency (USEPA) has published methods for reducing the environmental footprint of remediation activities (*Green Remediation Best Management Practices: Mining Sites*, 2012). This includes design-specific methodologies and generalized approaches for remediation methods to reduce the impacts of vehicle energy usage (from transport and completed work), remediation-based waste, nutrient loading, and the introduction of new contamination to existing resources.

A.2 Supplemental Methods

A.2.1 Data Analysis

Software and Tools Setup
Spatiotemporal migration analyses were conducted in Python 3 (Van Rossum & Drake, 2009) using Spyder 6 as the integrated development environment. Data processing and numerical operations were performed using the Pandas (McKinney, 2010) and NumPy (Harris et al., 2020) libraries, which supported dataset merging, filtering, and summary statistic calculations across the eight-year monitoring record. Visualizations were generated using Plotly Express and Plotly Graph Objects (Plotly Technologies Inc., 2015), enabling the production of interactive and publication-quality figures including the three-dimensional spatiotemporal concentration surfaces. File and directory management was handled using the Pathlib and OS modules from the Python standard library to maintain consistent input/output paths across scripts. Version control was managed through GitHub, with iterative commits tracking script development and figure revisions throughout the analysis period.

Analytical Workflow

River distances were manually assigned to each monitoring station as a numeric field in the master dataset using *pandas*, with stations classified into three basin regions (Potosí Mining Zone, Bolivia Non-Mining, and Argentina/Paraguay) via a dictionary-based mapping applied through *pandas.Series.map()*. Three-dimensional contamination surface models were generated for each priority metal using *plotly.graph_objects*. Water column concentration data were restructured into a regular 50x50 spatiotemporal grid using *scipy.interpolate.griddata()* (Virtanen et al., 2020) with *method='linear'*, and *numpy.where()* was applied to replace NaN edge values with nearest-neighbor fill via a secondary *griddata()* call with *method='nearest'*. The resulting interpolated arrays were passed to *plotly.graph_objects.Surface()* with station observation points overlaid as *plotly.graph_objects.Scatter3d()* markers colored by region.

Seasonal dynamics were characterized by assigning each sampling record a season label using a dictionary mapping calendar months to wet or dry categories, applied via *pandas.Series.map()*. Wet season months were defined as November through April and dry season months as May through October, derived empirically from precipitation records. Median metal concentrations were then calculated per season, region, and matrix combination using *pandas.DataFrame.groupby()* followed by *.median()*, and results were visualized as grouped bar plots using *plotly.graph_objects.Bar()*.

For sediment-water partitioning analysis, sediment and water sample tables were merged on station identifier and campaign date fields using *pandas.DataFrame.merge()* with *how='inner'* to retain only temporally and spatially matched pairs. Where multiple grain size fractions were present for the same station and campaign, *pandas.DataFrame.groupby().median()* was applied across sieve fractions prior to merging to produce a single representative sediment concentration per sampling event. K_d values were calculated as a vectorized column operation dividing the sediment-phase concentration (mg/kg) by the aqueous-phase concentration (mg/L), with water concentrations converted from µg/L to mg/L by dividing by 1000 prior to the operation. Spearman rank correlations were computed using *scipy.stats.spearmanr()* (Virtanen et al., 2020), returning both the correlation coefficient and two-tailed p-value for each metal-parameter pair. All geochemical variables were log-transformed prior to correlation analysis using *numpy.log10()*, with a small constant added where zero values were present to avoid undefined values. A mechanism was considered verified where the Spearman correlation with K_d was statistically significant and positive at $p < 0.05$.

Hypoxic event analysis used *pandas.DataFrame.query()* to isolate dissolved oxygen records below the 4 mg/L threshold and identify event dates. Connectivity-restricted station subsets were filtered using *pandas.DataFrame.isin()* applied to a manually defined list of confirmed hydrologically connected stations for each event. Pre- and post-event sample windows were defined by computing the absolute difference in days between each sample date and the event date using pandas datetime operations, retaining records within a 180-day window on either side. Enrichment factors were calculated as the ratio of post-event median to pre-event median concentration per metal using *pandas.DataFrame.groupby().median()*. Differences between pre- and post-event distributions were evaluated using *scipy.stats.mannwhitneyu()* with *alternative='two-sided',* with statistical power limitations explicitly acknowledged where window sample sizes were n = 1 to 2.

Ranking Plots

For each water and sediment parameter, all individual samples can be ranked either by their measured concentration or their 95th percentile for highest hazard quotient (demonstrating exceedance against regulatory limits). Station-level plots summarize conditions at each monitoring station across time by filtering for the most recent sampling data (within five years) to focus on recent conditions, then take the 95th percentile for hazard quotient data as the score for the station. The top ten stations are shown along with a distribution of the samples collected within the last five years as a boxplot.

Parameter ranking plots identify a top ten list of which contaminants most severely surpass the related regulatory limits using hazard quotients. For each parameter, sampled data were isolated for recency (using a five-year window) and taking 95th percentile hazard quotient values for that parameter. A boxplot of sample concentrations over the last five years are shown.

Sieve-size plots focus on how contaminant levels vary by sediment grain size. For each sieve size, the 95th percentile hazard quotient value was selected across the last five years of sampled data and all locations. Hazard quotients for all sieve sizes were then plotted to show which grain-size fractions tend to hold the highest contaminant burdens.

A.2.2 Reference Standards

Regulatory standards were sourced from Bolivian Law 1333, the USEPA, USGS, and WHO to develop hazard quotient values and compare for regulatory compliance (Table A.2.2a; see Table 3.2.1a in the main text for sources).

Table A.2.2a Bolivian Law 1333 water quality standards. All units are in mg/L.

Contaminant	A	B	C	D	Contaminant	A	B	C	D
Aluminum	0.2	0.5	1	1	Mercury	0.001	0.001	0.001	0.001
Ammonia	0.05	1	2	4	Nickel	0.05	0.05	0.5	0.5
Arsenic	0.05	0.05	0.05	0.1	Nitrate	20	50	50	50
BOD_5	2	5	20	30	Nitrogen	5	12	12	12
Boron	1	1	1	1	Phosphates	0.4	0.5	1	1
COD	5	10	40	60	Selenium	0.01	0.01	0.01	0.05
Cadmium	0.005	0.005	0.005	0.005	Silver	0.05	0.05	0.05	0.05
Calcium	200	300	300	400	Sodium	200	200	200	200
Chlorides	250	300	400	500	Sulfate	300	400	400	400
Chromium	0.05	0.05	0.05	0.05	TDS	1000	1000	1500	1500
Copper	0.05	1	1	1	Zinc	0.2	0.2	5	5
Iron	0.3	0.3	1	1	pH	6–8.5	6–9	6–9	6–9
Lead	0.05	0.05	0.05	0.1					
Magnesium	100	100	150	150					
Manganese	0.5	1	1	1					

Table A.2.2b Regulatory standards for water quality from various international agencies. USEPA: United States Environmental Protection Agency. FAO: Food and Agriculture Organization of the United Nations. WHO: World Health Organization

Contaminant	Regulator	Value	Contaminant	Regulator	Value
Antimony	USEPA	0.006	Lead	USEPA	0.010
Arsenic	USEPA	0.010	Manganese	WHO	0.080
Arsenic	WHO	0.010	Mercury	USEPA	0.002
Arsenic	FAO	0.010	Mercury	WHO	0.006
Boron	WHO	2.400	Nickel	WHO	0.070
Cadmium	USEPA	0.005	Nitrate	USEPA	10.000
Cadmium	FAO	0.003	Nitrate	WHO	50.000
Cadmium	WHO	0.003	Nitrite	USEPA	1.000
Chromium	USEPA	0.100	Nitrite	WHO	3.000
Chromium	WHO	0.050	Selenium	USEPA	0.050
Copper	USEPA	1.300	Selenium	WHO	0.040
Copper	WHO	2.000	Silver	WHO	0.100
Iron	WHO	2.000	Thallium	USEPA	0.002
Lead	WHO	0.010			

Table A.2.2c Regulatory standards for sediment quality from various US agencies. TEL: Threshold Effects Level. PEL: Probable Effects Level. All units in mg/kg.

Contaminant	Regulator	TEL	PEL
Arsenic	USEPA	7.24	-
Cadmium	USEPA	0.676	-
Lead	USEPA	35	-
Mercury	USEPA	0.13	-
Arsenic	USGS	5.9	17
Cadmium	USGS	0.596	3.53
Chromium	USGS	37.3	90
Copper	USGS	35.7	197
Lead	USGS	35	91.3
Mercury	USGS	0.174	0.486
Nickel	USGS	18	36
Zinc	USGS	123	315

A.2.3 EJI Methods

Table A.2.3a summarizes all census-based indicators considered for the Environmental Justice Index (EJI), grouped by module (Health Vulnerability: Health; Environmental Burden: Exposure; Socioeconomic Disadvantage: Socioeconomic). The *Census Variable* column shows the names as they appear in the original census data, providing a reference for future work.

Table A.2.3a Summary of census-based indicators used in the EJI analysis by module, including variable names and definitions. Indicators marked with an asterisk (*) were excluded from the final model due to collinearity.

Module	**Indicator**	**Census Variable**	**Definition**
Health	Vulnerable Age Groups	p26_edad	Proportion under 6 years of age or 65 years and older.
	No Health Insurance	p31_afiliado	Proportion without formal health insurance.
	Informal healthcare only	p30e_tradic, p30f_autome, p30g_casera	Proportion relying solely on informal healthcare (i.e. traditional medicine or at-home remedies).
	Child Loss	p54_hvtot, p55_hstot	Proportion of women who have experienced child loss (children born – children surviving).
	Premature Mortality	m213_edad	Proportion of deaths occurring under 50 years of age.
Exposure	Farming	p51_actec_2d	Proportion employed in farming/agriculture.
	Mining	p51_actec_2d	Proportion employed in mining.
	Manufacturing	p51_actec_2d	Proportion employed in manufacturing.
	Construction	p51_actec_2d	Proportion employed in construction.
	Agricultural Participation*	p45_agro	Proportion of population participating in any agricultural activity.
	Subsistence Agriculture	p45_agro, p46_dest	Proportion producing agricultural products for

			household consumption.
	Unprotected Water Source	v07_aguapro	Proportion of households relying on unprotected water sources (rain, wells, rivers, other).
	No Piped Water*	v08_aguadist	Proportion of households without piped water.
	Unsafe Solid Waste Disposal	v11_basura	Proportion of households disposing solid waste via open dump, burning, burying, or other unsafe methods.
	Unsafe Liquid Waste Disposal	v16_desague	Proportion of households with septic/pit/open defecation disposal methods.
	Home Structural Vulnerability	v03_pared, v05_techo, v06_piso	Proportion of homes with vulnerable construction materials (adobe, wood, metal, straw, dirt floors).
	Unsafe Sanitation	v15_servsan, v16_desague	Proportion of households without private toilets, shared toilets with unsafe drainage, or bathrooms draining to open environment.
Socioeconomic	Indigenous Identity	p32_pueblo_per	Proportion of population identifying as indigenous.
	Disability	p42a_ver, p42b_oir, p42c_camina, p42d_comuni	Proportion of population with moderate/severe impairments in vision, hearing, mobility, or communication.
	Illiteracy*	p40_lee	Proportion of population over 15 years who cannot read.
	Education level	p41a_nivel	Proportion of population over

			15 whose highest educational attainment is primary school or below.
	Unemployed	condact_19	Proportion of economically active population over 15 years who are unemployed (first-time or previously employed).
	Internet Access	v19e_inetfijo, v19f_inetmovil	Proportion of households without fixed or mobile internet access.

The correlation matrices in Figure A.2.3a illustrate relationships among indicators within each EJI module (Health Vulnerability: Health; Environmental Burden: Exposure; Socioeconomic Disadvantage: Socioeconomic). High correlations between certain indicators, including *Agricultural Participation* with *Subsistence Agriculture*, *No Piped Water* with *Unprotected Water Source*, and *Illiteracy* with *Low Education*, led to the exclusion of redundant variables to mitigate collinearity effects.

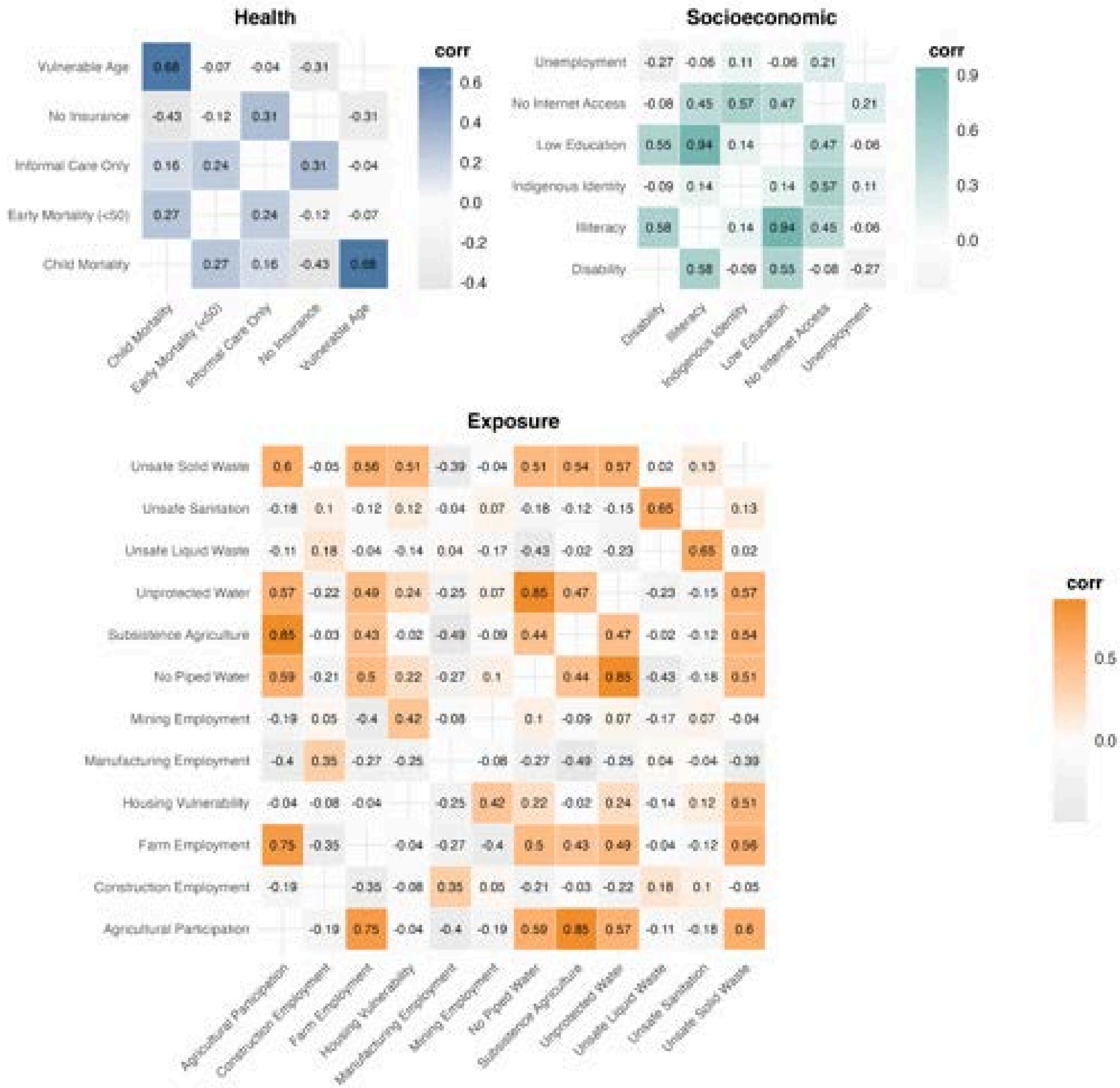

Figure A.2.3a Correlation matrices showing Spearman rank correlations among indicators within each EJI module: Health Vulnerability (top left), Socioeconomic Disadvantage (top right), and Environmental Burden (bottom). Colored cells represent the strength and direction of correlations (negative correlations shown in greyscale), with darker shades indicating stronger correlations, as shown by the adjacent color scales.

A.2.4 Principal Component Analysis

Explaining Principal Component Analysis

Principal Component Analysis takes a multitude of variables and combines them into a few composite variables (principal components) which explain the maximum amount of the data's variance. PCA analysis determines how much each individual variable contributes to the composite variable and how correlated the variables are to each other. Two plots are provided in PCA: one displays each variable as a vector plotted against the first two principal components, while the other describes the total variation in data described by the PCA (Figure A.2.4a). This indicates how much the variable contributes to the principal components. The direction of the vectors in relation to each other indicates how correlated the variables are. The second plot displays how much of the data's variance is explained by each principal component; if the top two dimensions explain the majority of variance in the data, then the resulting PCA plot is highly representative of the relationships between the variables.

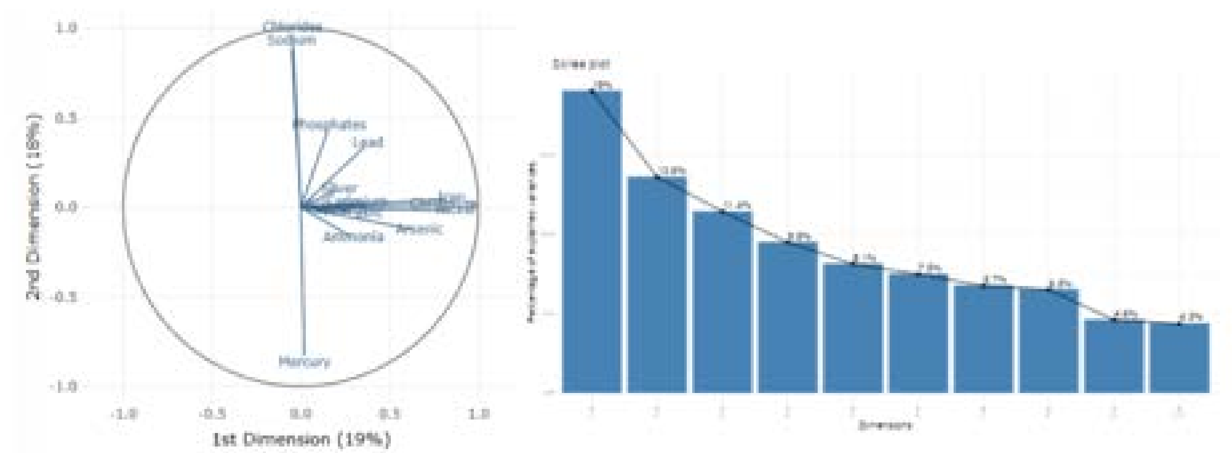

Figure A.2.4a Example Principal Component Analysis plot (left) and screeplot (right) from the water quality data. (Left) Vector length describes relative influence on total dataset variance. Vector angle describes correlation between contaminants. (Right) Screeplot describes total variance explained by each dimension. It should be noted that the screeplot and PCA plot presented here were not developed for the same dataset.

A PCA plot can be read to help understand how the concentrations of different chemicals are correlated. Aligned arrows represent positive correlation between analytes, opposing arrows represents negative correlation between analytes (a high concentration is one will likely mean a low concentration in the other), and perpendicular means they overall demonstrate a non-correlated relationship.

A.3 Supplemental Results

A.3.1 Priority Contaminants

Silver

Silver (Ag) experienced a large spike in reported concentration data within water-based samples in 2024 (see *4.1 Data Limitations*). The reason behind the recent spike in silver concentrations was not determined by the team, but current concentrations should be reviewed to determine past and future transport methods. The Trinational Commission was notified about these results but have yet to respond as of this publication.

Outside of 2024 data which experienced anomalous spikes, silver concentrations remained extremely low throughout the basin (Table A.3.1a). No regulatory level was found for silver in sediments and therefore were not analyzed. The Bolivian Law 1333 sets a 50 µg/L limit for drinking water. Water-based silver concentrations exceeded this limit in 1% of all samples, with a maximum of 6% exceedances in 2016.

Table A.3.1a Silver concentrations in water samples averaged across all stations by year.

Year	Samples	Exceedance (%)	Median Ag (µg/L)	Max (µg/L, station)
2016	66	6%	0.5	84.0 (Pilcomayo arriba Tacobamba)
2017	88	0%	0.5	27.0 (Canutillos 3)
2018	48	0%	10.0	33.1 (Cotagaita)
2019	60	0%	5.3	10.0 (Villamontes)
2020	29	0%	2.5	9.0 (Tarapaya)
2021	48	0%	2.5	2.5 (Chuquiago)
2022	70	0%	5.0	5.0 (San Josecito)
2023	117	0%	2.5	12.3 (El Puente)
All Years	526	1%	2.5	84.0 (Pilcomayo arriba Tacobamba)

Stations were characterized by their aggregate hazard quotient and plotted (Figures A.3.1a). El Puente had the most extreme silver concentration loading, but all reported concentrations were below HQ=1.

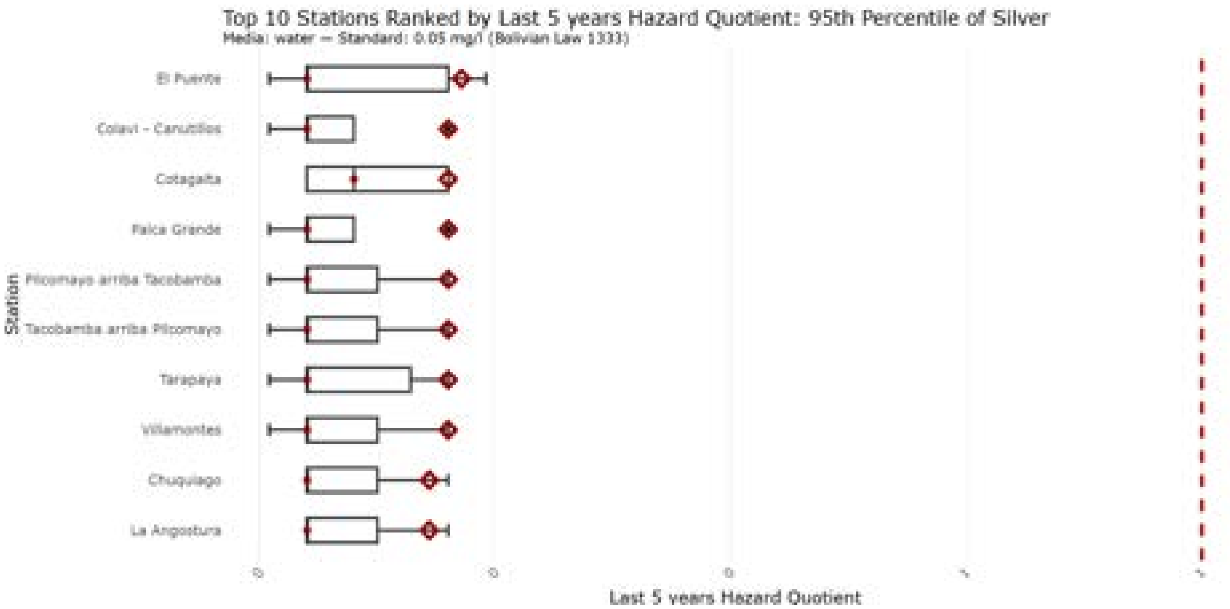

Figure A.3.1a The top ten water-based silver concentrations aggregated by station and calculated using the 95th percentile for the most recent 5 years of data for each station. Red dot signifies median value, red diamond signifies 95th percentile value. Red line signifies HQ=1, where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic. All reported silver concentrations were within regulatory limits (HQ<1).

Copper

Copper (Cu) experienced concerning concentrations within both water and sediment samples. Hazard quotients for water quality were calculated using the strictest regulatory limit, which is given by Bolivian Law 1333 at 50 µg/L (Table A.3.1b). Water-based copper concentrations exceeded this limit in 36% of all samples. Cu concentrations in sediments exceeded the strictest regulatory standard (set by the USGS as 35.7mg/kg) in 24% of samples (Table A.3.1c). Compared to water samples, sediment samples have a much lower hazard quotient, and therefore water-based concentrations are most concerning as they are mobilized along the Pilcomayo basin.

Table A.3.1b Copper concentrations in water samples averaged across all stations by year.

Year	Samples	Exceedance (%)	Median Cu (µg/L)	Max (µg/L, station)
2016	79	27%	12.5	2841 (Tarapaya)
2017	102	44%	19.5	5741 (Canutillos 3)
2018	63	52%	64.0	1465 (Cotagaita)
2019	82	44%	42.5	1458 (Tarapaya)
2020	35	49%	50.0	624 (Tarapaya)
2021	58	26%	5.0	516 (Colavi - Canutillos)
2022	90	22%	7.5	3047 (Colavi - Cantillos)
2023	148	32%	8.5	2457 (El Puente)
All Years	657	36%	15.0	5741 (Canutillos 3)

Table A.3.1c Copper concentrations in sediment samples averaged across all stations by year.

Year	Samples	As Exceedance (%)	Median Cu (mg/kg)	Max As (mg/kg, Station Sampled)
2016	109	21%	19.7	780 (Colavi - Canutillos)
2017	120	26%	17.8	198 (Tarapaya)
2018	83	14%	20.0	191 (Colavi - Canutillos)
2019	118	14%	19.1	215 (Tarapaya)
2020	19	32%	21.3	135 (Tacobamba arriba Pilcomayo)
2021	63	21%	21.6	111 (Tarapaya)
2022	74	26%	15.1	1074 (Tarapaya)
2023	100	44%	16.5	405 (Tumusla)
All Years	686	24%	19.6	1074 (Tarapaya)

Each station was reviewed to understand general loading patterns of each parameter of interest stated above. Using the 95th percentile method, stations were characterized by their aggregate hazard quotient (Figures A.3.1b & A.3.1c). Colavi - Canutillos had the most extreme copper concentration loading in water, while Tumusla had the most extreme loading for sediments. Copper was relatively evenly distributed between particle sizes (Figure A.3.1d).

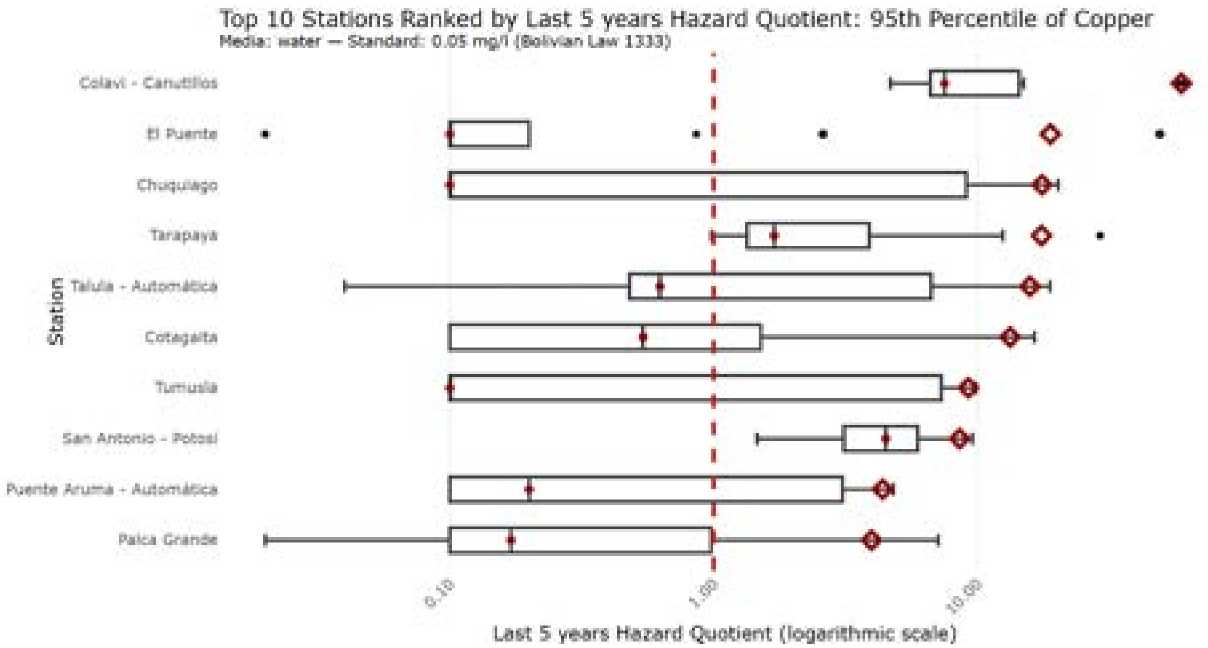

Figure A.3.1b The top ten water-based copper concentrations aggregated by station and calculated using the 95th percentile for the most recent 5 years of data for each station.The red dot signifies median value, and the red diamond signifies 95th percentile value. The red line signifies HQ=1, where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic. All samples taken at Colavi - Canutillos were above HQ=1.

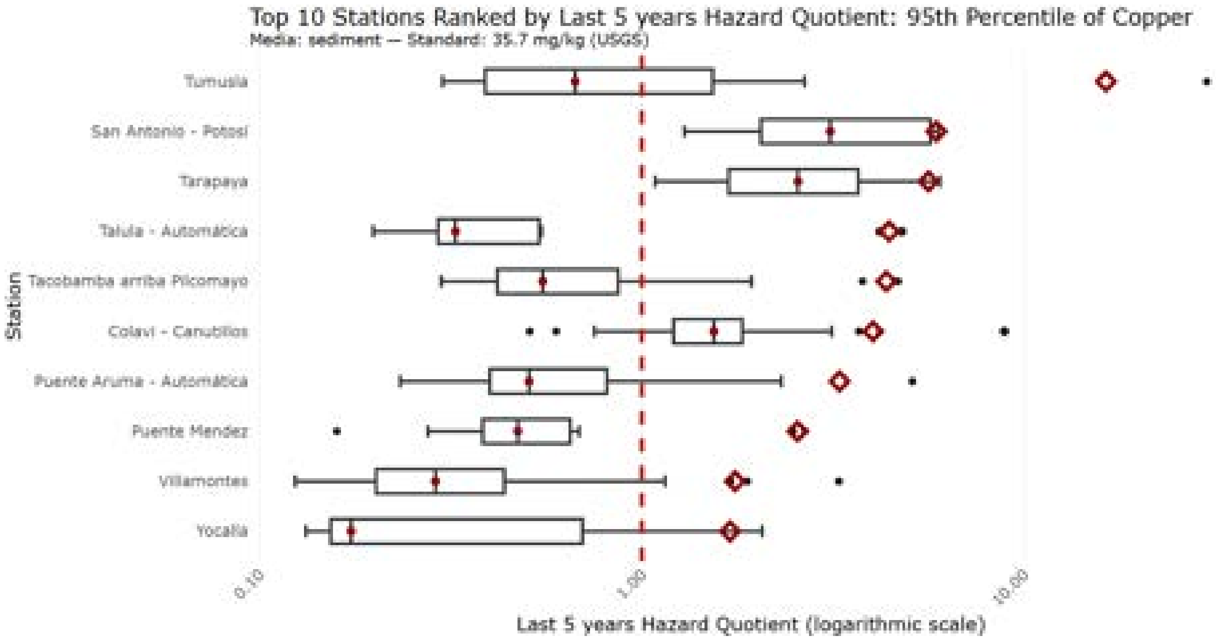

Figure A.3.1c The top ten sediment-based copper concentrations aggregated by station and calculated using the 95th percentile for the most recent 5 years of data for each station. The red dot signifies median value and the red diamond signifies 95th percentile value. The red line signifies HQ=1, where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic.

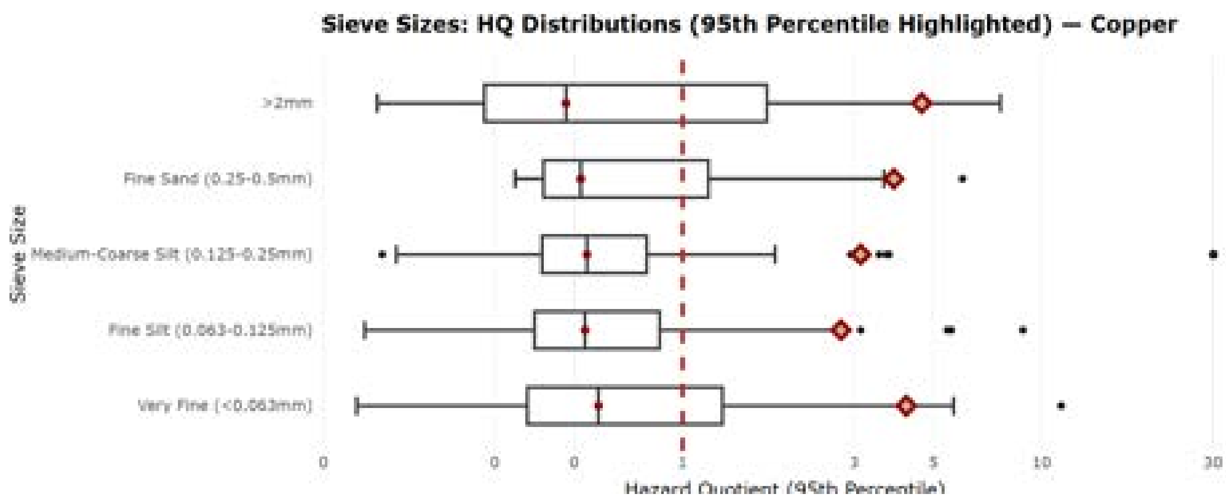

Figure A.3.1d 95th percentile copper hazard quotients reported within sediments at each station for each sieve size. The red dot signifies median value, red diamond signifies 95th percentile value. The red dotted line signifies where lead levels surpass regulatory standards (HQ=1). X-axis values are untransformed but the scale is logarithmic. All reported grain sizes experienced 95th percentile values above HQ=1.

Cadmium
Cadmium (Cd) was the third and fourth most concerning parameter in reported sediment and water samples, respectively. Hazard quotients using the strictest regulatory limit, which is given by the WHO at 3 µg/L for water matrices and the USGS standard of 0.596 mg/kg. Water-based cadmium concentrations exceeded this limit in 34% of all samples, and 22% of sediment-based concentrations surpassed regulatory levels (Tables A.3.1d and A.3.1e). Sediment samples generally experienced a lower exceedance rate. Tarapaya often held the highest reported concentration of cadmium in sediments (2018-2023), and San Antonio - Potosí had the highest reported samples in water-based samples between 2017-2022 (Figures A.3.1e and A.3.1f). Cadmium was sorbed fairly evenly to all sediment sizes (Figure A.3.1g); additional analysis can be conducted to determine whether there is significance in concentrations between each sieve size.

Table A.3.1d Cadmium concentrations in water samples averaged across all stations by year.

Year	Samples	Exceedance (%)	Median Cd (µg/L)	Max Cd (µg/L, station)
2016	72	26%	0.5	330 (Tarapaya)
2017	90	31%	1.0	38 (Tarapaya)
2018	51	65%	10.0	264 (San Antonio - Potosí)
2019	61	59%	10.0	132 (San Antonio - Potosí)
2020	35	40%	2.0	83 (San Antonio - Potosí)
2021	56	20%	0.25	29 (San Antonio - Potosí)
2022	79	22%	2.5	33 (San Antonio - Potosí)
2023	147	30%	0.6	57 (Talula - Automatica)
All Years	591	34%	1.0	330 (Tarapaya)

Table A.3.1e Cadmium concentrations in sediment samples averaged across all stations by year.

Year	Samples	Exceedance (%)	Mean Cd (mg/kg)	Max Cd (mg/kg, Station)
2016	109	19%	0.17	104 (Pilcomayo arriba Tacobamba)
2017	120	22%	0.21	13.4 (San Antonio - Potosí)
2018	83	10%	0.22	5.0 (Tarapaya)
2019	118	9%	0.19	13.9 (Tarapaya)
2020	19	21%	0.20	4.1 (Tarapaya)
2021	63	13%	0.21	8.2 (Tarapaya)
2022	74	58%	0.98	48.3 (Tarapaya)
2023	100	29%	0.45	32.0 (Tarapaya)
All Years	686	22%	0.24	104 (Pilcomayo arriba Tacobamba)

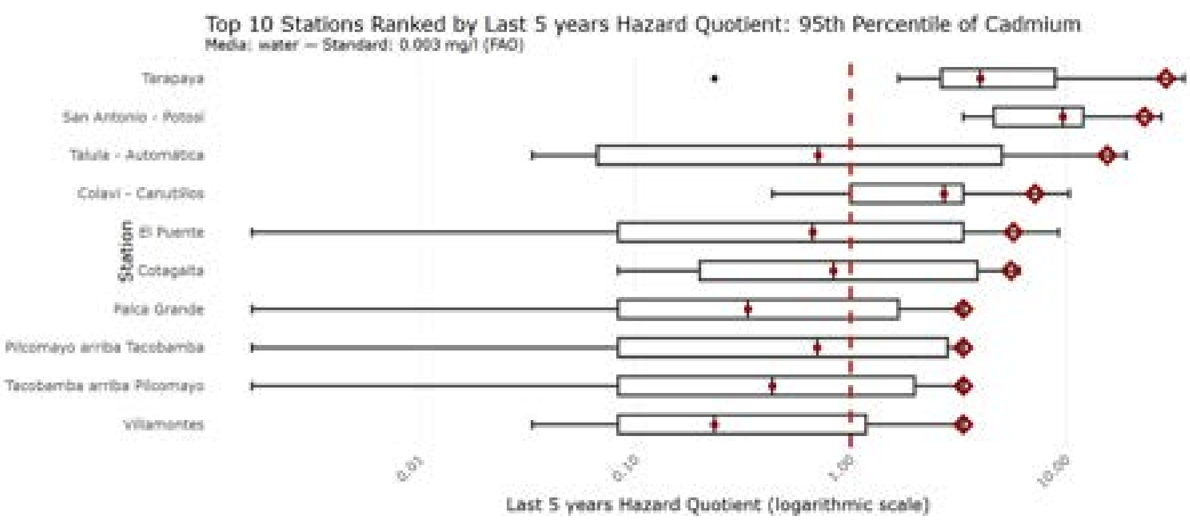

Figure A.3.1e. The top ten water-based cadmium concentrations aggregated by station and calculated using the 95th percentile for the most recent 5 years of data for each station. Red dot signifies median value, red diamond signifies 95th percentile value. Red line signifies HQ=1, where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic.

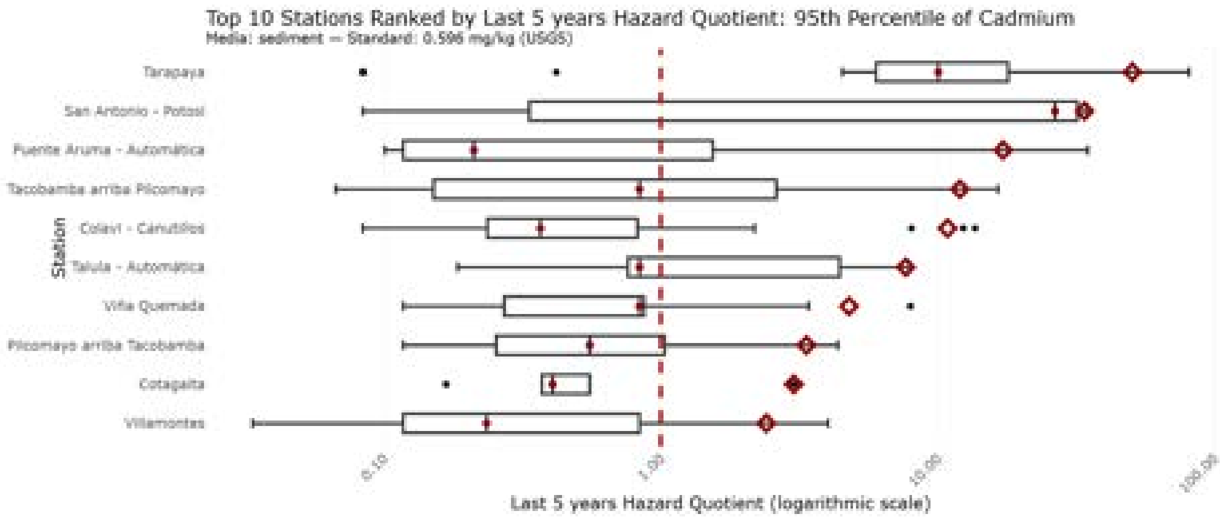

Figure A.3.1f The top ten sediment-based cadmium concentrations aggregated by station and calculated using the 95th percentile for the most recent 5 years of data for each station. Red dot signifies median value, red diamond signifies 95th percentile value. Red line signifies HQ=1, where sampled concentrations meet regulatory limits. X-axis values are untransformed but the scale is logarithmic.

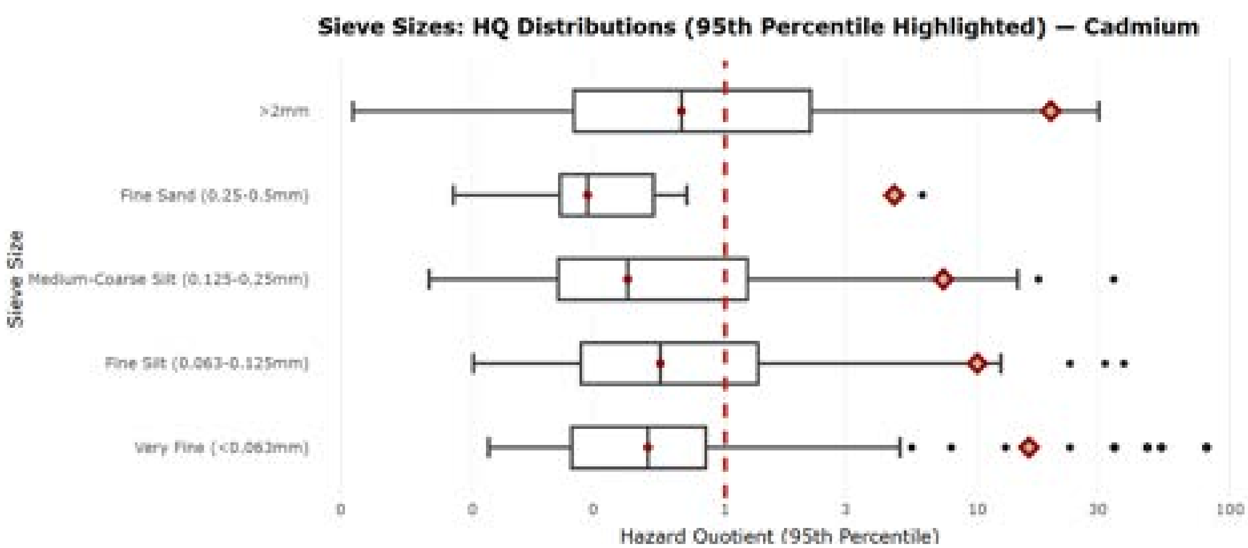

Figure A.3.1g Average hazard quotient for lead for each grain size. Red dotted line signifies where cadmium levels surpass regulatory standards (HQ=1).

A.3.2 Spatiotemporal Contaminant Migration

Wet-Season Hydrological Mobilization

Analysis of 1,369 water and sediment samples from 39 stations along a 500 km corridor (2016–2024) shows pulse-driven, wet-season arsenic migration (Figure A.3.2a). Water arsenic (83% exceedance; mean 550 µg/L vs. 50 µg/L limit) peaks at ~200 µg/L in Upper-Middle basi.

Effective remediation must therefore prioritize upstream source control, deploy real-time monitoring at Upper-Middle hotspots, and implement seasonally adaptive treatment systems to intercept arsenic pulses before they reach vulnerable middle- and lower-basin indigenous communities dependent on the river for drinking, irrigation, and fishing.

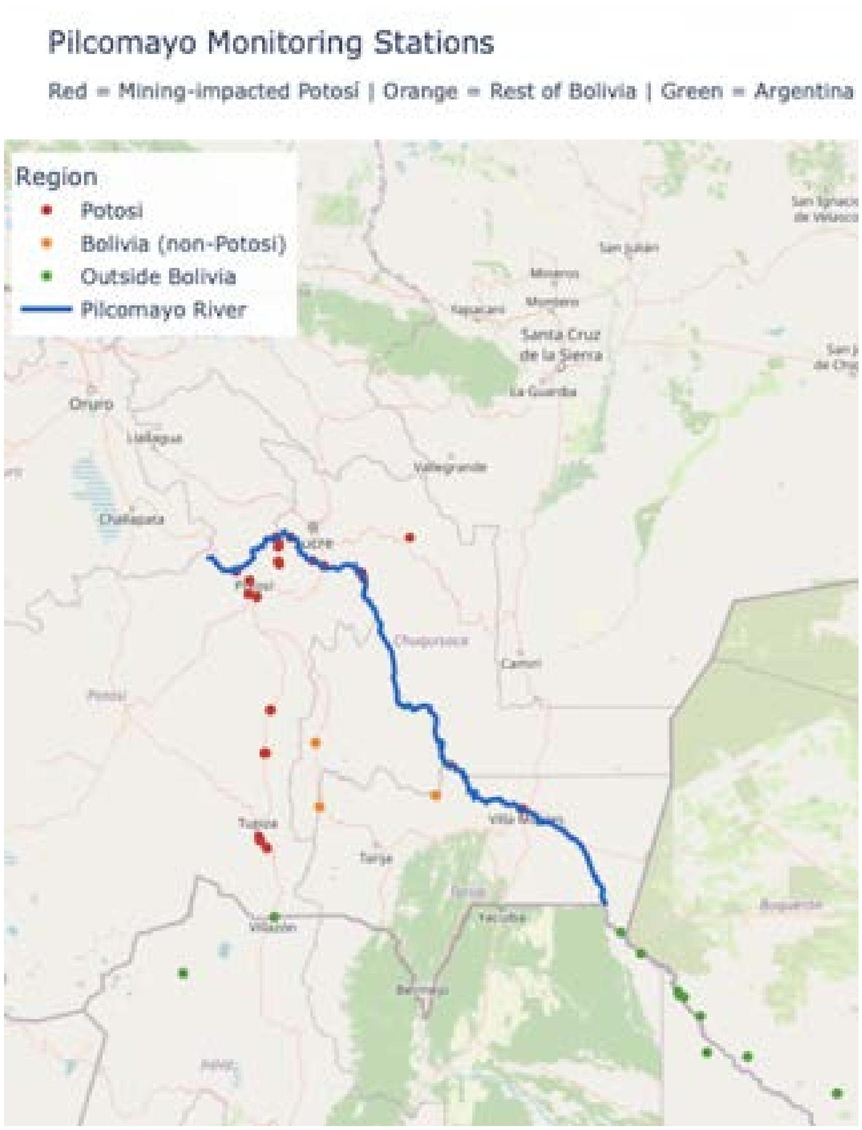

Figure A.3.2a Map of Pilcomayo Stations by Basin Segment. 39 monitoring stations along the ~500 km corridor are color-coded by sector: Upper (red), Upper-Middle (orange), Middle (green), Lower-Middle (blue), Lower (purple). Dense clustering near Potosí (Upper/Upper-Middle) reflects source proximity; downstream stations extend into Paraguay, capturing propagation into floodplains.

Table A.2.3b Distribution of the 39 unique monitoring stations across the five basin sectors of the Pilcomayo River (2016–2024). Upper and Upper-Middle sectors host most stations near Potosí sources; Lower and Lower-Middle capture downstream transport.

Basin Sector	Monitoring Stations
Lower	Agropil Bañado Las Garzas Cadete Pando Clorinda El Potrillo General Bruguez General Diaz Hacienda 9 de junio Montelindo en Hacienda Sta Ana Montelindo en ruta PY09 Negro Ruta PY09 Puente Loyola Ruta 11 Ruta 28-Vertedero Ruta 95 Salida Laguna Salada Tinfunke Verde Ruta PY09 Villa Hayes
Upper	Canutillos 3 Colavi - Canutillos Pilcomayo - Agua arriba confluencia Pilcomayo - Tacobamba Potosí - Naciente rio La Ribera Puente Mendez San Antonio - Potosi Tacobamba - Agua arriba confluencia Pilcomayo - Tacobamba Talula - Automática Tarapaya Viña Quemada Yocalla
Upper-Middle	Cotagaita Palca Grande Puente Aruma - Automática Tumusla
Middle	Chuquiago El Puente La Angostura San Josecito Villamontes
Lower-Middle	El Solitario La Quiaca Liviara Marca Borrada Maria Cristina Meyer Sitio 01 Misión La Paz - SNIH

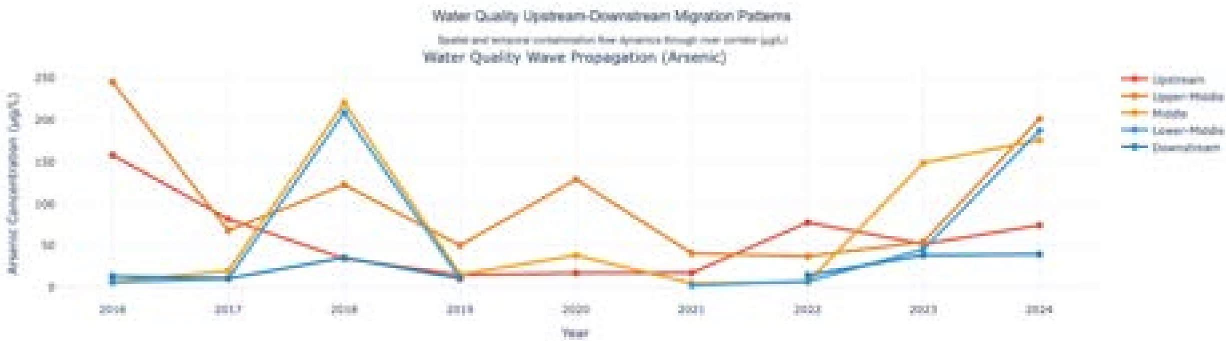

Figure A.3.2b Temporal Wave Propagation of Arsenic in Water Across River Segments (2016–2024). Annual mean water arsenic (µg/L) across five segments shows asynchronous peaks with 1–2-year downstream lag from Upper-Middle wet-season pulses.

Figure A.3.2c Longitudinal Profile of Mean Arsenic Concentration in Water by River Position. Average water arsenic (2016–2024) peaks at ~200 µg/L in Upper-Middle, declining downstream but exceeding WHO (10 µg/L) and Bolivian (50 µg/L) limits throughout.

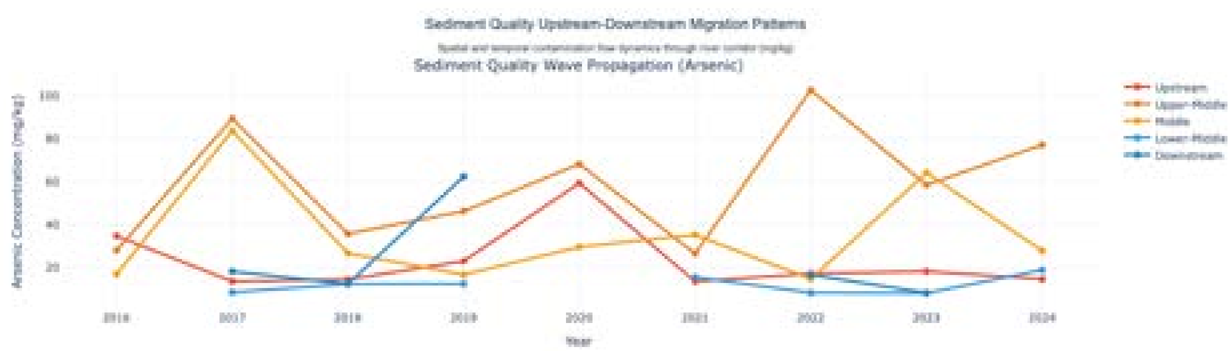

Figure A.3.2d Temporal Wave Propagation of Arsenic in Sediment Across River Segments (2016–2024). Annual mean sediment arsenic (mg/kg) shows slower, damped migration vs. water, with persistent Upper-Middle hotspots.

Figure A.3.2e Longitudinal Profile of Mean Arsenic Concentration in Sediment by River Position. Average sediment arsenic peaks at ~80 mg/kg in Upper-Middle, far above CCME ISQG (5.9 mg/kg), with gradual downstream attenuation.

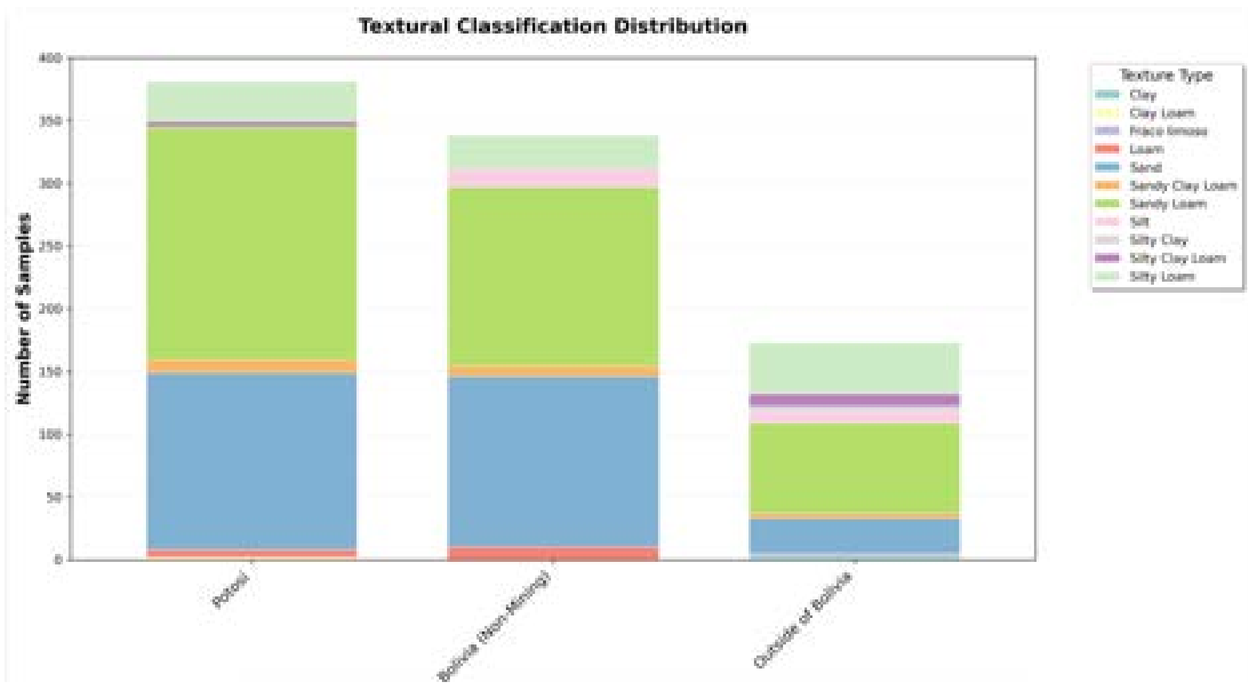

Figure A.3.2f Textural Classification Distribution in the Pilcomayo River Basin (2016—2024).

Water Contamination Timeseries

Arsenic contamination in water showed persistent elevation throughout the study period with episodic spikes. The Potosí Mining Zone exhibited baseline contamination punctuated by major concentration surges, including an early 2016 peak reaching 1,300 µg/L and a 2024 event reaching 450 µg/L. The Bolivia Non-Mining Zone maintained lower baseline concentrations (10-100 µg/L) but showed synchronized spikes following Potosí events with a characteristic 1-3 month temporal lag. The Argentina/Paraguay transboundary zone demonstrated dampened peak magnitudes with the maximum at approximately 200 μg/L.

Lead contamination exhibited relative stability for most of the study period followed by an extreme 2024 spike affecting all three regions simultaneously (Figure A.3.2g). This dramatic concentration surge reached 1,900 μg/L in Potosí , 1,500 μg/L in Bolivia, and 1,100 μg/L in Argentina/Paraguay. Prior to this 2024 event, baseline concentrations remained relatively stable (50-200 μg/L) with moderate seasonal oscillations (Figure A.3.2g). This concerning trend suggests a major source event, potentially a mine failure, tailings dam breach, or intensified mining activity, with clear transboundary impact propagating downstream.

Zinc demonstrated the highest absolute contamination levels among all metals studied. Potosí consistently maintained concentrations between 500-3,000 μg/L throughout the study period, with a 2023 mega-spike reaching 19,000 μg/L (Figure A.3.2g). This 2023 event affected all regions, generating concentrations of 8,000 μg/L in Bolivia and 3,000 μg/L in Argentina/Paraguay. Even baseline concentrations (500-1,000 μg/L) chronically exceeded WHO guidelines (3,000 μg/L) at Bolivia and Argentina monitoring sites.

Cadmium contamination paralleled lead patterns with a synchronized 2024 spike affecting all regions (400-1,200 μg/L across the basin). During non-spike periods, moderate baseline concentrations (20-100 μg/L) persisted across all regions (Figure A.3.2g).

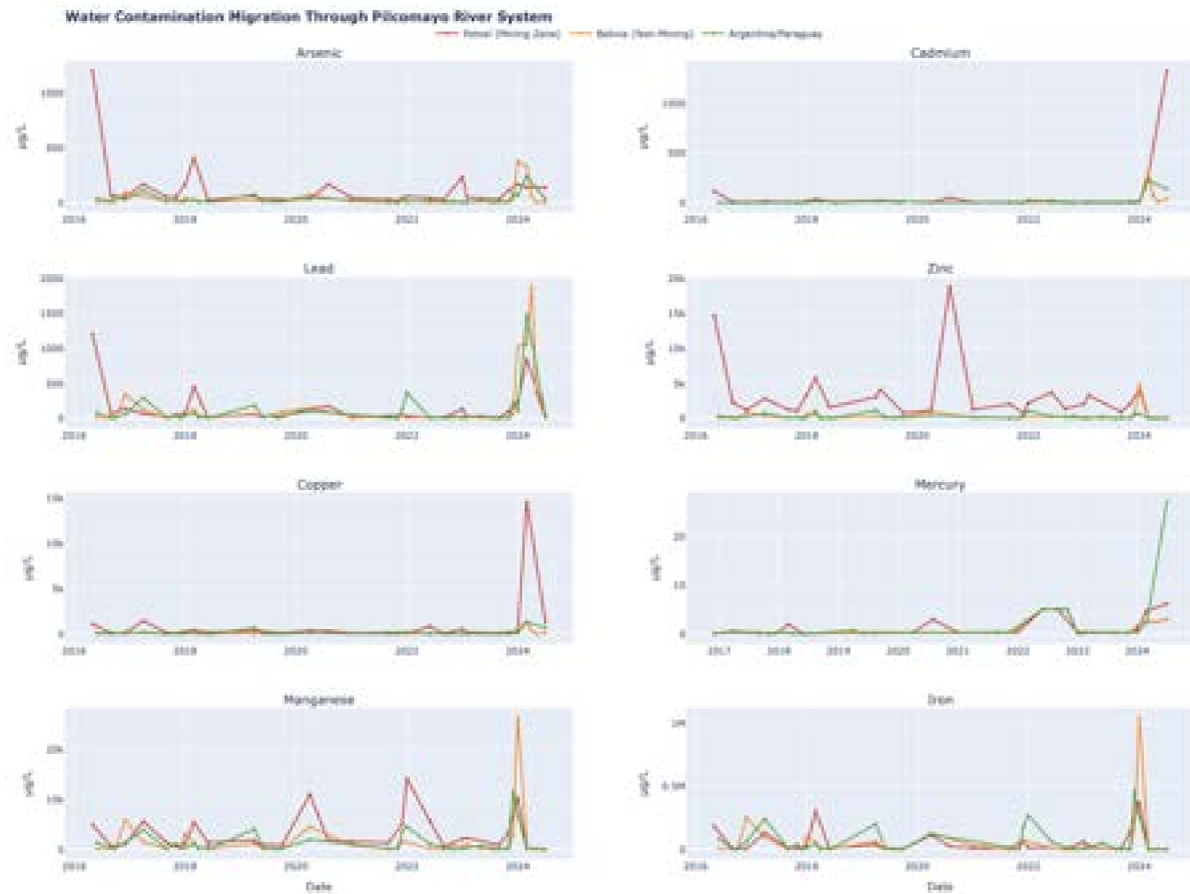

Figure A.3.2g Temporal migration of metal contamination through water phase of Pilcomayo River system (2016-2024). Time series reveal lagged downstream propagation of contamination pulses, with peaks appearing first in Potosí (red), followed by Bolivia (orange, 1-2 months later), then Argentina/Paraguay (green, 2-4 months later). The 2023-2024 period represents unprecedented multi-metal contamination affecting the entire basin.

Migration evidence emerged clearly from temporal pattern analysis. Contamination spikes appeared first in Potosí , followed by Bolivia (1-2 months later), then Argentina/Paraguay (2-4 months later), confirming downstream propagation through the river system. Peak magnitudes decreased by 30-60% moving downstream, indicating dilution and attenuation processes. The 2023-2024 period represents a contamination crisis with unprecedented multi-metal spikes affecting the entire basin, suggesting either escalating contamination from mining sources or extreme hydrological events mobilizing legacy contamination.

Sediment Contamination Timeseries

Arsenic in sediments showed persistent contamination at the Potosí source zone, maintaining baseline concentrations of 20-80 mg/kg with periodic spikes reaching above 200 mg/kg (Figure A.3.2h). A 2023 extreme event generated a sharp peak reaching 600 mg/kg, representing 30-fold exceedance of the 20 mg/kg sediment quality guideline. Downstream zones maintained lower baseline concentrations (5-20 mg/kg) but exhibited delayed spikes corresponding temporally to water phase peaks, confirming sediment as a secondary contamination pathway following aqueous transport.

Lead sediment concentrations demonstrated source zone dominance with early study period (2016-2017) peaks reaching 300 mg/kg in Potosí , which subsequently settled to 30-100 mg/kg during the middle study period (Figure A.3.2h). The 2023 contamination crisis generated a return to elevated concentrations (700 mg/kg in Potosí). Downstream zones in Bolivia and Argentina/Paraguay showed gradual concentration build-up over the eight-year period, indicating progressive legacy contamination accumulation in sediments distant from primary sources.

Zinc maintained consistent elevation in Potosí sediments (100-500 mg/kg) across the entire study period. The 2023 synchronized spike affected all regions with concentrations reaching 5,000 mg/kg in Potosí , 1,000 mg/kg in Bolivia, and 700 mg/kg in Argentina/Paraguay. The Argentina/Paraguay zone exhibited an upward concentration trend over eight years, suggesting cumulative loading from ongoing upstream inputs.

Mercury sediment contamination showed a dramatic 2023 spike with Potosí reaching 2,900 µg/kg, Bolivia reaching 2,500 µg/kg, and Argentina reaching 2,600 µg/kg (Figure A.3.2h). Unlike other metals, the mercury spike occurred simultaneously across all regions rather than showing lagged downstream propagation. This synchronized basin-wide response suggests either atmospheric deposition or widespread artisanal mining activity using mercury amalgamation techniques rather than simple downstream migration from a point source.

Figure A.3.2h Sediment contamination migration through Pilcomayo River system (2016-2024). Time series demonstrate temporal persistence of contamination with no evidence of natural attenuation. The 2023 synchronized multi-metal spike represents the most severe contamination period in the study. Mercury shows simultaneous basin-wide elevation, suggesting atmospheric deposition or artisanal mining activity rather than point-source transport.

Three key findings emerged from spatiotemporal analysis. First, all metals demonstrated temporal persistence throughout eight years of monitoring with no evidence of natural attenuation in either water or sediment matrices. Second, the 2023-2024 period represents the most contaminated interval in the study, characterized by multi-metal synchronized spikes affecting the entire basin. Third, sediments function as long-term contamination sinks, exhibiting gradual concentration build-up even in downstream zones located hundreds of kilometers from primary mining sources.

A.3.3 pH Temporal Dynamics

The Potosí Mining Zone exhibited predominantly acidic to neutral conditions with pH ranging from 5.5 to 8.0 and a median value near 7.0. Extreme acidification events occurred in 2020 (pH dropping to 5.5) and 2022 (pH reaching 6.5), indicating episodic acid mine drainage pulses. High temporal variability characterized this zone (standard deviation = 0.8 pH units), reflecting irregular releases from mining operations.

The Bolivia Non-Mining Zone demonstrated gradual pH neutralization with values ranging from 7.2 to 8.7 and a median near 8.2. These alkaline conditions suggest natural buffering through dissolution of carbonate minerals in the riverbed. Conditions remained relatively stable (standard deviation = 0.4) compared to the source zone, indicating effective natural attenuation of acidic inputs.

The Argentina/Paraguay transboundary zone maintained near-neutral to alkaline conditions with pH ranging from 6.7 to 8.5 and a median near 8.0. Seasonal oscillations were evident, with pH decreasing during wet season (dilution reducing buffering capacity) and increasing during dry season (concentration effect enhancing buffer strength). Despite upstream acidification events, downstream pH consistently recovered to 7.5-8.5, indicating effective natural buffering capacity of the river system.

Figure A.3.3a pH temporal dynamics across three regions (2016-2024). Potosí Mining Zone (red) shows high variability with acidification events (pH 5.5-6.5) in 2020 and 2022. Bolivia Non-Mining zone (orange) demonstrates natural buffering with consistently alkaline conditions. Argentina/Paraguay (green) maintains near-neutral to alkaline pH despite upstream inputs, confirming effective natural attenuation.

Critical implications emerge from pH dynamics. Low pH conditions (5.5-7.0) in the Potosí source zone enhance metal dissolution, preventing precipitation and facilitating downstream transport. pH-dependent speciation determines metal mobility, with arsenic existing as anionic arsenate ($HAsO_4^{2-}$) at pH above 6.5, while cationic metals (Cd^{2+}, Pb^{2+}, Zn^{2+}) dominate at pH below 8. For remediation consideration, pH adjustment through lime neutralization at Potosí sources could precipitate metals and potentially reduce downstream loading by 50-80%.

A.3.4 Principal Component Analysis

When comparing biological parameters, we found that ammoniacal nitrogen was highly positively correlated with Biochemical Oxygen Demand (BOD), likely indicating a shared source, such as domestic or human wastewater (Figure A.3.4a). Chemical oxygen demand was moderately positively correlated with ammonia nitrogen and BOD.

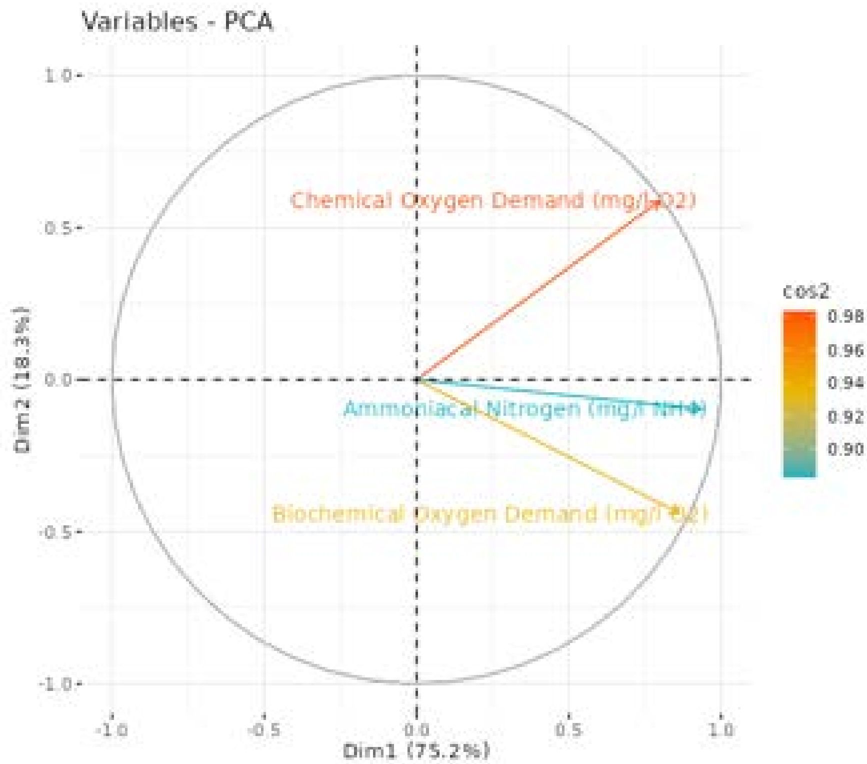

Figure A.3.4a Principal component analysis of chemical oxygen demand, ammoniacal nitrogen, and biochemical oxygen demand.

We performed a number of PCAs comparing these three biological parameters to total coliforms, fecal coliforms, dissolved oxygen, nitrite, and nitrate. They were not correlated with any of these parameters, however.

Principal Component Analysis (PCA) was conducted across the water quality parameters of highest concern, among other parameters (Figure A.3.4b). When considering the top 15 contaminants of concern across water and sediment (Figure A.3.4c), the top two dimensions explain little variance in the data (sum 37%), yet relationships between contaminants can still be seen. Chlorides and sodium are correlated to one another, yet negatively correlated with mercury. Sodium and chlorides are likely initially complexed together as a geogenic salt and dissociate together in the water column. Iron, chromium, cadmium, nickel, zinc, and copper are all correlated, but have little-to-no described correlation with the sodium/chloride pairing or with mercury.

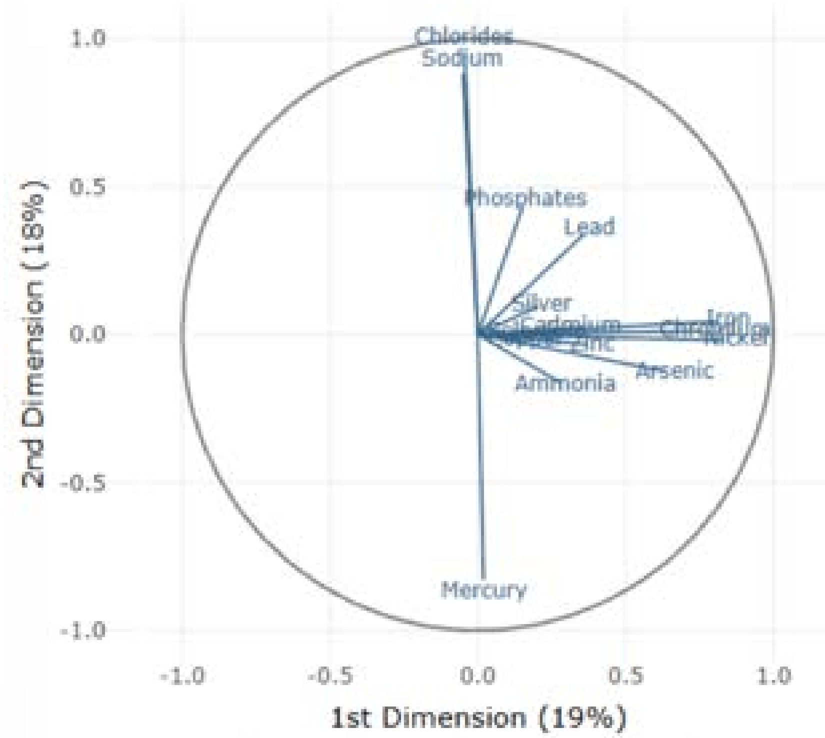

Figure A.3.4b PCA for parameters of highest concern mapped onto a unit circle. Total explained variance is low, but increases when fewer parameters are selected for analysis.

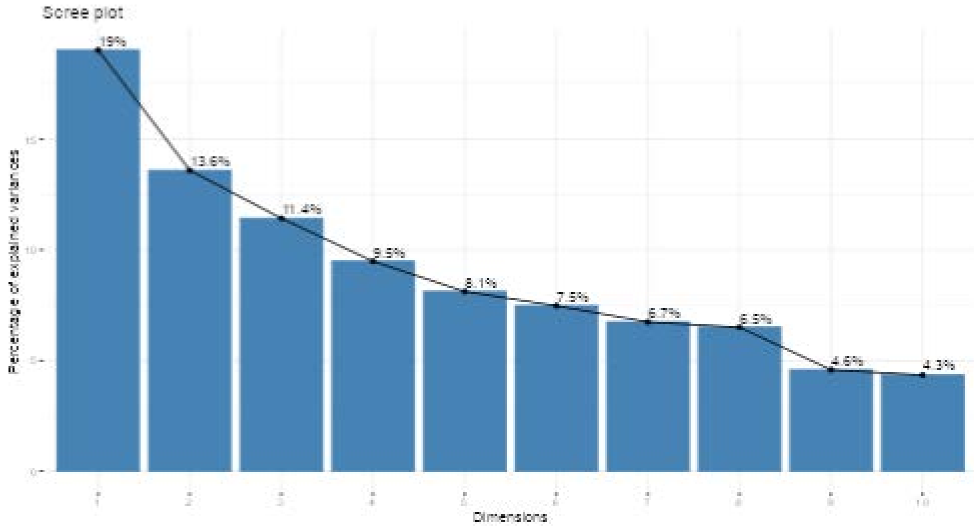

Figure A.3.4c Variance plot of the principal component analysis. The bars show how much variance is explained by each component. Many dimensions are required to describe a significant portion of variation within the data for the selected analytes.

Splitting the analysis by matrix helps increase the total variance explained, and accounts for differences in transport mechanism (Figures A.3.4d). Two sets of analytes were identified in reported water concentrations as being correlated with one another: (1) Nickel, iron, and chromium, and (2) cadmium, selenium, phosphates, ammonia, mercury, and copper. The two sets were not correlated with each other, as seen by the overall angle between the two sets. When sampling for these contaminants in the future, we expect these correlations to continue to be reported.

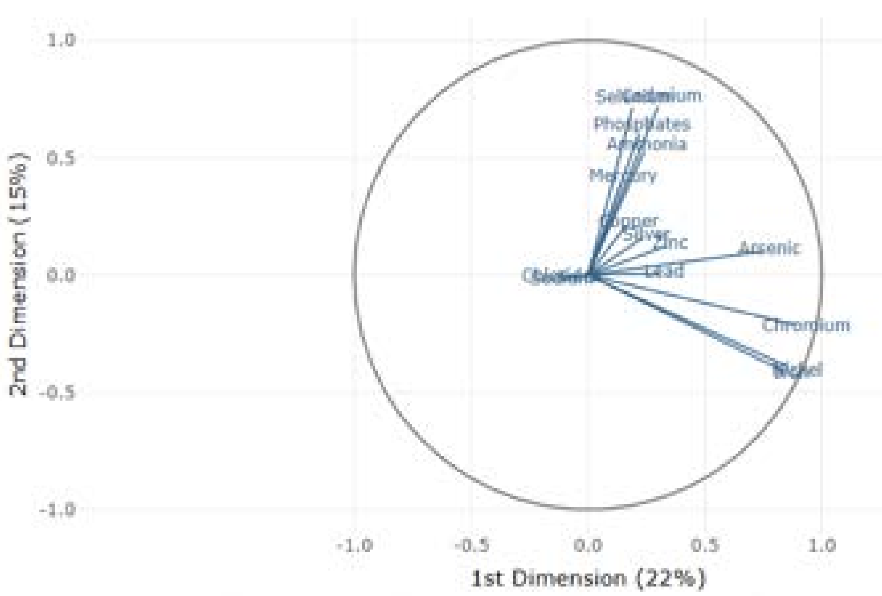

Figure A.3.4d Principal component analysis of top contaminants in water samples.

Sediment samples provided a different set of correlations (Figure A.3.4e). Chromium had a negative correlation with cadmium and silver; iron was negatively correlated to selenium; mercury, nickel and sodium were negatively correlated.

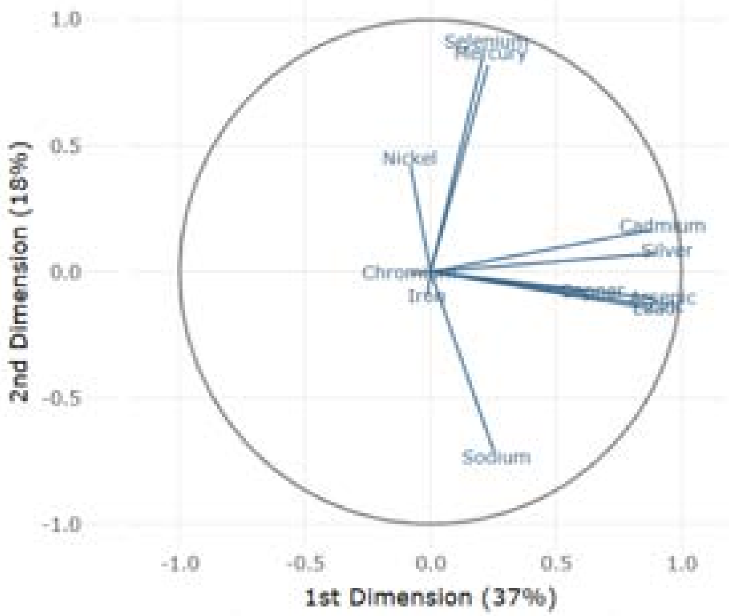

Figure A.3.4e Principal component analysis of heavy metal concentrations in sediment samples. Metals included: copper, cadmium, mercury, chromium, lead, arsenic, nickel, zinc.

The correlations served to discover subsets of the contaminants that have similarities, but this warrants further exploration. These could be due to these metals having similar binding or transport mechanisms, or they could be due to certain metals coming from similar sources or contamination events. Understanding these relationships may assist in narrowing down the sources and causes of contamination in the Pilcomayo River.

A.3.5 Sensitivity Analysis

Parameter and Station Rankings

Single-parameter sensitivity analyses revealed that station rankings are relatively robust to individual parameter choices, but parameter rankings show greater sensitivity to model configuration. This appendix details sensitivity analyses across HQ aggregation methods, temporal selection approaches, and their interactions.

Sensitivity of Station Rankings to HQ Aggregation

HQ aggregation methods were first tested to determine how its selection would affect station rankings (See Figure 4.3.5a). Spearman's correlations between station ranks were high across all aggregation methods (mean, median, maximum, 95th percentile: $\rho > 0.7$, $p < 0.01$), indicating similar relative ordering under most configurations. Comparing the set of top five priority stations by HQ method, the majority of methods (max, mean, 95th percentile) ranked a similar stations set, while the median method had limited alignment with the others (Figure A.3.5a). Mean HQ skewed toward higher values and yielded rankings similar to maximum and 95th percentile methods. These single-parameter tests do not capture interaction effects between temporal selection and HQ aggregation; uncertainty should be evaluated jointly across both dimensions.

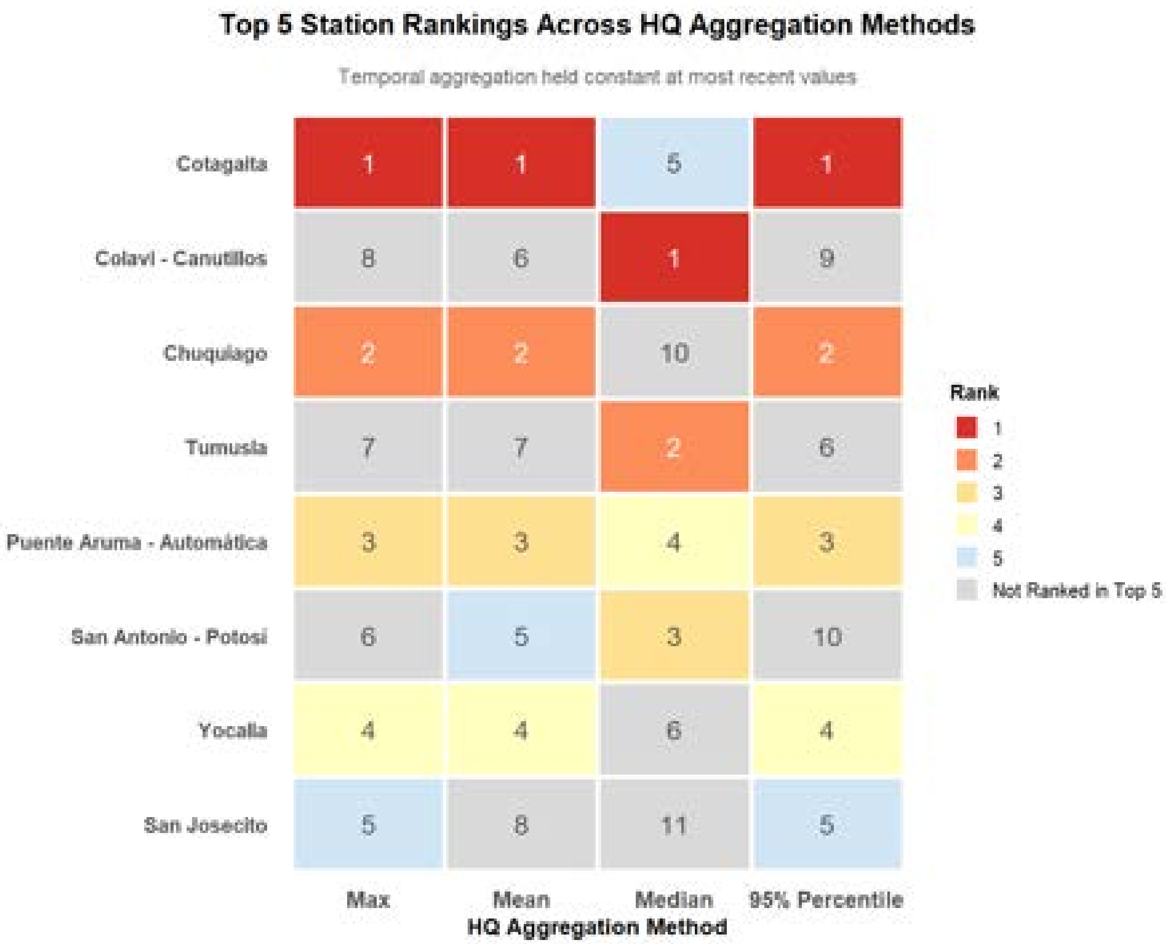

Figure A.3.5a Station rankings shown for each model configuration. Rankings are shown (1=highest priority, 5=lower priority) for each station. Gray boxes indicate the station was not ranked within the top five stations in that model configuration. Other than median HQ aggregation, models are generally aligned until the fourth priority station.

Sensitivity of Station Rankings to Temporal Selection

Varying temporal aggregation approaches were tested while holding HQ aggregation constant (95th percentile). Temporal methods included: (1) mean across all dates, (2) max across all dates, (3) most recent date only, and (4) recency-weighted aggregation. Temporal methods did not generate widely different priority station sets when using 95th percentile HQ aggregation (Figure A.3.6c). The top two stations were identical across all temporal configurations. Any configuration relying on recent-only data tended to have lower correlations than models using longer temporal windows (Figure A.3.5b). This suggests that recent-only models emphasize emerging conditions at the expense of longer-term patterns and should be selected deliberately based on project goals. Determining what is considered recent is discussed in the main text (*See 4.9.1 Water and Sediment Parameters*).

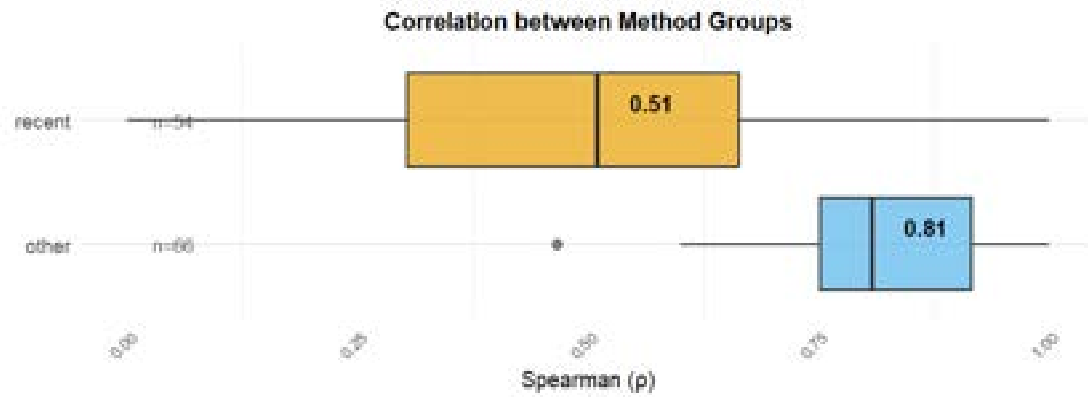

Figure A.3.5b Comparison of Spearman Correlation values between model runs. Models using the recent temporal aggregation method vary in correlation. Selecting a model using recency would provide drastically varied results, and should be considered based on merit rather than similarity.

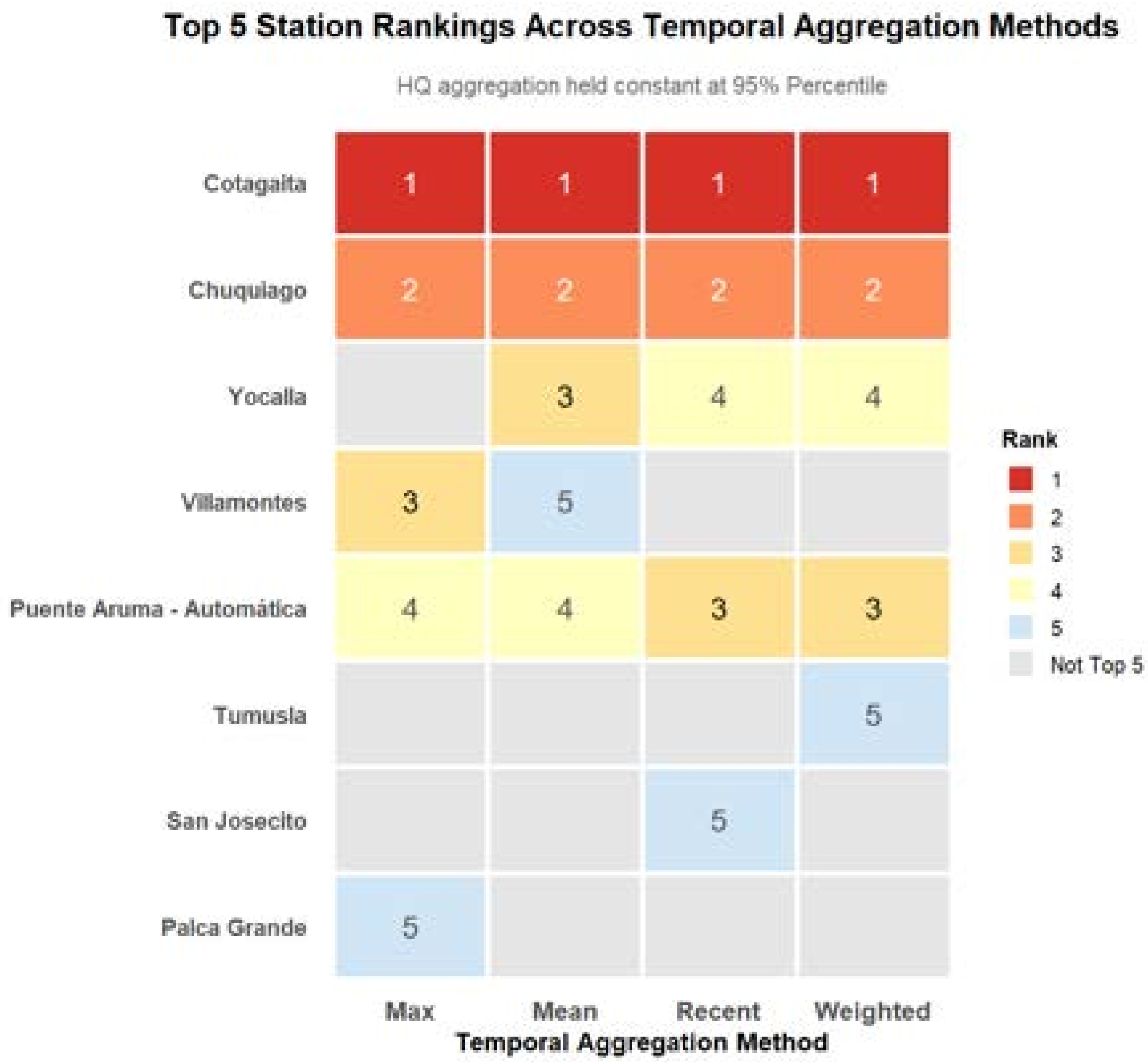

Figure A.5.6c Top five stations as recommended by various model configurations. Grey boxes indicate the station was not ranked within the top five for that configuration. The top two stations were aligned for all models using 95th percentile HQ aggregation.

Interaction Effects: Parameter Prioritization in Sediment Data
To assess whether interaction effects between HQ and temporal aggregation are more pronounced for parameter rankings (versus station rankings), a full factorial sensitivity analysis using sediment data was conducted. Spearman correlations revealed that models filtering for recent data were weakly correlated with other configurations (Figure A.3.5d). Several recent-only models showed strong negative correlations with historical models, suggesting shifts in dominant contaminants over time. Models filtering by sample year (recent and weighted) showed statistically non-significant correlations ($p > 0.05$), indicating temporal selection fundamentally alters which contaminants are prioritized.

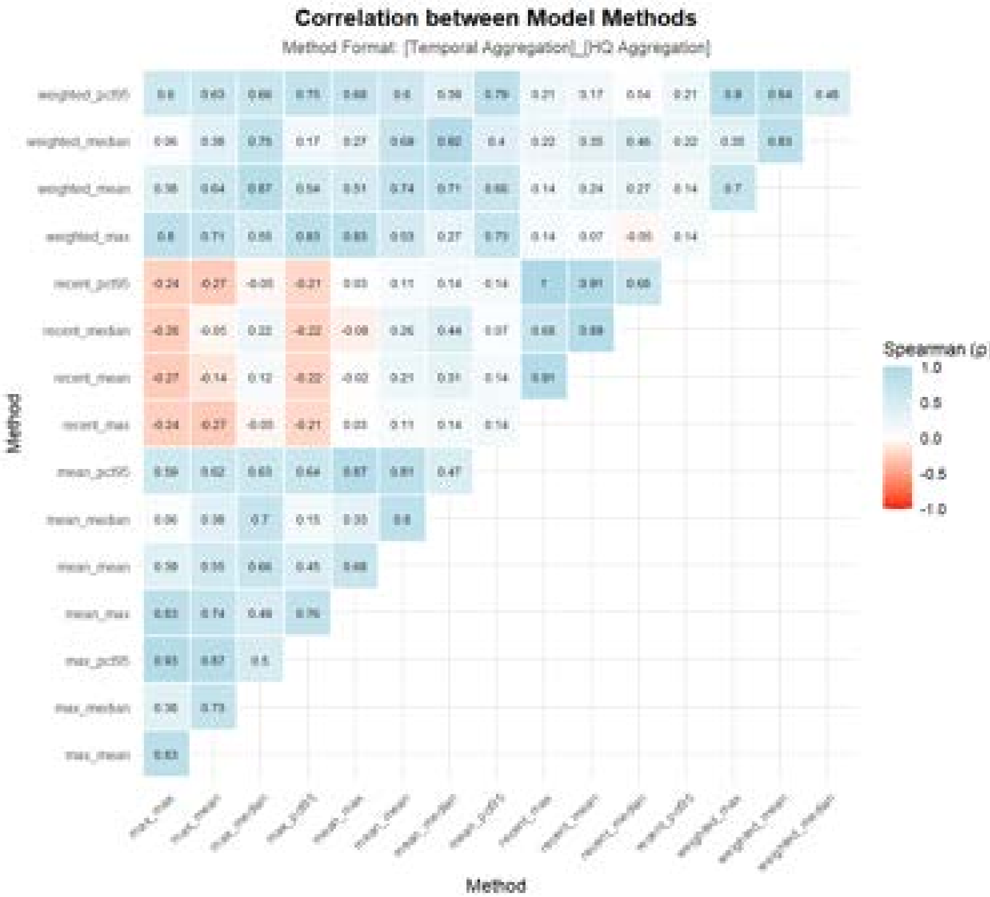

Figure A.3.5d Spearman correlations for station rankings for sediment quality data using various model parameters are shown. Models using the most recent sampling date are least correlated with other model outputs Weighted models utilized similar methods as the most recent option.

Station priority lists for sediment showed broad agreement on the top three sites (San Antonio - Potosí , Tarapaya, El Puente), but diverged substantially beyond these (Figure A.3.5e). Models using only recent data produced ties among several stations that were not prioritized by other configurations, reflecting emerging contamination patterns not captured in longer temporal windows.

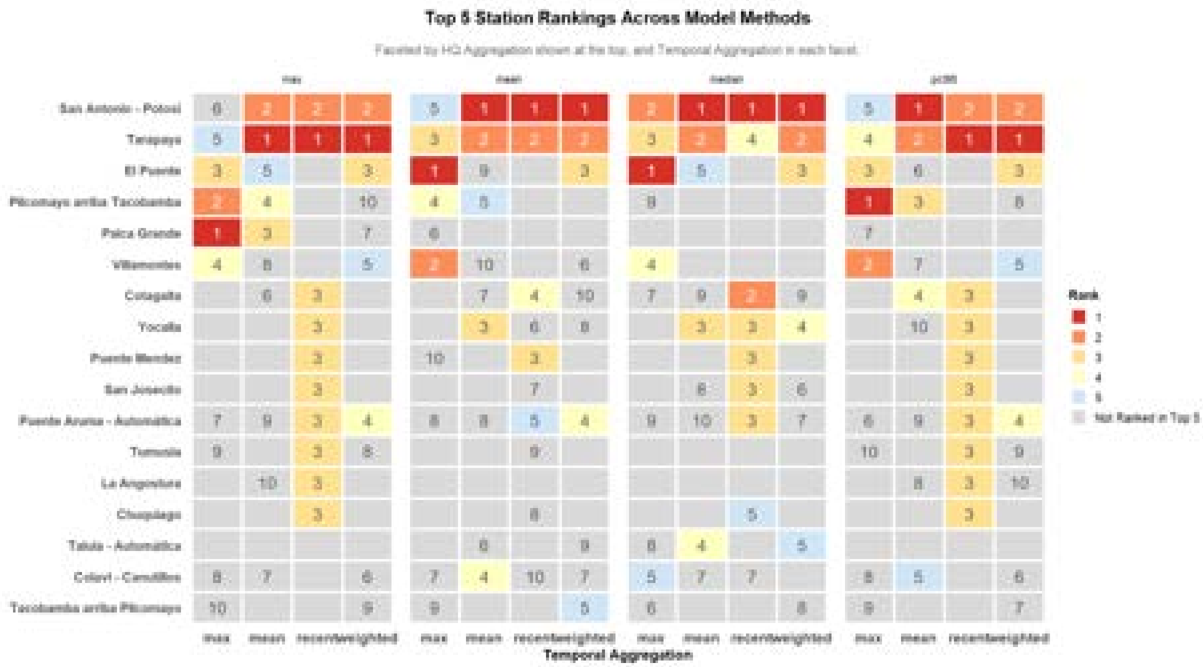

Figure A.3.5e Priority station rankings for sediment-based hazards. Only stations which were ranked within the top ten for at least one model are plotted. Models generally agree on the top two stations of concern (San Antonio - Potosí and Tarapaya). Eight other stations were mainly prioritized under the recency model, where only the most recent samples were considered.

Parameter Prioritization in Water Quality Data

We repeated the interaction analysis for water quality contaminants. Recent-median configurations showed the weakest correlations with other models, yet median-HQ models (excluding the recent-median configuration) did not demonstrate significant difference compared to other models (Figure A.3.5f). Selecting the recent-median model configuration would provide drastically varied results, and should be considered based on merit rather than similarity.

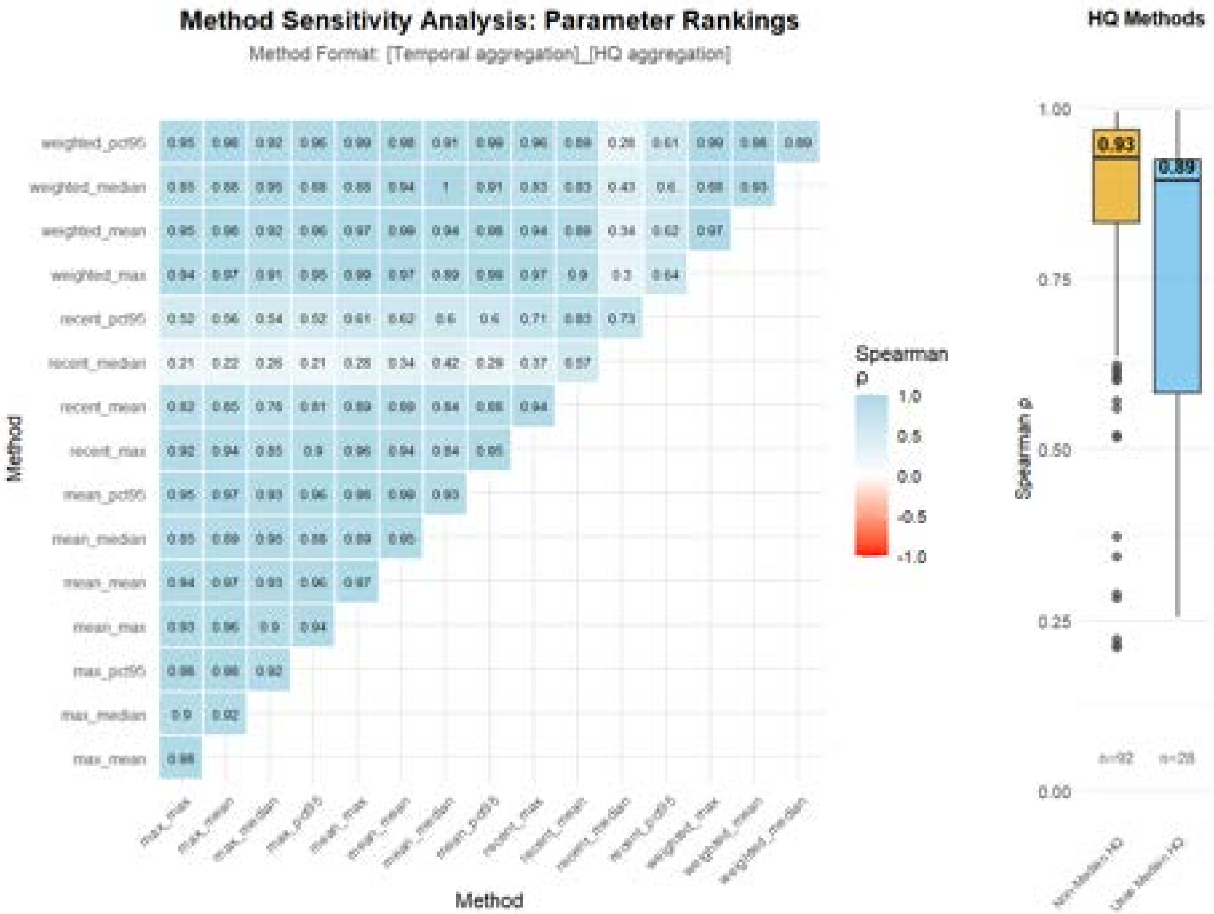

Figure A.3.5f (Left) Spearman correlation analysis plotted for each method pairing using water data. All combinations of temporal and HQ aggregation methods were compared. Recent years paired with median HQ values show weak correlations compared to other methodology pairings. (Right) Boxplot comparing Spearman's correlation in models using median HQ and others. Non-median HQ models are more similarly related and provide more consistent parameter selections when ranking all parameters than the wide correlation values provided by the median-HQ method.

Robustness to Top-N Truncation

We evaluated whether truncating results to the top ten set of parameters, rather than all parameters, changed the correlation outcomes. Differences between full-rank and top-10 analyses were minor, indicating that limiting output to top-priority contaminants does not materially alter the main conclusions. Spearman correlations computed using only these short priority lists (Figure A.3.5g) confirmed that the recent-95th percentile configuration had the weakest relationships with other models (ρ ranged between -0.214 & -0.057, $\rho_{median} = -0.095$). This configuration remained particularly weakly correlated with recent-median models, reinforcing that the choice of aggregation method, rather than list length, fundamentally shapes which contaminants are prioritized and must be selected deliberately based on project goals.

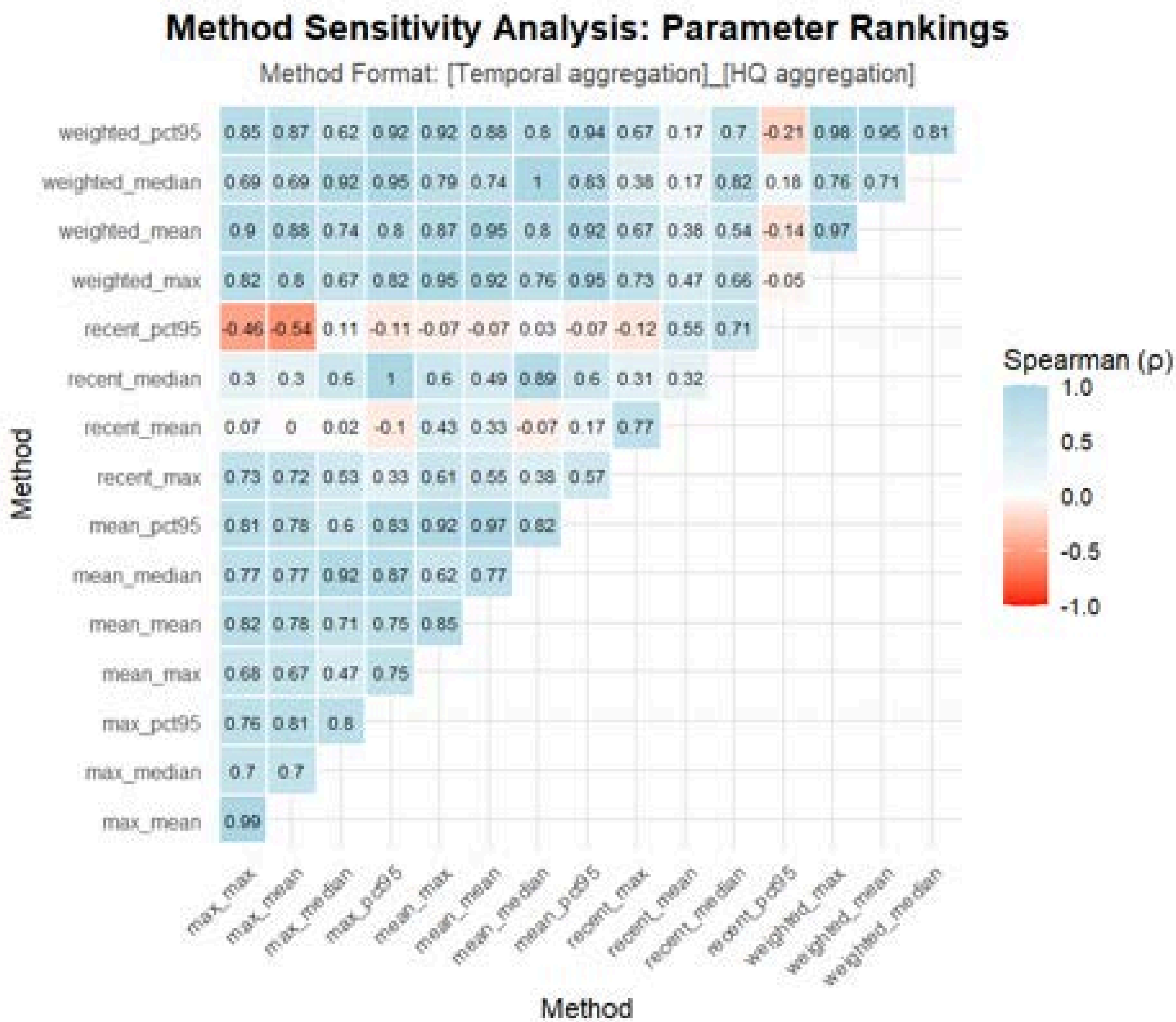

Figure A.3.5g Spearman correlations for water quality parameter models when only top 10 parameters are returned. Correlation results are similar to analysis where all parameters are returned, thus it was not found to make a meaningful impact on the final outcome of our analysis.

Sediment Parameter Prioritization

Correlation analyses for sediment contaminants showed no single configuration was uniformly uncorrelated with all others (Figure A.3.5h). The top five contaminants were generally stable across models, though some (e.g., silver, copper) ranked highly across all configurations while others (iron, cadmium, ammonia) appeared in top-five lists only under specific methods (Figure A.3.5i). No contaminant ranked in the top five for every configuration.

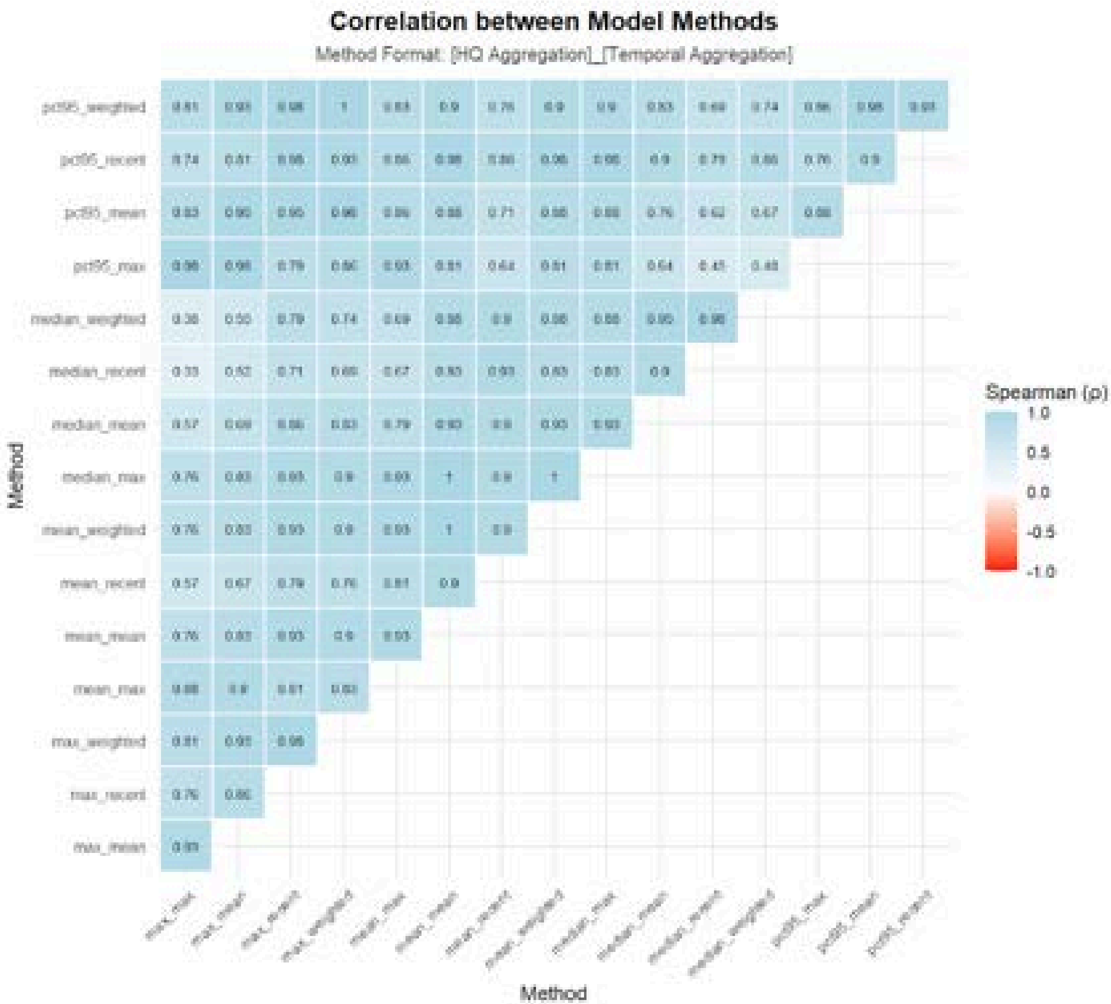

Figure A.3.5h Spearman correlations for sediment parameter models across all method combinations. No single configuration shows uniform lack of correlation with all others, indicating general stability in parameter prioritization.

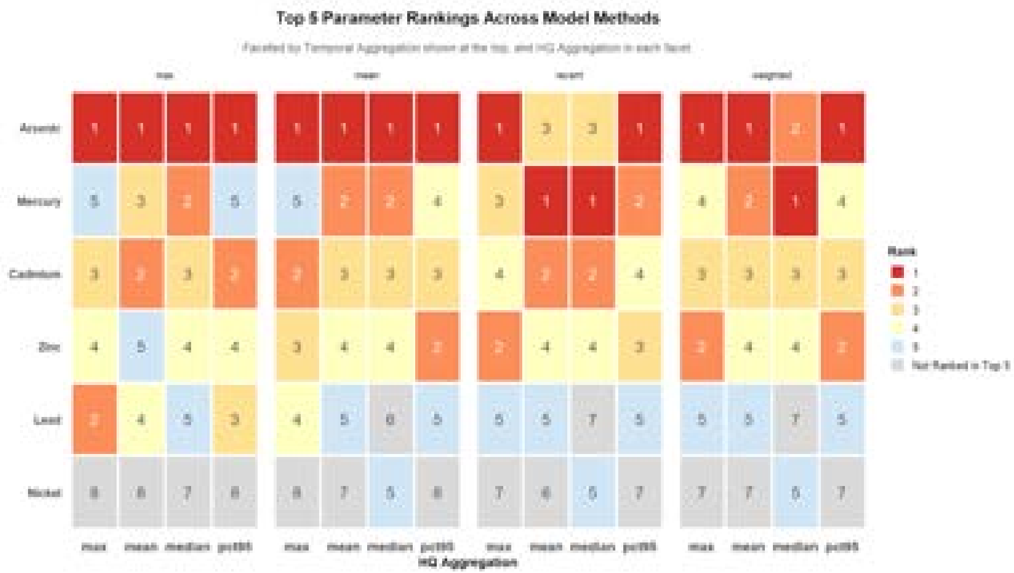

Figure A.3.5i Highest priority parameters for sediment data ranked using various model configurations. Tiles indicate the rank (1-5) of the parameter within the given model. Parameters are sorted top to bottom based on their top rank and frequency of high rankings. All parameters shown scored within the top 10 highest risk parameters in at least one model.

Validation of Selected Model Configuration (95th Percentile, 5-Year Window)

To confirm our selected configuration (95th percentile HQ, most recent 5 years) aligns with the broader model ensemble, we assessed correlations across both water and sediment data. For water quality contaminants, the 5-year, 95th percentile model was strongly correlated with most other configurations (Figure A.3.5j). Median models remained outliers, but the selected model showed robust agreement with max and mean approaches, validating its representativeness. Correlations remained high ($\rho > 0.8$) for non-median methods across all window sizes, but median HQ methods showed wider variation and lower correlation (Figure A.3.5k).

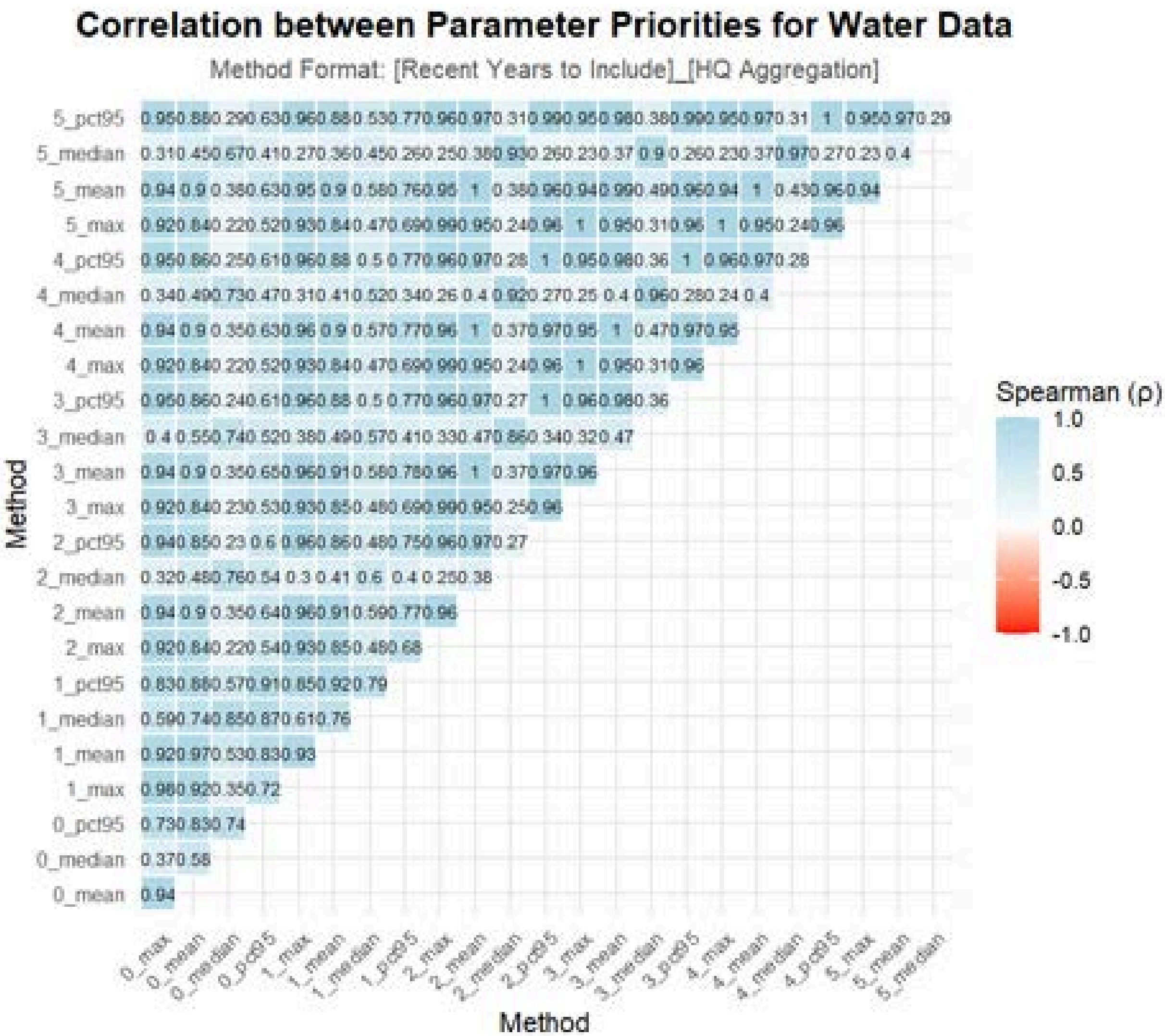

Figure A.3.5j Spearman correlations for models using water data to determine priority lists for contaminants. Median models were least correlated, but 5-year models using 95th percentile HQs were strongly correlated with other models.

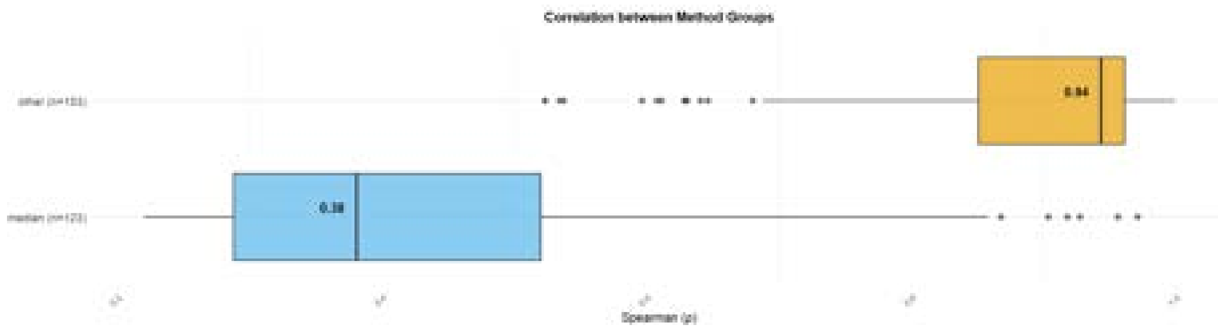

Figure A.3.5k Correlation ranges for models using median HQ aggregation versus those which do not. Not considering outliers, correlations are generally high ($\rho>0.8$) for models not using median, while median had a wider range of Spearman correlation values (IQR: $0.25<\rho<0.53$).

Air Particulate Concentration Modeling

Pollutant Species

The model was tested using different molecular weights and densities of the modelled pollutant species. Malachite ($Cu_2CO_3(OH)_2$) and Zinc Oxide (ZnO) – other oxidized forms of heavy metal minerals likely to be found in tailings – were used as the basis for HYSPLIT model inputs. Both have a lighter molecular weight and are less dense than Angelsite. These models had marginally smaller plumes of pollutant concentrations compared to the base model (Anglesite). Then, the model was tested using two extreme scenarios: 1) molecular mass of 50 g/mol and density of 6.3 g/cm^3 (density of Anglesite), and 2) molecular weight of 303.3 g/mol (Anglesite) and density of 10 g/cm^3. The model using 50 g/mol had slightly smaller plumes than the base model. The model using 10 g/cm^3 had slightly larger plumes of pollutant concentration. Overall, the effect of changing the physical properties of the pollutant were marginal.

The "shape" of pollutant particles was also tested. In the base model, shape was set to 1, which assumes that each pollutant particle is a perfect sphere. We tested the models using 0.75 and 0.5 as the shape parameter. We found that decreasing the shape parameter caused the plumes of pollutant concentration to increase in size marginally. Concentrations were unaffected and spatial trends were only marginally impacted.

Layer Height and Starting Height

We tested the effect of changing the layer height to 5m and 10m above ground level. Changing the layer height had noticeable effects on the shape and of pollutant plumes. This was what we expected since wind conditions vary at different heights and gravity causes pollutants to transport and accumulate differently at different heights. We also tested changing the starting height from 4m (height estimated for Los Cumpas) to 1m. There was some shift in the shape of plumes, but no change to the concentrations.

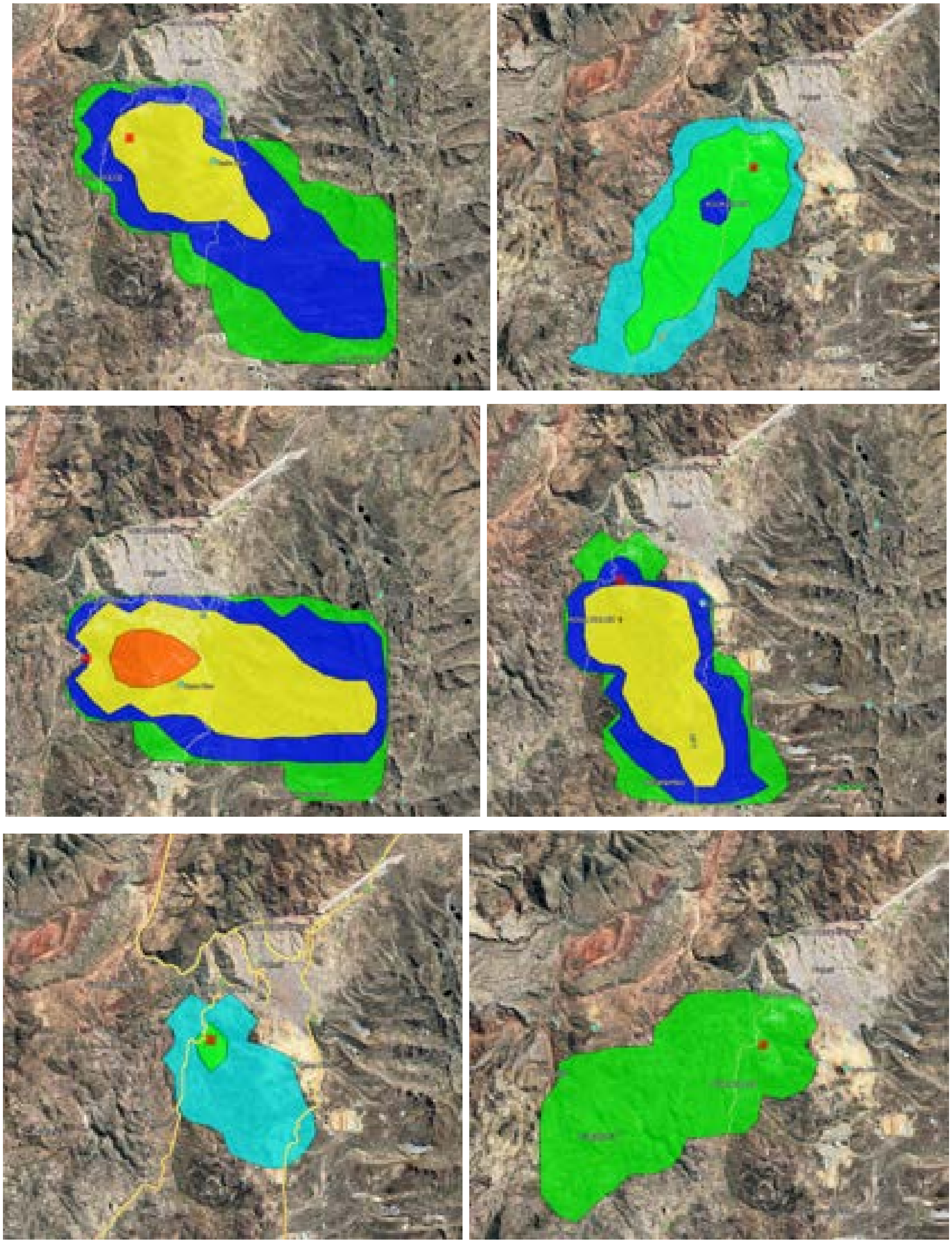

Figure A.3.5l HYSPLIT concentration outputs for the six days selected for modeling. Each image shows the model produced for 6:00 AM. Top left is May 5. Top right is June 3. Middle left is July 4. Middle right is July 5. Bottom left is August 10. Bottom right is September 8. Yellow = 0.001 g/m^3. Dark Blue = 1E-4 g/m^3. Green = 1E-5 g/m^3. Light Blue = 1E-6 g/m^3. Note that the largest plume of 0.01 g/m^3 concentrations by a significant degree was seen on July 4th.

Location

We examined the model outputs produced from each of the four point source locations. The shape, direction, and magnitude of concentration plumes were identical between the four when surface area was kept constant. The only difference between the four was the starting point of pollutant plumes. To verify the model and data, we tested a location far from Bolivia – Santa Barbara, California, United States. We expected that there would be differences in HYSPLIT outputs between Bolivia and California. When tested, we found that the concentration of plumes was very different and direction of plumes moderately different. This verifies that the GDAS data captures meteorological conditions at an appropriate resolution.

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